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(19) **United States**(12) **Patent Application Publication**
Parker et al.(10) **Pub. No.: US 2025/0276069 A1**(43) **Pub. Date: Sep. 4, 2025**(54) **HETEROBIFUNCTIONAL COMPOSITIONS
FOR TARGETED PROTEIN DEGRADATION
AND METHODS FOR THEIR USE****Related U.S. Application Data**

(60) Provisional application No. 63/156,593, filed on Mar. 4, 2021.

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(52) **U.S. Cl.**
CPC **A61K 47/55** (2017.08); **A61K 47/545**
(2017.08)(21) Appl. No.: **18/548,635**(22) PCT Filed: **Mar. 4, 2022**(86) PCT No.: **PCT/US22/18944**

§ 371 (c)(1),

(2) Date: **Sep. 1, 2023**(57) **ABSTRACT**

Compositions and methods for control and/or modification of endogenous protein degradation are described. The compositions are directed to heterobifunctional molecules having protein and enzyme system binding moieties linked together by an organic linker group. The compositions are selective for binding to certain endogenous proteins and function to recruit endogenous decomposition systems such as the polyubiquitin system for peptide cleavage and reas-simulation.

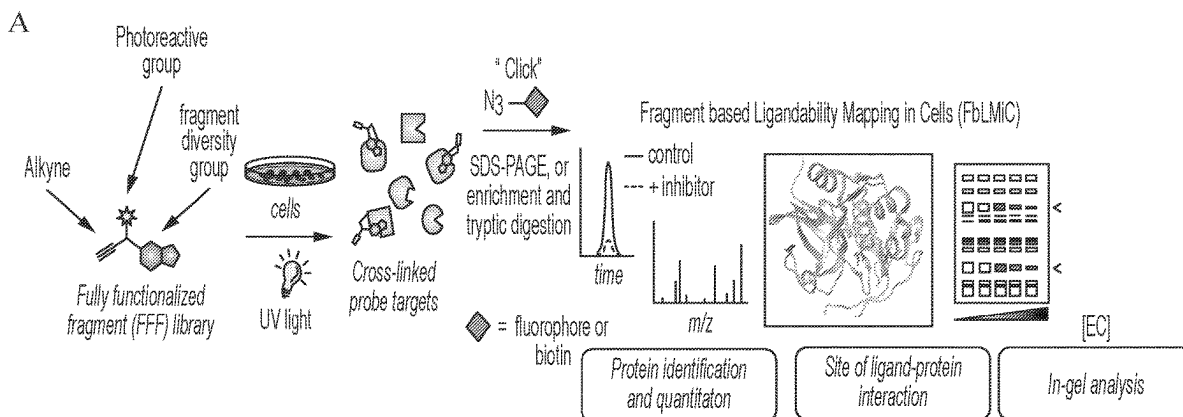


FIGURE 1A

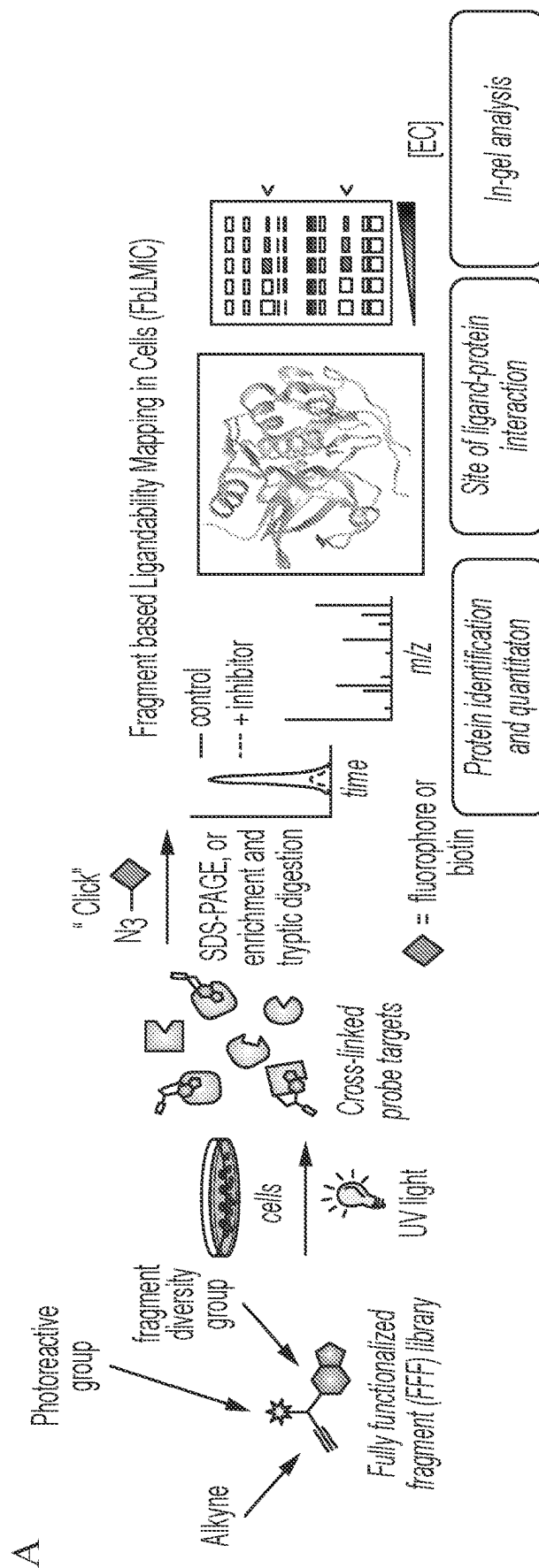
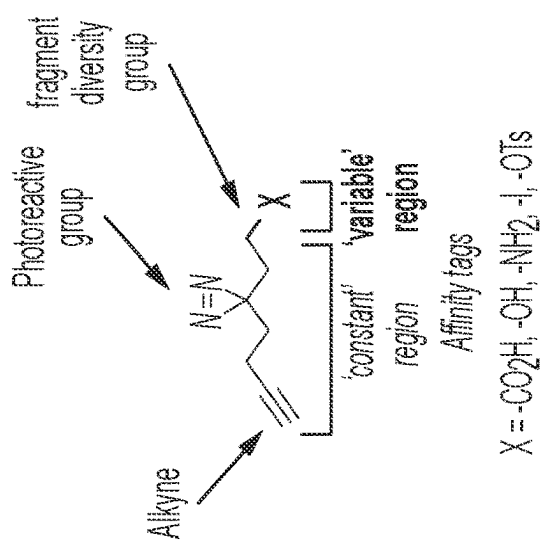
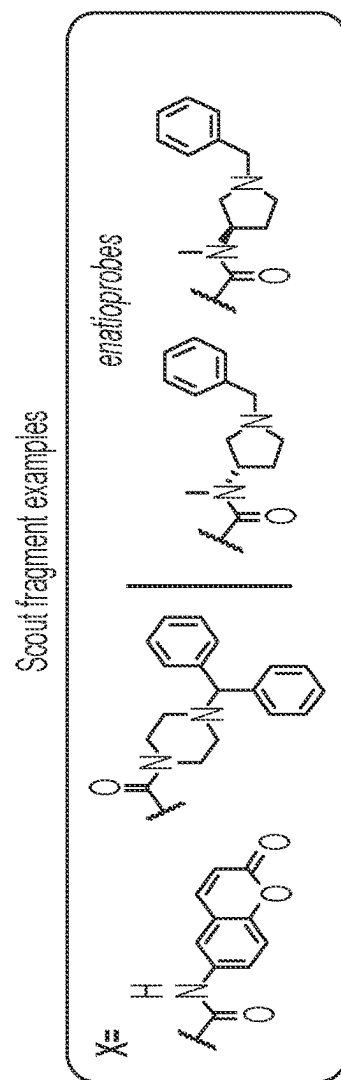


FIGURE 1B AND 1C

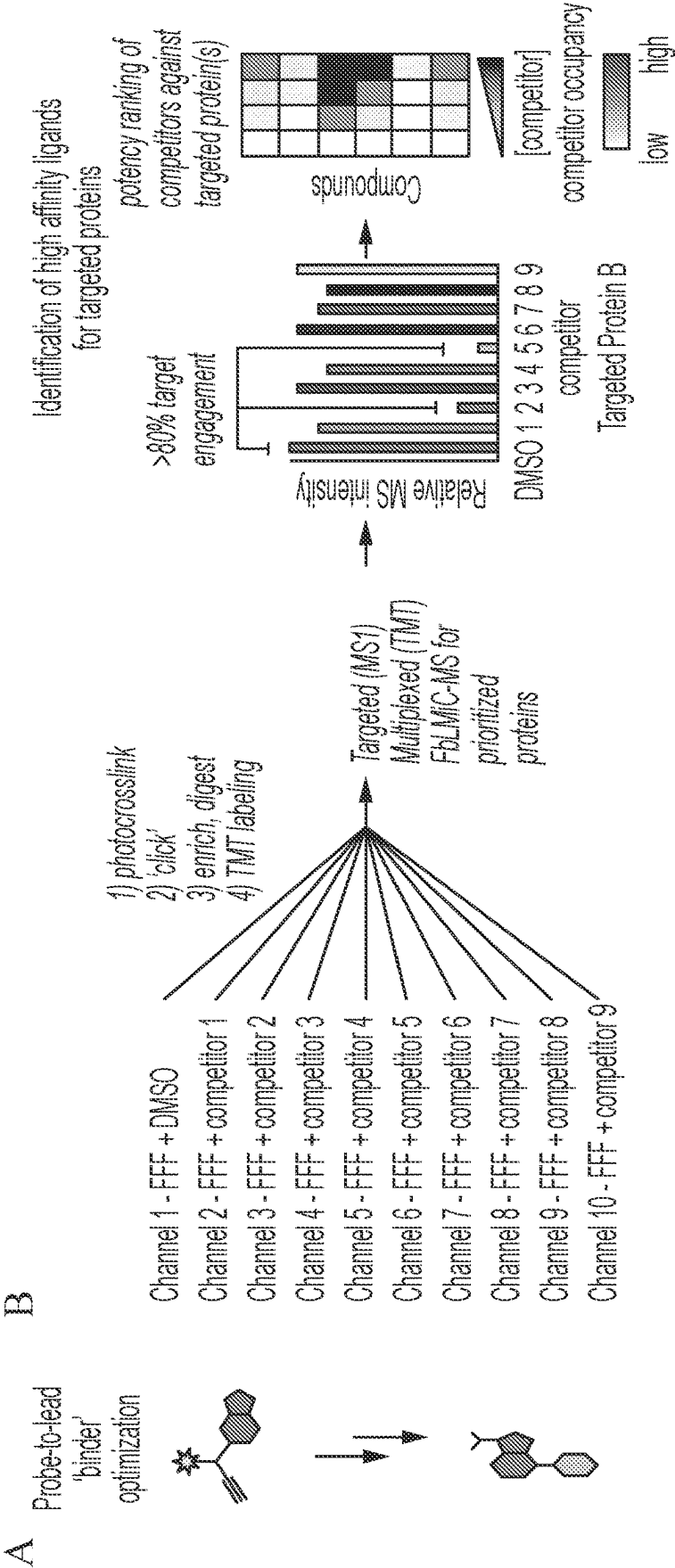
B



C



FIGURES 2A AND 2B



FIGURES 3A, 3B AND 3C

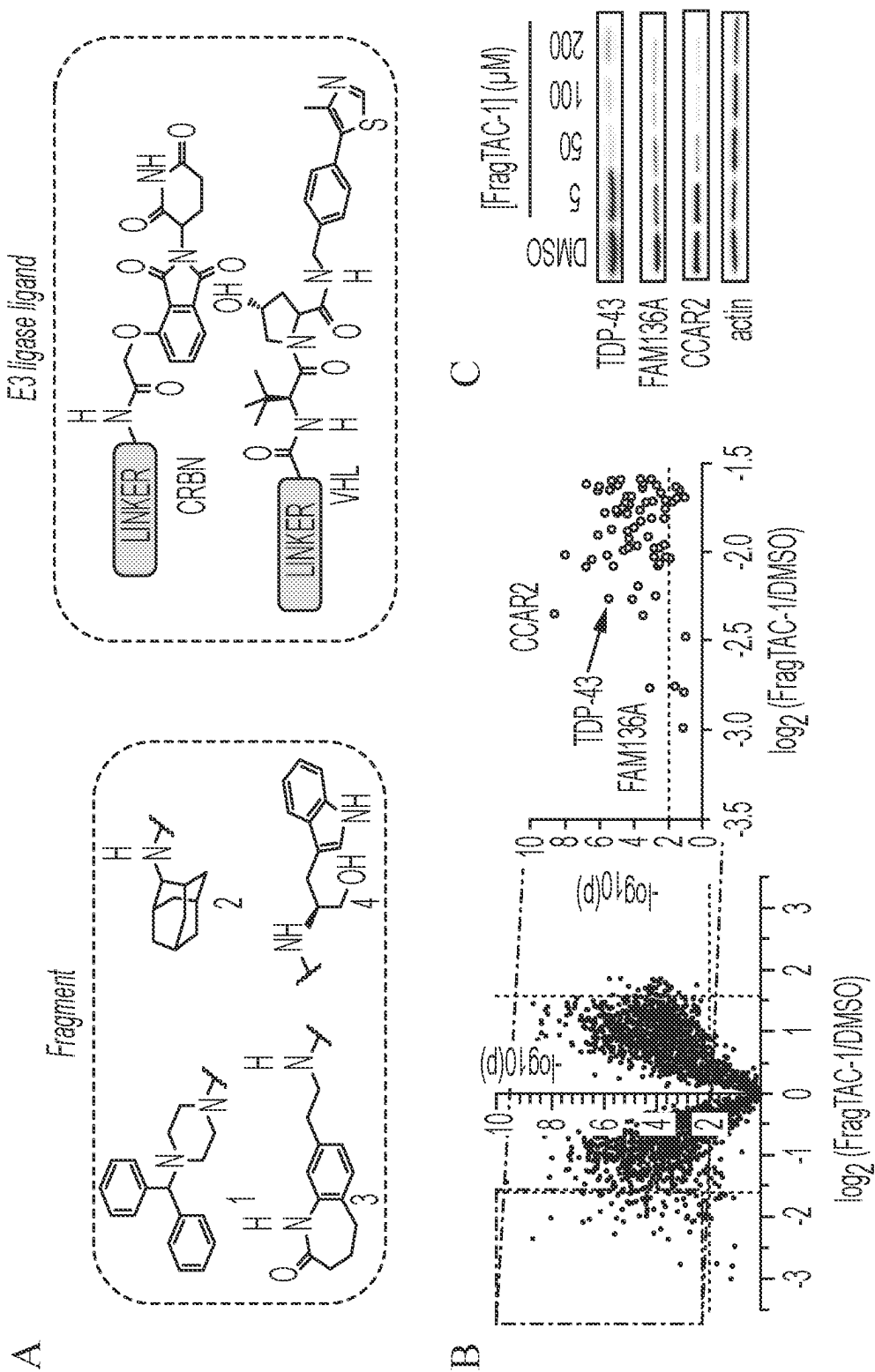
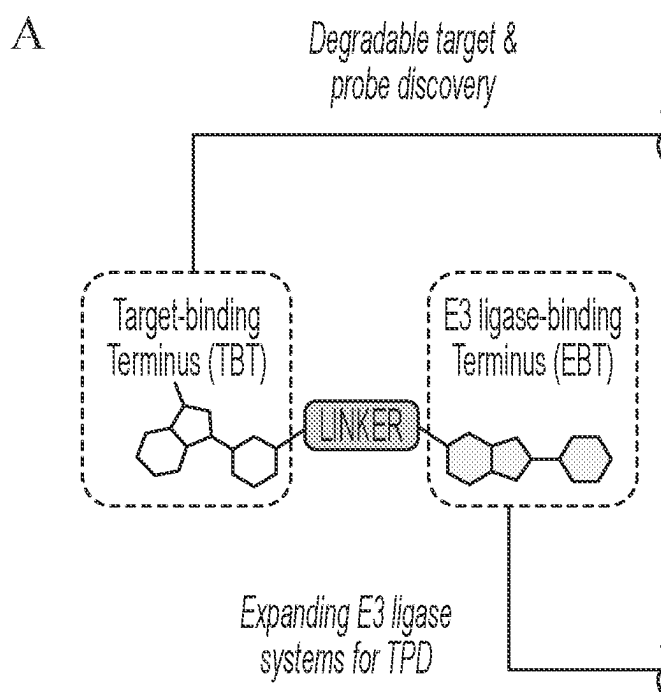
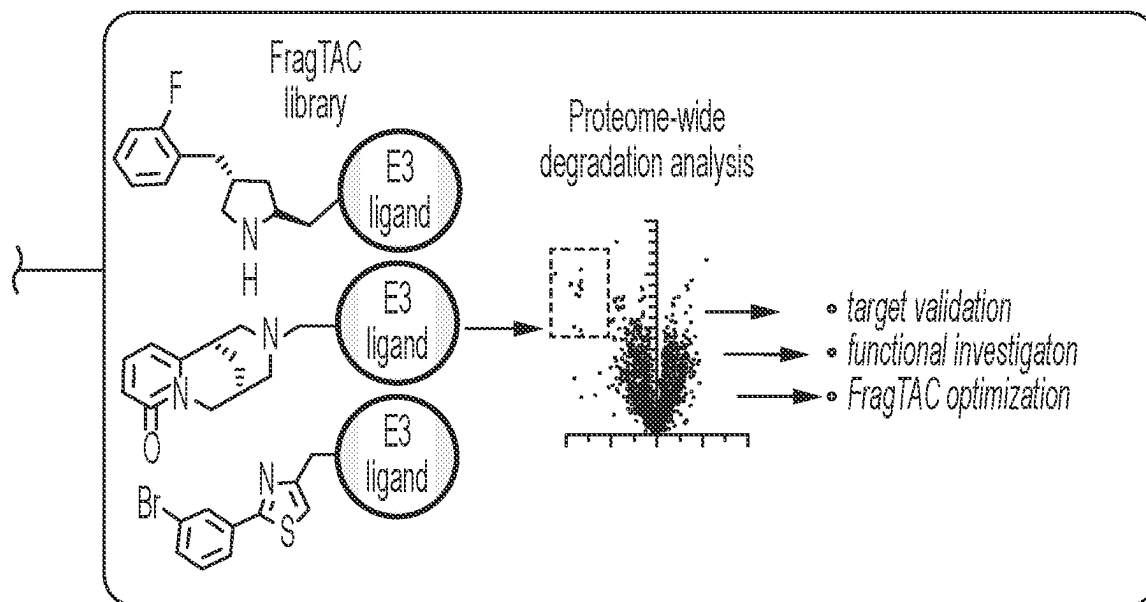


FIGURE 4A

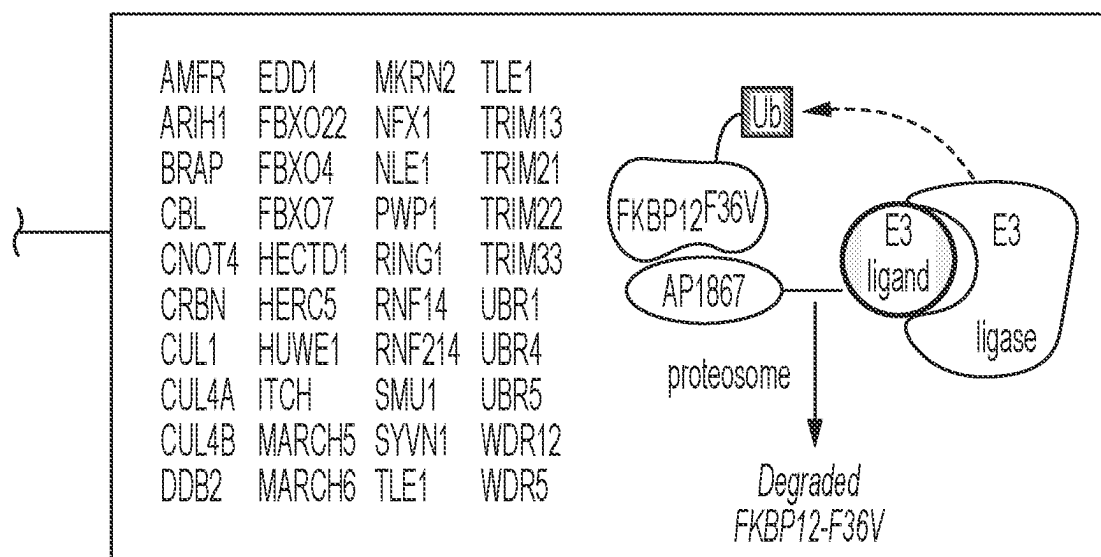


FIGURES 4B, 4C AND 4D

B



C



D

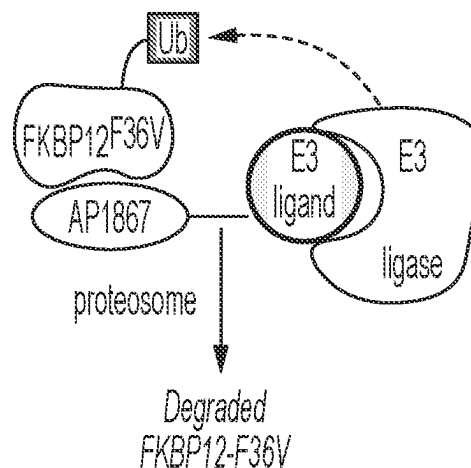


FIGURE 5

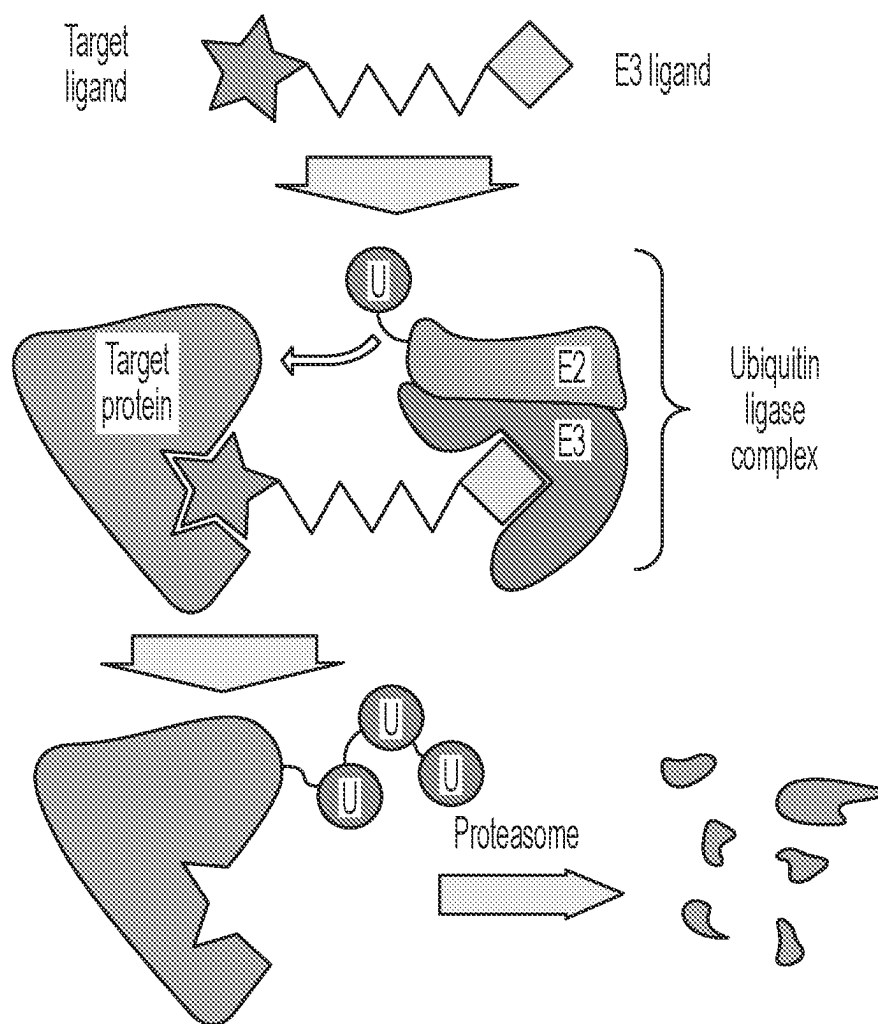


FIGURE 6

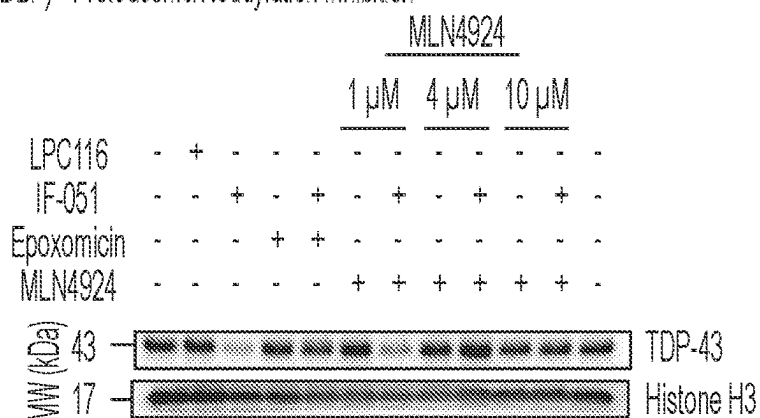
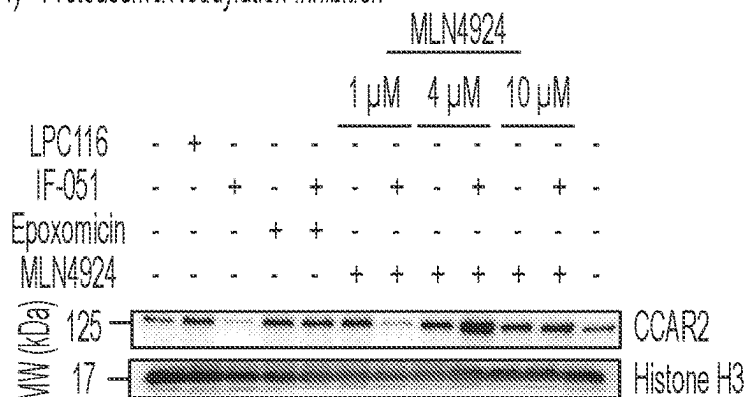
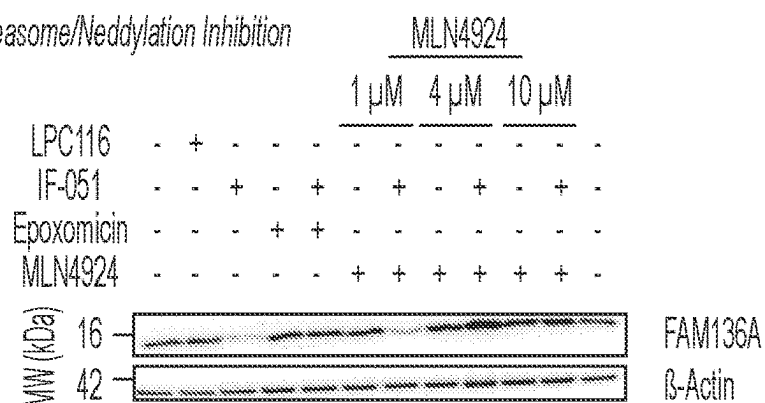
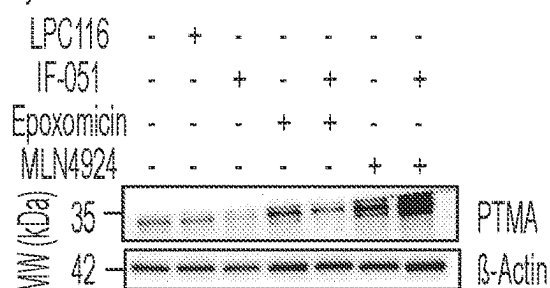
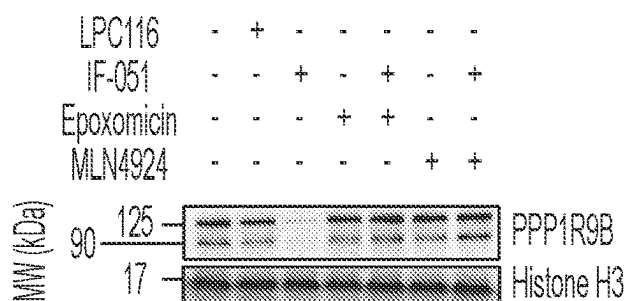
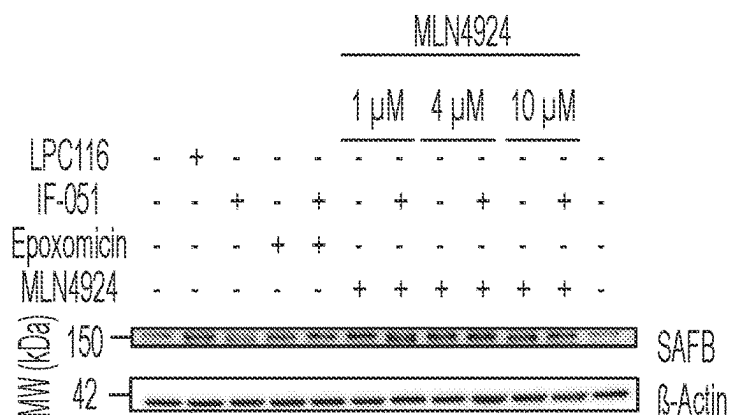
TDP-43 (TARDBP) - Proteasome/Neddylation Inhibition*CCAR2 (DBC1) - Proteasome/Neddylation Inhibition**FAM136A - Proteasome/Neddylation Inhibition**PTMA - Proteasome/Neddylation Inhibition*

FIGURE 6 (Cont.)

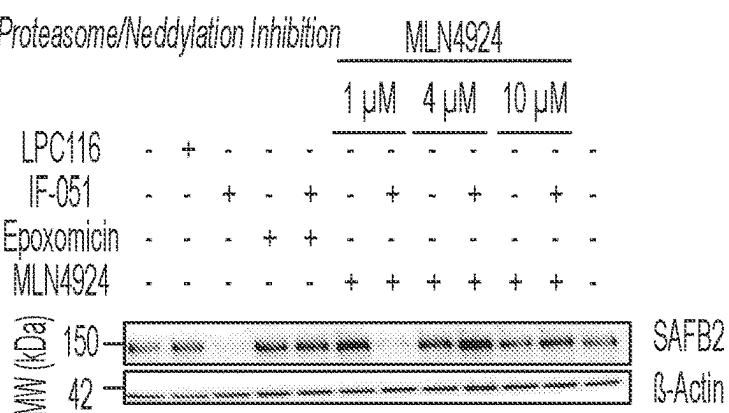
PPP1R9B (Spinophilin) - Proteasome/Neddylation Inhibition



SAFB - Proteasome/Neddylation Inhibition



SAFB2 - Proteasome/Neddylation Inhibition



MAVS - Proteasome/Neddylation Inhibition

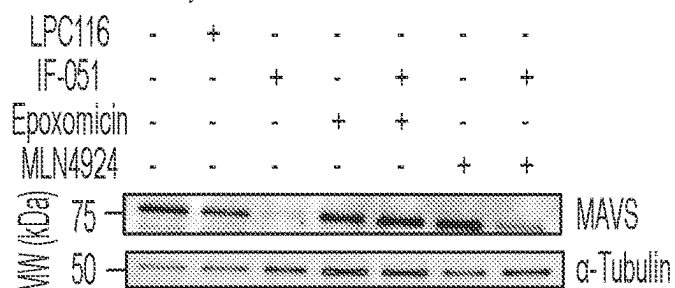


FIGURE 7

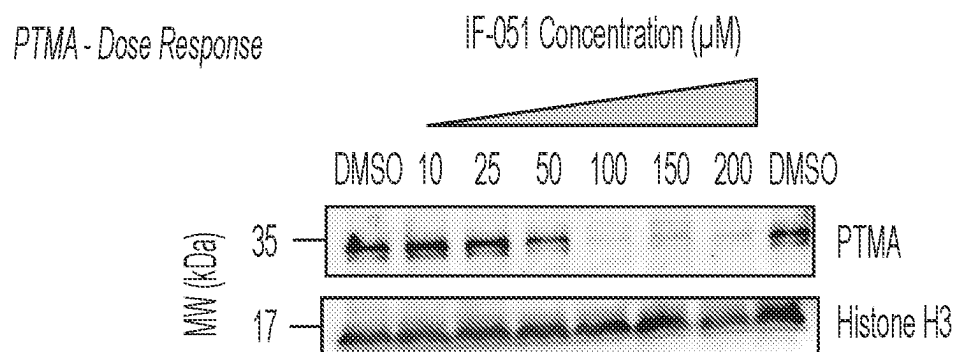
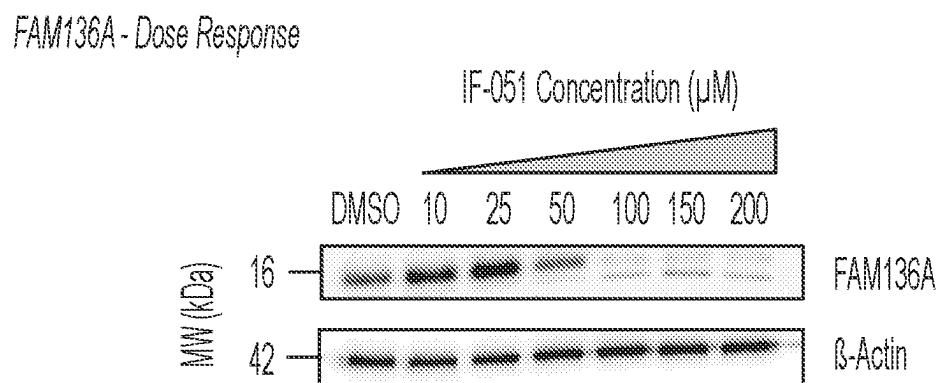
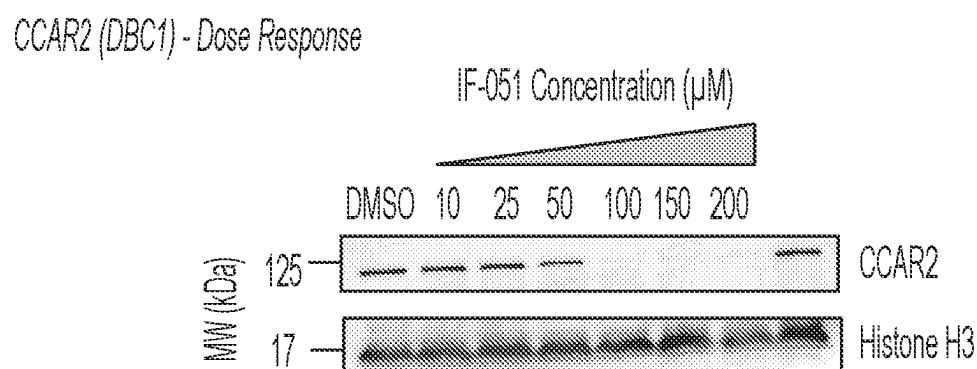
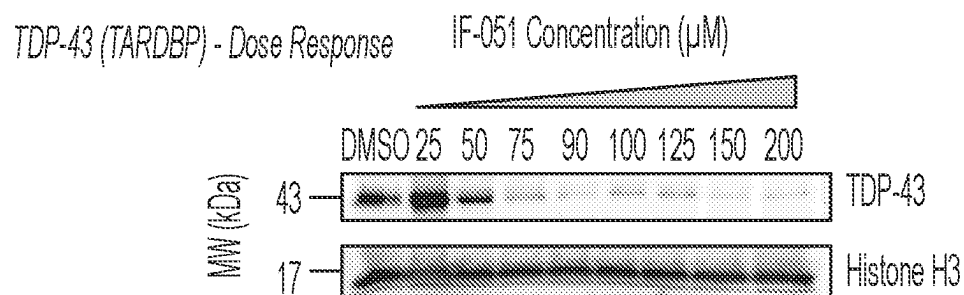


FIGURE 7 (Cont.)

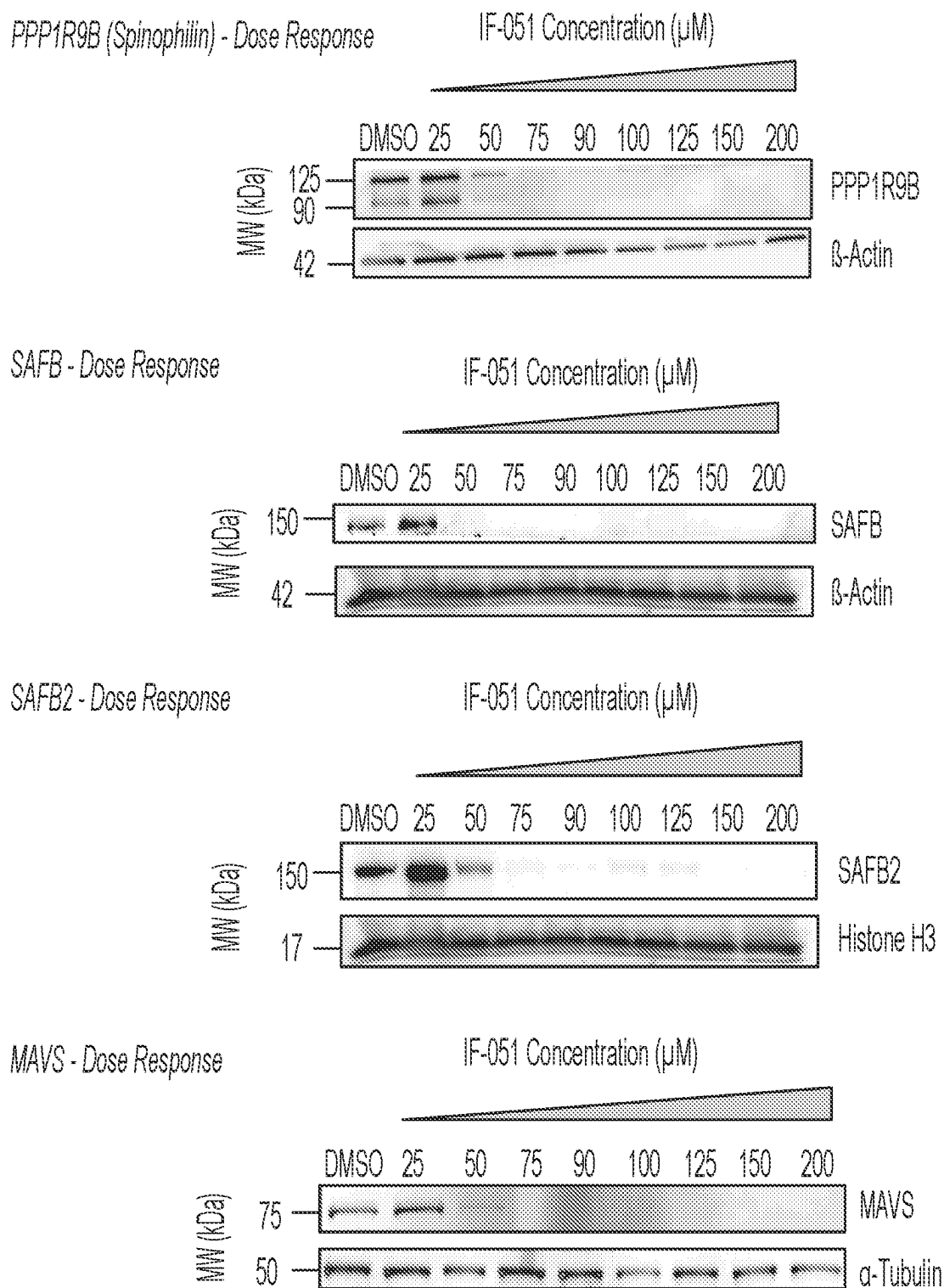
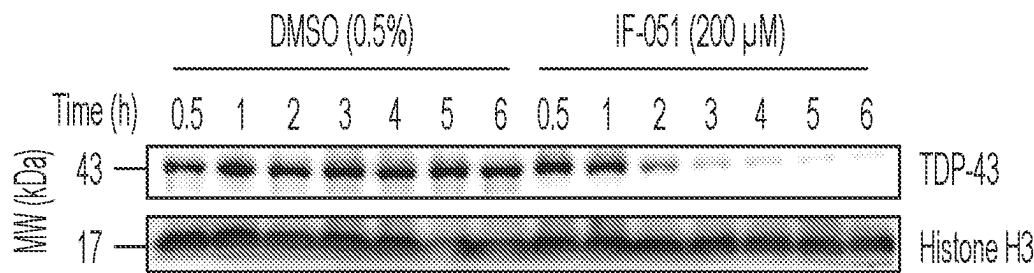
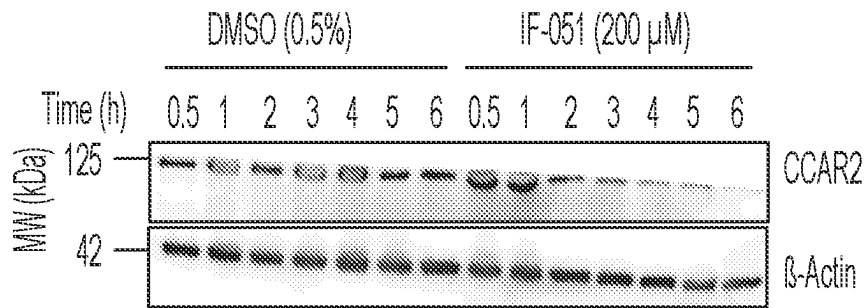


FIGURE 8

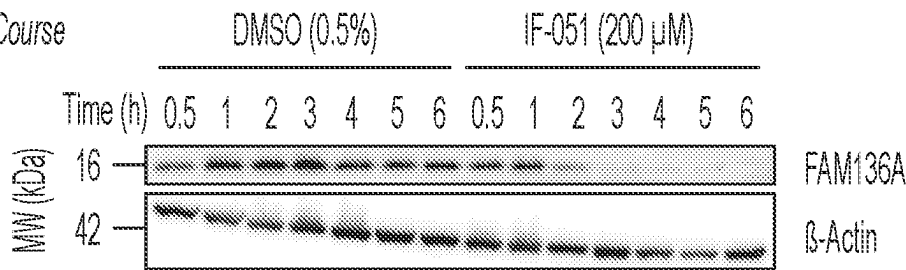
TDP-43 (TARDBP) - Time-Course



CCAR2 (DBC1) - Time-Course



FAM136A - Time-Course



PTMA - Time-Course

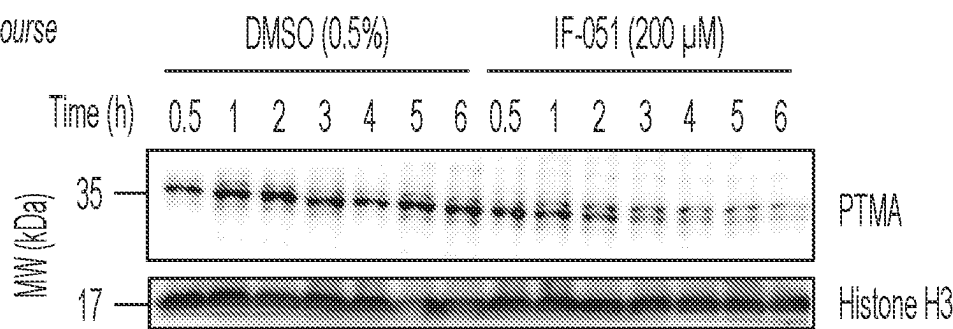
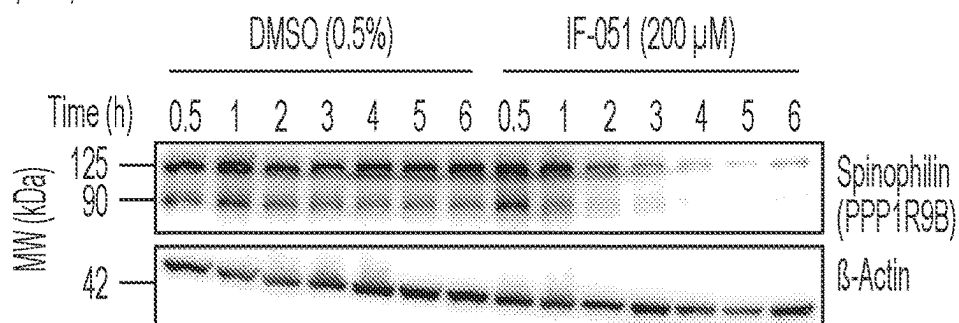
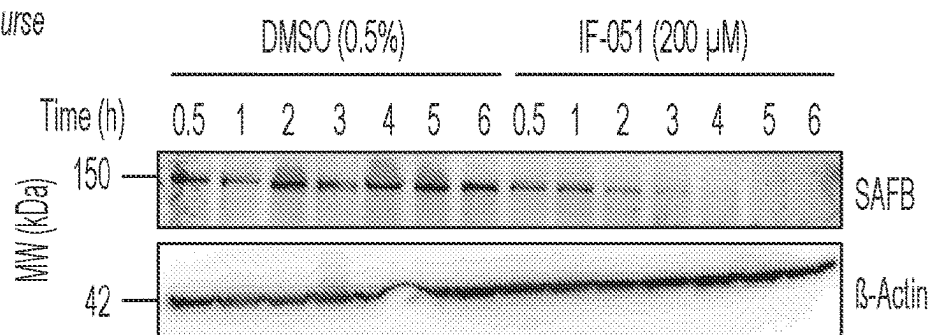


FIGURE 8 (Cont.)

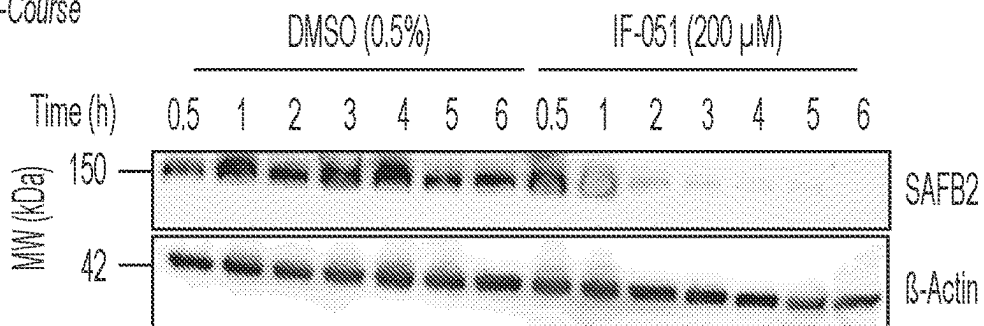
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SAFB - Time-Course



SAFB2 - Time-Course



MAVS - Time-Course

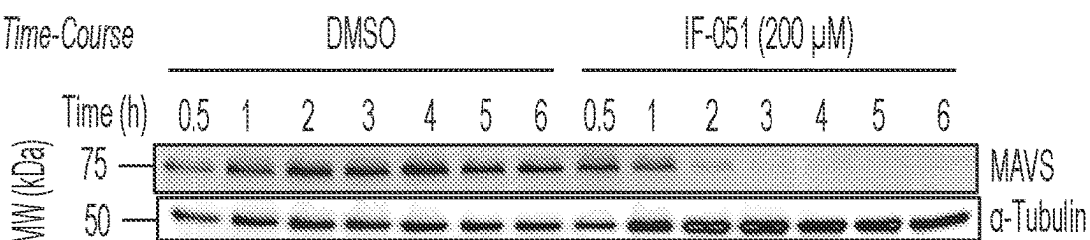


FIGURE 9

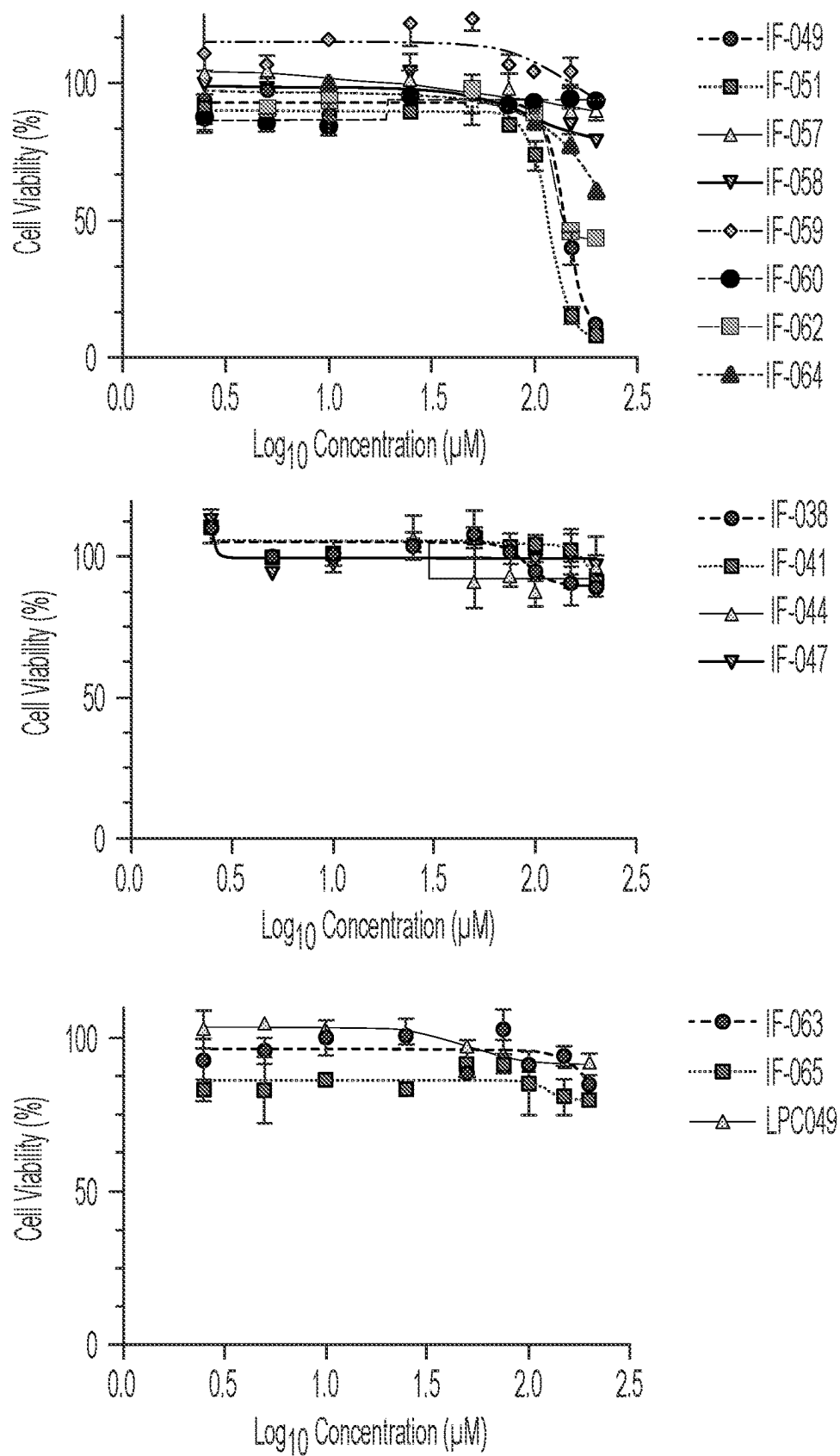
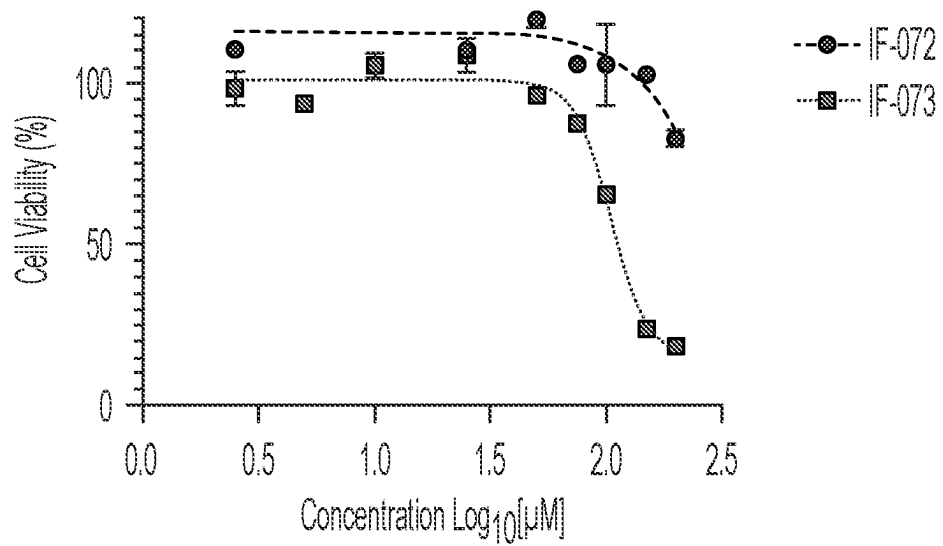
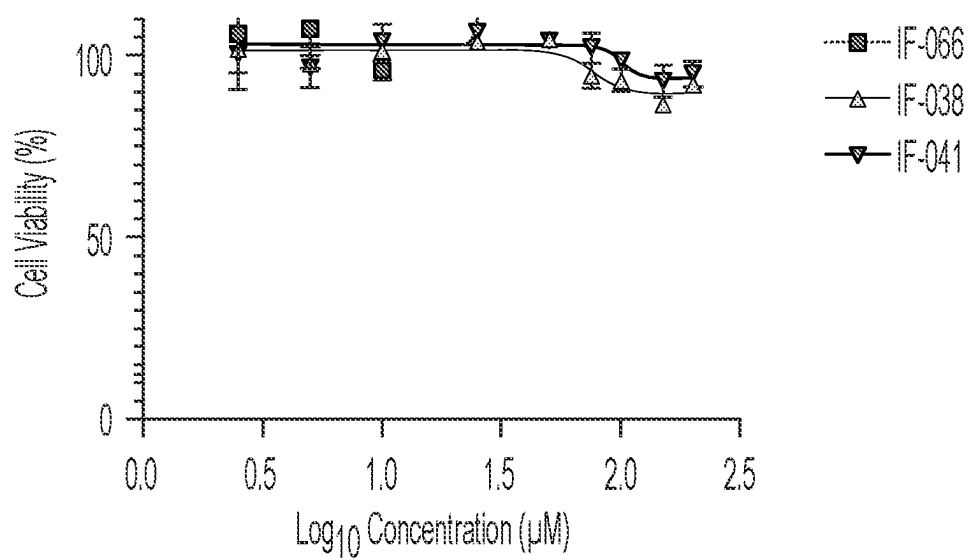


FIGURE 9 (Cont.)



HETEROBIFUNCTIONAL COMPOSITIONS FOR TARGETED PROTEIN DEGRADATION AND METHODS FOR THEIR USE

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] The subject patent application claims the benefit of priority to U.S. Provisional Patent Application Nos. 63/156,593 (filed Mar. 4, 2021; now pending). The full disclosure of the priority application is incorporated herein by reference in its entirety and for all purposes.

BACKGROUND

[0002] A powerful approach was recently developed that integrates fragment-based ligand discovery with chemical proteomics, called fragment-based ligand mapping in cells, to globally survey ligandable proteins and their ligandable sites (FbLMiC, FIG. 1A) (11). In FbLMiC, small-molecule probes, called fully functionalized fragment (FFF) probes are designed to contain: 1) a structurally minimized “constant” region bearing a photoactivatable diazirine group and alkyne handle, which together enable UV light-induced covalent modification and detection, enrichment, and identification of compound-bound protein targets; and 2) a “variable” recognition region consisting of structurally diverse small-molecule fragments (MW<300 Da) to promote interactions with a subset of the proteome. See FIGS. 1B and 1C.

[0003] Notable strengths of FbLMiC are: 1) fragments can be optimized into higher affinity ligands through FbLMiC-guided medicinal chemistry, resulting in lead chemical probes that can selectively modulate protein function (as has been shown with first-in-class chemical probes for PTGR2, SLC25A20 and PGRMC2, (11, 12)); 2) FFFs can map ligand binding site on endogenous protein targets, revealing fragments that interact at a variety of protein sites (e.g. active sites, cofactor binding sites, allosteric sites); 3) FFF probes engage proteins reversibly, overcoming the limitations of other chemical proteomic profiling techniques, such as activity-based protein profiling (ABPP), which require covalent reactions with amino acid side chains. Recently this platform has been expanded and has demonstrated that enantiomerically matched FFF's can be used to expedite the discovery of selective fragment-protein interactions (13). See enantioprobes; FIG. 1C.

[0004] To streamline the identification of high affinity and selectivity chemical probes against proteins with established FFF leads, a strategy has recently been developed and is termed competitive FbLMiC (FIGS. 2A and 2B). Here, the potency and selectivity of candidate small-molecule ligands is assessed across 100s-1000s of proteins. This method integrates targeting and tandem mass tagging (TMT) methods (14) to enable the optimization of chemical probes for protein targets directly within their endogenous biological environments. In competitive FbLMiC, up to nine different competitor analogs (and a DMSO control) are evaluated per experiment for blockade of FFF interactions with endogenous targets. Competitor libraries consist of members that share parent fragment core structures, either purchased or synthesized in-house. By performing targeted MS experiments, MS runtimes can be dramatically shortened, enabling the screening of ~150 compounds/day. Analogs that show the strongest blockade are optimized for potency and proteomic selectivity by an iterative cycle of MS-based FbLMiC-guided medicinal chemistry to furnish lead ‘binders.’ The original pilot FFF library consisted of only 13

members for proof of principle studies (11). For this work, a specialized library of ~150 “scout” FFF probes was synthesized and was based upon fragment (<250 Da) cores commonly found in bioactive compounds (e.g. drugs, natural products, human metabolites) (15) and possessed structures that are synthetically accessible for derivatization. Initial profiling of this library has demonstrated outstanding coverage with an unprecedented ligandability map of 5000+ proteins, including those that fall out of traditional “drug-gable” classes (e.g. adaptor proteins, transcription factors). However, the protein binding activity accomplished with members of the FFF library does not enable refined selection of the proteins of interest (POIs), modification of the POIs and/or their selective, programmed degradation.

[0005] A rudimentary solution involves reliance upon rationally designed heterobifunctional small molecules. These heterobifunctional molecules facilitate the study of an increasingly wide range of biological phenomena and have proven enabling in drugging challenging therapeutic targets and processes. Such molecules typically consist of two ligands or binders that are connected via a covalent linker, yielding a chimeric compound that can mediate the formation of ternary complexes between two unique proteins (6, 7). The ability to “recruit” enzymes to proteins of interest (POIs) endows these molecules with great potential as a generalizable strategy to influence PTMs.

[0006] Recently, it has been shown that one class of PTM enzymes, E3 ubiquitin ligases, can be recruited to target POIs via heterobifunctional molecules (8-10). These molecules, often referred to as proteolysis targeting chimeras (PROTACs), induce molecular proximity between an E3 ligase and a POI, leading to ubiquitination and targeted protein degradation (TPD). Despite the tremendous therapeutic potential of TPD, two major challenges overshadow the generalization of this approach, i) the small number of E3 ligase recruiters that have been identified, despite the excess of 600 predicted E3 ligases; and ii) the undetermined fraction of the proteome accessible to TPD, and correspondingly, the availability of small molecule ligands capable of binding to target POIs that can be coopted into such modalities.

[0007] Nearly all PROTACs utilize established small molecule ligands (e.g. inhibitors), limiting the scope of proteins that can be targeted and requiring synthetic ‘retrofitting’ that may be disruptive to binding to the POI. Further, due to this dearth of available ligands, the fraction of the proteome that can be targeted with PROTAC-type strategies is unknown. Therefore, a goal of the present invention is the development of PROTAC's that are based upon surveys of the entire proteome for proteins that may be tractable to targeted protein degradation in parallel to the identification of bifunctional degrader leads instead of choosing singular targets with a priori established ligands. Therefore, a goal of the present invention is the development of heterobifunctional compositions that functions to bind endogenous proteins and to modify them and/or to enable programmed selective degradation.

SUMMARY

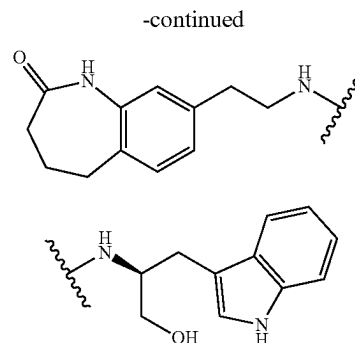
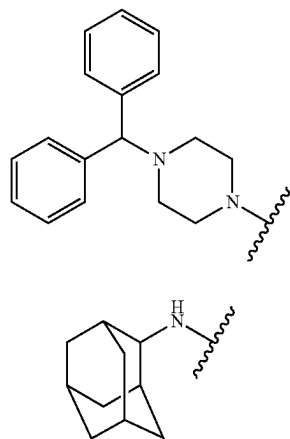
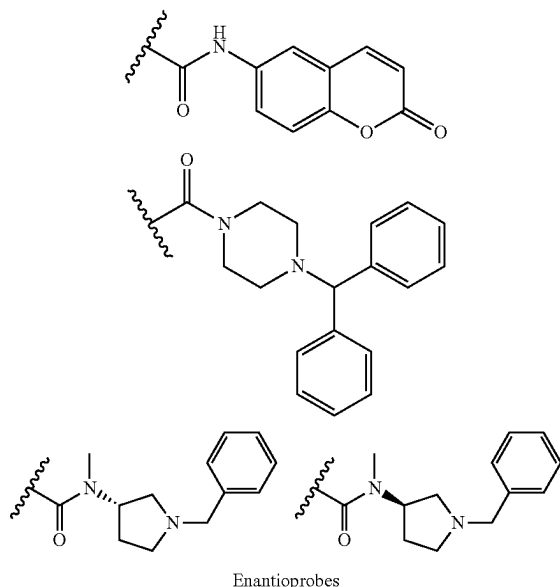
[0008] An aspect of the present invention is directed to heterobifunctional fragment based degrader molecules, FragTACS, having protein binding target moieties that are selected from unbiased whole proteome affinity interactions. Another aspect of the present invention is directed to methods for in vitro and/or in vivo endogenous protein degradation through the agency of heterobifunctional FragTACS.

[0009] Embodiments of the heterobifunctional FragTACS incorporate small molecule fragments that are preferably endogenous protein binding target moieties, small molecule recruiter moieties for preferably endogenous degradation enzymes and an organic group linking these two moieties together. The FragTACS may be depicted by a generic Formula I:

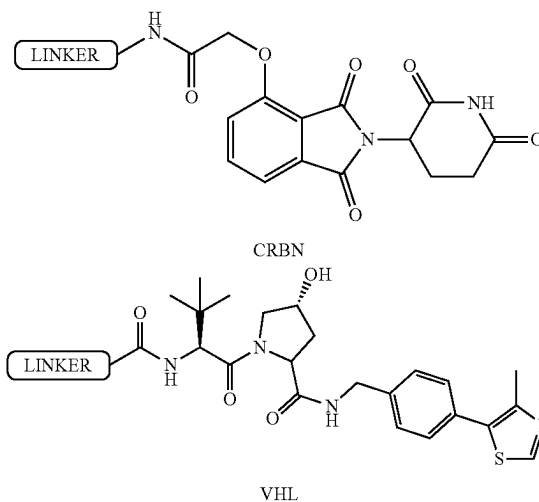
PBF-L-RBF

Formula I

[0010] The acronym PBF is a protein binding fragment of a small organic molecule moiety that enables selection of the protein of interest from a milieu of proteins, preferably endogenous proteins. The acronym RBF is a recruiter binding fragment of moiety of a small organic molecule moiety that is capable of recruiting in a cytoplasm context an enzyme that degrades, fragments and/or divided endogenous proteins into fragments for re-assimilation. The acronym L is an organic linker group having combinable, reactive functions at its termini that enable covalent attachment to the PBF and RBF. Exemplary PBF fragments include but are not limited to:



Exemplary RBF fragments include but are not limited to thalidomide derivatives for the CRBN ligase and the N-(4-thiazol-1-yl phenethyl) 4-hydroxyprolinamide for the VHL ligase



These examples of RBF fragments enable recruitment of E3 ubiquitin ligase activity. Exemplary linker groups include short and oligomeric polyol (PEG) and alkylenyl, moieties having amine and/or carboxyl termini for binding to the PBF and RBF fragments.

BRIEF DESCRIPTION OF THE DRAWINGS

[0011] FIGS. 1A, 1B and 1C depict Fragment-based Ligand-ability Mapping in Cells. FIG. 1A depicts Fully functionalized fragment (FFF) probes are composed of a drug-like fragment as well as a retrieval tag, enabling the covalent capture of fragment-bound protein targets directly in cells upon UV irradiation. Fragment targets, as well as the site of fragment interaction, can be identified and quantified by mass spectrometry- and gel-based methods. FIG. 1B depicts the general structure of FFF showing the constant affinity tag region (red), which consists of a photoreactive group (diazirine) and a latent affinity (alkyne) group, as well as the variable region (blue), which contains fragment recognition elements for binding to proteins in cells. FIG. 1C depicts example structures of fragment scout- and enantioprobes.

[0012] FIGS. 2A and 2B depict targeted competitive FbLMiC workflow for chemical probe development for prioritized proteins. FIG. 2A depicts optimization of scout probe to lead binder through iterative competitive FbLMiC. FIG. 2B depicts targeted MS-FbLMiC accelerates chemical probe development by enabling rapid, multiplexed analyses of a defined set of prioritized targets in native biological systems. Shown is a targeted MS-FbLMiC workflow that we have developed for assessing small molecule interactions in prioritized targets, where a cell preparation is treated with up to nine compounds (and a DMSO control) and a corresponding FFF scout probe, click conjugation to a desthiobiotin-azide tag, streptavidin enrichment, elution, and labeling of enriched peptides with tandem-mass tags (TMT). Samples are then combined and analyzed in a single LC-MS/MS/MS experiment, where peptides of interest in prioritized proteins are selected for sequential MS1, MS2, and MS3 analysis. MS3 signals are used to quantify peptides, where reductions in signal (e.g., >80%) mark compound-sensitive sites.

[0013] FIGS. 3A, 3B and 3C depict fragment-based discovery of degradable proteins. FIG. 3A depicts chemical structures of a preliminary set of fragment-based PROTACs (FragTACs), which consist of a small molecule fragment chemically linked to established E3 recruiting ligands. FIG. 3B shows representative data for a FragTAC (1, 100 μ M, 6 hrs) incubated with HCC1806 cells. A relative abundance of ~7000 proteins was determined using quantitative multiplexed proteomics. Vertical dashed lines mark 3-fold changes relative to DMSO and horizontal dashed lines mark p value=0.05. Red box (inset) shows targets with >3-fold decreased abundance. FIG. 3C depicts western blots confirming dose-dependent downregulation of 3 example targets (TDP-43, FAM136A, and CCAR2).

[0014] FIGS. 4A, 4B, 4C and 4D depict expanding the druggable landscape of for targeted protein degradation. FIG. 4A provides the strategy to expand the number of targets and E3 proteins available for TPD approaches. FIG. 4B shows a synthesizable library of ~300 FragTACs and examine their ability to modulate protein levels via unbiased proteome-wide proteomic analyses in primary human immune cells. FIG. 4C shows a representative subset of E3 ligases for which FbLDisC has identified hit ligands. Note that hit ligands have been discovered for members of most E3 subfamilies. FIG. 4D shows a model system for testing whether druggable sites on E3 ligases support TPD. Ligands for a given E3 are coupled to a compound AP11867 that binds to the FKBP12-F36V protein and assayed for inducing degradation of recombinant FKBP12-F36V in cells that endogenously express the E3 ligase.

[0015] FIG. 5 depicts in schematic style how a FragTAC binds to a target protein and recruits a fragmentation ligase, which in this example is the ubiquitin ligase complex. The ubiquitin ligase ubiquitinizes the protein to add ubiquitin peptide chains to the protein. This ubiquitination marks the protein for proteolysis through proteasome enzymatic degradation to yield amino acid fragments for re-assimilation.

[0016] FIG. 6 shows results of Western blot studies of proteasome/neddylation inhibition of several target proteins in HCC1806 cells.

[0017] FIG. 7 shows results of Western blot studies of dose-dependent response of several target proteins.

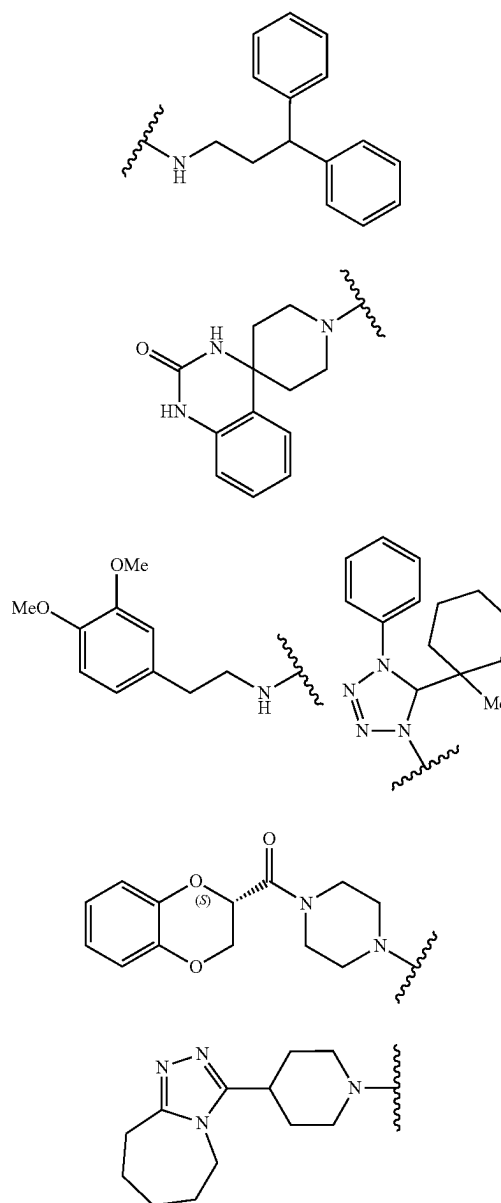
[0018] FIG. 8 shows results of Western blot studies of time course of the responses of several target proteins.

[0019] FIG. 9 shows results of cell viability assays of representative examples.

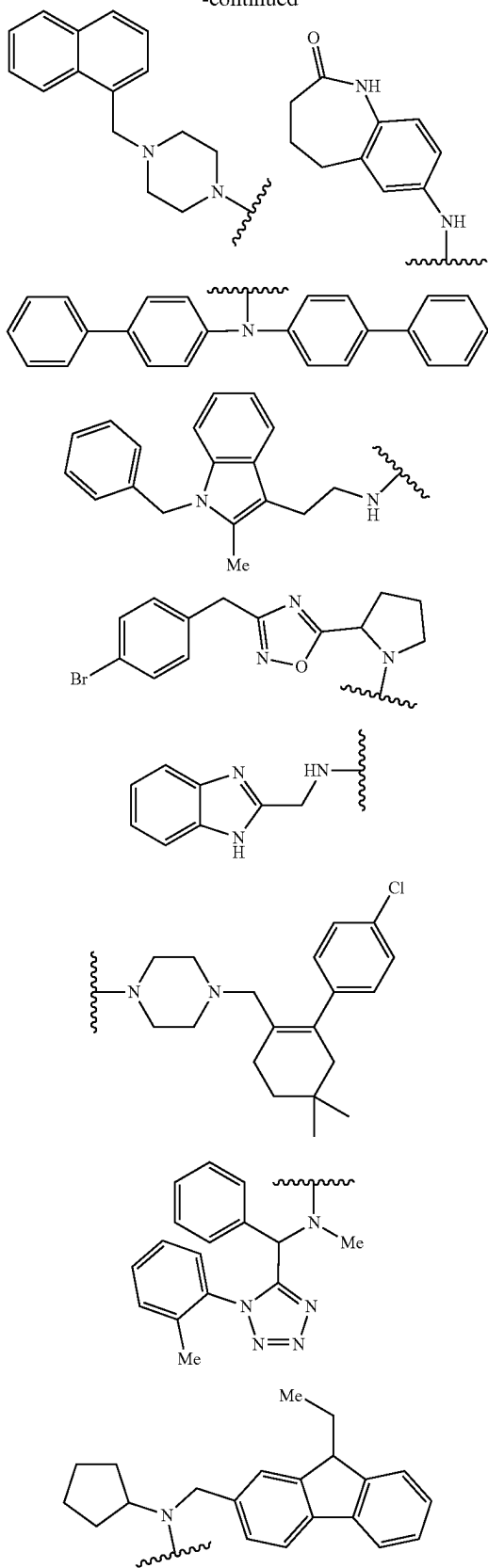
DETAILED DESCRIPTION

[0020] In general, the invention provides heterobifunctional FragTACs that contain a protein binding fragment (PBF) and a recruiter binding fragment (RBF) that are connected via a linker moiety (L). As exemplified herein, the PBF moiety can be any small molecule that targets one or more endogenous proteins, the RBF moiety can be any compound that recruits one or more endogenous degradation enzymes, and the L moiety can be any organic group that optimally links the two moieties with no or minimum impact on their biological functions.

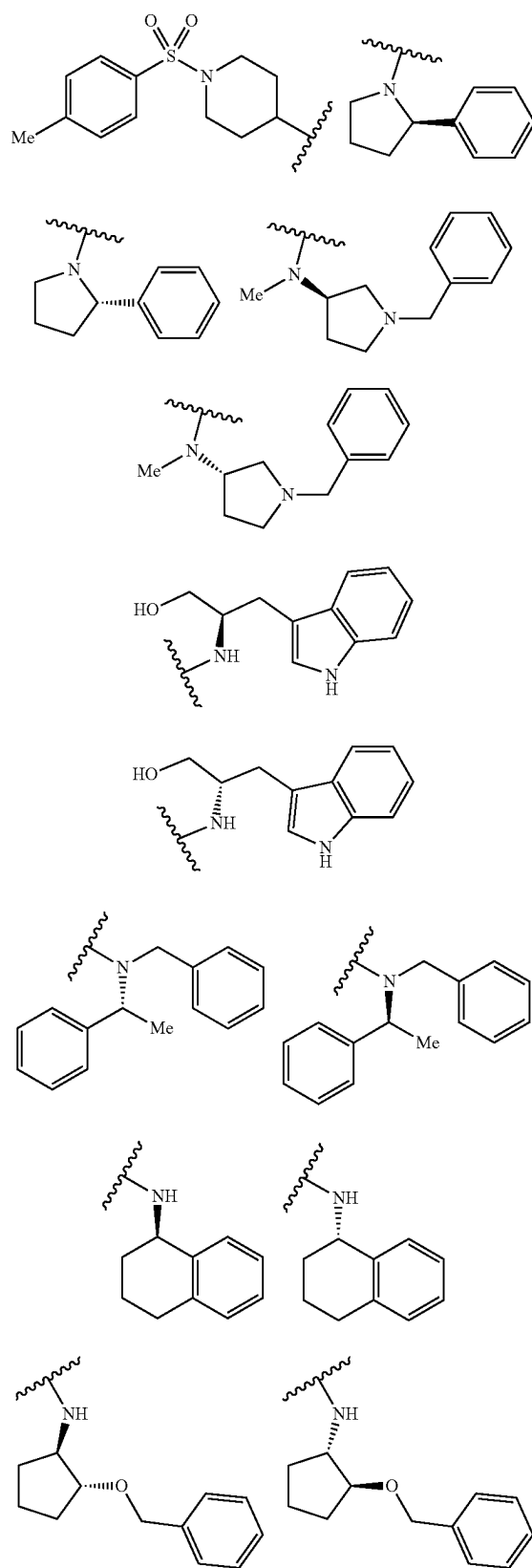
[0021] In various embodiments, the PBF moiety can be selected from any suitable commercially available fragments or synthetically accessible fragments, as described herein. Exemplary commercially available and synthetically derivatizable PBF fragments include but are not limited to:



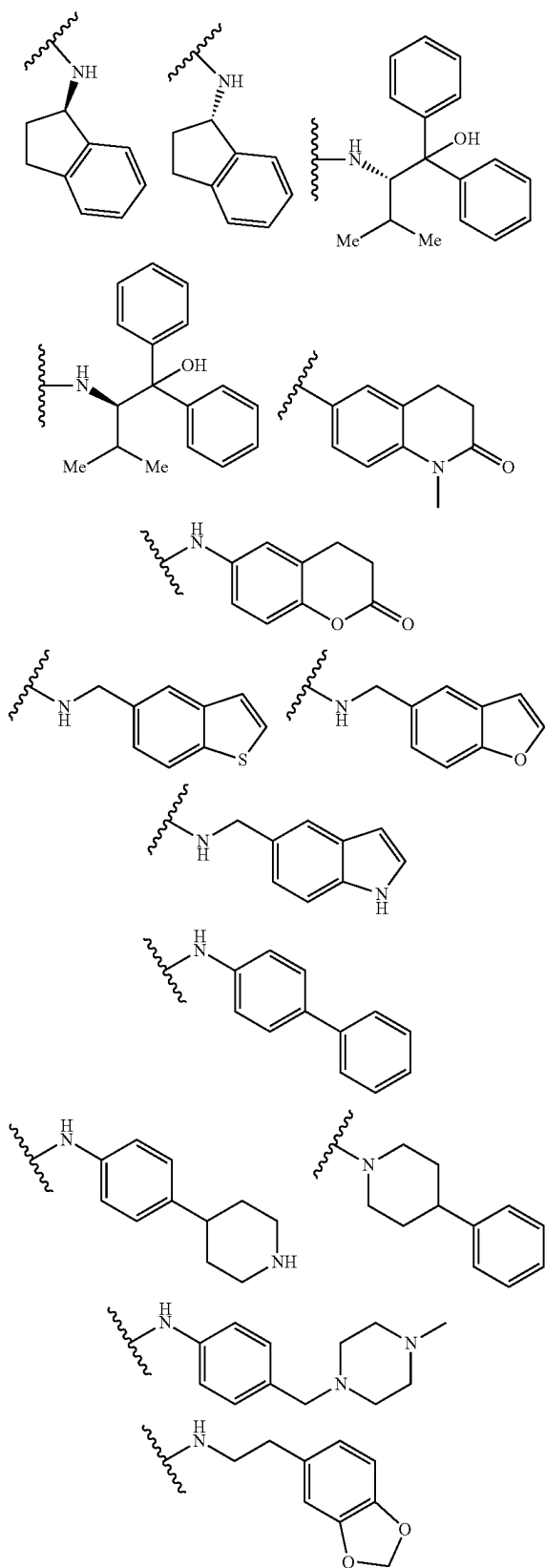
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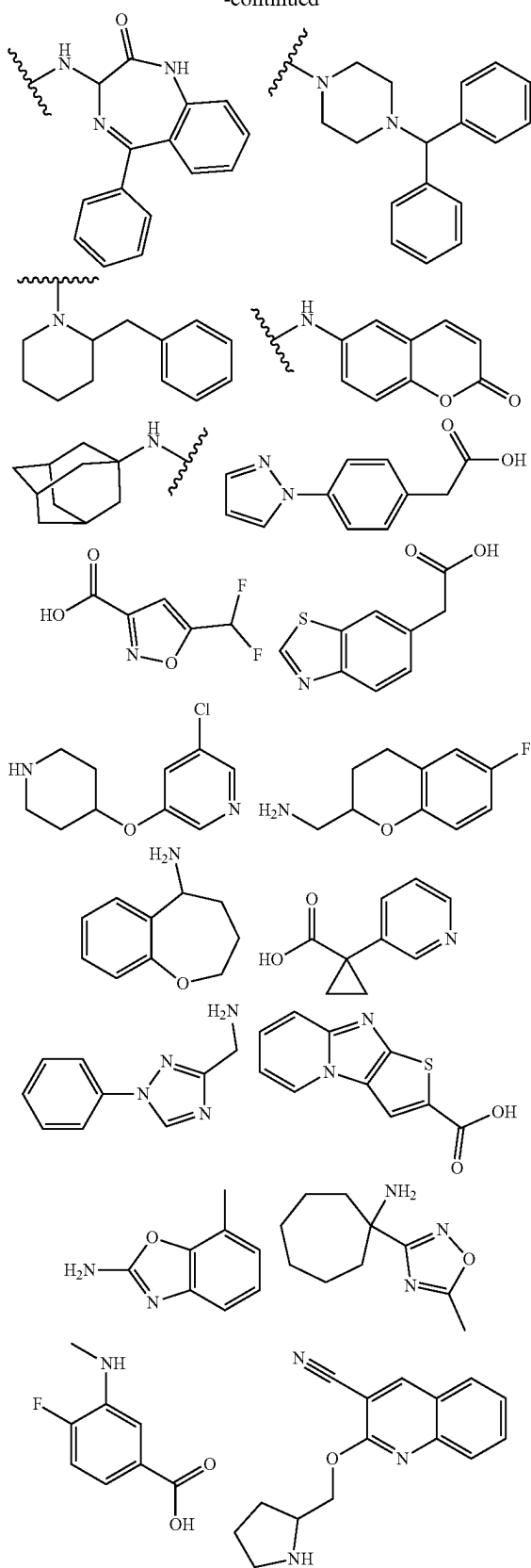
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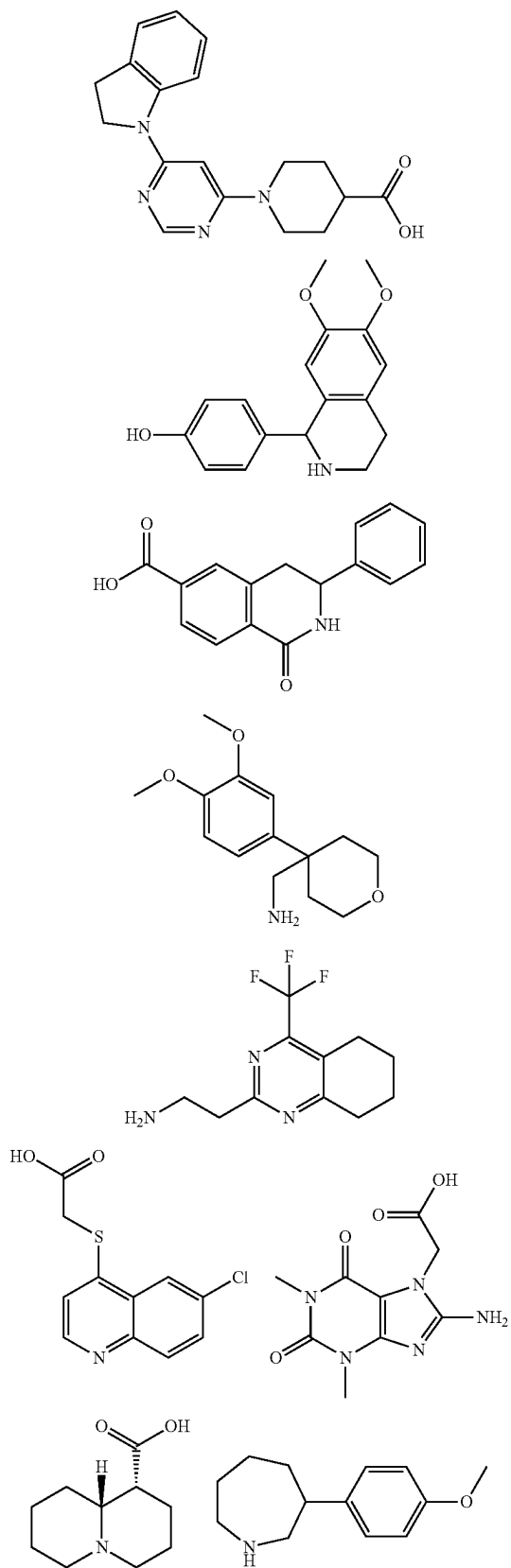
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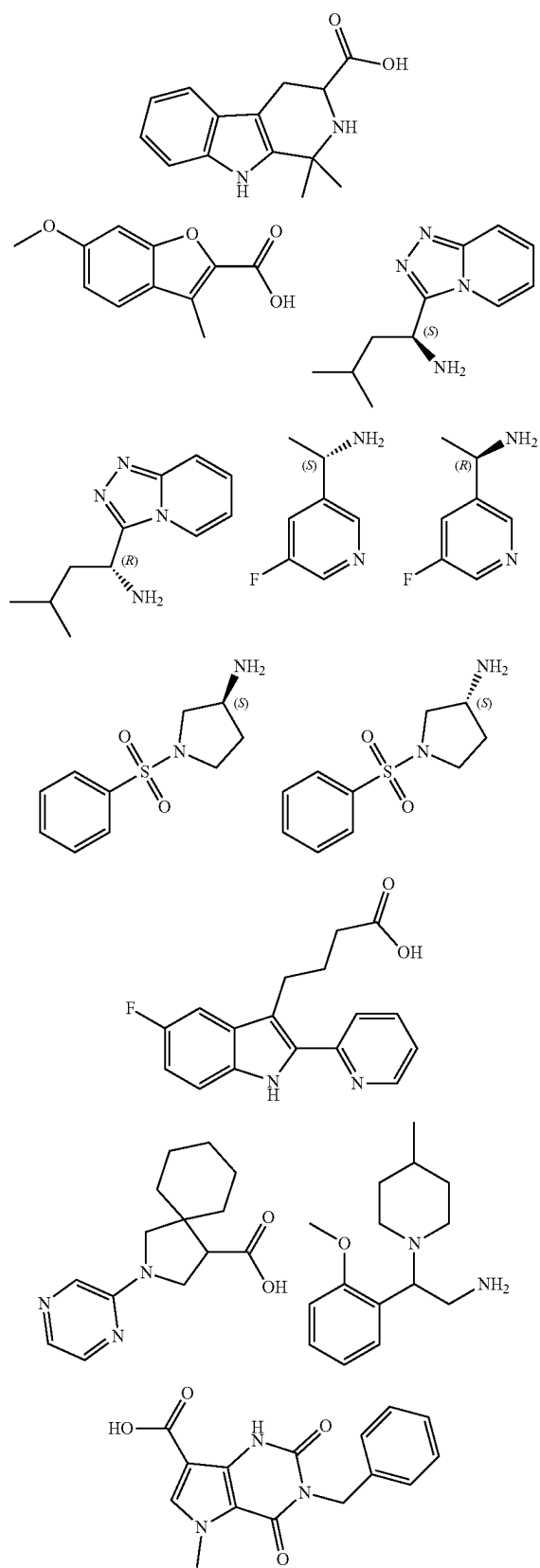
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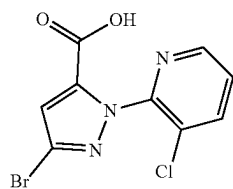
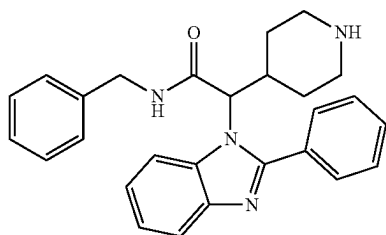
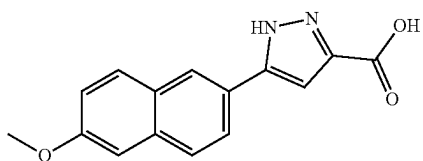
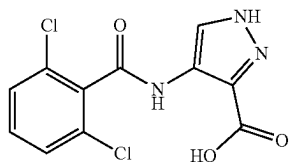
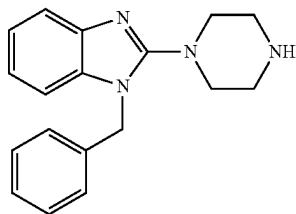
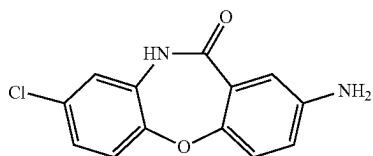
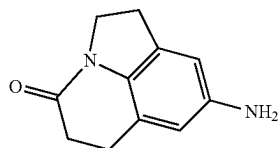
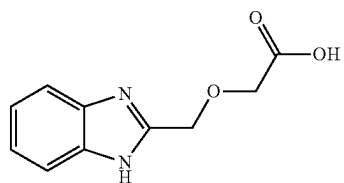
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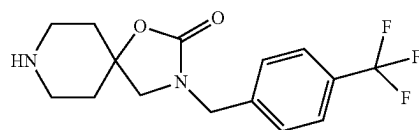
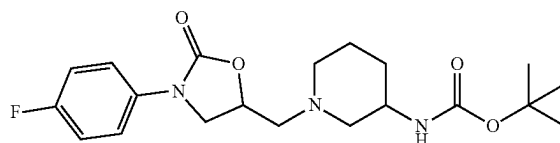
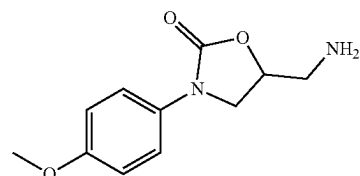
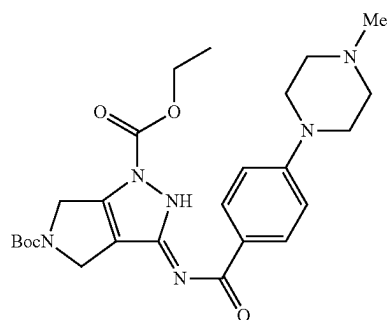
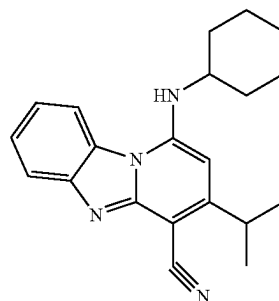
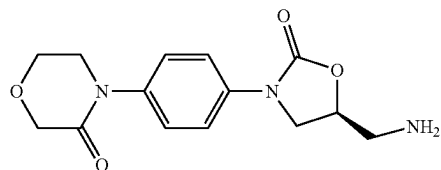
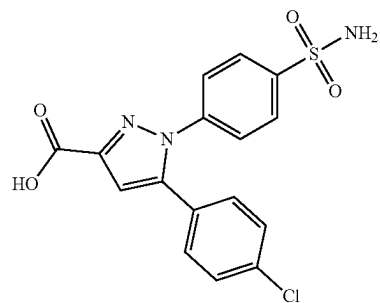
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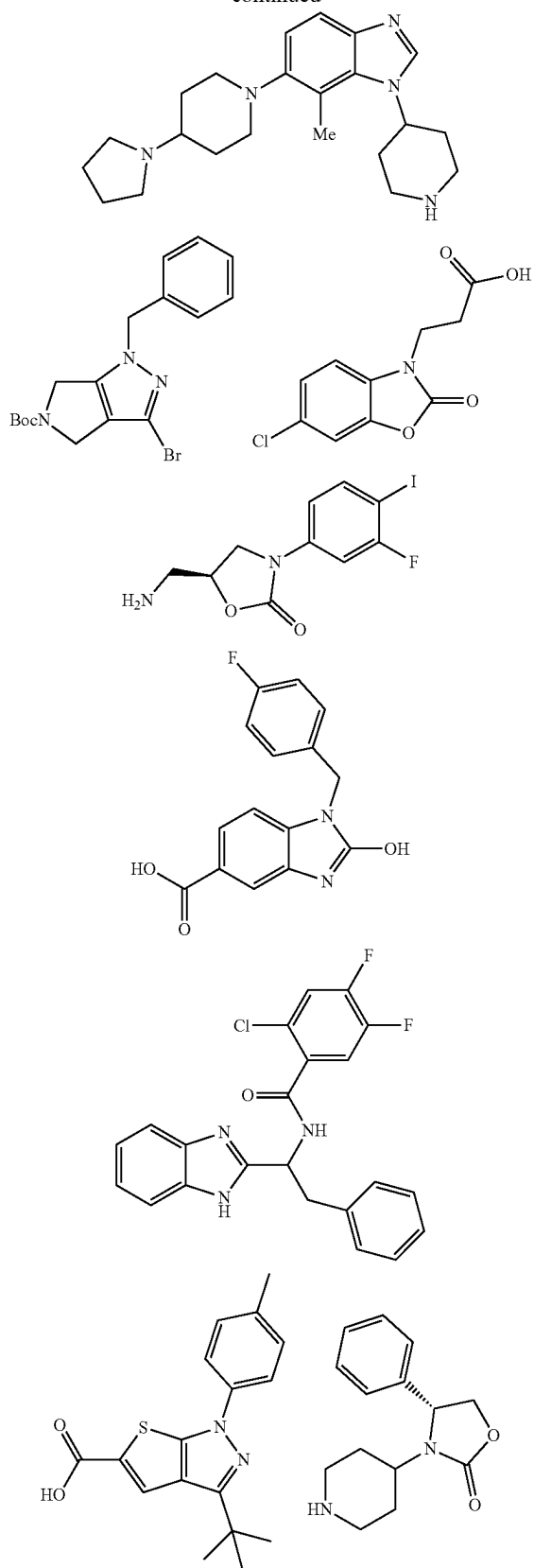
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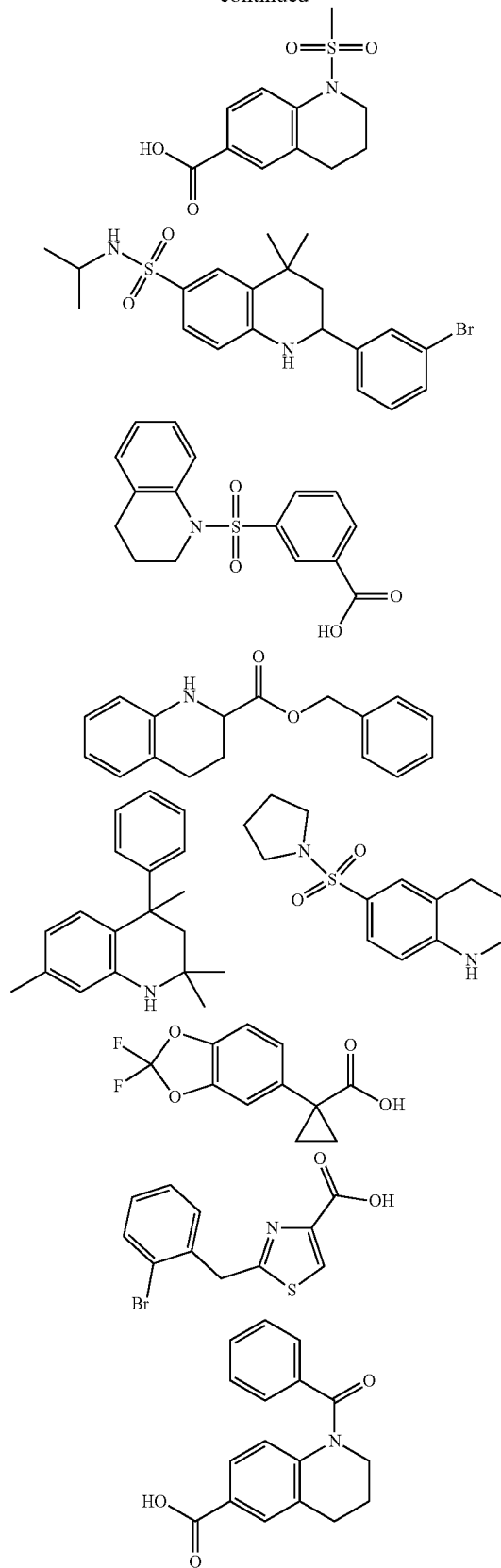
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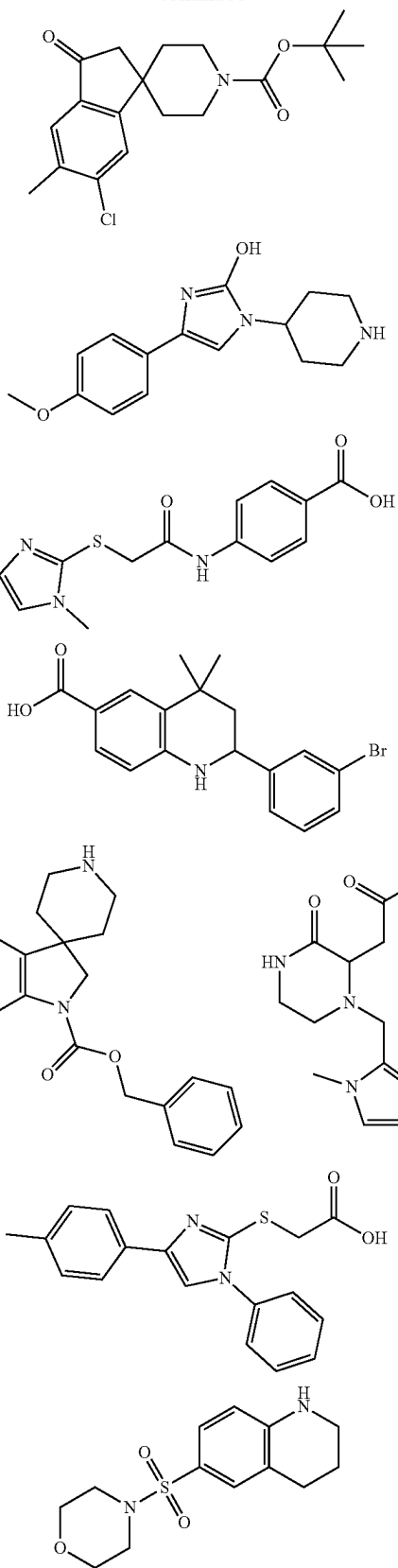
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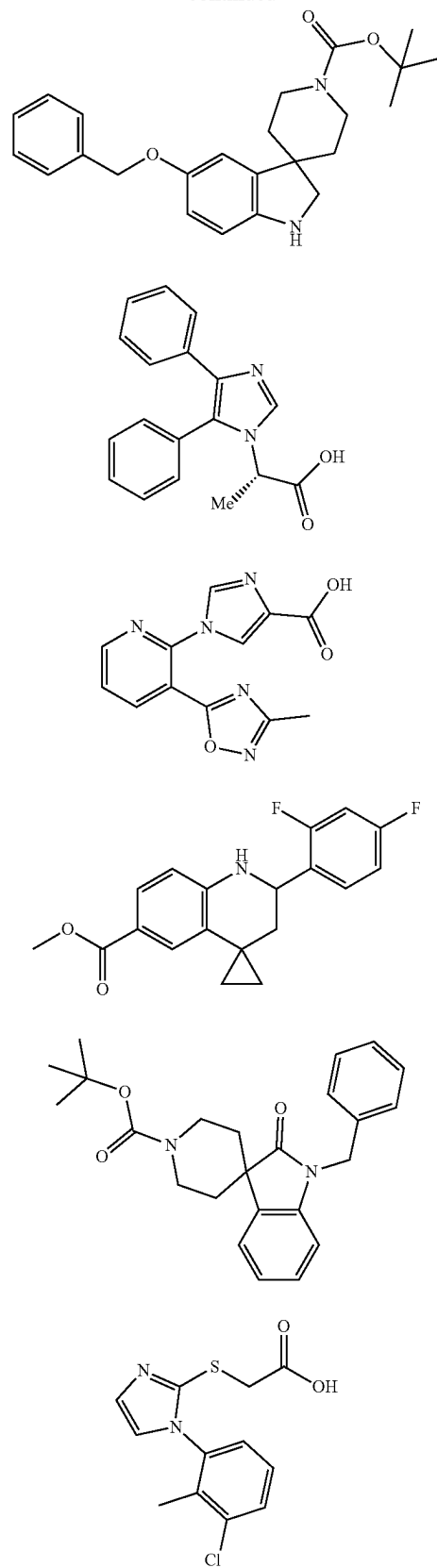
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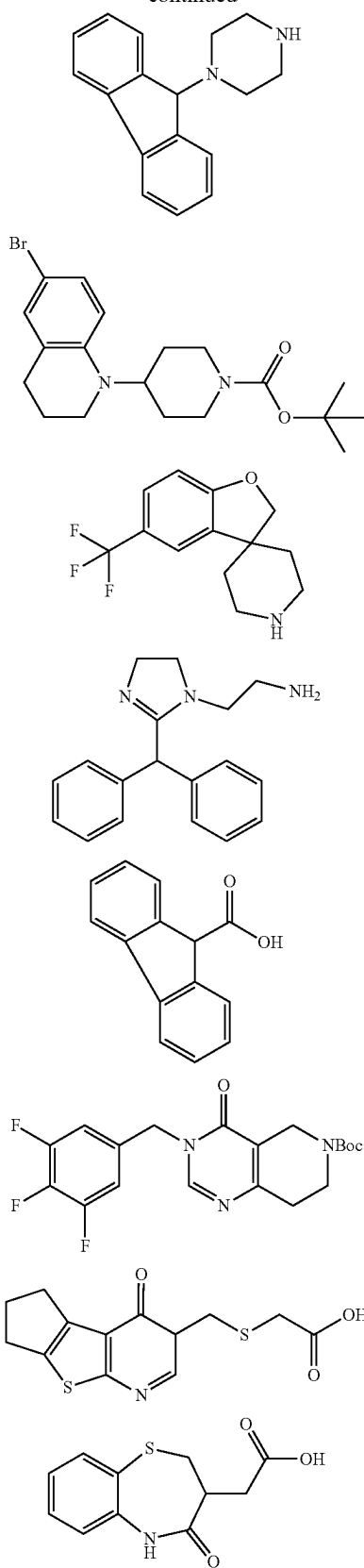
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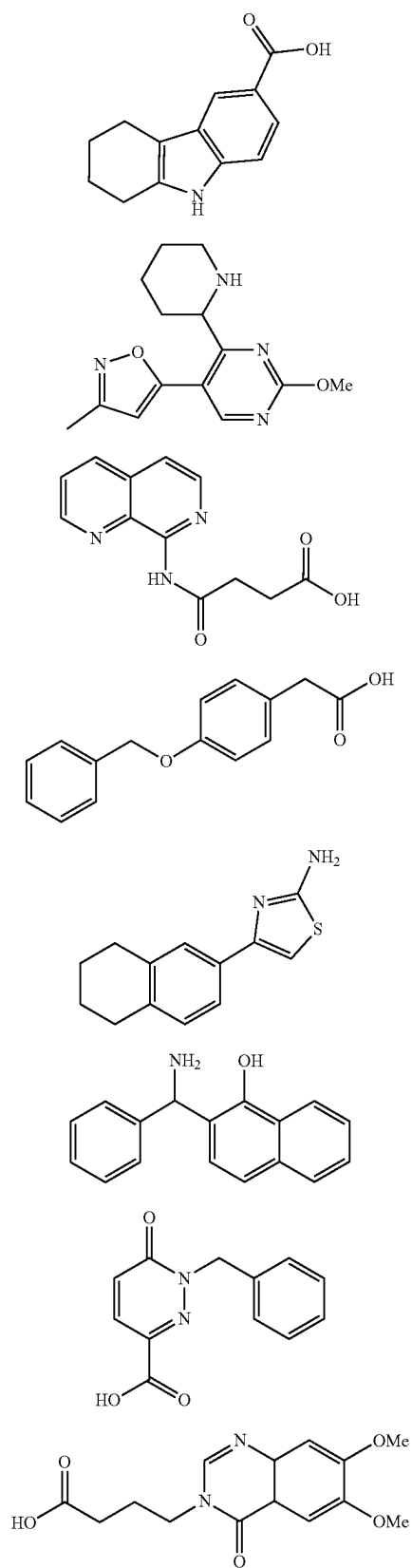
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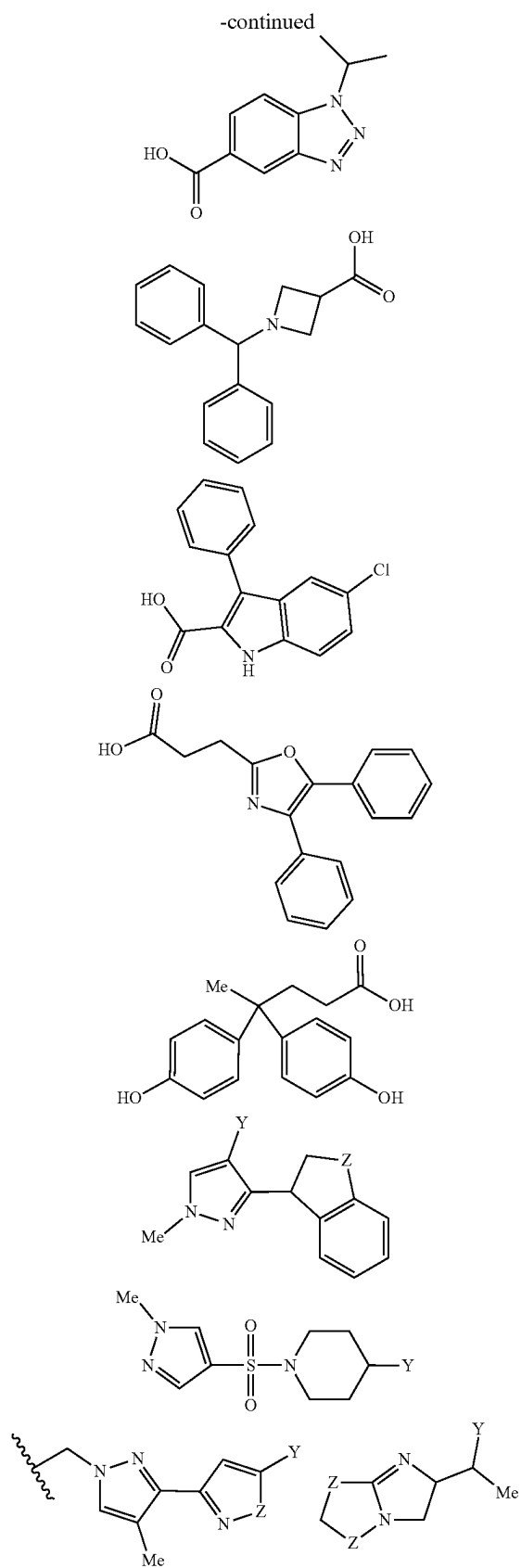
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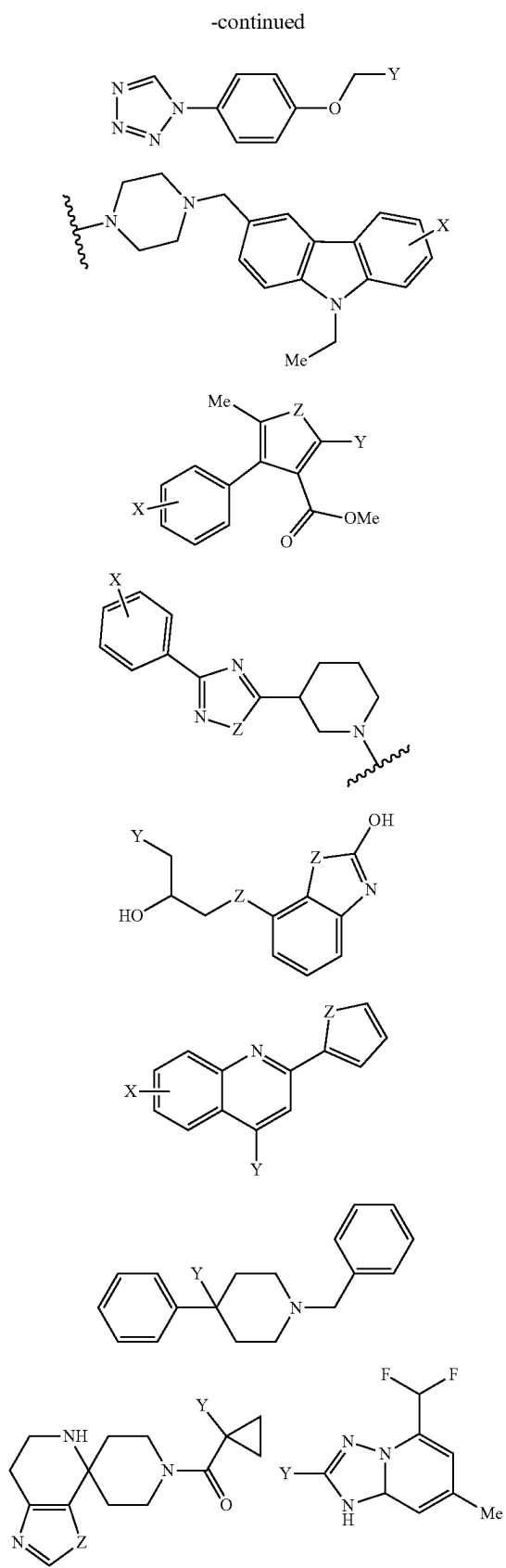
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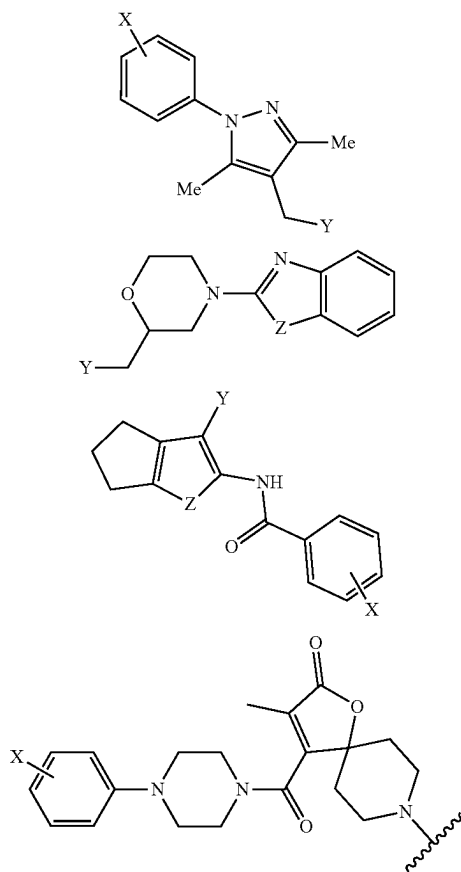
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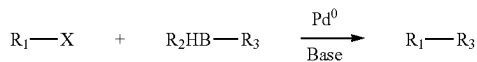


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[0022] In these compound structures, X is F, Cl, Br or I; Y is COOH or NH₂; Z is O, NH or S.

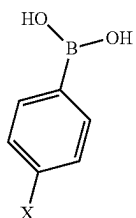
[0023] In some embodiments, the PBF moiety is a synthetically accessible fragment. Exemplary synthetically accessible PBF fragments include but are not limited to:



Suzuki-Miyaura Coupling

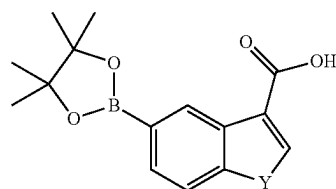
Organoboranes

General Examples

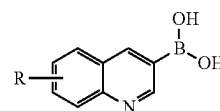


X = F, Cl, Br, I

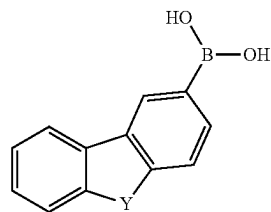
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Y = NH, S, O

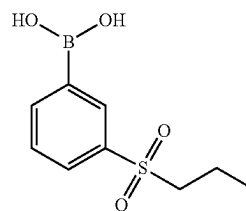
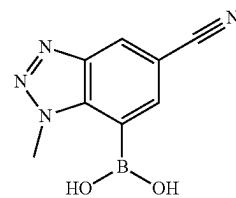
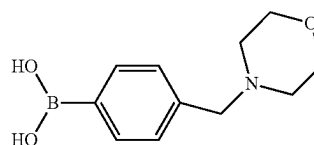
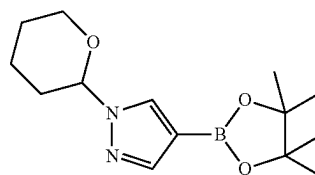


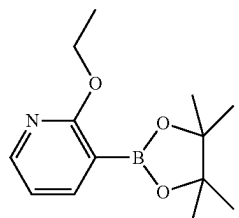
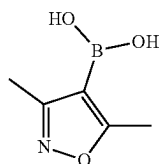
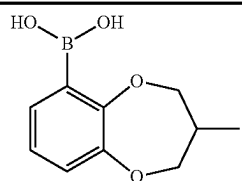
R = F, Cl, Br, I, alkyl, acyl



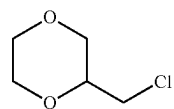
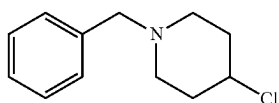
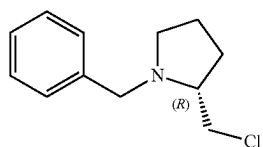
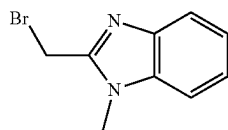
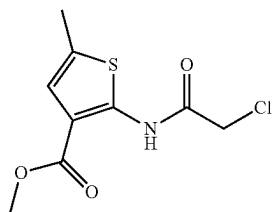
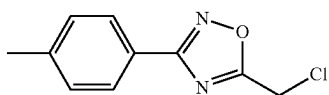
Y = NH, S, O

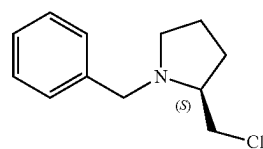
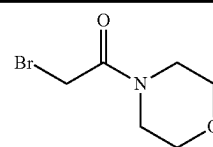
Specific Examples

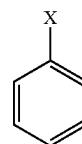


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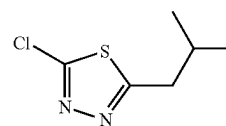
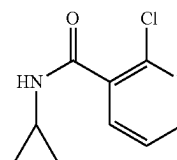
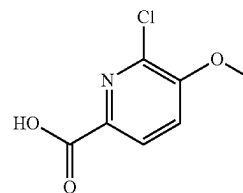
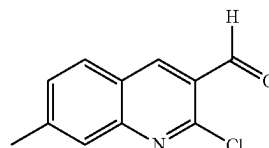
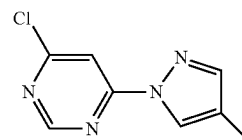
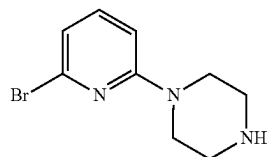
Organohalides

Aliphatic

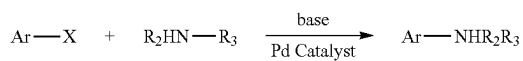
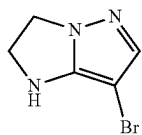
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Aromatic

X = Cl, Br, I



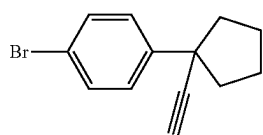
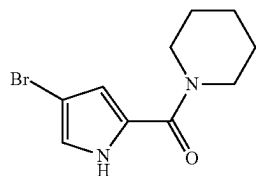
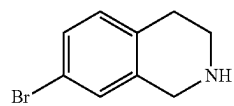
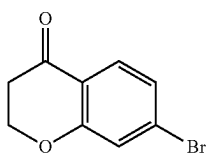
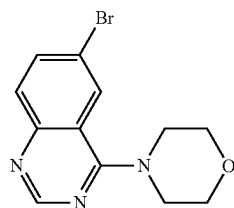
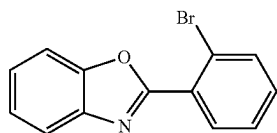
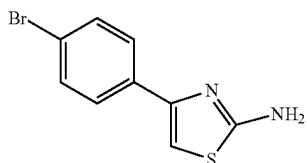
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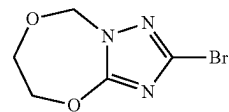
Buchwald-Hartwig Amination

Aryl Halides

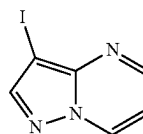
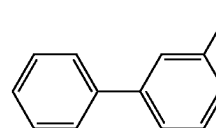
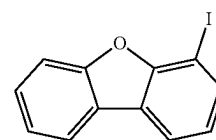
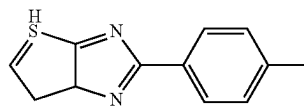
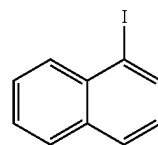
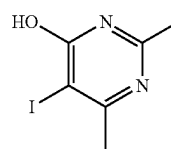
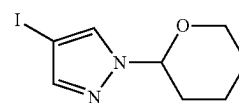
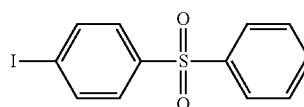
Bromides



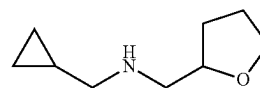
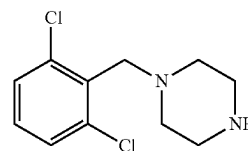
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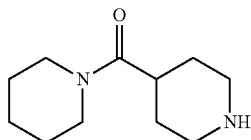
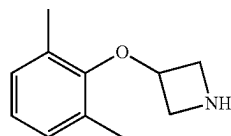
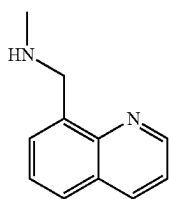
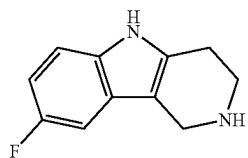
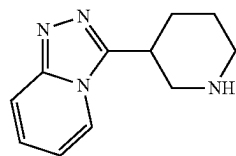
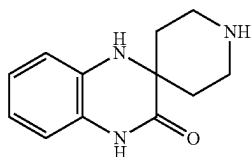
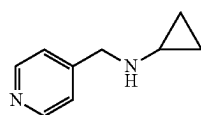
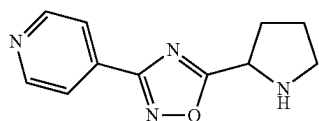
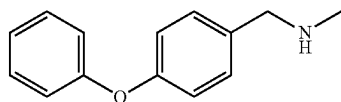
Iodides



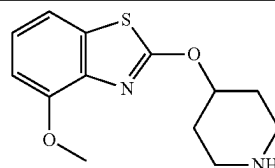
Secondary Amines



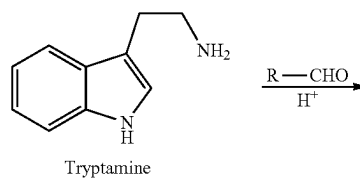
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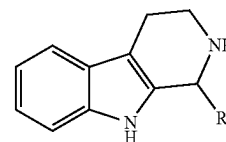
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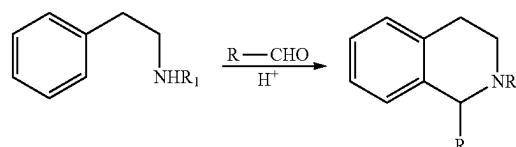
Pictet Spengler Reaction



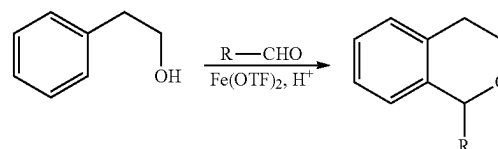
Tryptamine



Carboline

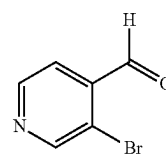
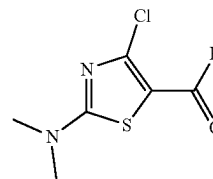
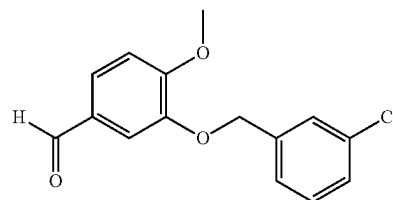


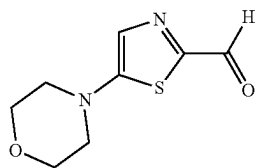
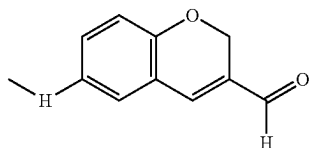
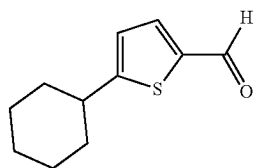
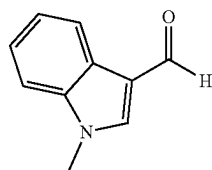
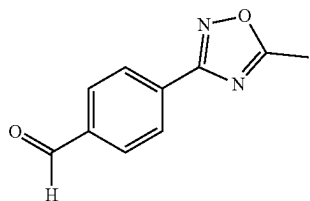
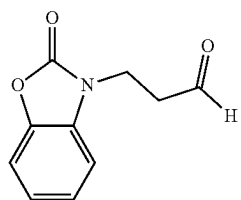
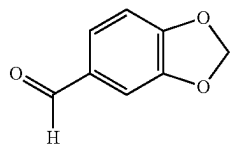
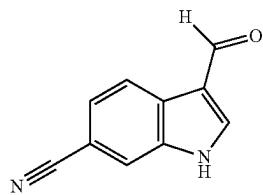
Isoquinoline

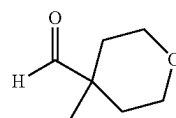
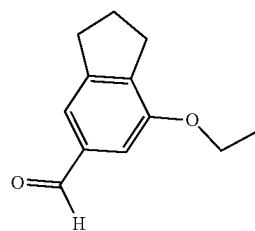
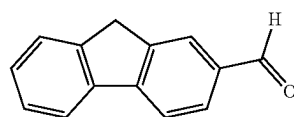
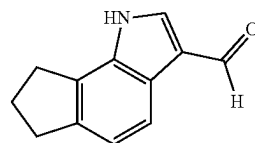
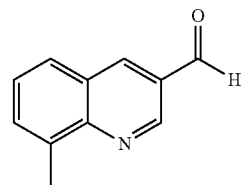
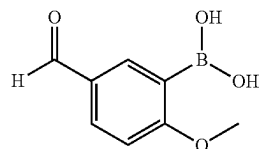
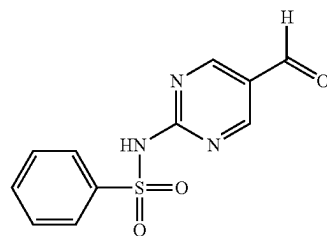


Isochroman

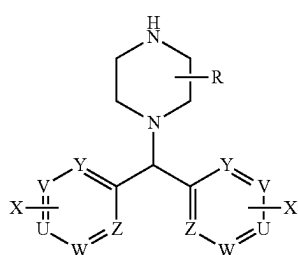
Aldehydes



-continued

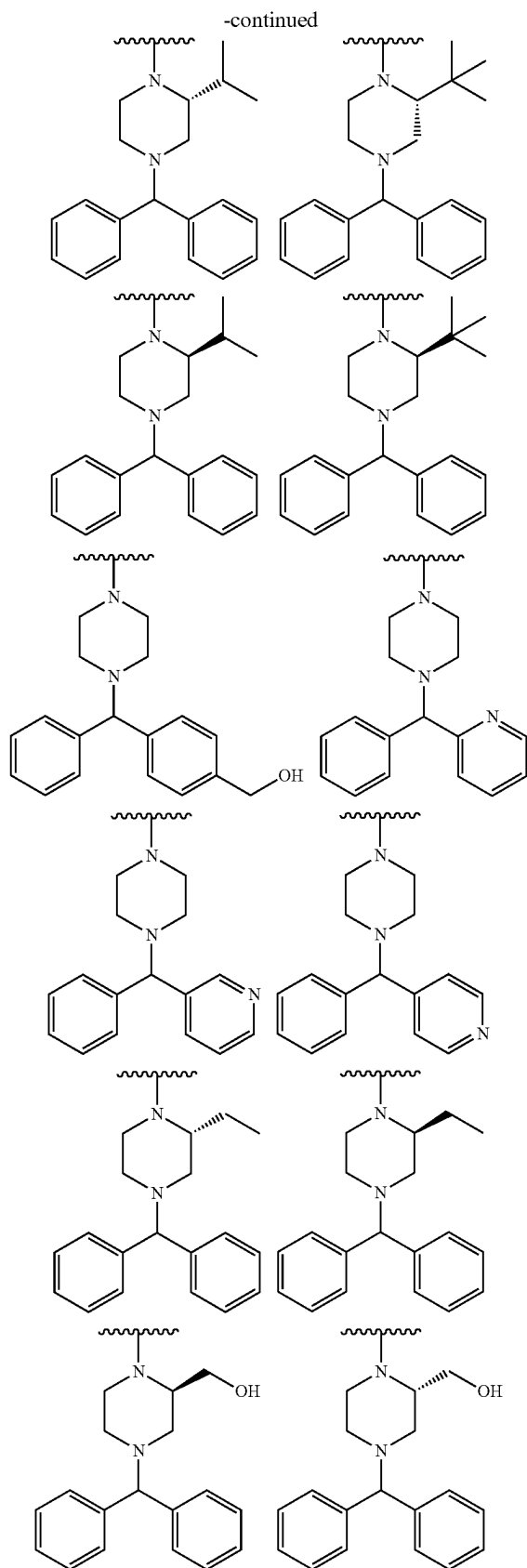
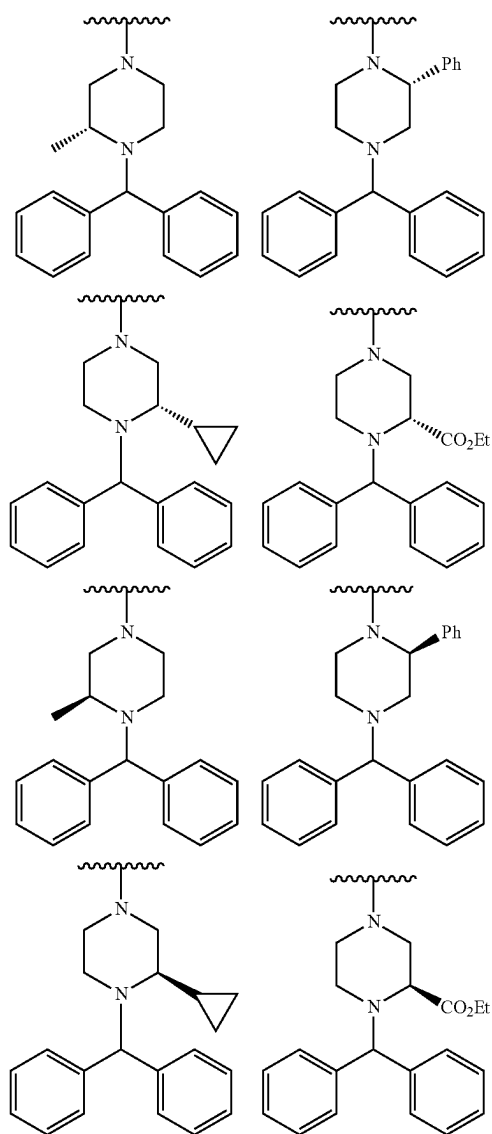
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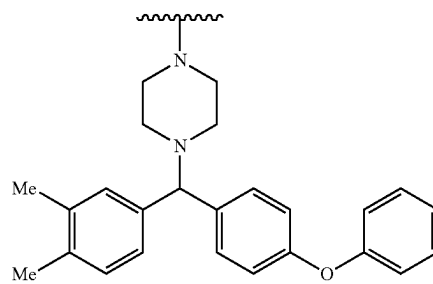
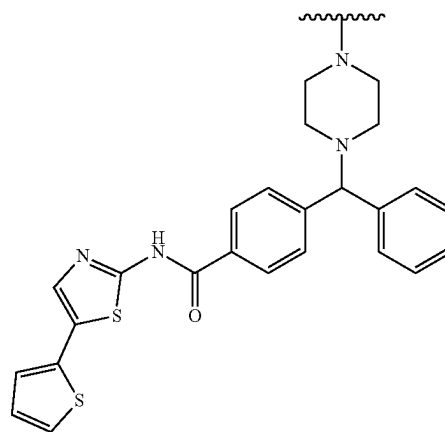
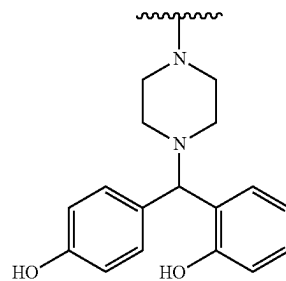
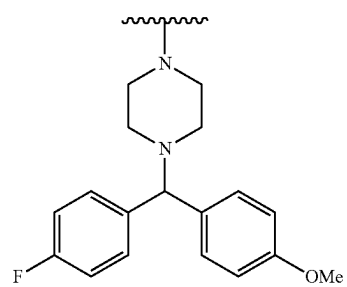
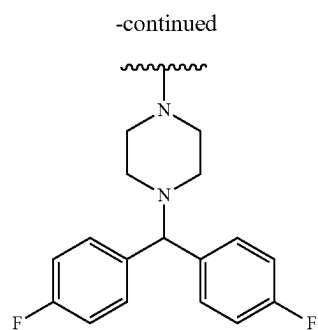
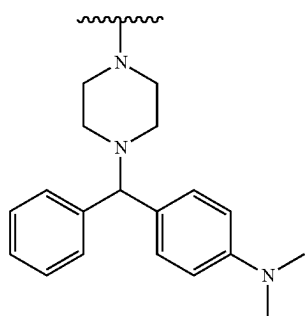
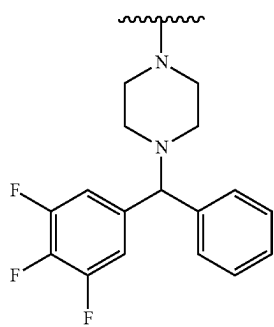
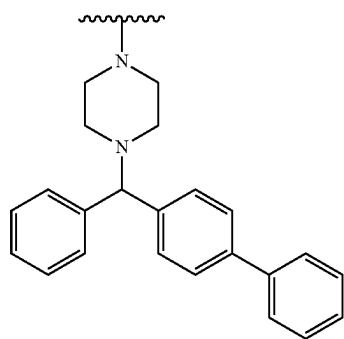
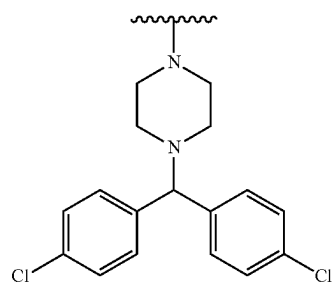
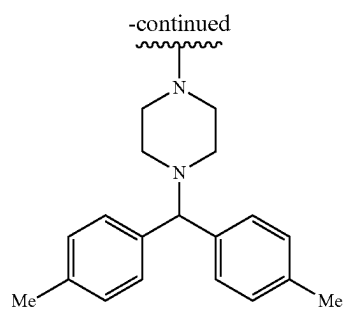
[0024] In some embodiments, the PBF moiety is a synthetically accessible benzhydrylpiperazine derived fragment with a structure shown in Formula XL VII.

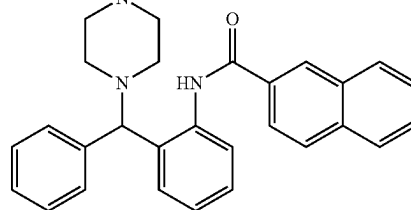
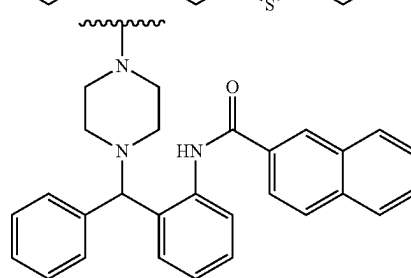
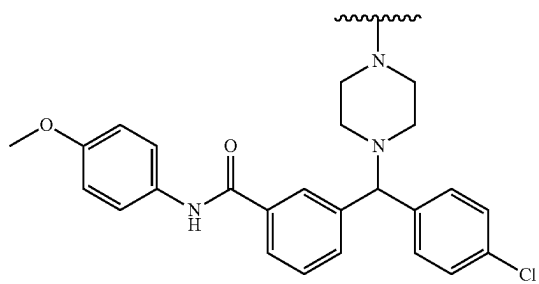
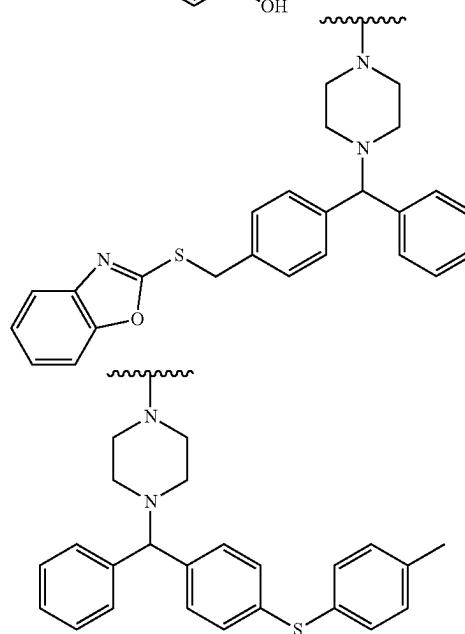
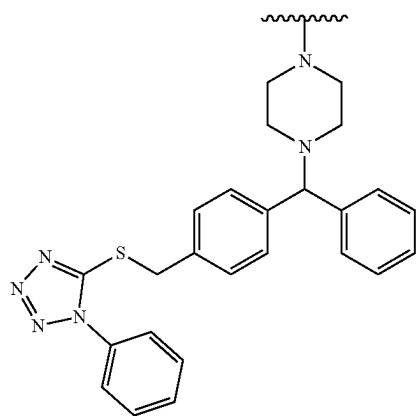
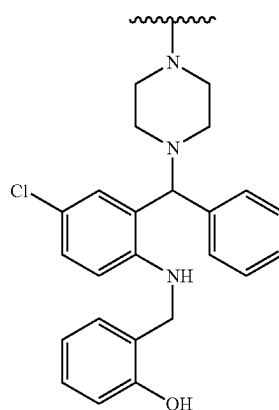
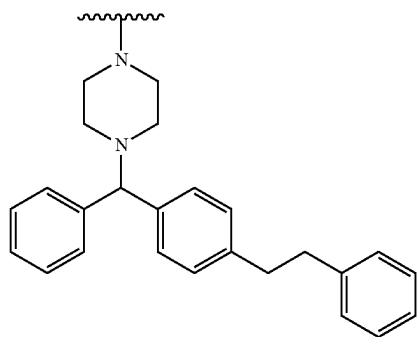
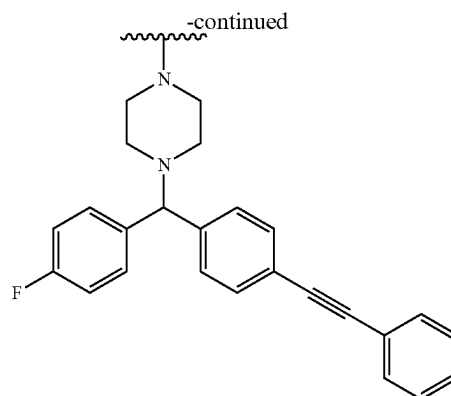
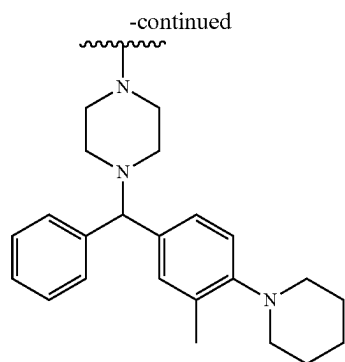


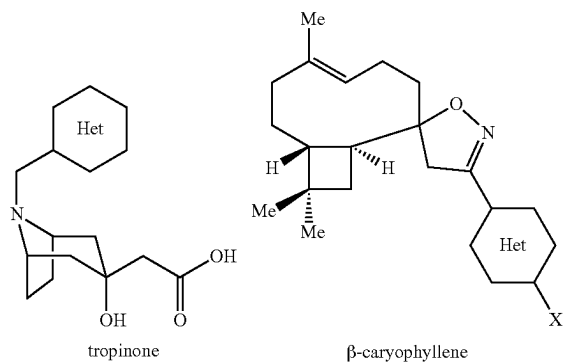
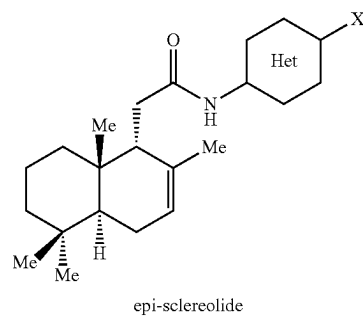
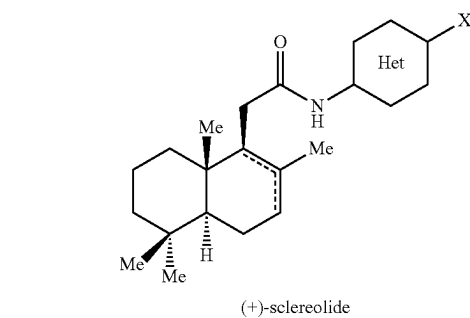
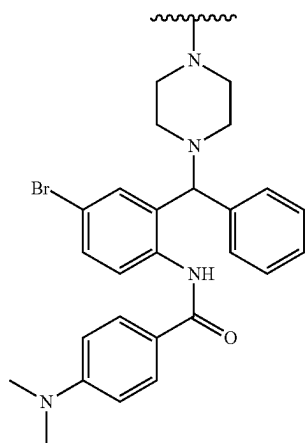
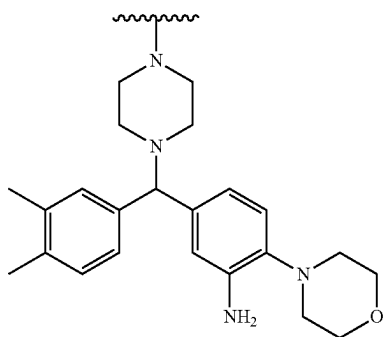
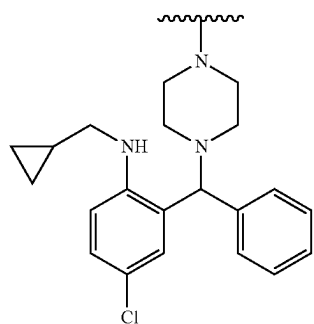
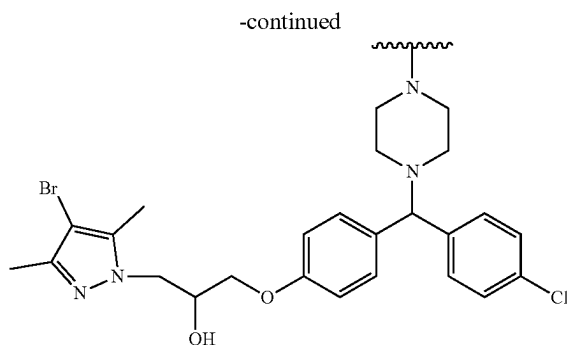
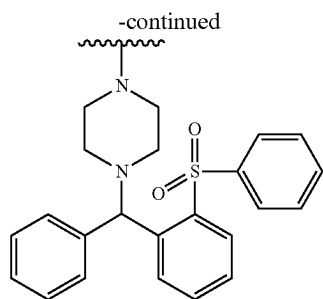
Formula XLVII

In the structure, R is alkyl group, aryl group, COOEt or H; U is CH or N; V is CH or N; W is CH or N; X is H, F, Cl, Br, I, alkyl group or aryl group; Y is CH or N; Z is CH or N. Exemplary synthetically accessible benzhydrylpiperazine derivative fragments include but are not limited to:

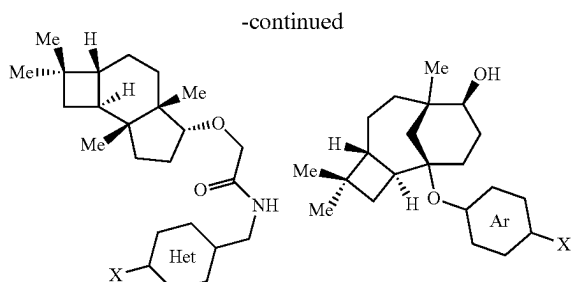




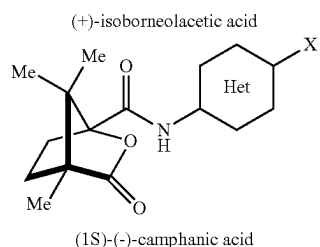
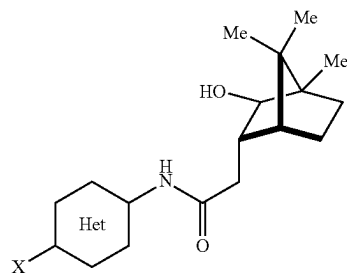
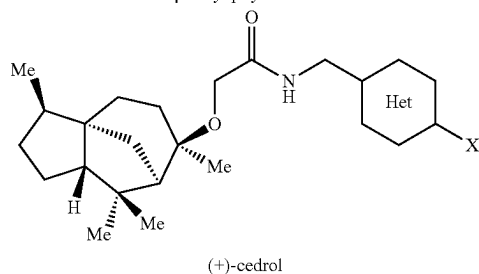




[0025] In some other embodiments, the employed PBF moiety is a synthetically accessible natural product-derived fragment. Exemplary synthetically accessible natural product-derived PBF fragments include but are not limited to:

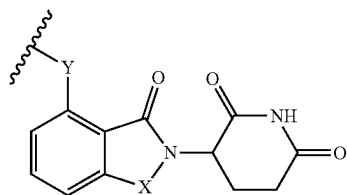


From β -caryophyllene- α -oxide

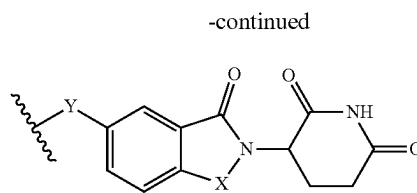


[0026] Like the PBF moiety, the RBF moiety in the FragTACs of the invention can also employ a number of suitable compounds. These include, e.g., CRBN ligands, VHL ligands, IAP ligands, MDM2 ligands, RNF ligands, DCAF ligands, KEAP1 ligands and FEMIB ligands. In some embodiments, the RBF moiety can be a CRBN ligand with a structure shown in any one of Formulae II-V below:

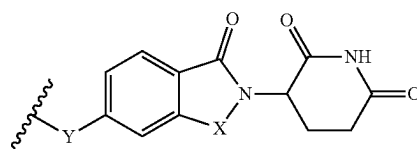
CRBN Ligands (IMiDs):



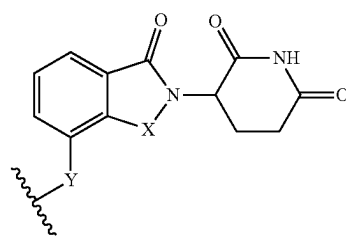
Formula II



Formula III

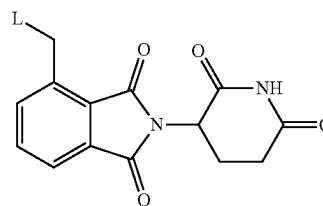
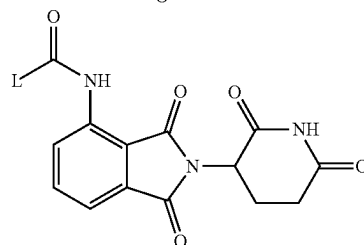
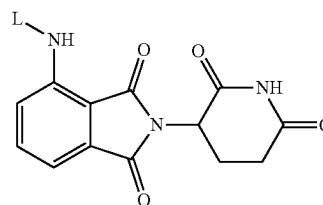
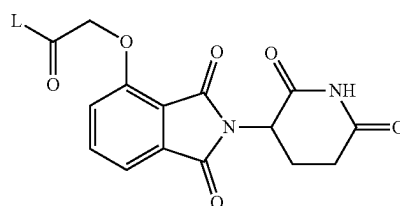


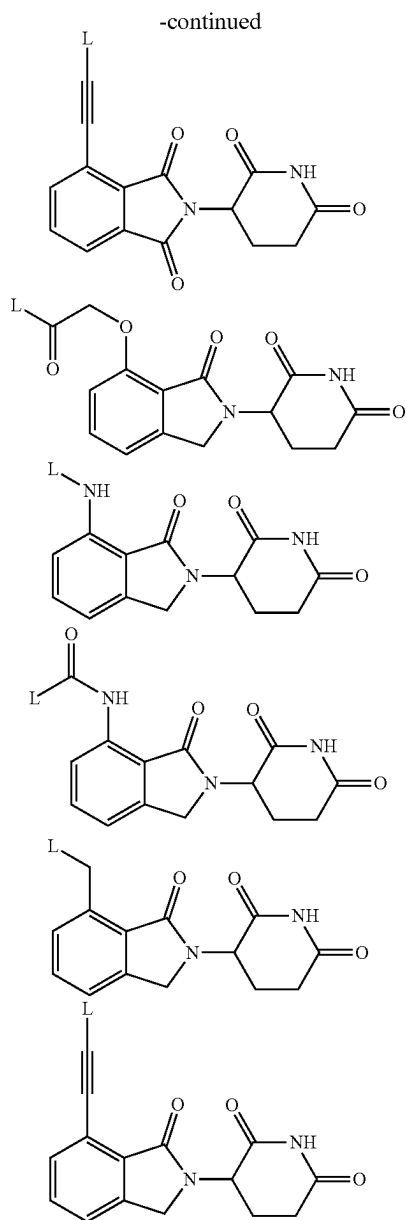
Formula IV



Formula V

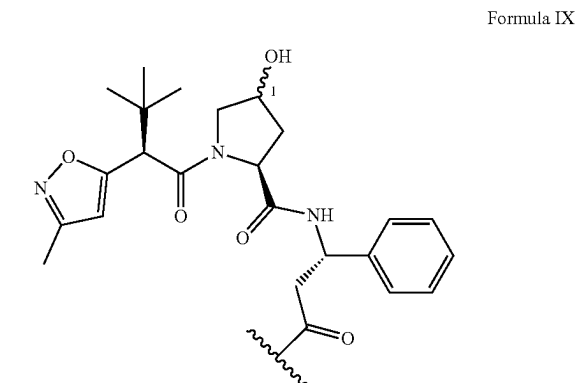
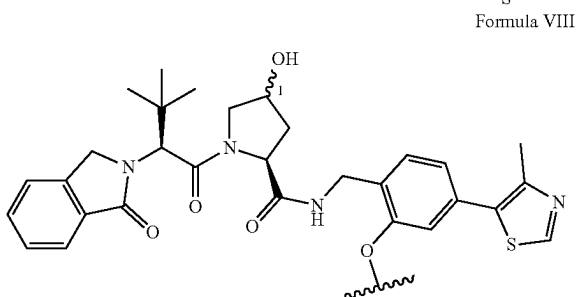
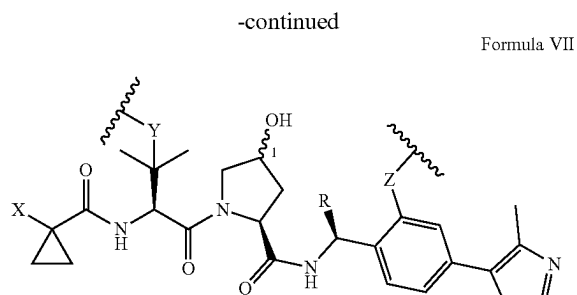
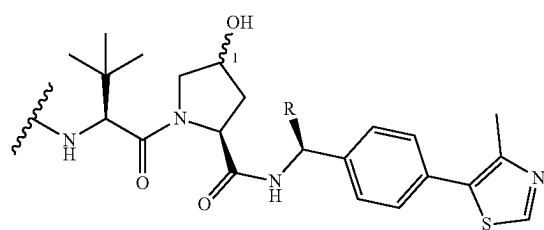
[0027] In these structures, X is CH₂ or CO; Y is NH, O, alkyne or CH₂. Exemplary CRBN-derived RBFs include but are not limited to:



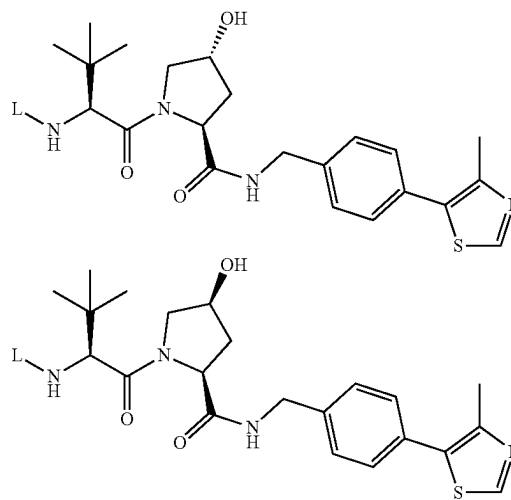


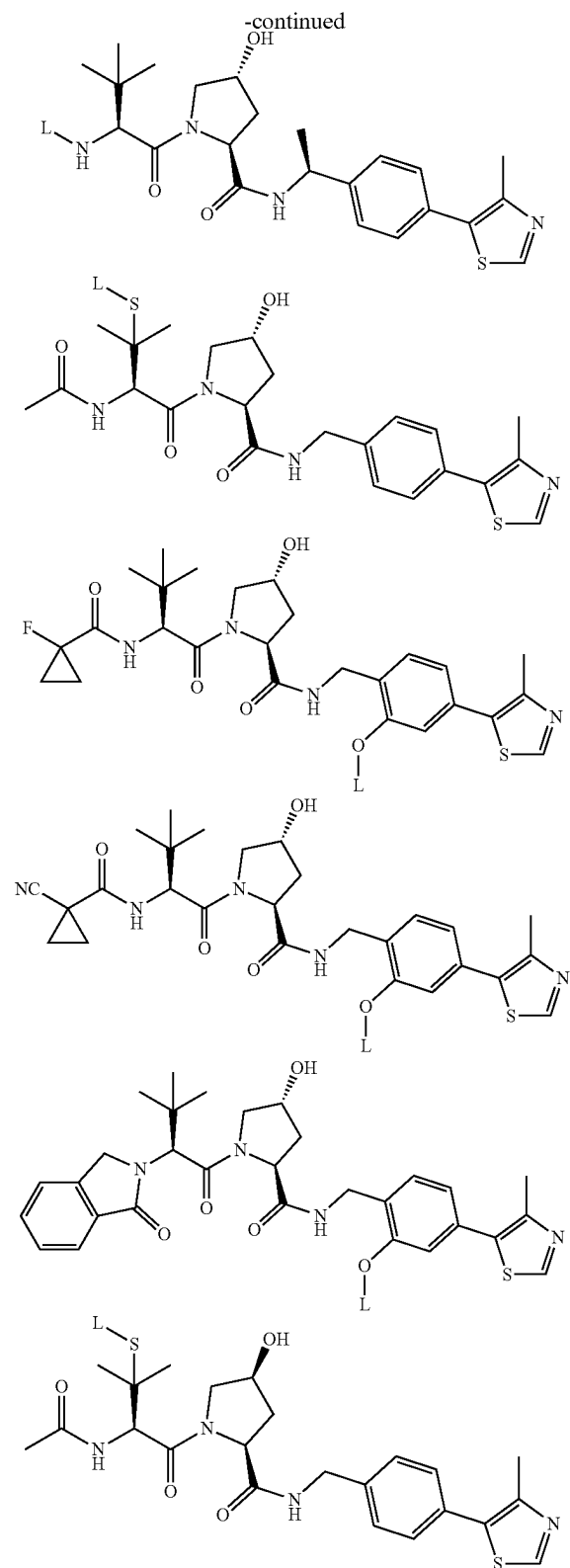
[0028] In some embodiments, the RBF moiety can be a VHL ligand with a structure shown in any one of Formulae VI-IX.

VHL Ligands (Von-Hippel Lindau):



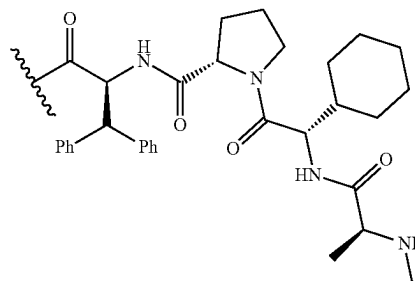
In these structures, stereochemistry of C1 is 1 or(S); R is H, CH₂ or CH₃; X is F or CN; Y is S or H; Z is O or H. Specific examples of VHL-derived RBFs include but are not limited to:



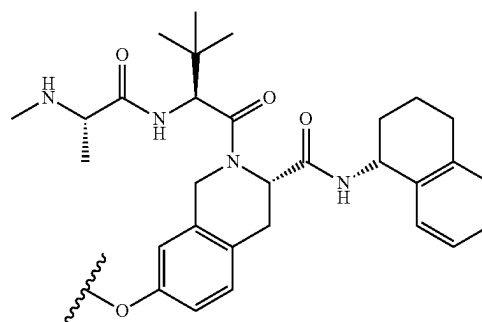


IAP Ligands (Inhibitor of Apoptosis Proteins):

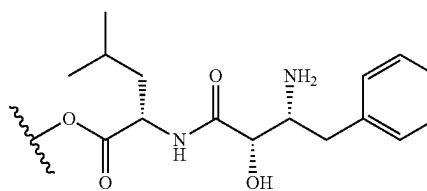
Formula X



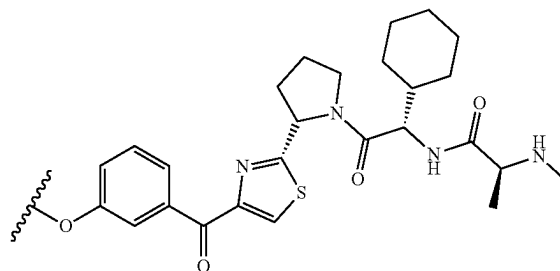
Formula XI



Formula XII



Formula XIII

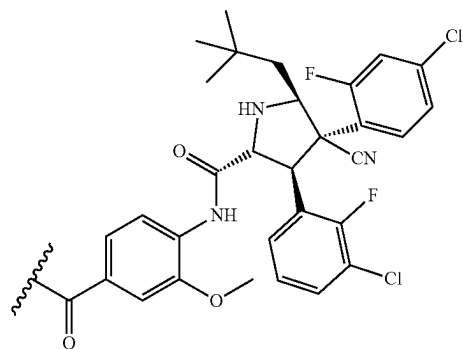
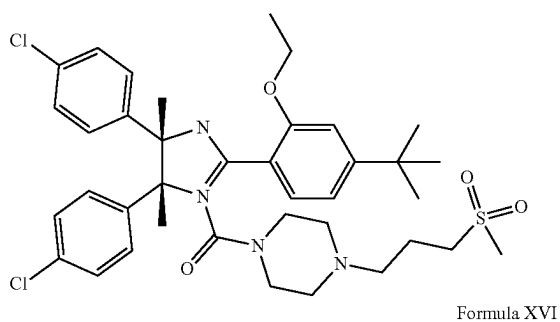
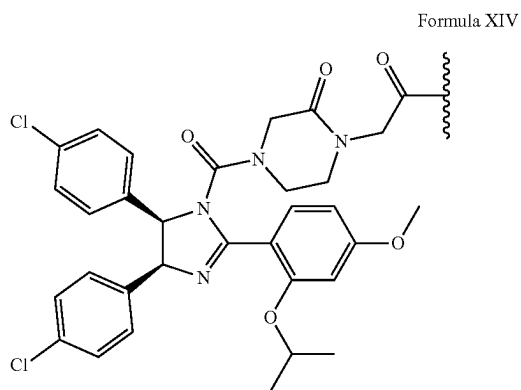


[0029] In some embodiments, the RBF moiety can be an IAP ligand with a structure shown in any one of Formulae X-XIII.

[0030] In some other embodiments, the RBF moiety can be a MDM2 ligand with a structure shown in any one of Formulae XIV-XVI.

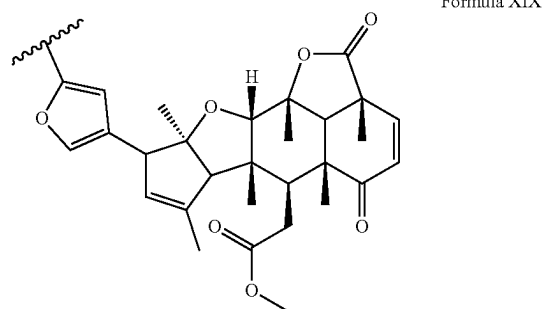
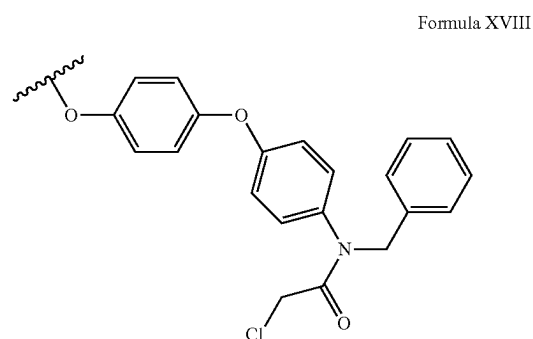
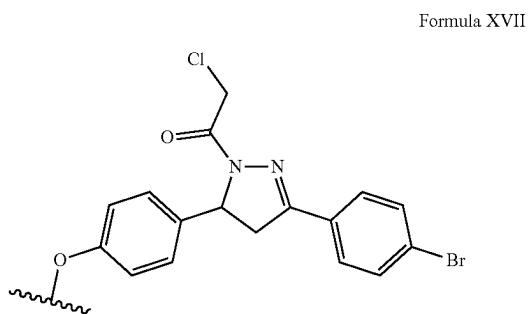
MDM2 Ligands:

-continued



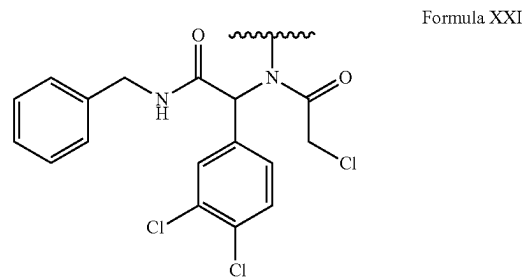
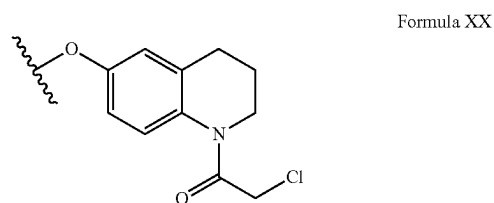
[0031] In still some other embodiments, the RBF moiety can be a RNF ligand with a structure shown in any one of Formulae XVII-XIX.

RNF Ligands (Ring Finger Proteins):



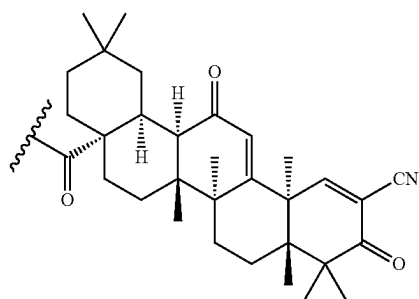
[0032] In some embodiments, the RBF moiety can be a DCAF ligand with a structure shown in any one of Formulae XX-XXI.

DCAF Ligands:

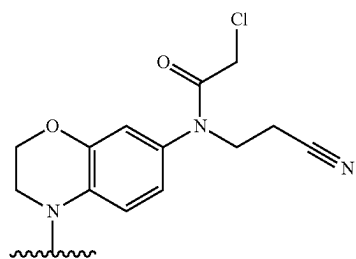


[0033] In some other embodiments, the RBF moiety can be a KEAP1 ligand or a FEM1B ligand with a structure shown in Formula XXII or XXIII, respectively.

KEAP1 Ligand:

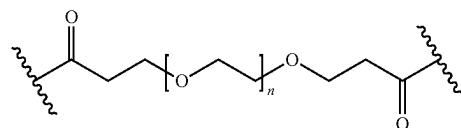


Formula XXII

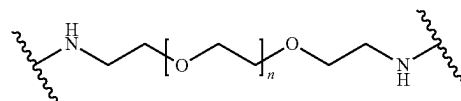


Formula XXIII

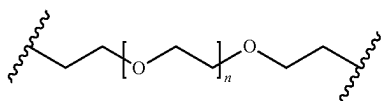
[0034] The linker (L) moiety in the FragTACs of the invention can employ any suitable compound or moiety that is capable of conjugating the PBF and RBF moieties without significant impact on their interactions with their cognate protein partners. In some embodiments, the L moiety can be a polyethylene glycol (PEG) linker with a structure shown in any one of Formulae XXIV-XXVI. In some other embodiments, the L moiety can be an aliphatic linker with a structure shown in any one of Formulae XXVII-XXX. In still some other embodiments, the L moiety can be a hybrid linker with a structure shown in any one of Formulae XXXI-XXXIX. In still some other embodiments, the L moiety can be an aryl-based linker with a structure shown in any one of Formulae XL and XLI. In some other embodiments, the L moiety can be a heterocycle-based linker with a structure shown in any one of Formulae XLII and XLIII. In some other embodiments, the L moiety can be a click chemistry-generated linker with a structure shown in any one of Formulae XLIV-XL VI. Polyethylene glycol (PEG) linkers:



Formula XXIV

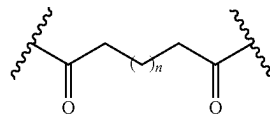


Formula XXV

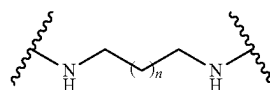


Formula XXVI

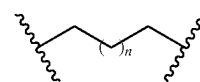
Aliphatic Linkers:



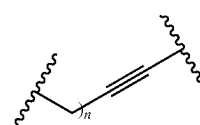
Formula XXVII



Formula XXVIII



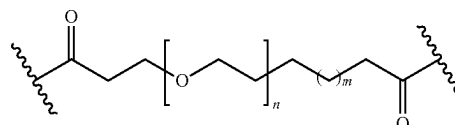
Formula XXIX



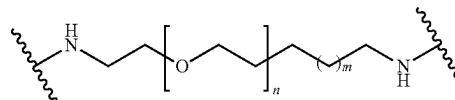
Formula XXX

In these structures, n is an integer between 0 and 10.

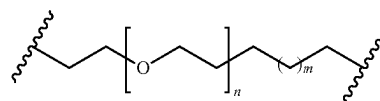
Hybrid Linkers:



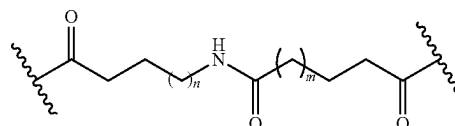
Formula XXXI



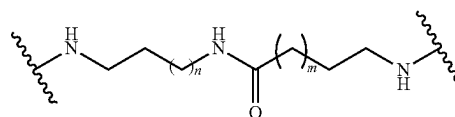
Formula XXXII



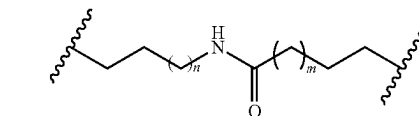
Formula XXXIII



Formula XXXIV



Formula XXXV

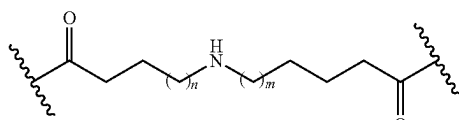


Formula XXXVI

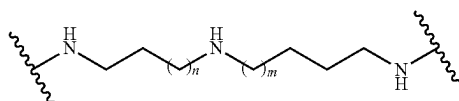
In these structures, n is an integer between 0 and 10.

-continued

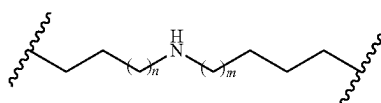
Formula XXXVII



Formula XXXVIII



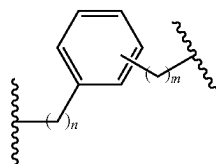
Formula XXXIX



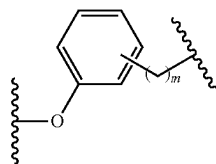
In these structures, n and m are each independently integers between 0 and 10.

Aryl-Based Linkers:

Formula XL



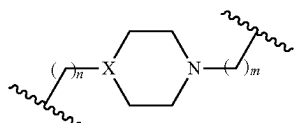
Formula XLI



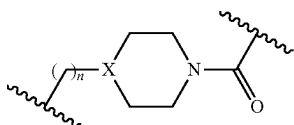
In these structures, n and m are each independently integers between 0 and 10.

Heterocycle-Based Linkers:

Formula XLII



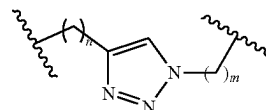
Formula XLIII



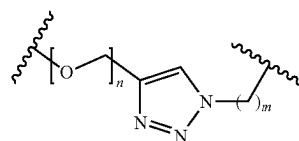
In these structures, n and m are each independently integers between 0 and 10; X is CH or N.

Click Chemistry-Generated Linkers:

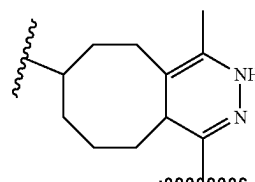
Formula XLIV



Formula XLV



Formula XLVI



In these structures, n and m are each independently integers between 0 and 10. Formulae XLIV and XLV are generated through the cycloaddition of an azide with an alkyne. Formula XLVI is generated through the cycloaddition of tetrazine and trans-cyclooctene.

[0035] One aspect of the present invention is directed to heterobifunctional compositions that use small molecule fragments appended to established E3 ligands. As illustrated in FIG. 5, these heterobifunctional compositions identify degradable targets and synthetically progressable ligands via unbiased whole proteome MS-based proteomics. This aspect of the invention has established that low target affinity interactions can lead to potent protein degradation.

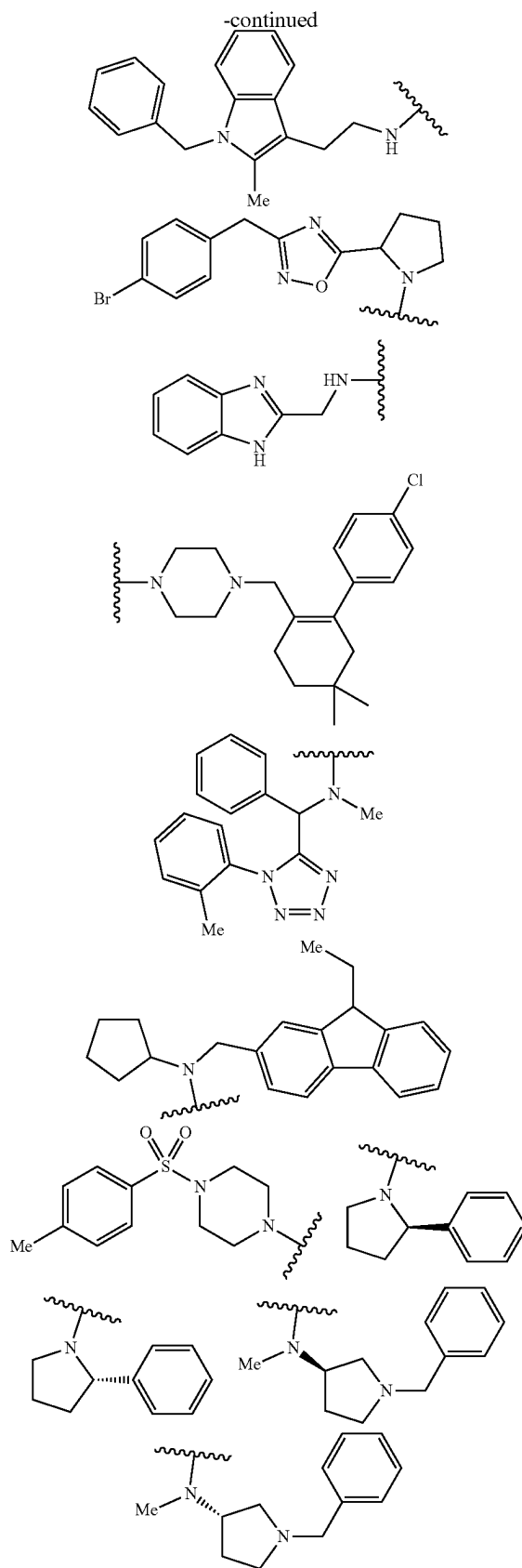
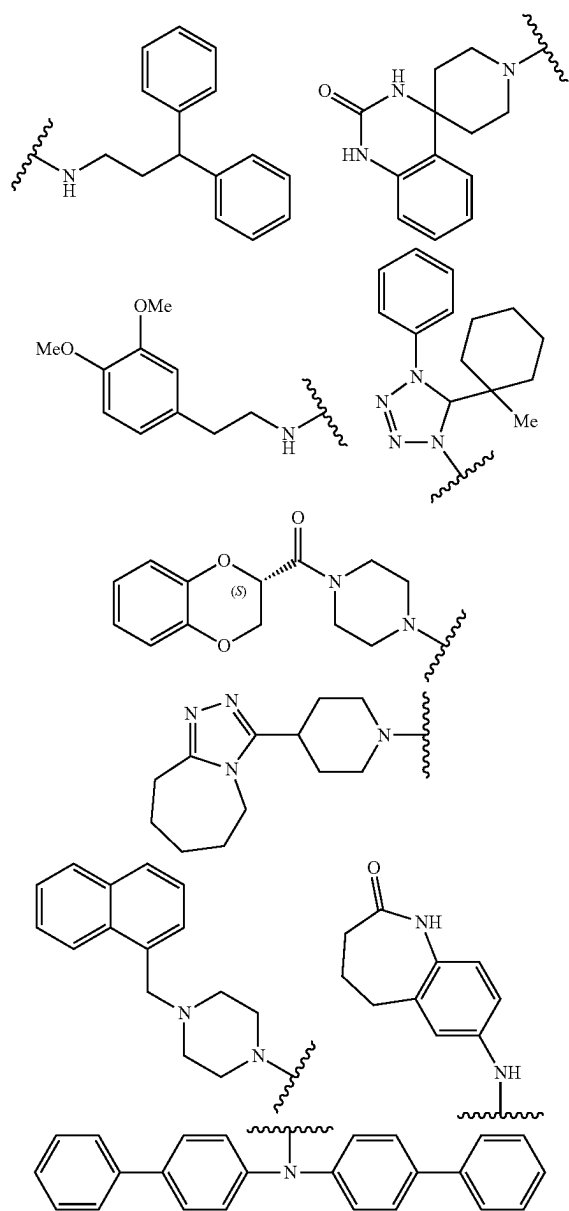
[0036] Embodiments of this aspect of the heterobifunctional compositions are directed to a small library which was designed and synthesized to incorporate of bifunctional, fragment-based degrader molecules ('FragTACs'). Generic Formula I, depicted above in the Summary, incorporates the protein binding terminus (PBT), the linker (L) and the recruiter binding terminus (RBT) as an embodiments of FragTACs. Exemplary embodiments of the PBT moiety include heterocyclic multicyclic organic molecular fragments listed below.

[0037] Exemplary embodiments of the L moiety include the linear organic chain fragments listed below. Exemplary embodiments of the RBT moieties include the organic molecular fragments listed below. These RBT moieties bind with the ligases as shown in the list including VHL, Cereblon, RNF-4, MDMR, RNF114, SNIPER and KEAP. Exemplary Cereblon and VHL ligand based FragTACs are listed below. The doses and percentages given in the list indicate the percent fragmentation at the given doses when the example is administered to a corresponding cell culture. See also FIG. 3A, eight members.

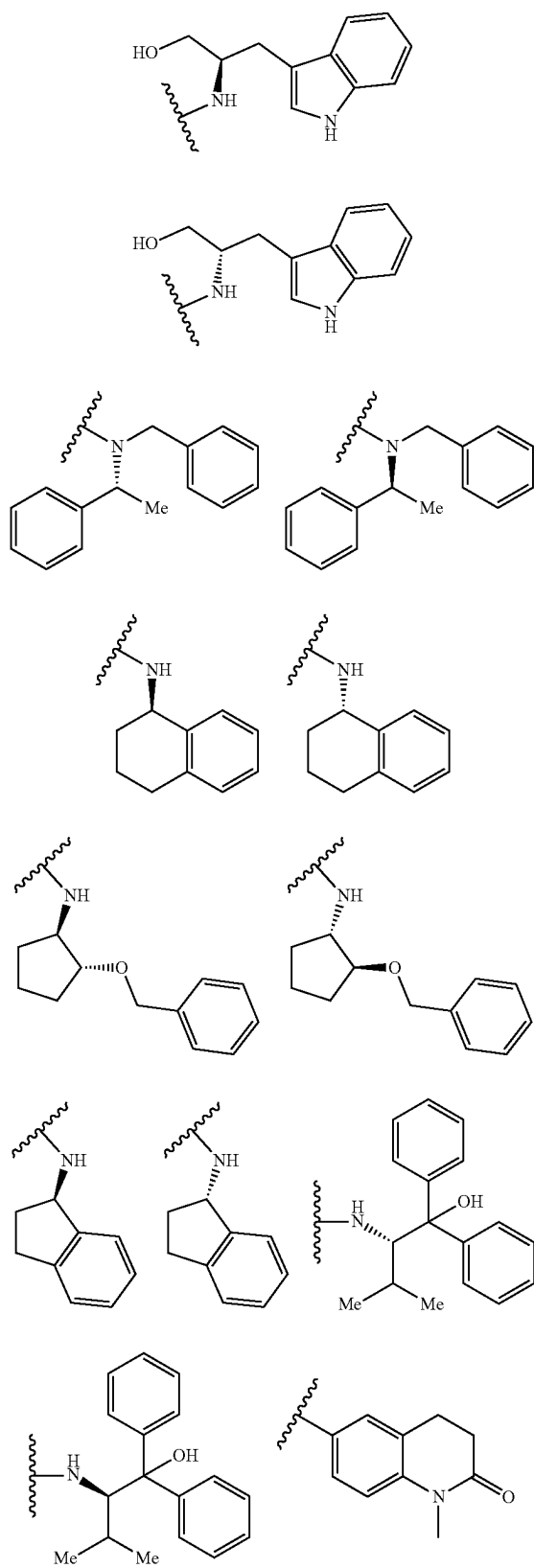
[0038] Focused embodiments of the heterobifunctional compositional FragTACs incorporate established von-Hippel Lindau (VHL) or Cereblon (CRBN) E3 ligands (18-20) which are chemically linked to any one of four fragment scaffolds. The fragment scaffolds were previously demonstrated to exhibit broad proteomic interactions (11). The ability of the FragTAC library to induce proteomic changes

in breast cancer cells was profiled and identified a combined ~120 downregulated proteins that span a broad range of classes (e.g. enzymes, transcription factors, chaperone proteins) as well as several members of protein complexes. FragTAC-1, in particular, substantially downregulated 43 proteins, including transcriptional repressor TDP-43, regulator protein CCAR2 and functionally uncharacterized protein FAM136A, which was confirmed to occur in a dose-dependent fashion (FIGS. 3B and 3C). Protein loss was blocked with the proteasome inhibitor MG132, suggesting that FragTAC-1 promotes proteasomal degradation (not shown). Notably, there are no known small molecule ligand for these proteins.

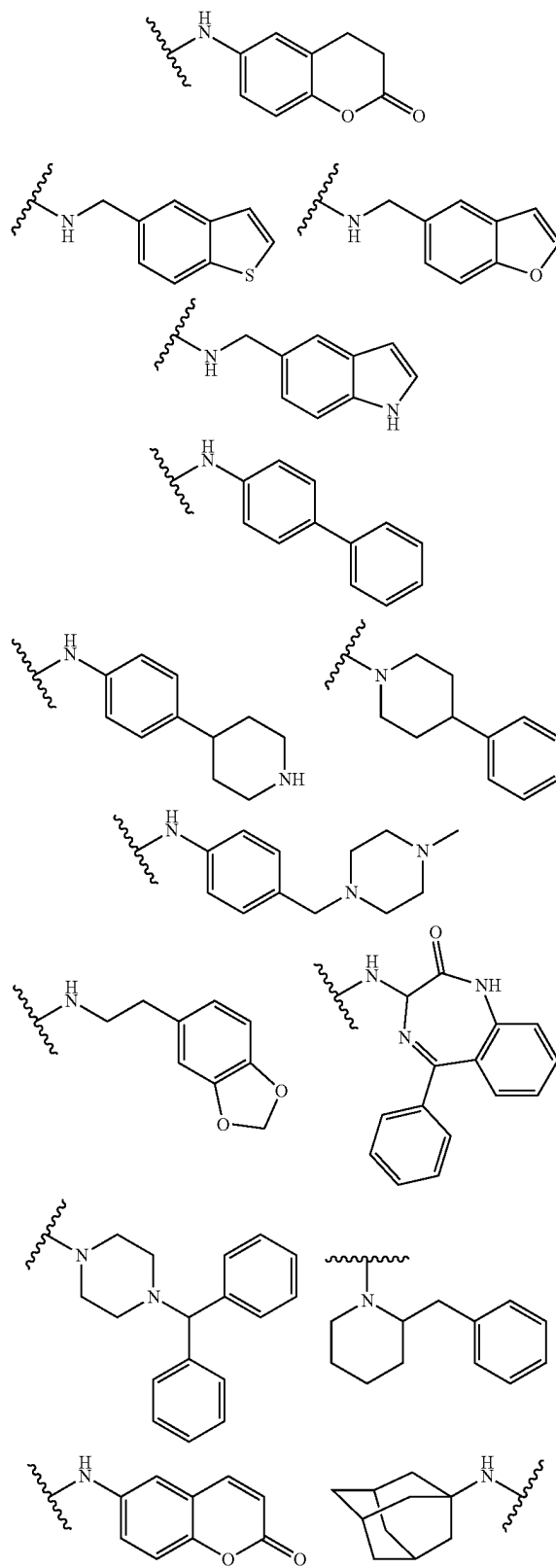
Examples of heterocyclic multicyclic organic molecular PBT fragments:



-continued

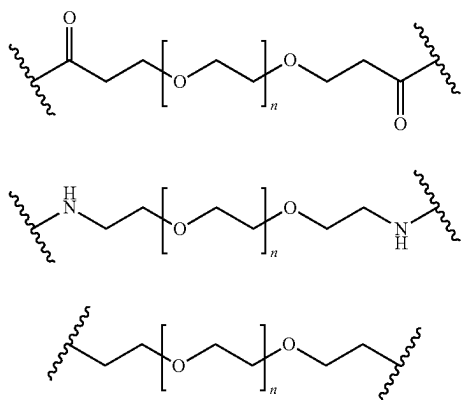


-continued

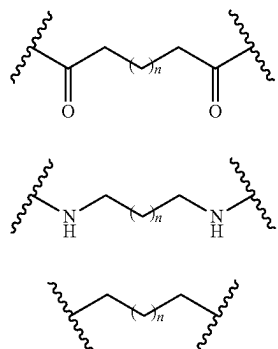


Examples of linear organic chain L moieties:

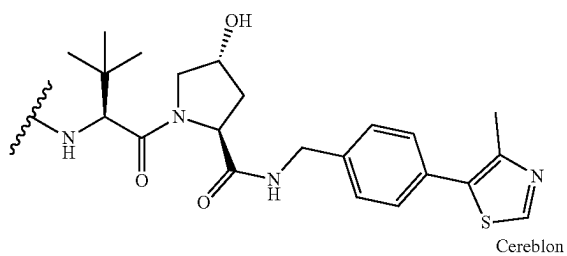
Peg Linkers



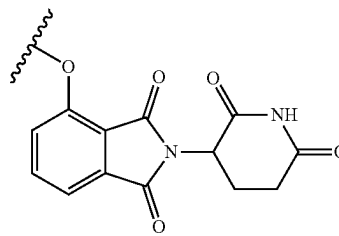
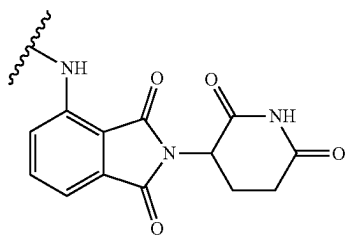
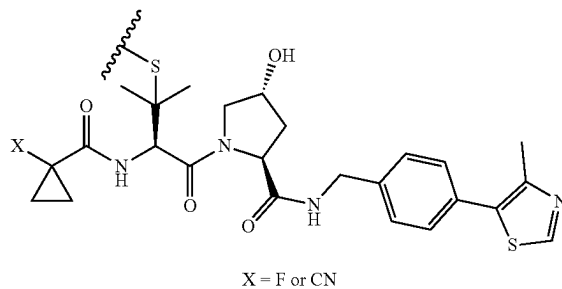
Aliphatic Linkers



Examples of organic molecular RBT moieties:

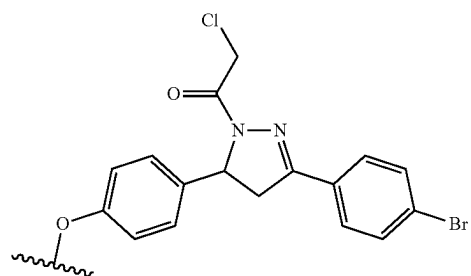


VHL

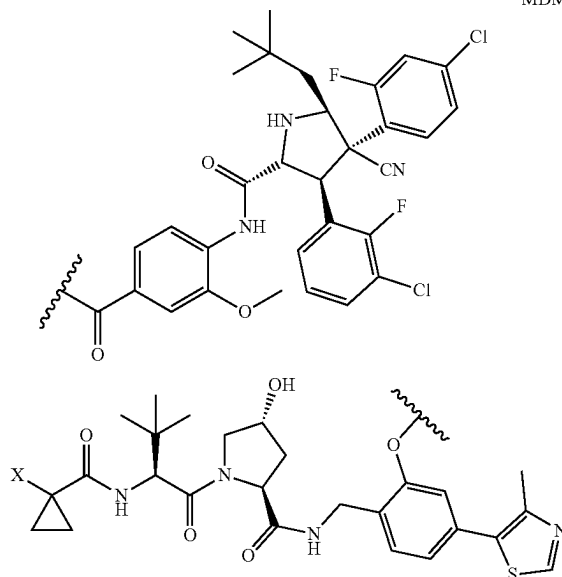


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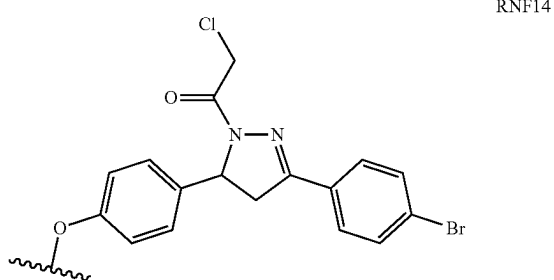
RNF4



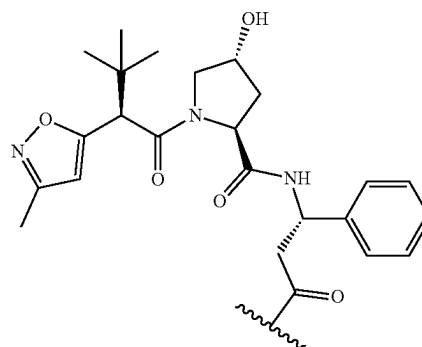
MDM2



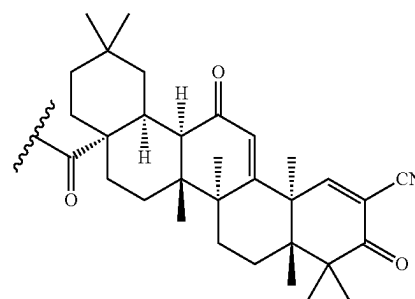
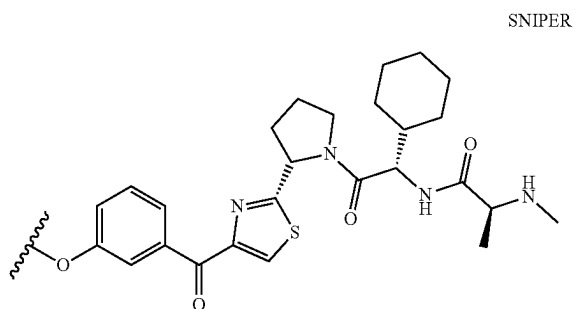
-continued



-continued

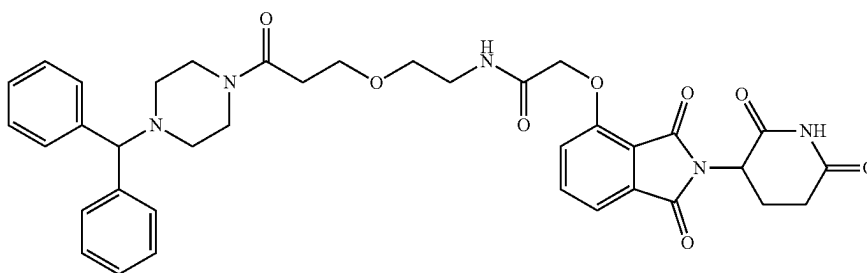


KEAP



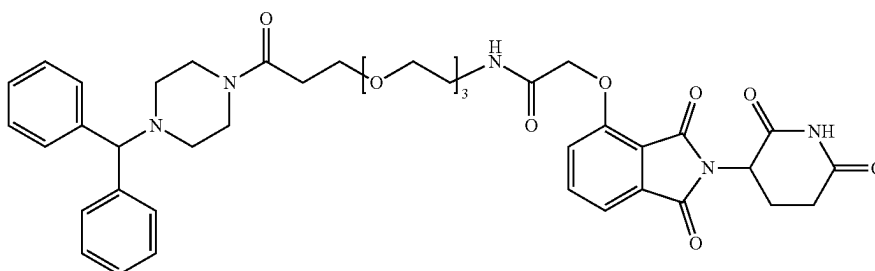
Cereblon Ligand-Based FragTAC Probes:

IF-014



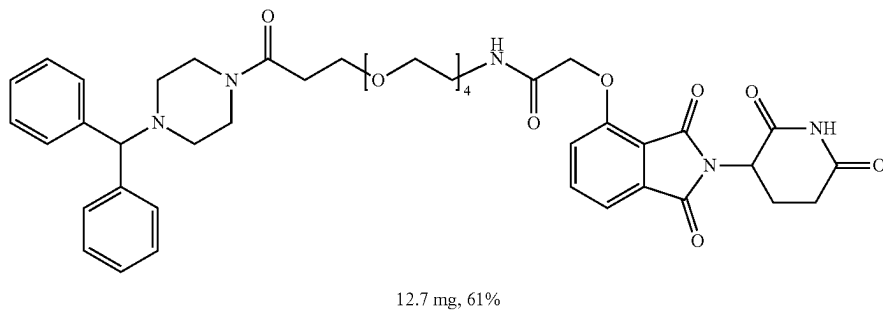
7.5 mg, 38%

IF-003



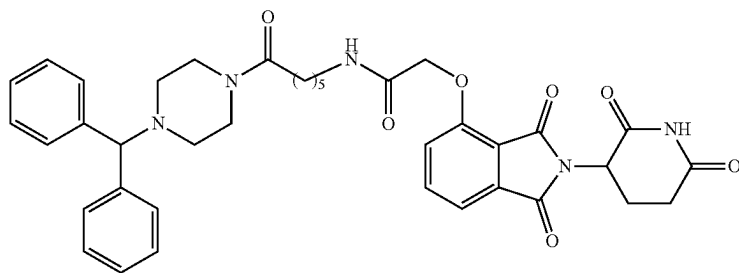
17.3 mg, 57%

IF-017



12.7 mg, 61%

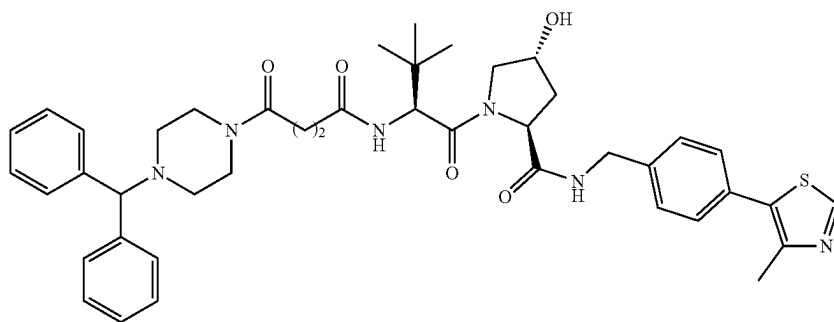
IF-027



25 mg, 99%

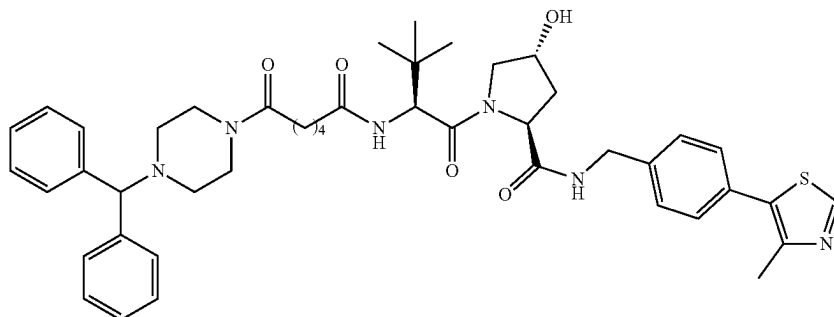
VHL Ligand-Based FragTAC Probes:

IF-038



4.2 mg, 44%

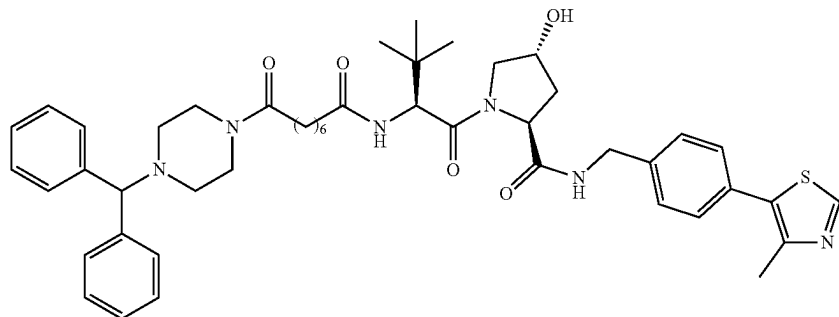
IF-041



3.3 mg, 26%

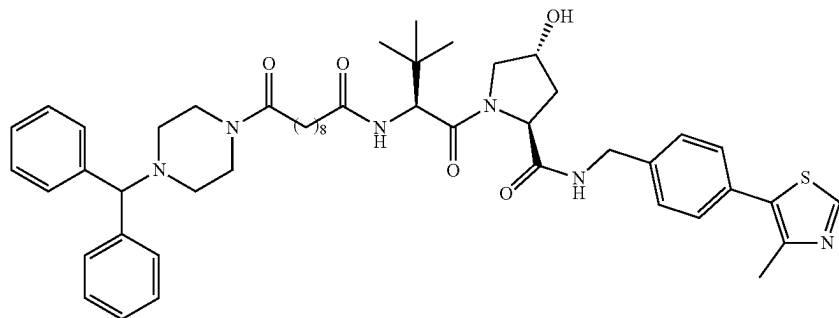
-continued

IF-044



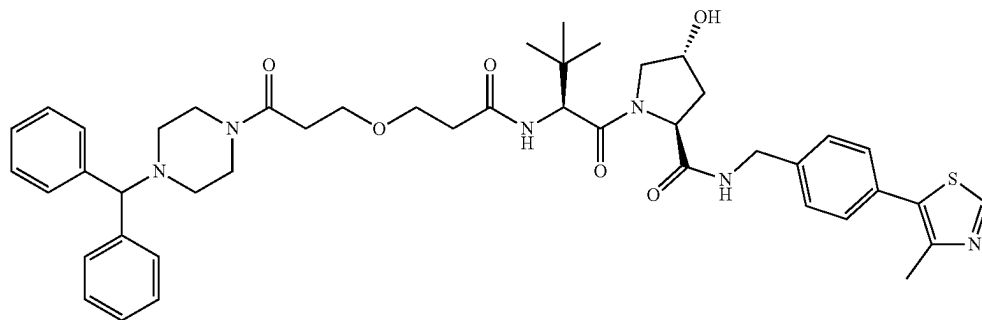
6.4 mg, 54%

IF-047



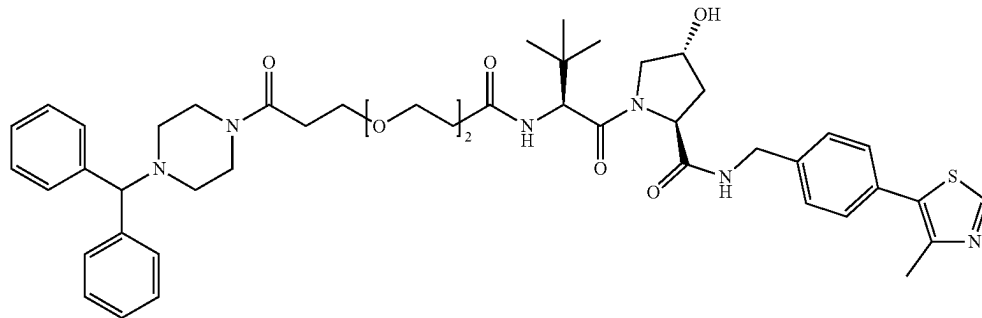
8.5 mg, 62%

IF-049

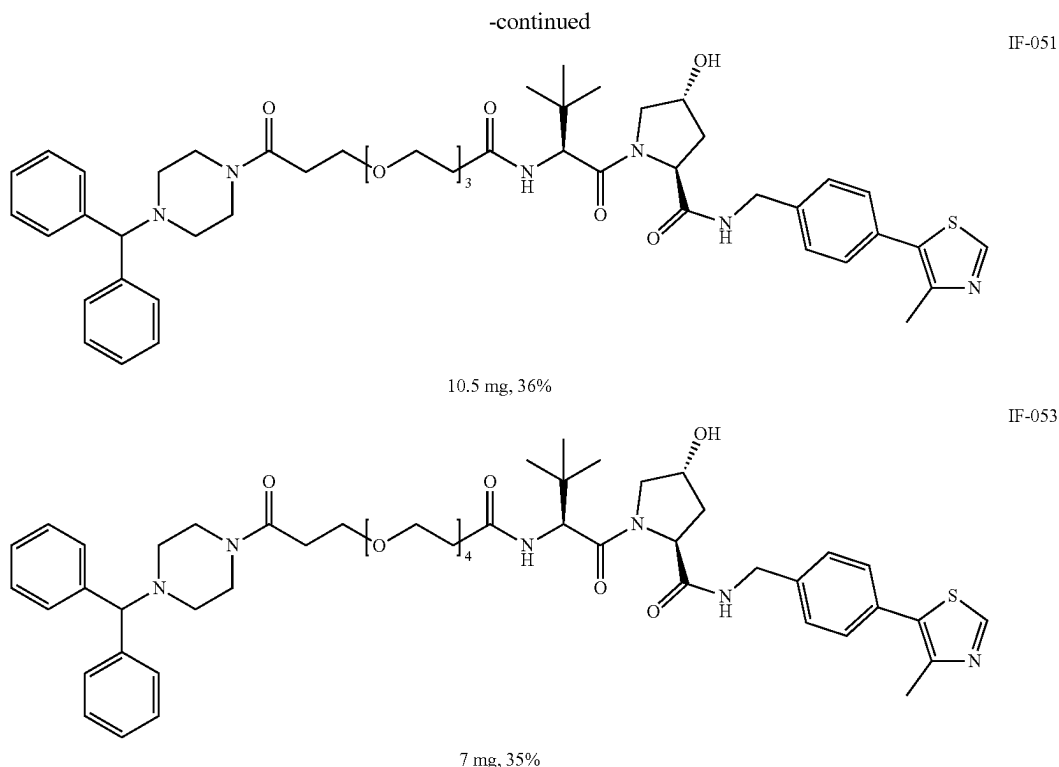


6.0 mg, 42%

IF-056



7.2 mg, 25%



De Novo Discovery of Degradable Protein Targets Using Fragment-Based PROTACs (FragTACs).

[0039] With the success in identifying degradable proteins using a small library of FragTACs, the strategy enveloping embodiments of the invention is expanded by synthesizing a larger FragTAC library (~300) and applying them to identify degradable targets in therapeutically relevant human immune model systems using multiplex proteomic workflows (FIGS. 4A-4D). The goal of this strategy enables 1) identification of characteristic chemical features of PROTACs that lead to successful TPD; 2) a gain of an understanding of the impact of ligand affinity and promiscuity in selectivity and efficiency; 3) comparatively assessment of routinely employed E3 ligands for their ability to induce TPD; 4) broad annotation of human immune targets that may be tractable to TPD; 5) establishment of a template to transition and optimize promiscuous fragment-based degraders to selective PROTACs for targets with compelling therapeutic or biological value.

[0040] Towards this end, a library may be prepared using ~30 small molecule fragments chemically linked to either CRBN or VHL E3 ligands (FIG. 4B). Fragments may be chosen based on their chemical diversity and non-overlapping proteomic interactions determined via the FbLDisC workflows (FIGS. 1A-1C) to maximize proteomic coverage. In initial studies, 4-5 unique linkers (e.g. PEG, aliphatic) may be utilized, enabling the generation of mini-libraries around each fragment that explore variables such as E3s and linker composition. This library may be screened in pooled primary peripheral blood mononuclear cells (PBMCs) from de-identified donors for their ability to induce TPD via unbiased quantitative TMT proteomics as established in our preliminary studies (FIG. 3B). Human PBMCs may be chosen as an initial model system, as they are composed of a diverse cell population (T cells, B cells, dendritic cells,

etc.) that contain uniquely expressed, immune-relevant targets, thus increasing the probability to uncover degradable, therapeutically translatable targets that serve critical roles in inflammation, infection, and cancer progression, for example. The variables discussed above may be encoded in the library to assess their independent contributions as well as establish a baseline of proteins susceptible to this TPD system. The FbLDisC workflows may be used to assess and optimize potency and selectivity of lead FragTACs, as needed, for prioritized targets. See FIGS. 2A-2B and FIGS. 3A-3C.

Chemoproteomic-Enabled Discovery of E3 Ligase-Binding Compounds that Support TPD in Human Cells.

[0041] Multiple E3 ligase components, including CRBN (31), VHL (32), RNF114 (33), and DCAF16 (34) have been shown to engage in tripartite complexes where bridging small molecules can direct specific protein substrates to ubiquitination and degradation. These findings, combined with the large number of E3 ligases (>600 (35)) in humans, have stimulated interest in the broader druggability potential of TPD (36). The recent proteome-wide ligandability maps using FbLDisC have identified hit ligands (or 'binders') for >40 E3 ligases (11, 13) (FIG. 4C). The capacity of these E3-binders to support TPD may be tested using a recently described dTAG model system (37) (FIG. 4D). E3 binders that induce degradation may be optimized into lead chemical probes using FbLDisC-guided medicinal chemistry as described in FIGS. 2A-2B. The chemical probes may be investigated for their ability to degrade targets with established ligands. Enzyme ligase E3 systems may be prioritized with restricted tissue expression, (e.g. in cancer or immune cells), as they may offer safer paths for drugs compared to broadly expressed E3 systems. To this end several fragment-based ligands for several E3 ligases have been validated. See Table 1.

[0042] Additional studies were performed to validate the activities of the FragTAC probes in HCC1806 cells via Western blot analyses. Results from these studies are shown in FIGS. 6-9.

TABLE 1

Examples E3s with newly validated ligands			
Gene	Type	Localization	Tissue
CUL1	Cullin		Lung fibroblasts
CUL4A	Cullin		
CUL4B	Cullin	Nucleus	
DDB2	DDB1-Adaptor	Nucleus; Accumulates at sites of damage following UV irradiation.	DNA Ubiquitously expressed; corneal endothelium
EDD1	HECT	Nucleus	Testis, brain, pituitary and kidney.
FBXO3	FBOX	Nucleus; Colocalizes with PML	
FBXO7	FBOX	Cytoplasm, Nucleus, Mitochondrion, Cytoplasm, Cytosol	
FBXO9	FBOX	Cytoplasm	
HUWE1	HECT	Cytoplasm, Nucleus	Heart, brain and placenta
PWP1	DDB1-Adaptor	Nucleus; Nucleolus; Chromosome	Placenta, skeletal muscle, kidney and pancreas
SYVN1	VBM	ER membrane	Ubiquitously expressed
TRIM21	RING	Cytoplasm; Cytoplasmic vesicle; Autophagosome; Nucleus; P-body	Heart and fetal lung.
UBR1	RING	Cytoplasm, Cytosol	Broadly, skeletal muscle, kidney and pancreas
UBR4	UIM	Membrane, Cytoplasm, Cytoplasm, Cytoskeleton, Nucleus	
UHRF1	RING	Nucleus; Localizes to replication foci	Thymus, bone marrow, testis, lung and heart. Overexpressed in breast cancer.

Results of Proteomics Studies

[0043] Examples of downregulated proteins targeted by FragTACs are listed in Table 2. The results show targets with

2 unique peptides or more and depleted 2-fold or more relative to DMSO control (P value <0.05). Compound structures can be found in the examples of FragTAC Probes described herein (Table 3).

TABLE 2

Types of proteins targeted by FragTAC probes		
Proteins	Accession Number	Probe Code
A2M	P01023	IF-051
AATF	Q9NY61	LPC020
ABCC5	O15440	LPC020
ABHD10	Q9NUJ1	LPC020
ABLM1	O14639	IF-051, CMC-6-123
ACAA2	P42765	LPC020
ACADM	P11310	LPC020
ACADSB	P45954	LPC020
ACAT1	P24752	LPC020
ACO2	Q99798	LPC020
ACOT1	Q86TX2	LPC020
ADAM15	Q13444	CMC-6-109, CMC-6-111, IF-072
ADIRF	Q15847	IF-051, IF-085
AGR2	O95994	LPC020
AHCTF1	Q8WYP5	LPC020
AHNAK	Q09666	IF-051
AHNAK2	Q8IVF2	IF-051
AHSG	P02765	LPC020, IF-085
AIDA	Q96BJ3	IF-051
AK2	P54819	LPC022, LPC020, IF-051, CMC-6-123, IF-049
AK3	Q9UIJ7	LPC020
AK4	P27144	LPC020
AKAP8	O43823	LPC020
ALAS1	P13196	CMC-6-123
ALCAM	Q13740	IF-051, LPC020, IF-049
ALDH1B1	P30837	LPC020
ALYREF	Q86V81	IF-051, LPC022, LPC020

TABLE 2-continued

Types of proteins targeted by FragTAC probes		
Proteins	Accession Number	Probe Code
ANAPC5	Q9UJX4	LPC020, LPC011
ANLN	Q9NQW6	CMC-6-109, CMC-6-111, IF-072, IF-080
ANPEP	P15144	IF-072
API5	Q9BZZ5	IF-051
APOB	P04114	LPC020
ARID1B	Q8NFD5	IF-051
ARMC10	Q8N2F6-2	IF-051, IF-049
ARPC5L	Q9BPX5	IF-051
ATAD2	Q6PL18	LPC011
ATP1A2	P50993	LPC020
ATP2B1	P20020	IF-051
ATP2B4	P23634	IF-051
ATP5F1A	P25705	LPC020
ATP5F1B	P06576	LPC020
ATP5F1C	P36542	LPC022, LPC020
ATP5F1D	P30049	LPC022, LPC020
ATP5PD	O75947	LPC020
ATP5PO	P48047	LPC020
ATPAF1	Q5TC12	LPC020
B4GALT1	P15291	LPC020
BACH1	O14867	IF-051, IF-049
BAZ1B	Q9UIG0	CMC-6-109, CMC-6-111, IF-072
BBX	Q8WY36	IF-051
BCAM	P50895	IF-051, CMC-6-109
BCKDK	O14874	IF-051
BCL2L12	Q9HB09	LPC020
BCLAF1	Q9NYF8	IF-051, LPC020
BICRA	Q9NZM4	IF-051
BOD1L1	Q8NFC6	IF-051, IF-049
BOLA3	Q53S33	IF-051
BPTF	Q12830	LPC020
BRD2	P25440	IF-051, IF-065
BRD8	Q9H0E9	IF-051
BRD9	Q9H8M2	IF-051
BTF3L4	Q96K17	IF-051
BUD13	Q9BRD0	IF-051, LPC020, IF-049
BUD31	P41223	IF-051
C2orf49	Q9BVC5	IF-051
C9orf78	Q9NZ63	IF-051
CACTIN	Q8WUQ7	IF-051
CAMLG	P49069	IF-051
CARD19	Q96LW7-2	IF-051, CMC-6-123
CASK	O14936	IF-051
CBLL1	Q75N03	IF-051
CBR4	Q8N4T8	LPC020
CBX1	P83916	IF-051, CMC-6-109, IF-072
CBX3	Q13185	IF-051, CMC-6-109, IF-072
CBX5	P45973	IF-051
CCAR2	Q8N163	LPC020
CCDC12	Q8WUD4	IF-051, LPC022, LPC020, CMC-6-123
CCDC124	Q96CT7	IF-051
CCDC137	Q6PK04	IF-051
CCDC58	Q4VC31	IF-051, LPC022, LPC020, CMC-6-123
CCDC90B	Q9GZT6	LPC011
CCNE2	O96020	LPC011
CCNL1	Q9UK58	LPC011
CCNT1	O60563	IF-051
CD109	Q6YHK3	IF-051, LPC020
CD151	P48509	IF-072
CD3EAP	O15446	IF-051
CD46	P15529	IF-051
CD58	P19256	IF-051, CMC-6-123
CD59	P13987	IF-051
CD74	P04233-2	CMC-6-109, CMC-6-111, IF-072
CD9	P21926	IF-049
CDC40	O60508	IF-051, LPC022, LPC020, IF-049
CDC5L	Q99459	IF-051, LPC022, LPC020
CDH1	P12830	IF-051, LPC022, LPC020, IF-049
CDH13	P55290	CMC-6-109, CMC-6-111, IF-072
CDH2	P19022	IF-051
CDH3	P22223	CMC-6-109, CMC-6-111, IF-072
CDK12	Q9NYV4	IF-051, IF-049

TABLE 2-continued

Types of proteins targeted by FragTAC probes		
Proteins	Accession Number	Probe Code
CDK2AP1	O14519	IF-051
CDK2AP2	O75956	IF-051, IF-049
CDK5RAP1	Q96SZ6	CMC-6-109, CMC-6-111, IF-072, LPC020
CDK5RAP1	Q96SZ6-3	LPC022
CENPC	Q03188	IF-051
CFL2	Q9Y281	IF-051
CHAF1B	Q13112	IF-051
CHAMP1	Q96JM3	CMC-6-109, CMC-6-111, IF-072
CHCHD1	Q96BP2	IF-051
CHCHD2	Q9Y6H1	IF-051, LPC020, IF-072, CMC-6-123
CHCHD2P9	Q5T1J5	IF-051
CHCHD4	Q8N4Q1	IF-051
CHD6	Q8TD26	IF-051
CHD7	Q9P2D1	LPC020
CHTOP	Q9Y3Y2	IF-051
CIRBP	Q14011	IF-051
CKAP2	Q8WWK9	LPC020
CKMT1A	P12532	IF-051, LPC022, LPC020
CLCC1	Q96S66	LPC022, LPC020
CLIC4	Q9Y696	IF-051
CLPB	Q9H078	IF-051, LPC020, CMC-6-123
CLPB	Q9H078-2	IF-051
CNPY2	Q9Y2B0	LPC020
CNST	Q6PJW8	IF-051, IF-049
COA7	Q96BR5	LPC022, LPC020, IF-072, CMC-6-123, IF-085
COL17A1	Q9UMD9	IF-051
COQ9	O75208	LPC020
COX20	Q5RI15	LPC020
COX6B1	P14854	LPC020
COX7A2	P14406	LPC020
COX7A2L	O14548	LPC022, LPC020
COX7C	P15954	CMC-6-123, LPC020
CPOX	P36551	LPC022, LPC020, IF-072, IF-051, IF-049
CPSF7	Q8N684	IF-051
CRYZ	Q08257	LPC020
CSRP1	P21291	IF-051
CSTF2	P33240	IF-051, LPC022, LPC020
CSTF2T	Q9H0L4	LPC022, LPC020
CSTF3	Q12996	IF-051, LPC022, LPC020, IF-049
CT45A10	P0DMU9	IF-051, IF-049
CTNNA1	P35221	IF-051
CTNNB1	P35222	IF-051
CTNND1	O60716-3	LPC020
CTSB	P07858	IF-051, LPC020, IF-072
CTSD	P07339	CMC-6-123, LPC020, IF-072
CTSH	P09668	IF-072, IF-051, CMC-6-123, CMC-6-111, CMC-6-109, IF-085
CTSS	P25774	IF-072
CTSV	O60911	IF-051
CTSZ	Q9UBR2	IF-051, IF-080, IF-085
CTTN	Q14247	IF-051
CUSTOS	Q96C57	IF-051
CUX1	P39880	IF-051, IF-049
CWC15	Q9P013	IF-051, LPC020
CYCS	P99999	IF-051, LPC022, LPC020, CMC-6-123
CYP1B1	Q16678	IF-073
DAP3	P51398	LPC020
DAZAP1	Q96EP5	IF-051, LPC020
DBN1	Q16643	IF-051
DDX21	Q9NR30	IF-051
DDX52	Q9Y2R4	LPC011
DFFA	O00273	IF-051
DHTKD1	Q96HY7	LPC020
DHX9	Q08211	IF-051
DIABLO	Q9NR28	LPC022, LPC020, IF-051, CMC-6-123, IF-049
DIDO1	Q9BTC0	IF-051, LPC022, LPC020
DNAJA3	Q96EY1	LPC020
DNAJB14	Q8TBM8	LPC020
DNAJC8	O75937	IF-051
DNASE2	O00115	LPC020, IF-072
DOLPP1	Q86YN1	LPC011
DPP7	Q9UHL4	CMC-6-123, IF-072

TABLE 2-continued

Types of proteins targeted by FragTAC probes		
Proteins	Accession Number	Probe Code
DSG2	Q14126	IF-051, IF-049
ECHS1	P30084	LPC020
ECI2	O75521	LPC020
ECSIT	Q9BQ95	LPC020
EDF1	O60869	IF-051
EFNB1	P98172	IF-051, IF-049
EFNB2	P52799	IF-051, CMC-6-123
EFR3A	Q14156	LPC020
EIF1	P41567	IF-051
ELAVL1	Q15717	LPC022, LPC020
ELOA	Q14241	IF-051
EMG1	Q92979	IF-051
EPN1	Q9Y613	IF-051, LPC022, LPC020
EPN2	O95208	IF-051
EPPK1	P58107	LPC020
ERAL1	O75616	LPC020
ERBIN	Q96RT1	LPC020
ERH	P84090	IF-051, LPC020
ERP29	P30040	IF-051, LPC020
ETFA	P13804	LPC020
ETFDH	Q16134	LPC020
EZR	P15311	IF-051
FAHD1	Q6P587	IF-051, LPC020, IF-049
FAM136A	Q96C01	IF-051, LPC022, LPC020, IF-049
FAM169A	Q9Y6X4	IF-051, IF-049
FAM177A1	Q8N128	IF-073
FAM49B	Q9NUQ9	IF-051
FAM83B	Q5T0W9	LPC020
FAM83H	Q6ZRV2	LPC020
FASTKD3	Q14CZ7	LPC020
FCF1	Q9Y324	IF-049
FDX1	P10109	LPC020
FDXR	P22570	LPC020
FECH	P22830	LPC020
FIP1L1	Q6UN15	IF-051, IF-049
FKBP1A	P62942	IF-051
FNBP4	Q8N3X1	IF-051
FSTL1	Q12841	IF-051
FTL	P02792	CMC-6-123, IF-085
FUBP3	Q96I24	LPC020
FUCA2	Q9BTY2	LPC020
FUS	P35637	IF-051, LPC022, LPC020
FYTDD1	Q96QD9	IF-051, LPC020, IF-049, CMC-6-123
FYTDD1	Q96QD9-2	LPC022, LPC020
GADD45GIP1	Q8TAE8	LPC020
GATAD2B	Q8WXI9	IF-051
GATB	O75879	LPC022, LPC020
GCSH	P23434	CMC-6-123, LPC020
GFER	P55789	IF-051, LPC020, IF-072, CMC-6-123
GFM2	Q96989	LPC020
GGH	Q92820	IF-072
GHITM	Q9H3K2	LPC022, LPC020, IF-072, IF-051, CMC-6-123, IF-060, IF-073, IF-062, IF-064, IF-085
GIPC2	Q8TF65	LPC011
GLA	P06280	IF-051
GLB1L2	Q8IW92	LPC020
GLO1	Q04760	IF-051
GLRX5	Q86SX6	LPC020
GLS	O94925-3	LPC022, LPC020
GLUD1	P00367	LPC020
GM2A	P17900	IF-051, CMC-6-123, IF-085
GMFB	P60983	IF-051
GNG12	Q9UBI6	LPC020
GPATCH4	Q5T3I0	IF-051
GPC1	P35052	IF-051, CMC-6-111, IF-072, LPC020
GPKOW	Q92917	CMC-6-109, IF-051, IF-072
GPT2	Q8TD30	LPC022, LPC020
GRPEL2	Q8TAA5	LPC022, LPC020
GRSF1	Q12849	LPC020
GSTK1	Q9Y2Q3	LPC020
GTF2B	Q00403	IF-051, IF-049

TABLE 2-continued

Types of proteins targeted by FragTAC probes		
Proteins	Accession Number	Probe Code
GTF2I	P78347	CMC-6-109, IF-072
H1FX	Q92522	CMC-6-109, CMC-6-111, IF-072
H3F3A	P84243	LPC011
HADHB	P55084	LPC020
HARS2	P49590	LPC020
HAX1	O00165	CMC-6-109, IF-072
HCCS	P53701	LPC020
HCLS1	P14317	IF-051
HEBP1	Q9NRV9	IF-051, LPC020
HEXB	P07686	CMC-6-123
HIBCH	Q6NVY1	LPC020
HINT1	P49773	IF-051
HINT2	Q9BX68	IF-051, LPC020
HIST1H1C	P16403	CMC-6-123
HIST1H1T	P22492	IF-049
HIST2H2AB	Q8IUE6	IF-051
HIST2H2BF	Q5QNW6	IF-051
HIST2H3A	Q71DI3	LPC011
HMGA1	P17096-2	IF-051
HMGB2	P26583	IF-051
HMGCL	P35914	LPC020
HMGNI	P05114	IF-051
HMGNI2	P05204	IF-051
HMGNI4	O00479	IF-051
HMGNI5	P82970	IF-051
HNRNPA0	Q13151	IF-051, LPC022, LPC020
HNRNPA1	P09651	IF-051, LPC020
HNRNPA1	P09651-2	LPC020
HNRNPA2B1	P22626	LPC022, LPC020
HNRNPA3	P51991	LPC022, LPC020
HNRNPAB	Q99729-2	IF-051
HNRNPAB	Q99729-3	IF-051, LPC020
HNRNPCL1	O60812	LPC020
HNRNPD	Q14103	IF-051
HNRNPDL	O14979	IF-051, LPC022, LPC020
HNRNPH1	P31943	LPC022, LPC020
HNRNPH2	P55795	LPC022, LPC020
HNRNPH3	P31942	LPC020
HNRNPK	P61978-2	IF-051, LPC020, LPC011
HNRNPL	P14866	IF-051, LPC022, LPC020
HNRNPM	P52272	LPC020
HNRNPR	O43390	IF-051, LPC020
HNRNPR	O43390-2	LPC020
HNRNPUL2	Q1KMD3	LPC020
HPCAL1	P37235	IF-051
HPDL	Q96IR7	LPC020
HSD17B10	Q99714	LPC020
HSDL2	Q6YN16	LPC020
HSPD1	P10809	LPC020
HSPE1	P61604	CMC-6-123
HTRA2	O43464	IF-051, LPC022, LPC020, IF-049
IBA57	Q5T440	LPC020
ICE1	Q9Y2F5	CMC-6-111, IF-072
IDH3G	P51553	IF-051, LPC020, IF-049
IFI30	P13284	IF-072, IF-051, CMC-6-111, CMC-6-109, IF-049
IGFBP7	Q16270	IF-051
IK	Q13123	IF-051, LPC020, IF-049
IL6ST	P40189	IF-051, LPC020, CMC-6-123
ILF2	Q12905	IF-051
ILF3	Q12906-7	IF-051
IMMT	Q16891	LPC020
INCENP	Q9NQS7	IF-080
INF2	Q27J81	IF-051, LPC020, IF-049, CMC-6-123
ISCA1	Q9BUE6	LPC020
ISCU	Q9H1K1	IF-051, CMC-6-123
ISG20L2	Q9H9L3	IF-051, CMC-6-123
ISOC2	Q96AB3	LPC020
ITGA3	P26006	CMC-6-109, CMC-6-111, IF-072
ITIH4	Q14624	IF-051
ITM2B	Q9Y287	LPC020
JUP	P14923	IF-051
KCTD5	Q9NXV2	LPC011

TABLE 2-continued

Types of proteins targeted by FragTAC probes		
Proteins	Accession Number	Probe Code
KDEL2	Q7Z4H8	LPC020
KDM2A	Q9Y2K7	LPC020, LPC011
KHSRP	Q92945	IF-051
KIAA0391	O15091	LPC020
KIAA1217	Q5T5P2	IF-051
KIF20A	O95235	IF-080
KIF22	Q14807	CMC-6-109, CMC-6-111, IF-072, IF-080
KIF23	Q02241	CMC-6-109, CMC-6-111, IF-072
KIFC1	Q9BW19	CMC-6-109, CMC-6-111, IF-072
KLF5	Q13887	LPC020
KRT1	P04264	CMC-6-111
KRT10	P13645	CMC-6-111, LPC034
KRT13	P13646	LPC020
KRT14	P02533	LPC020
KRT17	Q04695	LPC020
KRT18	P05783	LPC020
KRT19	P08727	IF-051, LPC020, IF-049
KRT2	P35908	CMC-6-111, LPC034
KRT5	P13647	LPC020
KRT6A	P02538	LPC020
KRT7	P08729	LPC020
KRT79	Q5XKE5	LPC020
KRT8	P05787	LPC020
KRT9	P35527	CMC-6-111
KTN1	Q86UP2	IF-051
L1RE1	Q9UN81	LPC011
L2HGDH	Q9H9P8	LPC020
LAMA3	Q16787-1	IF-051
LAMTOR2	Q9Y2Q5	LPC020
LARP1	Q6PKG0	IF-051
LASP1	Q14847	IF-051
LCOR	Q96JN0-3	IF-051, IF-049
LGALS1	P09382	IF-051
LGMN	Q99538	LPC022, LPC020, IF-072, IF-051, CMC-6-123, IF-049, IF-073, IF-085
LMNA	P02545	IF-051
LMNB1	P20700	IF-051
LMNB2	Q03252	IF-051
LMO7	Q8WWI1	IF-051, IF-072, CMC-6-123
LMTK2	Q8IWU2	IF-051
LNPEP	Q9UIQ6	IF-051
LPXN	O60711	CMC-6-109, CMC-6-111, IF-072
LRCH3	Q96I18	LPC011
LRIF1	Q5T3J3	LPC022, LPC020
LSM14A	Q8ND56	IF-051
LSM2	Q9Y333	IF-051
LSM6	P62312	IF-051
LSM7	Q9UK45	IF-051
LSM8	O95777	IF-051
LVRN	Q6Q4G3	CMC-6-109, CMC-6-111, IF-072
LYPD3	O95274	IF-051, IF-049
LYRM7	Q5U5X0	IF-049
MAGOH	P61326	IF-051
MALSU1	Q96EH3	LPC020
MAN1A2	O60476	IF-051
MAN2B1	O00754	IF-072
MANF	P55145	IF-051
MAP7	Q14244	LPC011
MARCH5	Q9NX47	LPC020
MATR3	P43243	IF-051, LPC022, LPC020, IF-049
MAVS	Q7Z434	IF-051, IF-049
MBP	P02686	IF-051
MCAT	Q8IVS2	LPC020
MCM3AP	O60318	LPC020
MDC1	Q14676	IF-051, LPC020
ME2	P23368	LPC020
MED1	Q15648	LPC020
MED11	Q9P086	LPC011
MED21	Q13503	IF-051
MED4	Q9NPJ6	LPC020
MEN1	O00255	CMC-6-109, CMC-6-111, IF-072

TABLE 2-continued

Types of proteins targeted by FragTAC probes		
Proteins	Accession Number	Probe Code
MESD	Q14696	IF-051
MESD6	Q6ZSS7	IF-049
MGA	Q81WI9	LPC020
MGME1	Q9BQP7	IF-049
MIC13	Q5XKP0	LPC022, LPC020
MICA	Q29983	IF-051
MICU1	Q9BPX6	LPC020
MIF	P14174	IF-051
MKI67	P46013	IF-072, IF-051, CMC-6-111, CMC-6-109, IF-049
MMP1	P03956	IF-051
MMTAG2	Q9BU76	LPC020
MNAT1	P51948	LPC020
MOB1B	Q7L9L4	IF-051
MPRIIP	Q6WCQ1	IF-051
MRPL1	Q9BYD6	LPC020
MRPL10	Q7Z7H8	LPC020
MRPL12	P52815	IF-051, CMC-6-123
MRPL13	Q9BYD1	LPC022, LPC020
MRPL14	Q6P1L8	LPC022, LPC020
MRPL15	Q9P015	LPC020
MRPL16	Q9NX20	LPC022, LPC020
MRPL17	Q9NRX2	LPC022, LPC020
MRPL20	Q9BYC9	LPC020
MRPL21	Q7Z2W9	LPC020
MRPL23	Q16540	LPC022, LPC020
MRPL24	Q96A35	LPC020
MRPL3	P09001	LPC020
MRPL33	O75394	CMC-6-123, LPC020
MRPL34	Q9BQ48	CMC-6-123
MRPL37	Q9BZE1	LPC022, LPC020
MRPL38	Q96DV4	LPC020
MRPL39	Q9NYK5	LPC020
MRPL4	Q9BYD3	LPC020
MRPL40	Q9NQ50	LPC020
MRPL41	Q8IXM3	LPC022, LPC020, IF-049, LPC011
MRPL42	Q9Y6G3	LPC020
MRPL45	Q9BRJ2	LPC022, LPC020
MRPL46	Q9H2W6	LPC020
MRPL47	Q9HD33	LPC020
MRPL49	Q13405	LPC020
MRPL50	Q8N5N7	LPC020
MRPL52	Q86TS9	LPC020
MRPL54	Q6P161	IF-049
MRPL9	Q9BYD2	LPC020
MRPS10	P82664	LPC020
MRPS11	P82912	LPC020
MRPS12	O15235	LPC022
MRPS15	P82914	LPC020
MRPS16	Q9Y3D3	LPC020
MRPS17	Q9Y2R5	LPC020
MRPS18B	Q9Y676	LPC020
MRPS18C	Q9Y3D5	LPC022, LPC020
MRPS2	Q9Y399	LPC020
MRPS22	P82650	LPC020
MRPS23	Q9Y3D9	LPC022, LPC020
MRPS25	P82663	LPC020
MRPS26	Q9BYN8	LPC020
MRPS28	Q9Y2Q9	LPC020
MRPS30	Q9NP92	LPC022, LPC020
MRPS31	Q92665	LPC020
MRPS34	P82930	LPC020
MRPS35	P82673	LPC020
MRPS6	P82932	LPC020
MRPS7	Q9Y2R9	LPC020
MRPS9	P82933	LPC020
MSLN	Q13421	IF-051, LPC020, CMC-6-123
MSLN	Q13421-2	IF-051, IF-049
MSN	P26038	IF-051
MT-CO2	P00403	LPC020
MT-ND1	P03886	LPC020
MT1X	P80297	IF-051
MTDH	Q86UE4	IF-051

TABLE 2-continued

Types of proteins targeted by FragTAC probes		
Proteins	Accession Number	Probe Code
MTERF3	Q96E29	LPC020
MTFR1	Q15390	IF-051, CMC-6-123, IF-085
MTG1	Q9BT17	LPC022, LPC020
MTHFD1L	Q6UB35	LPC020
MTHFD2	P13995	LPC020
MTREX	P42285	IF-051
MXRA7	P84157	IF-051, IF-049
MYDGF	Q969H8	IF-051
MYO19	Q96H55	IF-072
NAGA	P17050	IF-072
NCBP3	Q53F19	IF-051
NCL	P19338	IF-051
NCOA4	Q13772	LPC020
NCOA5	Q9HCD5	IF-051, IF-049
NDUFA10	O95299	LPC020
NDUFA13	Q9P0J0	LPC020
NDUFA2	O43678	LPC020
NDUFA4	O00483	LPC022, LPC020, IF-049
NDUFA5	Q16718	LPC020
NDUFA7	O95182	IF-049
NDUFA8	P51970	LPC020
NDUFA9	Q16795	LPC020
NDUFAB1	O14561	IF-051, IF-049
NDUFAF2	Q8N183	LPC020
NDUFAF4	Q9P032	LPC020
NDUFB10	O96000	LPC020
NDUFB4	O95168	LPC020
NDUFS1	P28331	LPC020
NDUFS2	O75306	LPC020
NDUFS3	O75489	LPC020
NDUFS6	O75380	LPC020
NDUFS8	O00217	LPC020
NDUFV1	P49821	LPC020
NDUFV2	P19404	LPC020
NECTIN1	Q15223	IF-051, LPC020, IF-049, CMC-6-123
NECTIN2	Q92692	IF-051, LPC020
NENF	Q9UMX5	IF-051
NFS1	Q9Y697	LPC020
NFU1	Q9UMS0	IF-051, IF-049
NGRN	Q9NPE2	LPC020
NOC4L	Q9BV14	LPC020
NOL12	Q9UGY1	IF-051, IF-049
NOL7	Q9UMY1	IF-051
NOLC1	Q14978-2	IF-051
NOLC1	Q14978-3	LPC011
NONO	Q15233	IF-051, LPC020
NOP53	Q9NZM5	LPC020
NPC2	P61916	IF-072, IF-051, CMC-6-123, IF-073, IF-085
NPM3	O75607	LPC020
NRAS	P01111	LPC020
NSD2	O96028	IF-051
NSUN5	Q96P11	LPC020
NT5E	P21589	IF-051
NUCB1	Q02818	CMC-6-123, IF-085
NUMA1	Q14980	IF-051, LPC020, IF-049
NUP37	Q8NFH4	LPC020
NUP50	Q9UKX7	IF-051
NUP54	Q7Z3B4	LPC020
NUP62	P37198	LPC020
NUSAP1	Q9BXS6	IF-051
NVL	O15381	IF-051, CMC-6-109, CMC-6-111, IF-072
NXF1	Q9UBU9	IF-051
NXT1	Q9UKK6	LPC020
OAS1	P00973	IF-051
OAT	P04181	LPC020
OCLN	Q16625	IF-051
OGDH	Q02218	LPC020
OVOL1	O14753	IF-051
OXA1L	Q15070	LPC020
P4HB	P07237	IF-051
PABPN1	Q86U42	IF-051, LPC020, IF-049
PAM16	Q9Y3D7	LPC020

TABLE 2-continued

Types of proteins targeted by FragTAC probes		
Proteins	Accession Number	Probe Code
PANX1	Q96RD7	LPC020
PARL	Q9H300	IF-051, LPC020
PATZ1	Q9HBE1	CMC-6-111, IF-072
PAWR	Q96IZ0	IF-051
PBRM1	Q86U86	LPC020
PCF11	O94913	IF-051
PCLAF	Q15004	IF-073
PDAP1	Q13442	IF-051
PDE6C	P51160	IF-051
PDHB	P11177	LPC020
PDIA3	P30101	IF-051
PDIA6	Q15084	LPC020
PEBP1	P30086	IF-051
PHB	P35232	LPC020
PHF14	O94880	IF-072, LPC011
PHF3	Q92576	IF-051, LPC022, LPC020
PHIP	Q8WWQ0	IF-072
PICALM	Q13492	IF-051
PKP1	Q13835	CMC-6-111, IF-072
PKP2	Q99959-2	IF-051
PLEC	Q15149-3	LPC020
PLEC	Q15149-4	IF-051
PLEKHA6	Q9Y2H5	IF-072
PLRG1	O43660	IF-051
PLSCR1	O15162	LPC020
PLSCR3	Q9NRY6	LPC020
POFUT1	Q9H488	IF-051, LPC020
POFUT2	Q9Y2G5-1	LPC020
POGZ	Q7Z3K3	IF-072
POLDIP2	Q9Y2S7	LPC020
POLR2D	O15514	LPC020
POLR2F	P61218	IF-051, CMC-6-123
POLR2G	P62487	IF-051
POLRMT	O00411	LPC020
POM121C	A8CG34	IF-051, LPC020
POSTN	Q15063	IF-051
PPIB	P23284	IF-051
PPIF	P30405	IF-051, IF-049
PPP1R9B	Q96SB3	IF-051, IF-049
PPT1	P50897	IF-051, IF-049, IF-072
PQBP1	O60828	IF-051, LPC020, IF-049
PRC1	O43663	IF-080
PRKN	O60260-5	LPC020
PRKRIP1	Q9H875	IF-051
PRNP	P04156	IF-051, LPC022, LPC020
PRPF19	Q9UMS4	IF-051
PRSS37	A4D1T9	LPC020
PSAP	P07602	IF-051, LPC020, IF-049
PSEN1	P49768	IF-051
PSIP1	O75475	IF-051
PSPC1	Q8WXF1	IF-051
PTBP1	P26599	IF-051, LPC020
PTBP3	O95758-1	LPC020
PTCD1	O75127	LPC020
PTK7	Q13308	IF-051, IF-049
PTMA	P06454	IF-051, IF-085
PTPMT1	Q8WUK0	LPC020
PTRH1	Q86Y79	LPC020, LPC011
PYM1	Q9BRP8	IF-051, CMC-6-123
QRSL1	Q9HOR6	LPC022
RACGAP1	Q9H0H5	IF-072, IF-080, IF-051, CMC-6-111, CMC-6-109, IF-049
RAD21	O60216	IF-051
RAIY	Q9UKM9	LPC020
RAVER1	Q8IY67	LPC020
RBBP6	Q7Z6E9	IF-051
RBBP7	Q16576	IF-051
RBM14	Q96PK6	LPC022, LPC020
RBM15	Q96T37	LPC020
RBM19	Q9Y4C8	IF-051, IF-049
RBM22	Q9NW64	IF-051
RBM27	Q9P2N5	LPC020
RBM3	P98179	CMC-6-109, IF-051

TABLE 2-continued

Types of proteins targeted by FragTAC probes		
Proteins	Accession Number	Probe Code
RBM4	Q9BWF3	LPC020
RBM6	P78332	IF-051
RBM8A	Q9Y5S9	IF-051
RBMX	P38159	IF-051, LPC020, IF-049
RBMXL1	Q96E39	IF-051
RDX	P35241	IF-051
REEP5	Q00765	IF-051, LPC022, LPC020, IF-049
REXO4	Q9GZR2	IF-051, LPC020, IF-049
RGS10	O43665	IF-051, IF-049
RIF1	Q5UIP0	IF-051, LPC020, IF-049
RMDN1	Q96DB5	IF-051, LPC020, CMC-6-123
RNASET2	O00584	CMC-6-123
RNF121	Q9H920	IF-051
RNMT	O43148	IF-051
RP9	Q8TA86	IF-051
RPF1	Q9H9Y2	LPC020, LPC011
RPLP1	P05386	IF-073
RPS21	P63220	IF-051
RPUSD3	Q6P087	LPC020
RREB1	Q92766	IF-051
RRN3	Q9NYV6	LPC011
RRP36	Q96EU6	IF-051
RSBN1	Q5VWQ0	IF-051
RSF1	Q96T23	IF-051
RSRC1	Q96IZ7	IF-051
S100A10	P60903	IF-051
S100A14	Q9HCY8	IF-051, LPC020, IF-049
SAFB	Q15424	IF-051, LPC020, IF-049
SAFB2	Q14151	IF-051, LPC022, LPC020, IF-049
SAMD1	Q6SPF0	IF-051, IF-049
SAMM50	Q9Y512	LPC020
SAP130	Q9H0E3	IF-051
SARNP	P82979	IF-051
SCAF11	Q99590	IF-051
SCAF8	Q9UPN6	IF-051
SCFD1	Q8WVM8	IF-051
SCO1	O75880	IF-051
SDC4	P31431	IF-051
SDF2L1	Q9HCN8	IF-051
SDHA	P31040	LPC020
SDHAF2	Q9NX18	LPC020
SDHB	P21912	LPC020
SELENOI	Q9C0D9	LPC011
SERPINH1	P50454	IF-051
SETX	Q7Z333	IF-073
SF1	Q15637-6	LPC020
SF3A1	Q15459	IF-051
SF3A3	Q12874	IF-051
SF3B2	Q13435	IF-051
SF3B4	Q15427	LPC020
SFPQ	P23246	IF-051
SFSWAP	Q12872	IF-051
SFXN1	Q9H9B4	LPC020
SFXN3	Q9BWM7	LPC020
SHMT2	P34897	LPC020
SLC16A1	P53985	IF-051
SLC1A4	P43007	IF-051
SLC1A5	Q15758	IF-051
SLC25A10	Q9UBX3	LPC020
SLC25A6	P12236	LPC020
SLC30A1	Q9Y6M5	LPC020, IF-072
SLC3A2	P08195	IF-051
SLC43A3	Q8NB15	IF-051
SLC48A1	Q6P1K1	LPC020
SLC6A6	P31641	LPC020
SLC9A1	P19634	IF-051
SLC9A3R1	O14745	IF-051
SLC9A3R2	Q15599	IF-051
SLIRP	Q9GZT3	LPC020
SLMAP	Q14BN4	IF-051, IF-049
SLU7	O95391	LPC022, LPC020
SMARCB1	Q12824	IF-051, LPC020

TABLE 2-continued

Types of proteins targeted by FragTAC probes		
Proteins	Accession Number	Probe Code
SMARCD1	Q96GM5	IF-051
SMARCD2	Q92925-2	LPC020
SMIM12	Q96EX1	IF-051
SMU1	Q2TAY7	IF-051
SNIP1	Q8TAD8	LPC020
SNRNP70	P08621	IF-051
SNRPA1	P09661	IF-051
SNRPC	P09234	IF-051
SNRPD1	P62314	LPC011
SNW1	Q13573	IF-051
SON	P18583-5	LPC020
SON	P18583-6	LPC020
SORL1	Q92673	IF-049
SORT1	Q99523	CMC-6-109, IF-072
SOX15	O60248	IF-051
SPECC1	Q5M775	IF-051
SPIN1	Q9Y657	IF-051
SPINT1	O43278	IF-051, IF-049
SPRY4	Q9C004	IF-051
SPRYD4	Q8WW59	IF-049
SRCAP	Q6ZRS2	IF-051
SREK1	Q8WXA9-2	IF-051
SRP9	P49458	IF-051
SRRT	Q9BXP5	IF-051
SRSF9	Q13242	LPC020
SSBP1	Q04837	LPC020
ST14	Q9Y5Y6	IF-051, LPC020, IF-049, IF-072
STK24	Q9Y6E0	IF-051
STOML2	Q9UJZ1	LPC020
STX12	Q86Y82	IF-051
STX18	Q9P2W9	IF-051
STX5	Q13190	IF-051
STX7	O15400	IF-051
SUGP1	Q8IWZ8	IF-051
SUMF2	Q8NB17	LPC020
SYAP1	Q96A49	IF-051
SYF2	O95926	IF-051
SYMPK	Q92797	IF-049
SYNE2	Q8WXH0	IF-051
TACC1	O75410	IF-051, IF-049
TACO1	Q9BSH4	LPC020
TAF11	Q15544	IF-049
TAF15	Q92804	IF-051, LPC020
TAGLN2	P37802	IF-051
TARDBP	Q13148	LPC022, LPC020
TBCA	O75347	IF-051
TBL1XR1	Q9BZK7	IF-051
TBP	P20226	LPC020
TCF20	Q9UGU0	IF-051
TCOF1	Q13428-3	IF-051
TCOF1	Q13428-7	LPC020
TEX264	Q9Y619	IF-051
TGFB1	Q15582	IF-051, CMC-6-123
THOC7	Q619Y2	LPC020
THRAP3	Q9Y2W1	LPC020
TIMM13	Q9Y5L4	IF-051, LPC022, LPC020
TIMM17A	Q99595	LPC020
TIMM23B	Q5SRD1	LPC022, LPC020
TIMM29	Q9BSF4	IF-051, IF-049
TIMM50	Q3ZCQ8	LPC020
TK1	P04183	CMC-6-109, CMC-6-111, IF-072
TM6IM6	P55061	LPC049, LPC011, IF-065, IF-063
TMEM106B	Q9NUM4	IF-051, CMC-6-123
TMEM126A	Q9H061	LPC020
TMEM214	Q6NUQ4	IF-072
TMEM237	Q96Q45	IF-049
TMEM30A	Q9NV96	IF-051
TMEM40	Q8WWA1	LPC020
TMPO	P42166	LPC020
TMSB4X	P62328	IF-051
TNFRSF12A	Q9NP84	LPC020
TOMM6	Q96B49	LPC020

TABLE 2-continued

Types of proteins targeted by FragTAC probes		
Proteins	Accession Number	Probe Code
TOMM7	Q9P0U1	LPC020
TOP2A	P11388	CMC-6-109, CMC-6-111, IF-072, LPC011
TOR1AIP2	Q8NFQ8	IF-051
TOR4A	Q9NXH8	LPC049
TP53BP1	Q12888	IF-051, LPC020
TPP1	O14773	LPC020, IF-072
TPST1	O60507	LPC020
TPX2	Q9ULW0	CMC-6-109, CMC-6-111, IF-072, IF-080
TRIM28	Q13263	CMC-6-109, CMC-6-111, IF-072
TRIM33	Q9UPN9	IF-072
TRMT10C	Q7L0Y3	LPC020
TSFM	P43897	LPC020
TSNAX	Q99598	LPC011
TST	Q16762	LPC020
TTC19	Q6DKK2	IF-051, IF-049, CMC-6-123
TUFM	P49411	LPC020
TXN2	Q99757	IF-051, LPC020
TXNDC12	O95881	IF-051
TXNDC5	Q8NBS9	IF-051
TXNIP	Q9H3M7	IF-073, IF-080, IF-085
TXNL4A	P83876	IF-051
TYW1	Q9NV66	LPC011
U2AF2	P26368	IF-051
U2SURP	O15042	LPC020
U2SURP	O15042-2	IF-051
UBE2G2	P60604	IF-051
UBE2J2	Q8N2K1	IF-051
UHRF1	Q96T88	CMC-6-109, CMC-6-111, IF-072, IF-080
URB1	O60287	LPC020
USP36	Q9P275	IF-051, IF-049
UTP15	Q8TED0	LPC011
UTP18	Q9Y5J1	LPC011
VAMP7	P51809	IF-051, IF-049
VDAC1	P21796	LPC020
VDAC2	P45880	LPC020
VIM	P08670	IF-051, CMC-6-123
VIRMA	Q69YN4	LPC020
VSIR	Q9H7M9	CMC-6-109, CMC-6-111, IF-072
VTI1B	Q9UEU0	IF-051
WDFY1	Q8IWB7	IF-072
WDR33	Q9C0J8	LPC020
WDR76	Q9H967	CMC-6-109, CMC-6-111, IF-072
WTAP	Q15007	IF-051, IF-049
XPNPEP3	Q9NQH7	LPC020
XRN2	Q9H0D6	IF-051
YAP1	P46937	IF-051
YARS2	Q9Y2Z4	LPC020
YIPF3	Q9GZM5	IF-051, LPC020, IF-049
YLPM1	P49750	IF-051, IF-049
ZBTB1	Q9Y2K1	IF-051, CMC-6-123, IF-085
ZBTB11	O95625	IF-080
ZC3H13	Q5T200	IF-051, LPC020
ZC3H14	Q6PJT7	IF-051, IF-072
ZC3H18	Q86VM9	IF-051
ZC3HC1	Q86WB0	LPC020
ZCCHC8	Q6NZY4	IF-051
ZFR	Q96KR1	LPC022, LPC020
ZMYM4	Q5VZL5	LPC022, LPC020
ZNF148	Q9UQR1	IF-051
ZNF185	O15231	IF-051, IF-049
ZNF217	O75362	IF-051, CMC-6-123, IF-085
ZNF512B	Q96KM6	IF-051
ZNF638	Q14966	IF-051, LPC020
ZNF800	Q2TB10	IF-072
ZNF830	Q96NB3	IF-051, LPC020

Biological Materials and Methods

[0044] Materials: Rabbit anti-SAFB1 (PA5-2135P, 1:3,000 dilution) and rabbit anti-FAM136A (PA5-56345, 1:1,000 dilution) were ordered from Life Technologies. Rabbit anti-SAFB2 (11642-1-AP, 1:3,000 dilution), rabbit anti-TARDBP (10782-2-AP, 1:1,000 dilution) were ordered from Proteintech. Rabbit anti-PTMA (PA5-71580, 1:1,000 dilution) was ordered from Invitrogen. Rabbit anti-LGMN (93627S, 1:1,000), rabbit anti-PPP1R9B (14136S, 1:1,000 dilution), rabbit anti-DBC1 (5693S, 1:1,000 dilution), rabbit anti-MAVS (3993S, 1:1,000 dilution), rabbit anti-Beta-actin (4967S, 1:1,000 dilution), rabbit anti-histone H3 (9715S, 1:1,000 dilution), mouse anti-HA (2367S, 1:2,000 dilution), rabbit anti-HA (3724S, 1:2,000 dilution), rabbit anti-FLAG (14793, 1:1,000 dilution), mouse anti-FLAG (8146S, 1:1,000 dilution), anti-rabbit HRP (7074P2, 1:10,000 dilution) were ordered from Cell Signaling Technology. Epoxomicin (A2606, proteasome inhibitor) was ordered from APEx Bio. MG132 (HY-13259, proteasome inhibitor) and MLN4924 (HY-70062, neddylation inhibitor) were ordered from MedChem Express.

[0045] Mammalian cell culture: MDA-MB-231 cells (ATCC) were maintained in DMEM supplemented with 10% (v/v) fetal bovine serum (FBS), 1% (v/v) penicillin/streptomycin, and 2 mM glutamine. HCC1806 cells (ATCC) were maintained in RPMI supplemented with 10% (v/v) fetal bovine serum (FBS), 1% (v/v) penicillin/streptomycin, and 2 mM glutamine. All cell lines were grown at 37° C. in a humidified 5% CO₂ atmosphere.

[0046] Treatment of Live Cells with FragTAC Probes: Separate 6 cm (for gel-based analyses) or 10 cm (for MS-based analyses) dishes of MDA-MB-231 cells or HCC1806 cells were grown to 80-95% confluency with DMEM media and RPMI media respectively. The growth medium was aspirated, and the cells were washed with Dulbecco's phosphate buffered saline (DPBS). The cells were incubated with serum-free media containing fragment probes (200 µM) or control probes (200 µM) for 6 hours at 37° C. under a 5% CO₂ atmosphere. The cultures were scraped, washed with cold DPBS, and collected into 15 mL centrifuge tubes, then transferred to 1.5 mL Eppendorf tubes. The cell suspensions were centrifuged (1,400 g, 3 min) and the pellets were stored at -80° C. until the next stage of processing.

[0047] Sample preparation for TMT MS Quantitative Proteomics: For treatment of cell lysates, cells were collected, cell pellets were resuspended in 100 µL DPBS containing 1× Halt protease inhibitor cocktail, and lysed by sonication (15 ms on, 40 ms off, 15% amplitude, 1 s total). Protein concentrations were normalized (2 mg/mL in 100 µL with cold DPBS) using the Lowry Protein Assay (Pierce). Urea (48 mg) was weighed for each sample in 2 mL LoBind Eppendorf tubes and cell lysates (100 µL=200 µg) were added (final urea concentration=8 M). Pellets were resuspended in a freshly prepared 1:1 solution (50 µL) TCEP (200 mM in DPBS) and K₂CO₃ (600 mM in DPBS) and incubated (30 min, 37° C.) while shaking. After reaction, a solution of freshly prepared iodoacetamide (70 µL, 400 mM in DPBS) was added and incubated for 30 min at room temperature while protected from light. After reaction, 1.8 mL of cold 3:1:2 MeOH/CHCl₃/H₂O solution was added to each tube, and the samples were centrifuged (10,000×g, 10 min, 4° C.), forming a disc. The supernatant was carefully removed, MeOH (600 µL) was added, samples were centrifuged as previously described and the supernatant was removed. Cell pellets were resuspended in TEAB (160 µL, 100 mM, pH 8.5) and sonicated as described previously.

Endoproteinase LysC (20 µL, ½ vial, 10 µg dissolved in 220 µL of 100 mM TEAB pH 8.5) was added to each and incubated at 37° C. for 2 h with shaking. Sequencing-grade modified porcine trypsin (20 µL, 1 vial, 20 µg dissolved in 220 µL of 100 mM TEAB pH 8.5), a solution of Protease Max (2 µL, 1% (w/v) in 100 mM TEAB pH 8.5) and a solution of CaCl₂ (2 µL, 100 mM) were added to the samples and incubated at 37° C. overnight (14 h) with shaking. The digest was separated by centrifugation (12,000×g, 10 min, 4° C.). Peptide concentration was determined using Pierce Quantitative Fluorometric Peptide Assay Kit (Thermo Fisher Scientific, 23290) according to manufacturer's instructions. MS-grade acetonitrile (12.5 µL) was added to each and samples were labeled with respective TMT 10 plex isotope (8 µL, 20 µg/µL) for 1 h with occasional vortexing at RT. To quench, hydroxylamine (3 µL, 5% v/v) was added to each sample, vortexed, and incubated for 15 min at RT. Formic acid (5 µL) was added to each tube to acidify and the samples were dried under vacuum centrifugation. The samples were combined by redissolving the contents of one tube in a solution of trifluoroacetic acid (TFA, 400 µL, 0.1% in water) and transferred into each sample until all samples were redissolved. The stepwise process was repeated with formic acid (Buffer A, 200 µL, 0.1% in water) for a final volume of 600 µL. The samples were fractionated using a fractionation kit (Pierce high pH Reversed-Phase Fractionation Kit, Thermo Fisher Scientific 84868) according to manufacturer's instructions. The peptide fractions were eluted from the spin column with consecutive solutions of 0.1% triethylamine combined with MeCN (5-75% MeCN). The fractions were combined pairwise (fraction 1 and fraction 10, fraction 2 and fraction 11, etc.), dried via vacuum centrifugation, and stored at -80° C. until ready for mass spectrometer injection.

[0048] LC-MS analysis of TMT samples: TMT labeled samples were redissolved in MS buffer A (20 µL, 0.1% formic acid in water). 3 µL of each sample was loaded onto an Acclaim PepMap 100 precolumn (75 µm×2 mm) and eluted on an Acclaim PepMap RSLC analytical column (75 µm×15 cm) using the UltiMate 3000 RSLCnano system (Thermo Fisher Scientific). Buffer A was prepared as described above and buffer B (0.1% formic acid in MeCN) were used in a 220 min gradient (flow rate 0.3 mL min, 35° C.) of 2% buffer B for 10 min, followed by an incremental increase to 30% buffer B over 192 min, 60% buffer B for 5 min, 60-95% buffer B for 1 min, hold at 95% buffer B for 5 min, followed by descent to 2% buffer B for 1 min followed by re-equilibration at 2% for 6 min. The elutions were analyzed with a Thermo Fisher Scientific Orbitrap Fusion Lumos mass spectrometer with a cycle time of 3 s and nano-LC electrospray ionization source applied voltage of 2.0 kV. MS1 spectra were recorded at a resolution of 120,000 with an automatic gain control (AGC) value of 1×10⁶ ions, maximum injection time of 50 ms (dynamic exclusion enabled, repeat count 1, duration 20 s). The scan range was specified from 375 to 1,500 m/z. Peptide fragmentation MS2 spectra was recorded via collision-induced diffusion (CID) and quadrupole ion trap analysis (AGC 1.8×10⁴, 30% collision energy, maximum inject time 120 ms, isolation window 1.6). MS3 spectra was generated by high-energy collision-induced dissociation (HCD) with collision energy of 65%. Precursor selection included up to 10 MS² ions for the MS³ spectrum.

[0049] TMT proteomics data analysis: Proteomic analysis was performed with the processing software Proteome Discoverer 2.4 (Thermo Fisher Scientific). Peptide sequences were identified by matching proteome databases with

experimental fragmentation patterns via the SEQUEST HT algorithm. Fragment tolerances were set to 0.6 Da, and precursor mass tolerances set to 10 ppm with one missed cleavage site allowed. Spectra were searched against the *Homo Sapiens* proteome database (42,358 sequences) using a false discovery rate of 1% (Percolator). MS³ peptide quantitation was performed with a mass tolerance of 20 ppm. Identified proteins were required to have at least two unique peptides. TMT ratios obtained by Proteome Discoverer were transformed with log₂(x), and p-values were calculated via Student's two-tailed t-tests with two biological replicates.

[0050] Cell viability assays: Cells were seeded in white-opaque 96-well plates in full growth media at a density of 2,000 cells/well (50 μ L) and were allowed to grow for 14 h at 37° C. in a humidified 5% CO₂ atmosphere. The cells were then treated with compounds in triplicate and incubated at 37° C. in a humidified 5% CO₂ atmosphere for 6 hours. Cell viability was determined using the luciferase-based Cell Titer-Glo Luminescent Cell Viability Assay (Promega) following manufacturer's guidelines. Data represents the average and standard deviation of triplicates in measured luminescence.

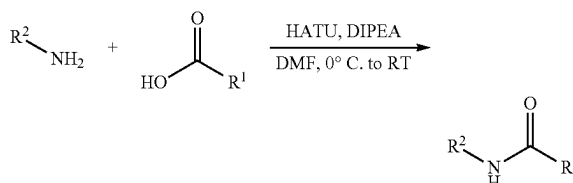
[0051] Immunoblots: After cells were harvested, cell pellets were resuspended in 100 μ L DPBS containing 1 $\times$ Halt protease inhibitor cocktail, and lysed by sonication (15 ms on, 40 ms off, 15% amplitude, 1 s total). Protein concentrations were normalized (2 mg/mL in 100 μ L with cold DPBS) as previously described. 4X SDS gel loading buffer (33 μ L) was added to the solution of protein lysate and the resulting mixture was heated at 95° C. for 15 min. Proteins (15 μ g total protein loaded per gel lane) were resolved by SDS-PAGE (10% acrylamide) made in-house, and transferred to PVDF membrane (0.2 μ M, 1620177, Bio-Rad). The membrane was blocked with 5% BSA in Tris-buffered saline with Tween (TBST) buffer (0.1% Tween 20, 20 mM Tris-HCl 7.6, 150 mM NaCl) at room temperature for 1 h. The antibody was diluted with fresh 5% BSA in TBST buffer (dilutions were performed following manufacturer's guidelines) and incubated with membrane overnight (14 h) at 4° C. Membrane was washed three times with TBST buffer, left 5 minutes between each wash on a rocker and then incubated with secondary antibody in 5% dry milk in TBST at room temperature for 2 h on a rocker. Membrane was washed three times with TBST buffer and visualized by in-gel fluorescence on a Bio-Rad ChemiDoc MP Imaging System. The images were processed using Image Lab (version 5.2.1) software.

[0052] Quantification and Statistical Analysis: All data fitting and statistical analysis were performed using Graph-Pad Prism version 9.00 for Windows and Mac. Statistical significance was defined as p<0.05 and determined by 2-tailed Student's t-tests.

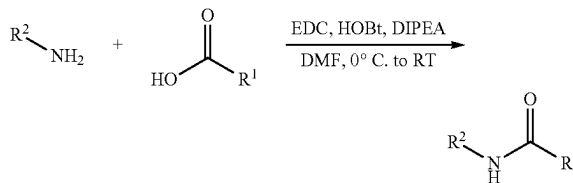
Compound Synthesis and Characterization

[0053] General Synthetic Information: All commercial reagents acquired from Sigma-Aldrich, Fisher Scientific, Combi-Blocks, MedChemExpress, AstaTech, Matrix Scientific and BroadPharm were used without further purification. Distilled water was used for all water necessities in synthetic procedures (e.g., reagent, solvent, work-up). All reactions were run with anhydrous solvents. TLC analyses were completed with EMD Millipore silica gel coated (250 μ M) F254 glass plates and visualized with UV light (254 nm) or by employing diverse stains (e.g., iodine, CAM, PMA, Ninhydrin, DNP, KMnO₄), followed by gentle heating. Flash column chromatography was performed with silica gel

60. Automated purifications were performed on Biotage Isolera One purification system using manually packed columns of 5G, 10G or 25G of silica. Preparative Thin Layer Chromatography (pTLC) was carried out using EMD Millipore silica gel coated (250 μ M) F254 glass plates or glass backed plates 1000-2000 μ m thickness (Analtech). ¹H- and ¹³C-NMR spectra were recorded on a Bruker AVANCE NEO 400 MHz instrument with SampleXpress, Bruker AVANCE NEO 500 MHz NMR instrument equipped with 5 mm BBFO SP probe, and SampleCase-24 sample changer, and Bruker AVANCE III HD 600, instrument equipped with a 5 mm CPDCH CryoProbe. Data was collected at ambient temperature unless otherwise stated using standard pulse methods as supplied by Bruker software. Coupling constants are quoted to the nearest 0.1 Hz and multiplicities are given by the following abbreviations and combinations: m (multiplet), s (singlet), d (doublet), dd (doublet of doublets), ddd (doublet of doublet of doublets), t (triplet), td (triplet of doublets), tt (triplet of triplets), q (quartet), br (broad). All NMR data was processed in MestReNova v12.0.2. Chemical shifts for proton and carbon resonances are reported in parts per million (ppm) on the δ scale relative to the residual protons of the deuterated solvent of relevance. Mass spectrometry data were collected on a Thermo Scientific ISQ single-quadrupole instrument (ESI; low resolution), Agilent 6125 and 6135 single-quadrupole instruments (ESI; low resolution), and Agilent 6230 single-quadrupole TOF (ESI-TOF; high resolution).



[0054] General Procedure 1: Synthesis of amides using HATU as coupling reagent To a solution of amine (1.1 eq.), carboxylic acid (1.0 eq.) and HATU (1.5 eq.) in dry DMF at 0° C. was added DIPEA (3.0 eq.). The reaction mixture was stirred for 2-18 h and then quenched with DI H₂O and diluted with Ethyl acetate (EtOAc). The aqueous layer was extracted with EtOAc (3 times), and the combined organic layers were washed with H₂O (2 times), a saturated aqueous solution of NH₄Cl (2 times) and a saturated aqueous solution of NaCl (2 times) before being dried over anhydrous Na₂SO₄ and filtered. Volatiles were removed by rotary evaporation and the crude product was purified by automated column chromatography, pTLC, or organic solvent washes (i.e., the organic solvent was added into the vial and mixed with the solid decanting).



[0055] General Procedure 2: Synthesis of amides using EDC as coupling reagent. To a solution of amine (1.1 eq.), carboxylic acid (1.0 eq.), EDC (1.5 eq.) and HOBT (1.5 eq.) in dry DCM at 0° C. was added DIPEA (3.0 eq.). The reaction mixture was stirred overnight, then quenched with DI H₂O. The aqueous layer was extracted with CH₂Cl₂ (3 times), and the combined organic layers were washed with

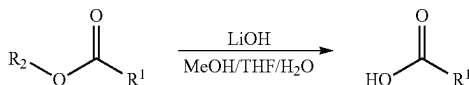
H₂O (2 times), a saturated aqueous solution of NH₄Cl (2 times) and a saturated aqueous solution of NaCl (2 times) before being dried over anhydrous Na₂SO₄ and filtered. Volatiles were removed by rotary evaporation and the crude product was purified by automated column chromatography, pTLC, or organic solvent washes (i.e., the organic solvent was added into the vial and mixed with the solid decanting).



[0056] General Procedure 3: Tert-butyl ester deprotection of carboxylic acids using TFA. To a solution of t-Butyl protected acid (1.0 eq.) in dry CH₂Cl₂ was added a solution of TFA (20-50%). The reaction mixture was stirred at RT for 0.5-12 h. The crude product was concentrated through a nitrogen flow, diluted in CH₂Cl₂ and evaporated again. No purification was needed.

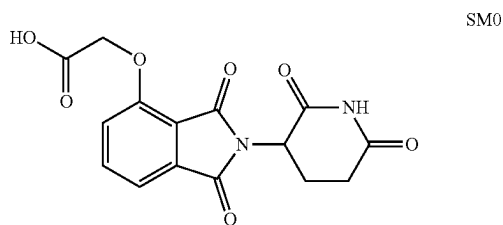


[0057] General Procedure 4: Boc deprotection of amines using TFA. To a solution of Boc protected amine (1.0 eq.) in dry CH₂Cl₂ was added a solution of TFA (20-50%). The reaction mixture was stirred at RT for 0.5-12 h. The crude product was concentrated through a nitrogen flow, diluted in CH₂Cl₂ and evaporated again. No purification was needed.

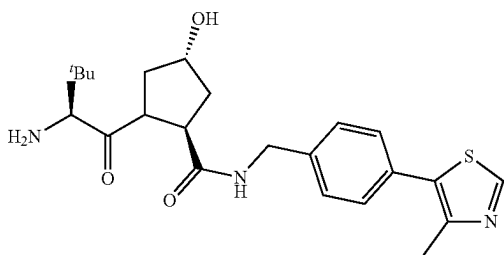


[0058] General Procedure 5: Ester hydrolysis. The ester (1.0 eq.) was dissolved in a mixture of THF/MeOH/H₂O (3:1:2) and LiOH, H₂O (7.0 eq.) was added. The reaction mixture was stirred at RT for 3-12 h. The solvents were evaporated, and the residue was acidified with 1N aqueous until pH=1, before being extracted with EtOAc, dried over anhydrous Na₂SO₄ and filtered. The organic solvent was evaporated, and no purification was needed.

Starting Materials:



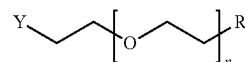
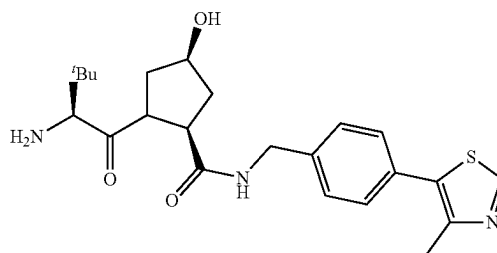
SM0



SM1

-continued

SM1'



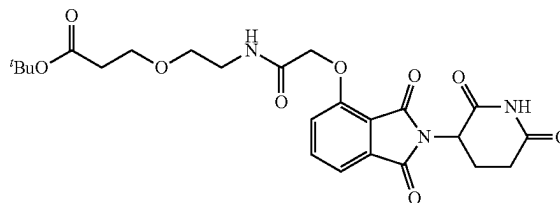
- L1: n = 1, R = NH₂, Y = COO'Bu
 L2: n = 1, R = COOH, Y = COO'Bu
 L3: n = 2, R = NH₂, Y = COO'Bu
 L4: n = 2, R = COOH, Y = COO'Bu
 L5: n = 3, R = NH₂, Y = COO'Bu
 L6: n = 3, R = COOH, Y = COO'Bu
 L7: n = 3, R = NH₂, Y = NHBoc
 L8: n = 3, R = COOH, Y = NHBoc
 L9: n = 4, R = COOH, Y = COO'Bu



- L10: m = 2, R = COOH, Y = COO'Bu
 L11: m = 4, R = COOH, Y = COO'Bu
 L12: m = 5, R = NH₂, Y = COO'Bu
 L13: m = 6, R = COOH, Y = COOMe
 L14: m = 6, R = COOH, Y = NHBoc
 L15: m = 6, R = NH₂, Y = NHBoc
 L16: m = 8, R = COOH, Y = COOMe

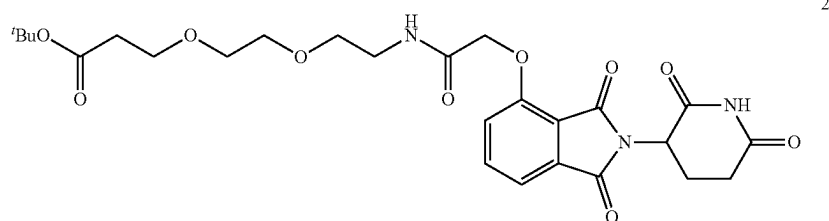
Characterization Data:

1



Tert-butyl 3-(2-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)oxy) acetamido) ethoxy) propanoate (1)

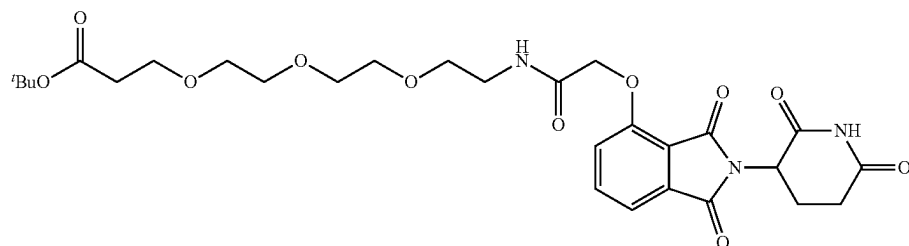
[0059] General Procedure 1. Reaction scale: 10.5 mg (31.60 μmol, 1.0 eq.) of SM0 and 6.4 μL (33.55 μmol, 1.1 eq.) of L1. Purified by pTLC (10% MeOH/CH₂Cl₂) to afford 1 as a white foam (14.7 mg, 97%). R_f=0.63 (10% MeOH/CH₂Cl₂, UV-active); LC-MS (ESI+) calc'd for C₂₄H₂₉N₃O₉ 503.2, found [M+H]⁺ 504.0. ¹H NMR (400 MHz, CDCl₃) δ 8.69 (s, 1H), 7.77-7.69 (t, 1H), 7.65 (d, J=5.9 Hz, 1H), 7.54 (d, J=7.3 Hz, 1H), 7.18 (d, J=8.4 Hz, 1H), 5.04-4.96 (m, 1H), 4.64 (s, 2H), 3.71 (s, 2H), 3.67-3.54 (m, 4H), 3.50 (ddt, J=14.6, 7.8, 3.1 Hz, 1H), 2.89 (t, J=11.3 Hz, 2H), 2.56 (dd, J=7.9, 6.1 Hz, 2H), 2.21-2.09 (m, 2H), 1.42 (d, J=1.7 Hz, 9H). ¹³C NMR (101 MHz, CDCl₃) δ 171.05, 168.21, 166.82, 166.68, 154.43, 136.96, 133.69, 119.30, 118.11, 117.29, 80.99, 77.36, 77.04, 76.94, 76.73, 69.11, 67.84, 66.68, 49.24, 38.98, 38.63, 36.09, 31.34, 29.72, 28.08, 22.78.



2

Tert-butyl 3-(2-(2-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)oxy) acetamido) ethoxy) propanoate (2)

[0060] General Procedure 1. Reaction scale: 20 mg (60.0 μmol , 1.0 eq.) of SM0 and 15 μL (65 μmol , 1.1 eq.) of L3. Purified by pTLC (10% MeOH/ CH_2Cl_2) to afford 2 as a yellow foam (19.9 mg, 59%). $R_f=0.57$ (10% MeOH/ CH_2Cl_2 , UV-active); LC-MS (ESI+) calc'd for $\text{C}_{26}\text{H}_{33}\text{N}_3\text{O}_{10}$ 547.2, found $[\text{M}+\text{H}]^+$ 548.4. ^1H NMR (400 MHz, CDCl_3) δ 8.70 (s, 1H), 7.73 (dd, $J=8.4$, 7.4 Hz, 1H), 7.59 (br. t, 1H), 7.55-7.48 (m, 1H), 7.18 (d, $J=8.5$ Hz, 1H), 5.04-4.92 (m, 1H), 4.65 (s, 2H) 3.73-3.52 (m, 10H), 2.89-2.70 (m, 4H), 1.42 (s, 11H). ^{13}C NMR (101 MHz, CDCl_3) δ 171.72, 171.45, 168.54, 167.24, 167.13, 166.17, 154.88, 137.40, 134.14, 119.69, 118.47, 117.71, 81.24, 77.79, 77.68, 77.48, 77.16, 70.67, 70.59, 69.93, 68.30, 67.31, 49.71, 39.47, 39.06, 36.62, 31.85, 30.15, 28.53, 23.12.

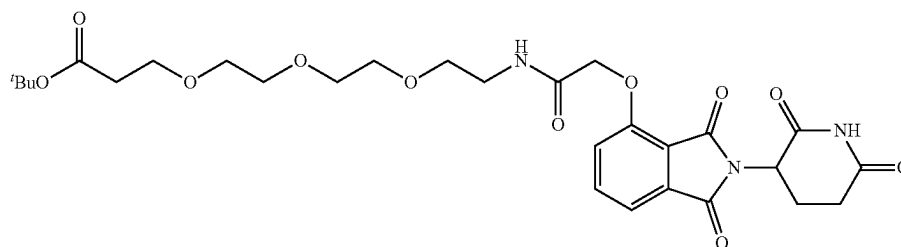


3

Tert-butyl 1-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)oxy)-2-oxo-6,9,12-trioxa-3-azapentadecan-15-oate (3)

[0061] General Procedure 1. Reaction scale: 20 mg (60.2 μmol , 1.0 eq.) of SM0 and 15 μL (66.2 μmol , 1.1 eq.) of L5. Purified by pTLC (10% MeOH/ CH_2Cl_2) to afford 3 as a yellow foam (21.5 mg, 61%). $R_f=0.76$ (8% MeOH/ CH_2Cl_2 , UV-active); LC-MS (ESI-) calc'd for $\text{C}_{28}\text{H}_{37}\text{N}_3\text{O}_{11}$ 591.2, found $[\text{M}-\text{H}]^-$ 590.0. ^1H NMR (400 MHz, CDCl_3) δ 8.92

(s, 1H), 7.72 (dd, $J=8.4$, 7.3 Hz, 1H), 7.63 (br. t, 1H), 7.53 (d, $J=7.3$ Hz, 1H), 7.18 (d, $J=8.3$ Hz, 1H), 4.99-4.88 (m, 1H), 4.64 (s, 2H), 3.70-3.57 (m, 14H), 2.92-2.80 (m, 2H), 2.48 (t, $J=6.6$ Hz, 2H), 1.42 (s, 11H). ^{13}C NMR (101 MHz, CDCl_3) δ 171.10, 170.97, 168.15, 166.88, 166.67, 165.80, 154.49, 136.94, 133.70, 119.39, 118.09, 117.28, 80.62, 77.37, 77.25, 77.05, 76.93, 76.73, 70.39, 70.35, 70.32, 70.24, 69.43, 67.96, 66.87, 49.32, 39.13, 38.62, 36.15, 31.42, 29.69, 28.09, 22.71.



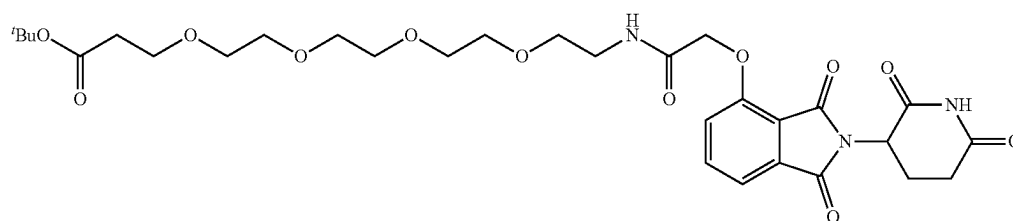
4

Tert-butyl 1-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)oxy)-2-oxo-6,9,12-trioxa-3-azatetradecan-14-yl) carbamate (4)

[0062] Reaction scale: 100 mg (301.0 μmol , 1.0 eq.) of SM0 and 96.8 mg (331.1 μmol , 1.1 eq.) of L7. Purified by Biotage Isolera (2-14% MeOH/ CH_2Cl_2 gradient, 16 CV) to afford 4 as a yellow foam (113 mg, 62%). $R_f=0.63$ (8% MeOH/ CH_2Cl_2 , UV-active); LC-MS (ESI+) calc'd for $\text{C}_{28}\text{H}_{38}\text{N}_3\text{O}_{11}$ 606.3, found $[\text{M}+\text{H}]^+$ 607.2. ^1H NMR (500 MHz, CDCl_3) δ 9.23 (s, 1H), 7.69 (t, $J=7.9$ Hz, 1H), 7.58 (t, $J=5.6$ Hz, 1H), 7.49 (d, $J=7.3$ Hz, 1H), 7.15 (s, 1H), 4.96-4.87 (m, 1H), 4.61 (s, 2H), 3.62-3.48 (m, 17H), 2.87-2.76 (m, 2H), 1.38 (s, 11H).

Tert-butyl 6-(2-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)oxy) acetamido) hexanoate (6)

[0064] General Procedure 1. Reaction scale: 15 mg (45.1 μmol , 1.0 eq.) of SM0 and 10.0 μL (49.7 μmol , 1.1 eq.) of L13. Purified by pTLC (8% MeOH/ CH_2Cl_2) to afford 6 as an off-white solid (19.6 mg, 87%). $R_f=0.79$ (8% MeOH/ CH_2Cl_2 , UV-active); LC-MS (ESI+) calc'd for $\text{C}_{25}\text{H}_{31}\text{N}_3\text{O}_8$ 501.2, found $[\text{M}+\text{H}]^+$ 502.0. ^1H NMR (400 MHz, CDCl_3) δ 9.66 (s, 1H), 7.75-7.71 (m, 1H), 7.61 (s, 1H), 7.55 (d, $J=7.3$ Hz, 1H), 7.39 (d, $J=1.4$ Hz, 2H), 7.28 (d, $J=7.3$ Hz, 2H), 7.20 (d, $J=1.4$ Hz, 1H), 4.98-4.91 (m, 1H), 4.72-4.55 (m, 3H), 3.63 (q, $J=3.8$ Hz, 3H), 3.45 (t, $J=5.1$ Hz, 2H), 2.35 (t, $J=4.9$

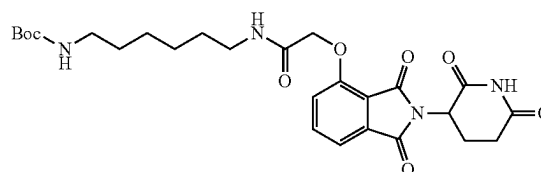


5

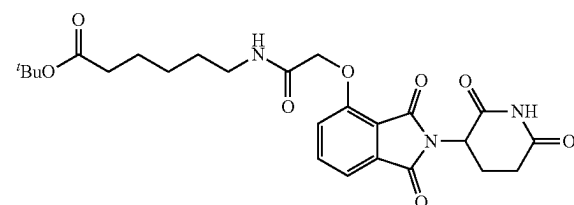
Tert-butyl 1-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)oxy)-2-oxo-6,9,12,15-tetraoxa-3-azaoctadecan-18-oate (5)

[0063] General Procedure 1. Reaction scale: 10 mg (32.5 μmol , 1.0 eq.) of SM0 and 10.18 μL (35.8 μmol , 1.1 eq.) of L9. Purified by pTLC (10% MeOH/ CH_2Cl_2) to afford 5 as a white foam (15.7 mg, 76%). $R_f=0.53$ (10% MeOH/ CH_2Cl_2 , UV-active); LC-MS (ESI+) calc'd for $\text{C}_{30}\text{H}_{41}\text{N}_3\text{O}_{12}$ 635.3, found $[\text{M}+\text{H}]^+$ 636.0. ^1H NMR (400 MHz, CDCl_3) δ 8.99 (s, 1H), 7.72 (dd, $J=8.4$, 7.4 Hz, 1H), 7.63 (t, $J=5.2$ Hz, 1H), 7.53 (d, $J=7.3$ Hz, 1H), 7.18 (d, $J=8.4$ Hz, 1H), 5.03-4.86 (m, 1H), 4.64 (s, 2H), 3.69-3.59 (m, 18H), 2.88-2.78 (m, 2H), 2.47 (t, $J=6.5$ Hz, 3H), 1.42 (s, 11H). ^{13}C NMR (101 MHz, CDCl_3) δ 171.34, 168.34, 167.10, 166.70, 154.44, 136.97, 133.65, 119.42, 117.25, 80.61, 77.42, 77.10, 76.79, 70.55, 70.38, 70.28, 69.50, 67.87, 66.79, 60.40, 49.31, 39.08, 38.63, 36.19, 31.46, 29.70, 28.08, 22.68, 21.05, 14.19.

Hz, 4H), 1.25 (s, 9H). ^{13}C NMR (101 MHz, CDCl_3) δ 171.77, 171.13, 168.15, 166.66, 166.41, 166.08, 154.62, 141.96, 136.86, 133.40, 128.43, 127.68, 126.98, 120.06, 118.32, 117.43, 77.16, 77.05, 76.84, 76.62, 76.52, 75.77, 68.48, 51.92, 51.44, 49.17, 45.70, 41.64, 39.03, 32.85, 31.22, 29.53, 28.69, 26.59, 25.16, 22.67.



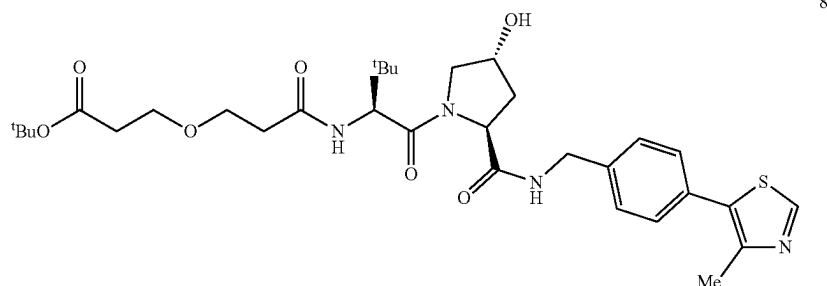
7



6

Tert-butyl (6-(2-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)oxy) acetamido) hexyl) carbamate (7)

[0065] ^1H NMR (400 MHz, CDCl_3) δ 7.74 (m, 1H), 7.55 (d, $J=7.3$ Hz, 1H), 7.46 (m, 1H), 7.19 (d, $J=8.3$ Hz, 1H), 4.99 (m, 1H), 4.71-4.60 (m, 3H), 3.46-3.28 (m, 2H), 3.17-3.04 (m, 2H), 2.93-2.76 (m, 3H), 2.14 (m, 1H), 1.64-1.56 (m, 2H), 1.48-1.33 (m, 16H).



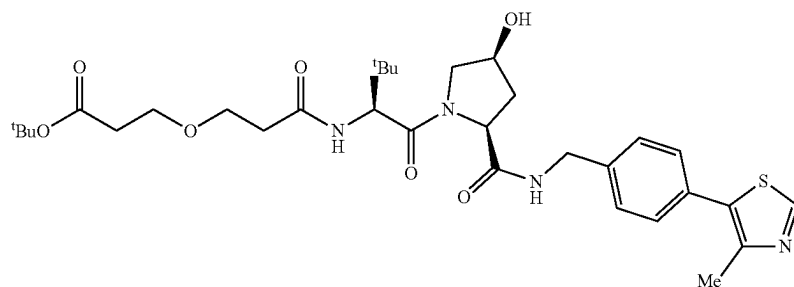
8

Tert-butyl 3-(3-(((S)-1-((2S,4R)-4-hydroxy-2-((4-(4-methylthiazol-5-yl)benzyl) carbamoyl) pyrrolidin-1-yl)-3,3-dimethyl-1-oxobutan-2-yl)amino)-3-oxopropoxy) propanoate (8)

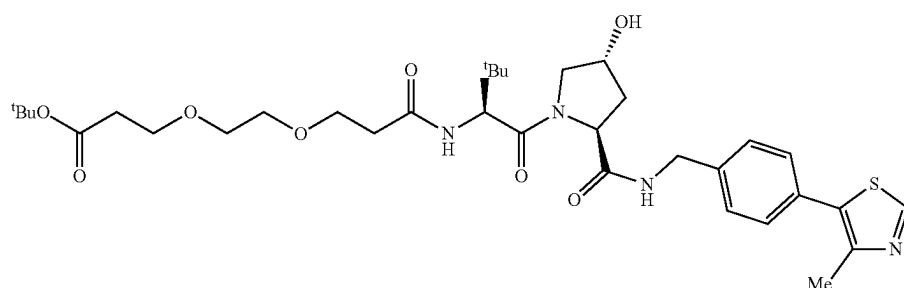
[0066] General Procedure 1. Reaction scale: 20 mg (46.5 μmol , 1.0 eq.) of SM1 and 11.2 mg (51.1 μmol , 1.1 eq.) of L2. Purified by pTLC (8% MeOH/ CH_2Cl_2) to afford 8 as a clear oil (22.6 mg, 77%). $R_f=0.59$ (8% MeOH/ CH_2Cl_2 , UV-active); LC-MS (ESI+) calc'd for $\text{C}_{32}\text{H}_{46}\text{N}_4\text{O}_7\text{S}$ 630.3, found $[\text{M}+\text{H}]^+$ 631.1. ^1H NMR (600 MHz, CDCl_3) δ 8.67 (s, 1H), 7.43 (t, $J=6.0$ Hz, 1H), 7.36-7.30 (m, 4H), 6.95 (d, $J=8.3$ Hz, 1H), 4.72 (t, $J=7.9$ Hz, 1H), 4.55 (dd, $J=15.0$, 6.7 Hz, 1H), 4.50 (dt, $J=4.3$, 2.1 Hz, 1H), 4.44 (d, $J=8.2$ Hz, 1H), 4.32 (dd, $J=15.0$, 5.3 Hz, 1H), 4.09-4.07 (m, 1H), 3.71-3.64 (m, 4H), 3.59 (dd, $J=11.3$, 3.7 Hz, 1H), 2.54-2.44 (m, 7H), 2.13-2.07 (m, 1H), 1.42 (s, 9H), 0.92 (s, 9H). ^{13}C NMR (151 MHz, CDCl_3) δ 172.08, 171.82, 171.19, 170.79, 170.63, 150.34, 148.45, 138.16, 131.63, 130.92, 129.52, 128.12, 128.09, 80.85, 77.26, 77.05, 76.84, 70.10, 66.82, 66.79, 66.66, 66.29, 60.42, 58.39, 57.77, 56.62, 55.99, 43.22, 36.69, 36.08, 35.85, 34.79, 31.93, 30.17, 29.71, 29.67, 29.37, 29.33, 28.93, 28.10, 28.08, 26.64, 26.41, 22.70, 21.27, 21.07, 17.45, 16.04, 14.14.

Tert-butyl 3-(3-(((S)-1-((2S,4S)-4-hydroxy-2-((4-(4-methylthiazol-5-yl)benzyl) carbamoyl) pyrrolidin-1-yl)-3,3-dimethyl-1-oxobutan-2-yl)amino)-3-oxopropoxy) propanoate (9)

[0067] General Procedure 1. Reaction scale: 20 mg (46.5 μmol , 1.0 eq.) of SM1' and 11.3 mg (51.1 μmol , 1.1 eq.) of L2. Purified by pTLC (8% MeOH/ CH_2Cl_2) to afford 9 as a white foam (0.6 mg, 77%). $R_f=0.57$ (8% MeOH/ CH_2Cl_2 , UV-active); LC-MS (ESI+) calc'd for $\text{C}_{32}\text{H}_{46}\text{N}_4\text{O}_7\text{S}$ 630.3, found $[\text{M}+\text{H}]^+$ 631.3. ^1H NMR (600 MHz, CDCl_3) δ 8.67 (s, 1H), 7.43 (t, $J=6.0$ Hz, 1H), 7.36-7.30 (m, 4H), 6.95 (d, $J=8.3$ Hz, 1H), 4.72 (t, $J=7.9$ Hz, 1H), 4.55 (dd, $J=15.0$, 6.7 Hz, 1H), 4.50 (dt, $J=4.3$, 2.1 Hz, 1H), 4.44 (d, $J=8.2$ Hz, 1H), 4.32 (dd, $J=15.0$, 5.3 Hz, 1H), 4.09-4.07 (m, 1H), 3.71-3.64 (m, 4H), 3.59 (dd, $J=11.3$, 3.7 Hz, 1H), 2.54-2.44 (m, 7H), 2.13-2.07 (m, 1H), 1.42 (s, 9H), 0.92 (s, 9H). ^{13}C NMR (151 MHz, CDCl_3) δ 172.08, 171.82, 171.19, 170.79, 170.63, 150.34, 148.45, 138.16, 131.63, 130.92, 129.52, 128.12, 128.09, 80.85, 77.26, 77.05, 76.84, 70.10, 66.82, 66.79, 66.66, 66.29, 60.42, 58.39, 57.77, 56.62, 55.99, 43.22, 36.69, 36.08, 35.85, 34.79, 31.93, 30.17, 29.71, 29.67, 29.37, 29.33, 28.93, 28.10, 28.08, 26.64, 26.41, 22.70, 21.27, 21.07, 17.45, 16.04, 14.14.



9



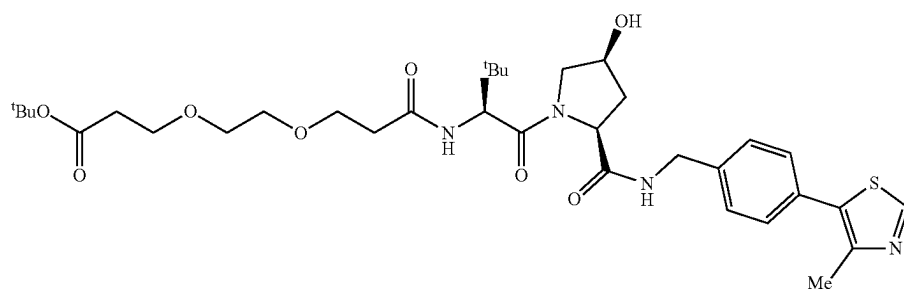
10

Tert-butyl 3-(2-(3-(((S)-1-((2S,4R)-4-hydroxy-2-((4-(4-methylthiazol-5-yl)benzyl) carbamoyl) pyrrolidin-1-yl)-3,3-dimethyl-1-oxobutan-2-yl)amino)-3-oxopropoxy) ethoxy) propanoate (10)

[0068] General Procedure 1. Reaction scale: 20 mg (46.5 μmol , 1.0 eq.) of SM1 and 13.5 mg (51.1 μmol , 1.1 eq.) of L4. Purified by pTLC (10% MeOH/ CH_2Cl_2) to afford 10 as a yellow oil (14.9 mg, 51%). $R_f=0.54$ (10% MeOH/ CH_2Cl_2 , UV-active); LC-MS (ESI+) calc'd for $\text{C}_{34}\text{H}_{50}\text{N}_4\text{O}_8\text{S}$ 674.3, found $[\text{M}+\text{H}]^+$ 675.2. ^1H NMR (400 MHz, CDCl_3) δ 8.67 (s, 1H), 7.48 (t, $J=6.0$ Hz, 1H), 7.38-7.30 (m, 4H), 7.05 (d, $J=8.2$ Hz, 1H), 4.73 (t, $J=8.0$ Hz, 1H), 4.56 (dd, $J=15.0$, 6.6 Hz, 1H), 4.50 (dt, $J=4.4$, 2.1 Hz, 1H), 4.44 (d, $J=8.2$ Hz, 1H), 4.32 (dd, $J=15.0$, 5.3 Hz, 1H), 4.09 (dt, $J=11.5$, 1.8 Hz, 1H), 3.72-3.65 (m, 4H), 3.61-3.56 (m, 5H), 2.52-2.45 (m, 8H), 2.13-2.06 (m, 1H), 1.42 (s, 9H), 0.93 (s, 9H).

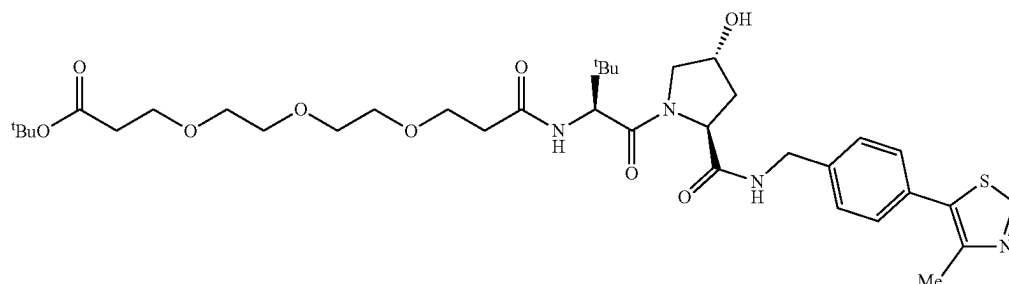
Tert-butyl 3-(2-(3-(((S)-1-((2S,4S)-4-hydroxy-2-((4-(4-methylthiazol-5-yl)benzyl) carbamoyl) pyrrolidin-1-yl)-3,3-dimethyl-1-oxobutan-2-yl)amino)-3-oxopropoxy) ethoxy) propanoate (11)

[0069] General Procedure 1. Reaction scale: 20 mg (46.5 μmol , 1.0 eq.) of SM1' and 13.4 mg (51.1 μmol , 1.1 eq.) of L4. Purified by pTLC (8% MeOH/ CH_2Cl_2) to afford 11 as a clear oil (19.3 mg, 62%). $RS=0.55$ (10% MeOH/ CH_2Cl_2 , UV-active); LC-MS (ESI+) calc'd for $\text{C}_{34}\text{H}_{50}\text{N}_4\text{O}_8\text{S}$ 674.3, found $[\text{M}+\text{H}]^+$ 675.1. ^1H NMR (400 MHz, CDCl_3) δ 8.67 (s, 1H), 7.49 (t, $J=6.0$ Hz, 1H), 7.39-7.29 (m, 4H), 7.07 (d, $J=8.1$ Hz, 1H), 4.72 (t, $J=8.0$ Hz, 1H), 4.60-4.41 (m, 3H), 4.32 (dd, $J=15.0$, 5.3 Hz, 1H), 4.14-4.05 (m, 1H), 3.68 (q, $J=6.1$ Hz, 4H), 3.60 (s, 5H), 2.55-2.42 (m, 8H), 2.11 (dd, $J=13.4$, 8.1 Hz, 1H), 1.42 (s, 9H), 0.93 (s, 9H). ^{13}C NMR (101 MHz, CDCl_3) δ 172.23, 171.79, 170.98, 170.88, 150.32, 148.44, 138.23, 131.64, 130.88, 129.49, 128.11, 80.70, 77.37, 77.25, 77.05, 76.73, 70.43, 70.17, 70.10, 67.14, 66.88, 58.40, 57.82, 56.63, 43.19, 36.66, 36.15, 35.92, 34.76, 29.71, 28.10, 26.41, 16.05.



11

12



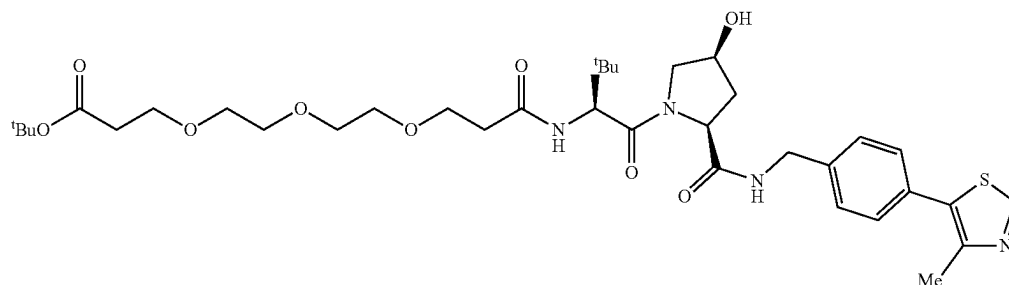
Tert-butyl(S)-15-((2S,4R)-4-hydroxy-2-((4-(4-methylthiazol-5-yl)benzyl) carbamoyl) pyrrolidine-1-carbonyl)-16,16-dimethyl-13-oxo-4,7,10-trioxa-14-azaheptadecanoate (12)

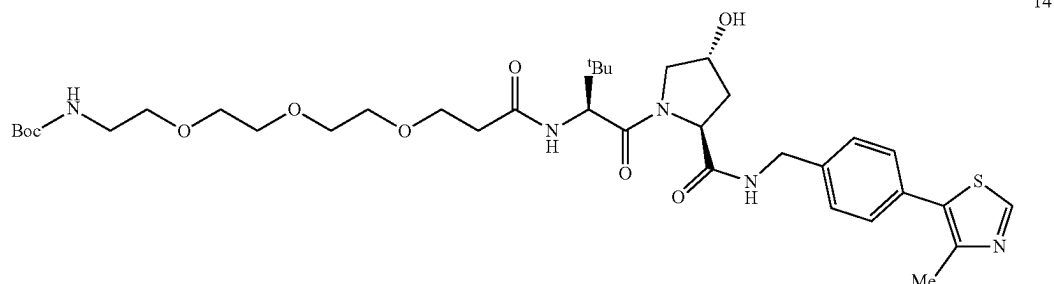
[0070] General Procedure 1. Reaction scale: 70 mg (162.6 μmol , 1.0 eq.) of SM1 and 54.8 mg (178.8 μmol , 1.1 eq.) of L6. Purified by Biotage Isolera (2-15% MeOH/ CH_2Cl_2 gradient, 16 CV) to afford 12 as a white foam (81 mg, 70%). R/=0.56 (10% MeOH/ CH_2Cl_2 , UV-active); LC-MS (ESI-) calc'd for $\text{C}_{36}\text{H}_{54}\text{N}_4\text{O}_9\text{S}$ 718.4, found $[\text{M}+\text{HCOO}]^-$ 763.2. ^1H NMR (400 MHz, CDCl_3) δ 8.60 (s, 1H), 7.48 (t, J=6.0 Hz, 1H), 7.26 (s, 4H), 7.02 (d, J=8.5 Hz, 1H), 4.60 (t, J=8.0 Hz, 1H), 4.45 (dd, J=15.4, 7.2 Hz, 3H), 4.25 (dd, J=15.1, 5.4 Hz, 1H), 3.94 (d, J=11.2 Hz, 1H), 3.60 (td, J=6.1, 3.8 Hz, 4H), 3.57-3.49 (m, 9H), 2.44-2.34 (m, 7H), 2.31 (td, J=8.3, 4.1 Hz, 1H), 2.04 (ddt, J=13.0, 8.1, 1.9 Hz, 1H), 1.35 (s, 9H), 0.87 (s, 9H). ^{13}C NMR (101 MHz, CDCl_3) δ 172.00, 171.50, 171.20, 170.94, 150.35, 148.36, 138.30, 131.65, 130.76, 129.41, 128.21, 128.01, 80.60, 77.43, 77.31, 77.11, 76.79, 70.43, 70.39, 70.36, 70.26, 69.98, 67.10, 66.84, 58.69, 57.64, 56.73, 50.39, 45.73, 43.09, 36.67, 36.33, 36.20, 35.11, 29.66, 28.08, 26.38, 16.00, 8.54.

Tert-butyl(S)-15-((2S,4S)-4-hydroxy-2-((4-(4-methylthiazol-5-yl)benzyl) carbamoyl) pyrrolidine-1-carbonyl)-16,16-dimethyl-13-oxo-4,7,10-trioxa-14-azaheptadecanoate (13)

[0071] General Procedure 1. Reaction scale: 15 mg (34.8 μmol , 1.0 eq.) of SM1' and 11.7 mg (38.3 μmol , 1.1 eq.) of L6. Purified by Biotage Isolera (2-15% MeOH/ CH_2Cl_2 gradient, 16 CV) to afford 13 as a white foam (81 mg, 70%). R/=0.62 (10% MeOH/ CH_2Cl_2 , UV-active); LC-MS (ESI+) calc'd for $\text{C}_{36}\text{H}_{54}\text{N}_4\text{O}_9\text{S}$ 718.4, found $[\text{M}+\text{H}]^+$ 719.2. ^1H NMR (400 MHz, CDCl_3) δ 8.69 (s, 1H), 7.67-7.59 (m, 1H), 7.42-7.32 (m, 4H), 6.95 (d, J=8.7 Hz, 1H), 5.59 (s, 1H), 4.73 (d, J=9.0 Hz, 1H), 4.65 (dd, J=15.0, 7.0 Hz, 1H), 4.49 (d, J=8.7 Hz, 2H), 4.31 (dd, J=15.0, 5.1 Hz, 1H), 3.96 (dd, J=11.0, 4.2 Hz, 1H), 3.80 (dd, J=11.1, 1.5 Hz, 1H), 3.74-3.69 (m, 3H), 3.66 (d, J=12.6 Hz, 5H), 3.60 (q, J=1.7 Hz, 4H), 2.55-2.45 (m, 7H), 2.35 (d, J=14.2 Hz, 1H), 2.19 (ddd, J=14.1, 8.9, 4.8 Hz, 1H), 1.44 (s, 9H), 0.93 (s, 9H). ^{13}C NMR (101 MHz, CDCl_3) δ 172.72, 172.18, 171.57, 170.95, 150.40, 137.45, 131.18, 129.61, 128.17, 80.62, 77.35, 77.23, 77.03, 76.72, 71.15, 70.51, 70.47, 70.43, 70.32, 67.20, 66.88, 59.87, 58.62, 57.12, 43.48, 36.80, 36.23, 35.10, 34.89, 29.70, 28.10, 26.32, 16.03.

13

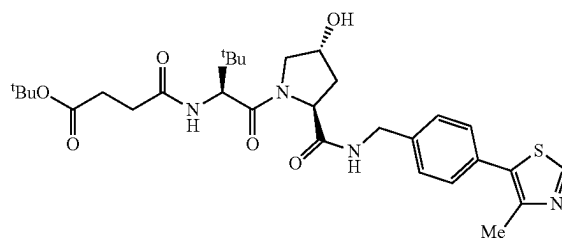




Tert-butyl ((S)-14-((2S,4R)-4-hydroxy-2-((4-(4-methylthiazol-5-yl)benzyl) carbamoyl) pyrrolidine-1-carbonyl)-15,15-dimethyl-12-oxo-3,6,9-trioxa-13-azahexadecyl) carbamate (14)

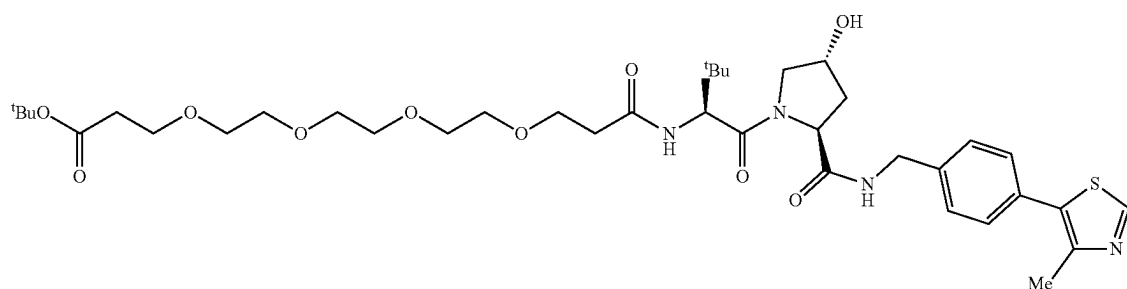
CH_2Cl_2 , UV-active); LC-MS (ESI+) calc'd for $\text{C}_{38}\text{H}_{58}\text{N}_4\text{O}_{10}\text{S}$ 762.4, found $[\text{M}+\text{H}]^+$ 763.3.

[0072] General Procedure 1. Reaction scale: 30 mg (69.9 μmol , 1.0 eq.) of SM1 and 24.6 mg (76.6 μmol , 1.1 eq.) of L8. Purified by Biotage Isolera (2-15% $\text{MeOH}/\text{CH}_2\text{Cl}_2$ gradient, 16 CV) to afford 14 as a yellow oil (40 mg, 78%). $R_f=0.52$ (10% $\text{MeOH}/\text{CH}_2\text{Cl}_2$, UV-active); LC-MS (ESI-) calc'd for $\text{C}_{36}\text{H}_{55}\text{N}_5\text{O}_9\text{S}$ 733.4, found $[\text{M}+\text{HCOO}]^-$ 778.2. ^1H NMR (400 MHz, CDCl_3) δ 8.67 (s, 1H), 7.43 (br. s, 1H), 7.36-7.29 (m, 4H), 7.02 (d, $J=8.4$ Hz, 1H), 5.11 (d, $J=7.5$ Hz, 1H), 4.70 (t, $J=8.0$ Hz, 1H), 4.57-4.45 (m, 3H), 4.32 (dd, $J=15.0, 5.3$ Hz, 1H), 4.07 (br. s, 1H), 3.70 (t, $J=5.7$ Hz, 2H), 3.60 (d, $J=15.4$ Hz, 10H), 3.50 (t, $J=5.2$ Hz, 2H), 2.52-2.41 (m, 7H), 2.15-2.07 (m, 1H), 1.42 (s, 9H), 0.93 (s, 9H). ^{13}C NMR (101 MHz, CDCl_3) δ 172.10, 171.73, 170.93, 156.08, 150.35, 148.42, 138.23, 131.65, 130.86, 129.47, 128.09, 79.26, 77.37, 77.26, 77.06, 76.74, 70.46, 70.23, 70.12, 70.06, 67.15, 60.40, 58.51, 57.71, 56.69, 43.18, 40.34, 36.64, 36.04, 34.92, 29.70, 28.44, 26.41, 21.06, 16.03, 14.20.



Tert-butyl 4-(((S)-1-((2S,4R)-4-hydroxy-2-((4-(4-methylthiazol-5-yl)benzyl) carbamoyl) pyrrolidin-1-yl)-3,3-dimethyl-1-oxobutan-2-yl)amino)-4-oxobutanoate (16)

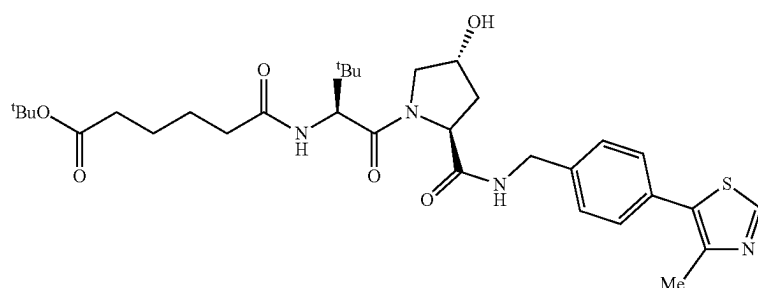
[0074] General Procedure 1. Reaction scale: 10 mg (23.2 μmol , 1.0 eq.) of SM1 and 4.5 mg (25.6 μmol , 1.1 eq.) of



Tert-butyl((S)-18-((2S,4R)-4-hydroxy-2-((4-(4-methylthiazol-5-yl)benzyl) carbamoyl) pyrrolidine-1-carbonyl)-19,19-dimethyl-16-oxo-4,7,10,13-tetraoxa-17-azaicosanoate (15)

[0073] General Procedure 1. Reaction scale: 20 mg (46.5 μmol , 1.0 eq.) of SM1 and 12.7 mg (51.1 μmol , 1.1 eq.) of L9. Purified by pTLC (10% $\text{MeOH}/\text{CH}_2\text{Cl}_2$) to afford 15 as a yellow foam (16.3 mg, 51%). $R_f=0.54$ (10% $\text{MeOH}/$

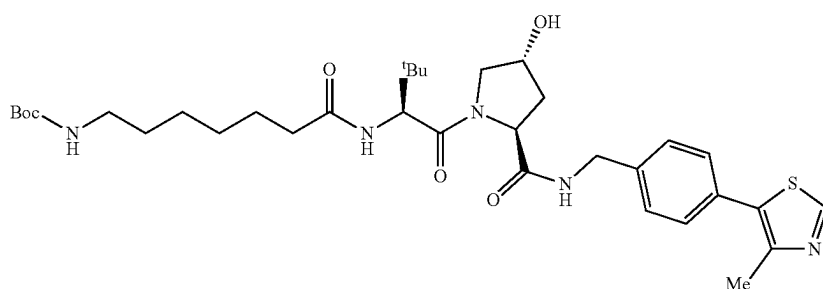
L10. Purified by pTLC (10% $\text{MeOH}/\text{CH}_2\text{Cl}_2$) to afford 16 as a white solid (7.5 mg, 53%). $R_f=0.54$ (10% $\text{MeOH}/\text{CH}_2\text{Cl}_2$, UV-active); LC-MS (ESI+) calc'd for $\text{C}_{30}\text{H}_{42}\text{N}_4\text{O}_6\text{S}$ 586.3, found $[\text{M}+\text{H}]^+$ 587.1. ^1H NMR (400 MHz, CDCl_3) δ 8.67 (s, 1H), 7.44 (t, $J=6.0$ Hz, 1H), 7.37-7.27 (m, 4H), 6.56 (d, $J=8.6$ Hz, 1H), 4.73 (t, $J=8.0$ Hz, 1H), 4.47 (d, $J=8.6$ Hz, 3H), 4.35 (d, $J=5.3$ Hz, 1H), 4.05 (d, $J=11.6$ Hz, 1H), 3.58 (dd, $J=11.4, 3.5$ Hz, 1H), 2.60-2.42 (m, 10H), 1.42 (s, 9H), 0.93 (s, 9H).



Tert-butyl 6-(((S)-1-((2S,4R)-4-hydroxy-2-((4-(4-methylthiazol-5-yl)benzyl) carbamoyl) pyrrolidin-1-yl)-3,3-dimethyl-1-oxobutan-2-yl)amino)-6-oxohexanoate (17)

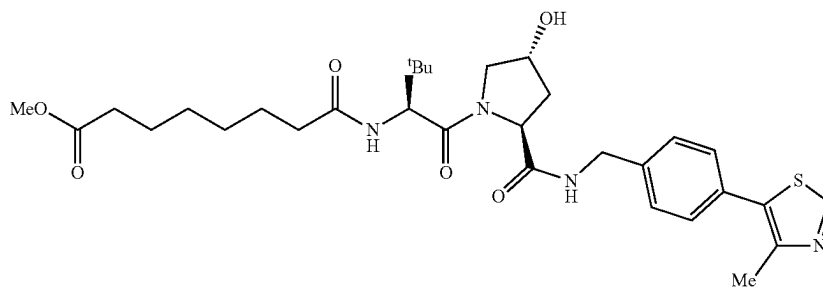
[0075] General Procedure 1. Reaction scale: 20 mg (46.5 μmol , 1.0 eq.) of SM1 and 10.3 mg (51.1 μmol , 1.1 eq.) of L11. Purified by pTLC (8% MeOH/ CH_2Cl_2) to afford 17 as

a white solid (24.6 mg, 86%). $R_f=0.55$ (8% MeOH/ CH_2Cl_2 , UV-active); LC-MS (ESI+) calc'd for $\text{C}_{32}\text{H}_{46}\text{N}_4\text{O}_6\text{S}$ 614.3, found $[\text{M}+\text{H}]^+$ 615.1. ^1H NMR (400 MHz, CDCl_3) δ 8.60 (s, 1H), 7.27 (t, $J=4.6$ Hz, 5H), 6.30 (d, $J=8.7$ Hz, 1H), 4.63 (t, $J=8.0$ Hz, 1H), 4.51-4.40 (m, 3H), 4.29-4.20 (m, 1H), 4.00 (d, $J=11.4$ Hz, 1H), 3.60-3.48 (m, 2H), 2.43 (s, 4H), 2.14-2.08 (m, 4H), 1.51 (m, 4H), 1.34 (s, 9H), 0.85 (s, 9H).



Tert-butyl (7-(((S)-1-((2S,4R)-4-hydroxy-2-((4-(4-methylthiazol-5-yl)benzyl) carbamoyl) pyrrolidin-1-yl)-3,3-dimethyl-1-oxobutan-2-yl)amino)-7-oxoheptyl) carbamate (18)

[0076] The protocols are the same as that described in Dolle et al., *J. Med. Chem.* 2021, 64, 15, 10682-10710.

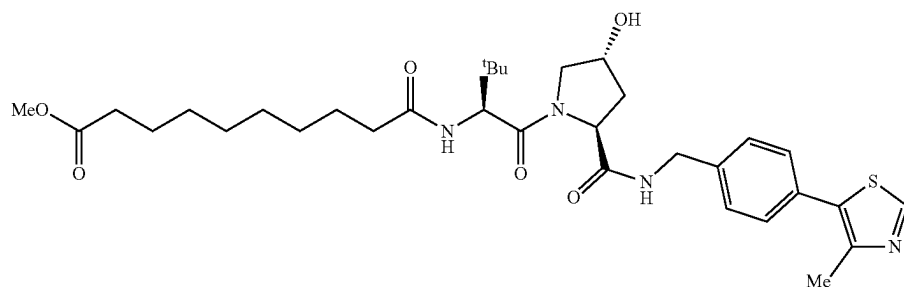


Methyl 8-(((S)-1-((2S,4R)-4-hydroxy-2-((4-(4-methylthiazol-5-yl)benzyl) carbamoyl) pyrrolidin-1-yl)-3,3-dimethyl-1-oxobutan-2-yl)amino)-8-oxooctanoate (19)

[0077] General Procedure 1. Reaction scale: 10.0 mg (23.2 μ mol, 1.0 eq.) of SM1 and 4.6 μ L (25.6 μ mol, 1.1 eq.) of L13. Purified by pTLC (10% MeOH/CH₂Cl₂) to afford 19 as a white solid (10.0 mg, 72%). R_f =0.61 (10% MeOH/CH₂Cl₂, UV-active); LC-MS (ESI+) calc'd for C₃₁H₄₄N₄O₆S 600.3, found [M+H]⁺ 601.1. ¹H NMR (400 MHz, CDCl₃) δ 8.68 (s, 1H), 7.40-7.29 (m, 5H), 6.15 (d, J=8.7 Hz, 1H), 4.71 (t, J=7.9 Hz, 1H), 4.61-4.46 (m, 3H), 4.33 (dd, J=14.9, 5.2 Hz, 1H), 4.15-4.03 (m, 1H), 3.64 (s, 3H), 2.51 (s, 3H), 2.27 (d, J=7.5 Hz, 2H), 2.19-2.15 (m, 2H), 1.63-1.56 (m, 4H), 1.31-1.25 (m, 7H), 0.93 (s, 9H).

N-(2-(3-(4-benzhydrylpiperazin-1-yl)-3-oxopropoxy)ethyl)-2-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)oxy) acetamide (21)

[0079] General Procedure 3 then 2. Reaction scale: 13.0 mg (29.1 μ mol) of 1 and 8.1 mg (32.0 μ mol) of benzhydrylpiperazine. Purified by pTLC (10% MeOH/DCM) to afford 21 as a white foam (7.5 mg, 38%). R_f =0.63 (10% MeOH/CH₂Cl₂, UV-active); LC-MS (ESI+) calc'd for C₃₇H₃₉N₅O₈ 681.3, found [M+H]⁺ 682.0. ¹H NMR (400 MHz, CDCl₃) δ 9.10 (s, 1H), 7.72 (d, J=1.1 Hz, 1H), 7.54

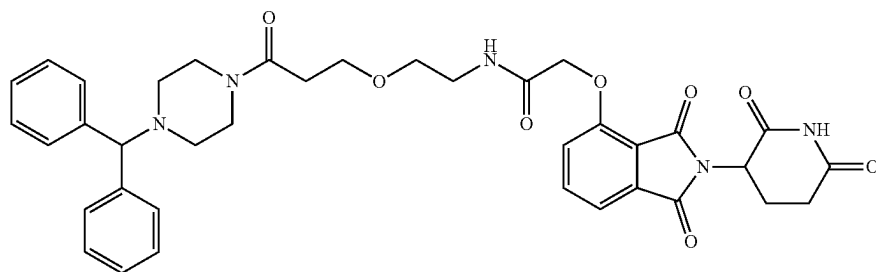


20

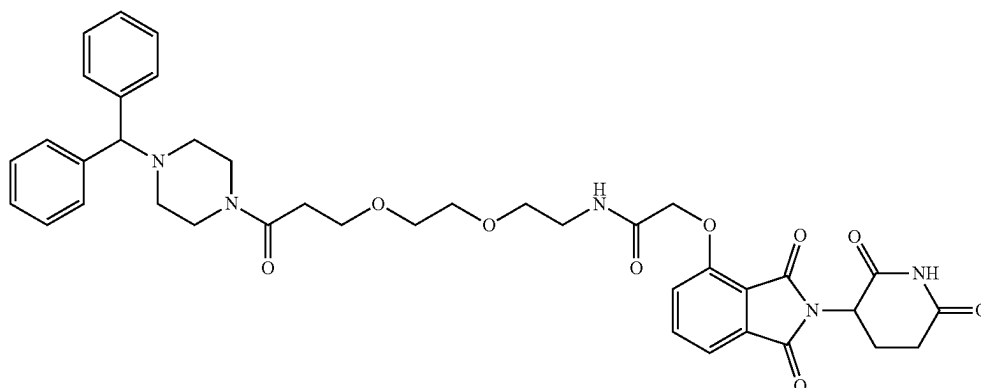
Methyl 10-(((S)-1-((2S,4R)-4-hydroxy-2-((4-(4-methylthiazol-5-yl)benzyl) carbamoyl) pyrrolidin-1-yl)-3,3-dimethyl-1-oxobutan-2-yl)amino)-10-oxodecanoate (20)

[0078] General Procedure 1. Reaction scale: 10.0 mg (23.2 μ mol, 1.0 eq.) of SM1 and 5.53 mg (25.6 μ mol, 1.1 eq.) of L16. Purified by pTLC (10% MeOH/CH₂Cl₂) to afford 20 as a white solid (10.3 mg, 71%). R_f =0.64 (10% MeOH/CH₂Cl₂, UV-active); LC-MS (ESI+) calc'd for C₃₃H₄₈N₄O₆S 628.3, found [M+H]⁺ 629.1. ¹H NMR (400 MHz, CDCl₃) δ 8.68 (s, 1H), 7.40-7.28 (m, 5H), 6.14 (d, J=8.7 Hz, 1H), 4.71 (t, J=7.9 Hz, 1H), 4.60-4.45 (m, 3H), 4.33 (dd, J=15.0, 5.2 Hz, 1H), 4.15-4.06 (m, 1H), 3.65 (d, J=4.8 Hz, 4H), 2.51 (s, 3H), 2.30-2.23 (m, 2H), 2.20-2.12 (m, 3H), 1.59 (d, J=9.7 Hz, 4H), 1.29-1.26 (m, 9H), 0.92 (s, 9H).

(dd, J=7.4, 0.6 Hz, 1H), 7.38 (s, 3H), 7.26 (d, J=1.4 Hz, 4H), 7.20-7.16 (m, 4H), 4.99-4.92 (m, 2H), 4.63 (d, J=2.9 Hz, 2H), 3.78 (p, J=1.5 Hz, 2H), 3.62 (t, J=5.1 Hz, 6H), 3.46 (d, J=4.8 Hz, 3H), 2.85-2.66 (m, 7H), 1.60 (m 3H). ¹³C NMR (101 MHz, CDCl₃) δ 171.14, 169.63, 168.26, 166.95, 166.63, 165.95, 154.60, 142.10, 136.96, 133.65, 128.63, 127.86, 127.18, 119.69, 118.25, 117.39, 77.34, 77.23, 77.02, 76.71, 75.90, 68.90, 68.19, 67.36, 60.41, 51.97, 51.51, 49.24, 45.85, 41.78, 39.37, 33.38, 31.29, 29.71, 29.37, 22.91, 14.21, 14.12.



21

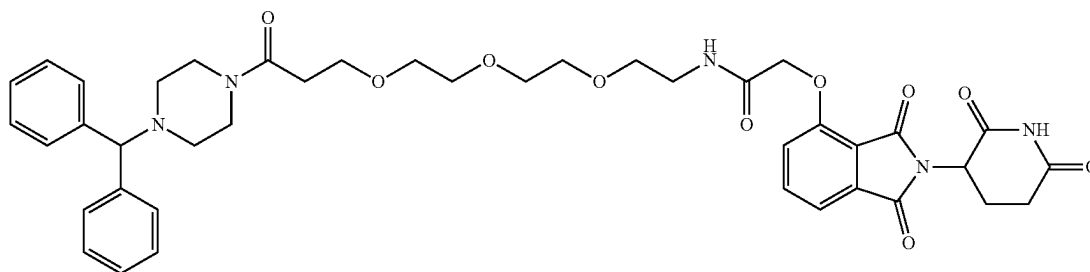


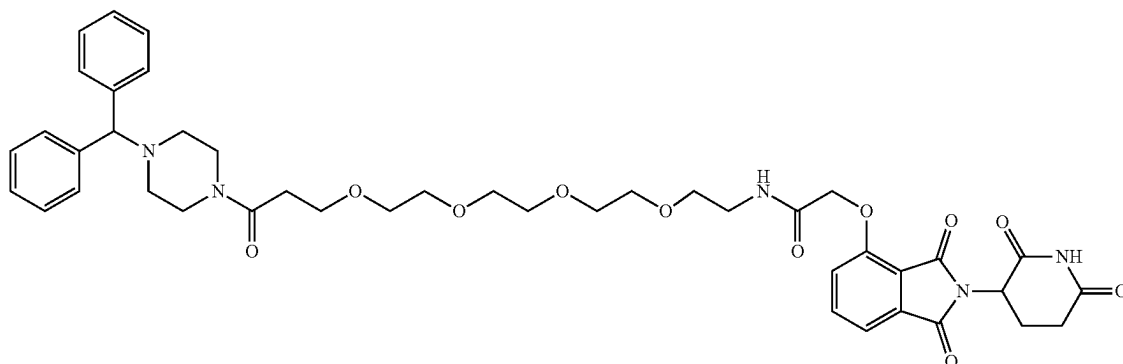
N-(2-(2-(3-(4-benzhydrylpiperazin-1-yl)-3-oxo-propoxy) ethoxy) ethoxy)ethyl)-2-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)oxy) acetamide (22)

[0080] General Procedure 3 then 2. Reaction scale: 19.9 mg (36.3 μ mol, 1 eq) of 2 and 10.1 mg (39.9 μ mol, 1.1 eq) of benzhydrylpiperazine. Purified by pTLC (10% MeOH/DCM) to afford 22 as a white foam (7.5 mg, 38%). R_f =0.61 (10% MeOH/ CH_2Cl_2 , UV-active); LC-MS (ESI+) calc'd for $\text{C}_{39}\text{H}_{43}\text{N}_5\text{O}_9$ 725.3, found $[\text{M}+\text{H}]^+$ 726.2. ^1H NMR (400 MHz, CDCl_3) δ 9.10 (s, 1H), 7.72 (dd, J =8.4, 7.4 Hz, 1H), 7.63 (t, J =4.9 Hz, 1H), 7.54 (dd, J =7.4, 0.6 Hz, 1H), 7.38 (m, 3H), 7.26 (m, 4H), 7.20-7.16 (m, 4H), 4.99-4.92 (m, 2H), 4.63 (s, 2H), 3.80 (td, J =6.9, 1.6 Hz, 3H), 3.78-3.62 (m, 13H), 3.46 (d, J =4.8 Hz, 3H), 2.85-2.66 (m, 5H). ^{13}C NMR (101 MHz, CDCl_3) δ 171.14, 169.63, 168.26, 166.95, 166.63, 165.95, 154.60, 142.10, 136.96, 133.65, 128.63, 127.86, 127.18, 119.69, 118.25, 117.39, 77.34, 77.23, 77.02, 77.02, 76.71, 75.90, 68.90, 68.19, 67.36, 60.41, 51.97, 51.51, 49.24, 45.85, 41.78, 39.37, 33.38, 31.29, 29.71, 29.37, 22.91, 14.21, 14.12.

N-(2-(2-(2-(3-(4-benzhydrylpiperazin-1-yl)-3-oxo-propoxy) ethoxy) ethoxy)ethyl)-2-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)oxy) acetamide (23)

[0081] General Procedure 3 then 2. Reaction scale: 16.11 mg (30.08 μ mol, 1.0 eq) of 3 and 8.35 mg (33.09 μ mol, 1.1 eq) of benzhydrylpiperazine. Purified by pTLC (8% MeOH/DCM) to afford 23 as a white foam (17.3 mg, 75%). R_f =0.58 (10% MeOH/ CH_2Cl_2 , UV-active); LC-MS (ESI+) calc'd for $\text{C}_{41}\text{H}_{47}\text{N}_5\text{O}_{10}$ 769.3, found $[\text{M}+\text{H}]^+$ 770.1. ^1H NMR (400 MHz, CDCl_3) δ 9.08 (s, 1H), 7.73 (dd, J =8.4, 7.3 Hz, 1H), 7.62 (t, J =5.2 Hz, 1H), 7.54 (d, J =7.3 Hz, 1H), 7.40 (d, J =7.0 Hz, 4H), 7.30-7.24 (m, 4H), 7.23-7.14 (m, 3H), 4.98-4.89 (m, 1H), 4.64 (s, 2H), 4.22 (s, 1H), 3.78 (t, J =6.8 Hz, 2H), 3.67-3.51 (m, 14H), 3.49-3.42 (m, 2H), 2.91-2.66 (m, 4H), 2.60 (t, J =6.8 Hz, 2H), 2.36 (q, J =5.4 Hz, 4H). ^{13}C NMR (101 MHz, CDCl_3) δ 169.3, 168.1, 166.8, 166.7, 165.8 (2C), 154.5, 142.2 (2C), 136.9, 128.6 (4C), 127.9 (4C), 127.2 (2C), 119.3, 117.3, 77.4, 77.0, 76.7, 75.9, 70.4, 70.3, 70.3, 69.5, 67.9, 67.2, 52.0, 51.5, 49.3, 41.8, 33.3, 31.5, 29.7, 29.4, 22.7.



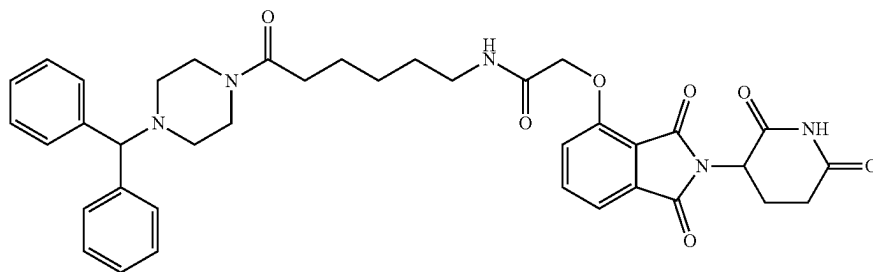


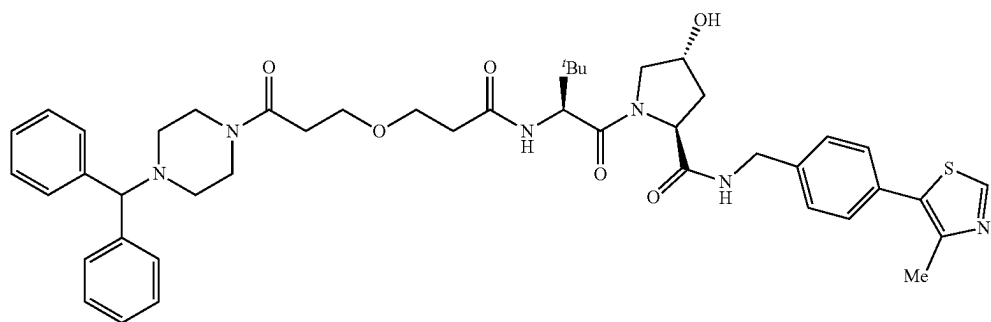
N-(15-(4-benzhydrylpiperazin-1-yl)-15-oxo-3,6,9,12-tetraoxapentadecyl)-2-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)oxy) acetamide (24)

[0082] General Procedure 3 then 2. Reaction scale: 15.0 mg (25.9 μmol , 1.0 eq) of 5 and 7.2 mg (28.5 μmol , 1.1 eq) of benzhydrylpiperazine. Purified by pTLC (10% MeOH/DCM) to afford 24 as a white foam (12.7 mg, 61%). $R_f=0.73$ (10% MeOH/ CH_2Cl_2 , UV-active); LC-MS (ESI+) calc'd for $\text{C}_{43}\text{H}_{51}\text{N}_5\text{O}_{11}$ 813.4, found $[\text{M}+\text{H}]^+$ 814.1. ^1H NMR (400 MHz, CDCl_3) δ 9.08 (s, 1H), 7.73 (dd, $J=8.4$, 7.3 Hz, 1H), 7.62 (t, $J=5.2$ Hz, 1H), 7.54 (d, $J=7.3$ Hz, 1H), 7.40 (d, $J=7.0$ Hz, 4H), 7.30-7.24 (m, 4H), 7.23-7.14 (m, 3H), 4.98-4.89 (m, 1H), 4.64 (s, 2H), 4.22 (s, 1H), 3.78 (t, $J=6.8$ Hz, 2H), 3.67-3.51 (m, 18H), 3.49-3.42 (m, 2H), 2.91-2.66 (m, 4H), 2.60 (t, $J=6.8$ Hz, 2H), 2.36 (q, $J=5.4$ Hz, 4H). ^{13}C NMR (101 MHz, CDCl_3) δ 171.11, 169.32, 168.16, 166.82, 154.46, 142.20, 136.93, 133.72, 128.61, 127.87, 127.15, 119.28, 117.27, 77.34, 77.23, 77.03, 76.71, 75.94, 70.59, 70.45, 70.35, 70.30, 69.47, 67.88, 67.26, 60.41, 52.01, 51.55, 49.34, 45.76, 41.73, 39.12, 33.41, 31.47, 29.71, 22.70, 14.21.

N-(6-(4-benzhydrylpiperazin-1-yl)-6-oxohexyl)-2-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)oxy) acetamide (25)

[0083] General Procedure 3 then 2. Reaction scale: 17 mg (38.17 μmol , 1.0 eq) of 6 and 10.4 mg (41.21 μmol , 1.1 eq) of benzhydrylpiperazine. Purified by pTLC (8% MeOH/DCM) to afford 25 as a white foam (25 mg, 99%). $R_f=0.67$ (8% MeOH/ CH_2Cl_2 , UV-active); LC-MS (ESI+) calc'd for $\text{C}_{38}\text{H}_{41}\text{N}_5\text{O}_7$ 679.3, found $[\text{M}+\text{H}]^+$ 680.0. ^1H NMR (400 MHz, CDCl_3) δ 9.08 (br. s, 1H), 7.73 (dd, $J=8.4$, 7.4 Hz, 1H), 7.54 (dd, $J=7.4$, 0.6 Hz, 1H), 7.39 (d, $J=7.3$ Hz, 4H), 7.26 (s, 4H), 7.20-7.17 (m, 3H), 4.98-4.93 (m, 1H), 4.63 (q, $J=3.1$ Hz, 2H), 4.21 (s, 1H), 3.78 (ddd, $J=9.1$, 7.1, 2.4 Hz, 2H), 3.48-3.27 (m, 7H), 2.90-2.76 (m, 1H), 2.73-2.58 (m, 3H), 2.36 (s, 3H), 1.60-1.30 (m, 6H). ^{13}C NMR (101 MHz, CDCl_3) δ 171.96, 171.31, 168.33, 166.84, 166.60, 166.27, 154.81, 142.14, 137.04, 133.58, 128.62, 127.87, 127.17, 120.24, 118.51, 117.61, 77.35, 77.23, 77.03, 76.81, 76.71, 75.96, 68.67, 52.11, 51.63, 49.35, 45.89, 41.83, 39.21, 33.04, 31.41, 29.71, 28.87, 26.78, 25.35, 22.86.





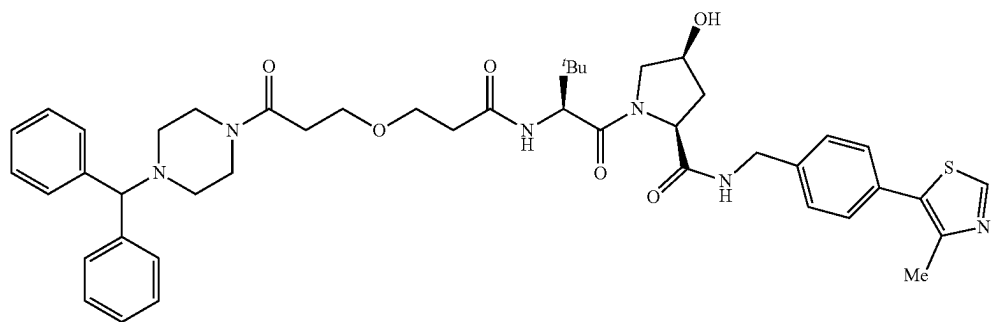
26

(2S,4R)-1-((S)-2-(3-(3-(4-benzhydrylpiperazin-1-yl)-3-oxopropoxy) propanamido)-3,3-dimethylbutanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl) pyrrolidine-2-carboxamide (26)

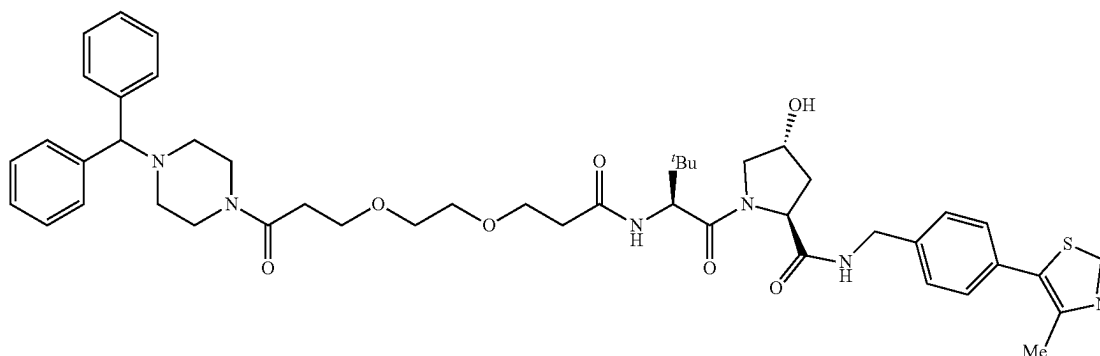
[0084] General Procedure 3 then 2. Reaction scale: 10.2 mg (17.8 μ mol, 1.0 eq) of 8 and 4.93 mg (19.52 μ mol, 1.1 eq) of benzhydrylpiperazine. Purified by pTLC (10% MeOH/DCM) to afford 26 as a pale yellow solid (6 mg, 42%). R_f =0.46 (10% MeOH/ CH_2Cl_2 , UV-active); LC-MS (ESI+) calc'd for $\text{C}_{45}\text{H}_{56}\text{N}_6\text{O}_6\text{S}$ 808.4, found $[\text{M}+\text{H}]^+$ 809. 3. ^1H NMR (400 MHz, CDCl_3) δ 8.68 (s, 1H), 7.47-7.30 (m, 11H), 7.20 (d, J =7.4 Hz, 2H), 6.95 (d, J =8.3 Hz, 1H), 4.73-4.65 (m, 1H), 4.59-4.52 (m, 1H), 4.45 (d, J =8.6 Hz, 1H), 4.29 (dd, J =15.1, 5.1 Hz, 1H), 3.79-3.49 (m, 10H), 2.52 (s, 7H), 2.03 (d, J =15.4 Hz, 1H), 1.61 (m, 9H), 0.90 (s, 9H).

(2S,4S)-1-((S)-2-(3-(3-(4-benzhydrylpiperazin-1-yl)-3-oxopropoxy) propanamido)-3,3-dimethylbutanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl) pyrrolidine-2-carboxamide (27)

[0085] General Procedure 3 then 2. Reaction scale: 11.4 mg (19.9 μ mol, 1 eq.) of 9 and 5.6 mg (22 μ mol) of benzhydrylpiperazine. Purified by pTLC (10% MeOH/DCM) to afford 27 as a white solid (6 mg, 42%). R_f =0.46 (10% MeOH/ CH_2Cl_2 , UV-active); LC-MS (ESI+) calc'd for $\text{C}_{45}\text{H}_{56}\text{N}_6\text{O}_6\text{S}$ 808.4, found $[\text{M}+\text{H}]^+$ 809.2. ^1H NMR (400 MHz, CDCl_3) δ 8.68 (s, 1H), 7.41-7.26 (m, 12H), 7.20 (d, J =7.4 Hz, 2H), 4.68-4.56 (m, 2H), 4.48 (d, J =8.9, 2H), 4.25 (dd, J =15.1, 5.1 Hz, 1H), 3.92 (dd, J =10.6, 4.1 Hz, 1H), 3.81-3.65 (m, 7H), 3.59 (s, 1H), 3.45 (d, J =5.3 Hz, 1H), 2.66-2.55 (m, 2H), 2.52 (s, 4H), 2.18 (dd, J =8.9, 5.4 Hz, 2H), 1.56 (s, 9H), 0.89 (s, 9H).



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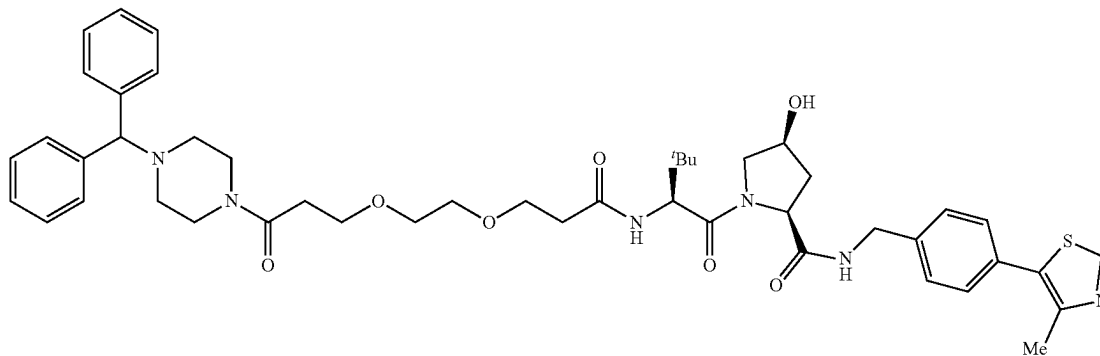


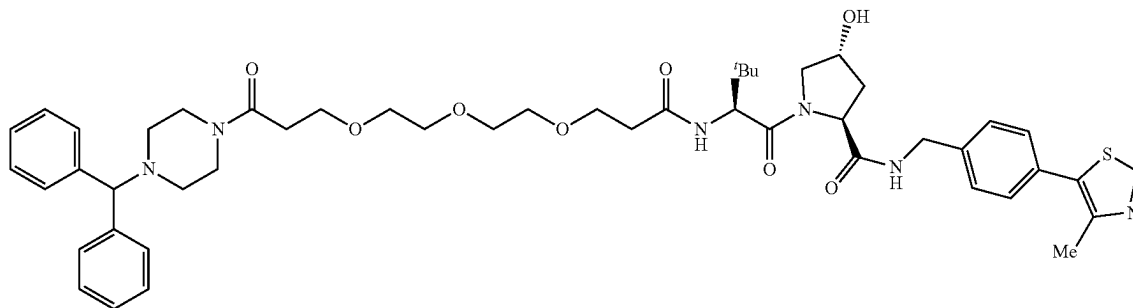
(2S,4R)-1-((S)-2-(3-(2-(3-(4-benzhydrylpiperazin-1-yl)-3-oxopropoxy) ethoxy) propanamido)-3,3-dimethylbutanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl) pyrrolidine-2-carboxamide (28)

[0086] General Procedure 3 then 2. Reaction scale: 7.5 mg (11.1 μmol , 1.0 eq) of 10 and 3.1 mg (12.2 μmol , 1.1 eq) of benzhydrylpiperazine. Purified by pTLC (10% MeOH/DCM) to afford 28 as a white foam (6.4 mg, 25%). $R_f=0.46$ (8% MeOH/ CH_2Cl_2 , UV-active); LC-MS (ESI+) calc'd for $\text{C}_{47}\text{H}_{60}\text{N}_6\text{O}_7\text{S}$ 852.4, found $[\text{M}+\text{H}]^+$ 853.3. ^1H NMR (400 MHz, CDCl_3) δ 8.67 (s, 1H), 7.52 (t, $J=6.0$ Hz, 1H), 7.41-7.35 (m, 5H), 7.34 (s, 3H), 7.28 (d, $J=7.1$ Hz, 3H), 7.21-7.16 (m, 2H), 7.03 (d, $J=8.3$ Hz, 1H), 4.71 (t, $J=8.1$ Hz, 1H), 4.57 (dd, $J=15.0$, 6.6 Hz, 1H), 4.50-4.42 (m, 2H), 4.32 (dd, $J=15.0$, 5.2 Hz, 1H), 4.20 (s, 1H), 4.12-4.06 (m, 1H), 3.80-3.65 (m, 5H), 3.59 (ddd, $J=10.3$, 6.3, 4.3 Hz, 7H), 3.49-3.41 (m, 3H), 2.56 (td, $J=6.8$, 1.3 Hz, 2H), 2.51 (s, 4H), 2.46 (q, $J=4.6$ Hz, 3H), 2.34 (dt, $J=10.7$, 5.2 Hz, 4H), 0.93 (s, 9H). ^{13}C NMR (101 MHz, CDCl_3) δ 172.14, 171.70, 170.88, 169.32, 150.27, 148.45, 142.08, 138.26, 131.63, 130.86, 129.47, 128.63, 128.10, 127.83, 127.19, 77.34, 77.22, 77.02, 76.70, 75.92, 70.48, 70.38, 70.13, 67.38, 67.16, 58.34, 57.70, 56.67, 51.97, 51.50, 45.75, 43.18, 41.72, 36.77, 35.95, 34.85, 33.38, 26.41, 16.08.

(2S,4S)-1-((S)-2-(3-(2-(3-(4-benzhydrylpiperazin-1-yl)-3-oxopropoxy) ethoxy) propanamido)-3,3-dimethylbutanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl) pyrrolidine-2-carboxamide (29)

[0087] General Procedure 3 then 1. Reaction scale: 16.2 mg (26.2 μmol , 1.0 eq) of 11 and 7.3 mg (28.8 μmol , 1.1 eq) of benzhydrylpiperazine. Purified by pTLC (10% MeOH/DCM) to afford 29 as a white foam (19.6 mg, 88%). $R_f=0.51$ (10% MeOH/ CH_2Cl_2 , UV-active); LC-MS (ESI+) calc'd for $\text{C}_{47}\text{H}_{60}\text{N}_6\text{O}_7\text{S}$ 852.4, found $[\text{M}+\text{H}]^+$ 853.3. ^1H NMR (600 MHz, CDCl_3) δ 8.59 (s, 1H), 7.44 (t, $J=6.0$ Hz, 1H), 7.32-7.28 (m, 4H), 7.26 (d, $J=1.2$ Hz, 4H), 7.21-7.17 (m, 5H), 7.12-7.08 (m, 2H), 6.98 (d, $J=8.4$ Hz, 1H), 4.63 (t, $J=8.1$ Hz, 1H), 4.48 (dd, $J=15.0$, 6.7 Hz, 1H), 4.43-4.38 (m, 2H), 4.23 (dd, $J=15.0$, 5.3 Hz, 1H), 4.12 (s, 1H), 4.06-3.99 (m, 1H), 3.69-3.57 (m, 5H), 3.51 (dddd, $J=15.6$, 12.6, 8.7, 5.3 Hz, 7H), 3.36 (dd, $J=6.2$, 4.0 Hz, 2H), 2.48 (td, $J=6.9$, 1.8 Hz, 2H), 2.43 (s, 3H), 2.38 (t, $J=5.6$ Hz, 2H), 2.26 (dt, $J=17.0$, 5.2 Hz, 4H), 0.85 (s, 9H). ^{13}C NMR (151 MHz, CDCl_3) δ 172.16, 171.69, 171.20, 170.96, 169.38, 150.32, 148.43, 142.09, 138.31, 131.67, 130.83, 130.08, 129.47, 129.41, 128.64, 128.09, 127.84, 127.21, 77.26, 77.04, 76.83, 75.93, 70.47, 70.38, 70.14, 67.37, 67.18, 60.42, 60.37, 58.45, 57.73, 56.71, 56.00, 51.98, 51.51, 45.76, 43.18, 41.74, 36.74, 36.06, 34.94, 33.38, 31.94, 29.72, 29.38, 26.42, 22.71, 21.08, 20.80, 16.07, 14.22, 14.14.



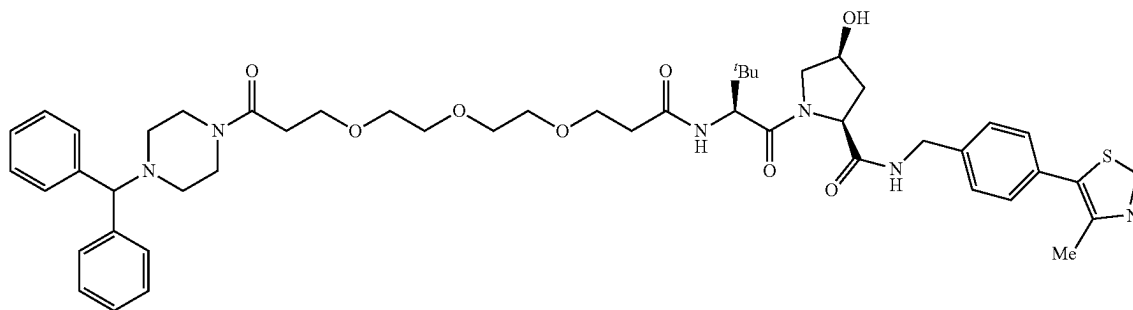


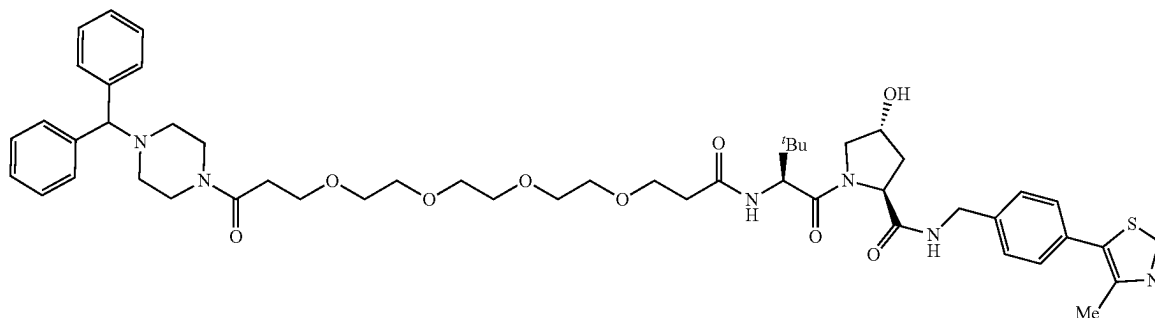
(2S,4R)-1-((S)-16-(4-benzhydrylpiperazin-1-yl)-2-(tert-butyl)-4,16-dioxo-7,10,13-trioxa-3-azahexadecanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide (30)

[0088] General Procedure 3 then 1. Reaction scale: 21.9 mg (33.0 μ mol, 1.0 eq) of 12 and 9.2 mg (36.35 μ mol, 1.1 eq) of benzhydrylpiperazine. Purified by pTLC (10% MeOH/DCM) to afford 30 as a white foam (10.5 mg, 35%). R_f =0.52 (10% MeOH/ CH_2Cl_2 , UV-active); LC-MS (ESI+) calc'd for $\text{C}_{49}\text{H}_{64}\text{N}_6\text{O}_8\text{S}$ 896.5, found $[\text{M}+\text{H}]^+$ 897.3. ^1H NMR (600 MHz, CDCl_3) δ 8.67 (s, 1H), 7.47 (t, J =6.0 Hz, 1H), 7.40-7.38 (m, 4H), 7.36-7.33 (m, 4H), 7.29-7.25 (m, 5H), 7.21-7.16 (m, 2H), 7.02 (d, J =8.4 Hz, 1H), 4.72 (t, J =8.1 Hz, 1H), 4.56 (dd, J =15.0, 6.7 Hz, 1H), 4.50-4.46 (m, 2H), 4.32 (dd, J =15.0, 5.3 Hz, 1H), 4.21 (s, 1H), 4.10 (d, J =11.0 Hz, 1H), 3.74 (tt, J =6.1, 3.1 Hz, 3H), 3.69 (dddd, J =9.8, 8.5, 5.0, 2.7 Hz, 3H), 3.60 (dd, J =13.1, 2.0 Hz, 9H), 3.45 (t, J =5.1 Hz, 2H), 2.57 (td, J =6.8, 4.1 Hz, 2H), 2.51 (s, 3H), 2.47 (ddd, J =7.2, 4.7, 3.1 Hz, 2H), 2.35 (dt, J =16.0, 4.1 Hz, 4H), 0.93 (s, 9H). ^{13}C NMR (151 MHz, CDCl_3) δ 172.19, 171.75, 170.91, 169.39, 150.32, 148.45, 142.12, 138.26, 131.65, 130.90, 130.88, 129.49, 128.64, 128.12, 127.85, 127.20, 77.26, 77.14, 77.04, 76.83, 75.94, 70.45, 70.37, 70.30, 70.19, 70.12, 67.17, 67.14, 58.39, 57.76, 56.70, 51.96, 51.52, 50.82, 45.73, 43.20, 41.74, 36.61, 35.98, 34.84, 33.71, 33.38, 30.37, 30.18, 29.72, 26.42, 16.08.

(2S,4S)-1-((S)-16-(4-benzhydrylpiperazin-1-yl)-2-(tert-butyl)-4,16-dioxo-7,10,13-trioxa-3-azahexadecanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide (31)

[0089] General Procedure 3 then 1. Reaction scale: 10 mg (15.1 μ mol, 1.0 eq) of 13 and 4.2 mg (16.6 μ mol, 1.1 eq) of benzhydrylpiperazine. Purified by pTLC (10% MeOH/DCM) to afford 30 as a white foam (10.5 mg, 35%). R_f =0.57 (10% MeOH/ CH_2Cl_2 , UV-active); LC-MS (ESI-) calc'd for $\text{C}_{49}\text{H}_{64}\text{N}_6\text{O}_8\text{S}$ 896.5, found $[\text{M}+\text{HCOO}]^-$ 941.1. ^1H NMR (400 MHz, CDCl_3) δ 8.68 (s, 1H), 7.46 (d, J =6.2 Hz, 1H), 7.42-7.37 (m, 4H), 7.34 (d, J =1.7 Hz, 4H), 7.29-7.24 (m, 5H), 7.21-7.16 (m, 2H), 7.05 (d, J =8.3 Hz, 1H), 4.72 (t, J =8.0 Hz, 1H), 4.57 (dd, J =15.0, 6.6 Hz, 1H), 4.47 (d, J =8.8 Hz, 2H), 4.32 (dd, J =15.0, 5.2 Hz, 1H), 4.21 (s, 1H), 4.14-4.09 (m, 1H), 3.72 (dt, J =17.1, 4.5 Hz, 5H), 3.60 (d, J =8.8 Hz, 9H), 3.45 (t, J =5.0 Hz, 2H), 2.61-2.54 (m, 2H), 2.51 (s, 3H), 2.46 (d, J =5.1 Hz, 2H), 2.35 (dt, J =10.0, 5.0 Hz, 4H), 0.93 (s, 9H). ^{13}C NMR (151 MHz, CDCl_3) δ 172.85, 172.00, 171.51, 169.28, 150.36, 148.50, 142.08, 129.73, 129.55, 128.63, 128.24, 128.11, 127.92, 127.83, 127.57, 127.19, 77.22, 77.01, 76.80, 75.94, 71.16, 70.51, 70.44, 70.32, 70.23, 67.25, 67.21, 59.82, 58.57, 57.01, 55.98, 51.97, 51.52, 45.72, 43.42, 41.70, 36.79, 35.22, 34.97, 33.69, 33.43, 31.92, 30.16, 30.03, 29.70, 29.65, 29.60, 29.47, 29.36, 29.32, 27.21, 26.70, 26.31, 25.53, 23.17, 22.69, 21.49, 16.06, 14.18, 14.12.



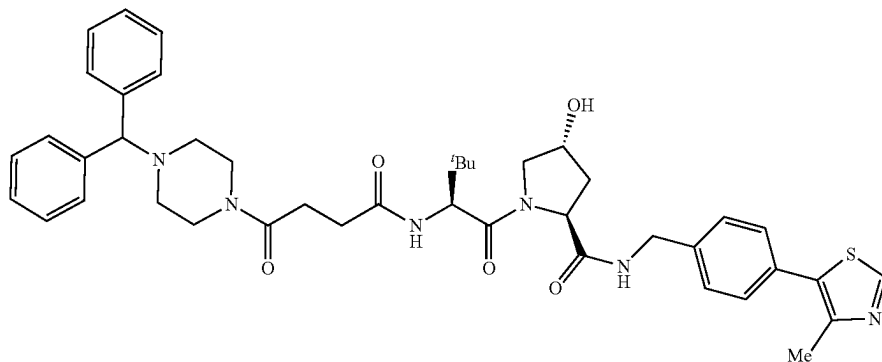


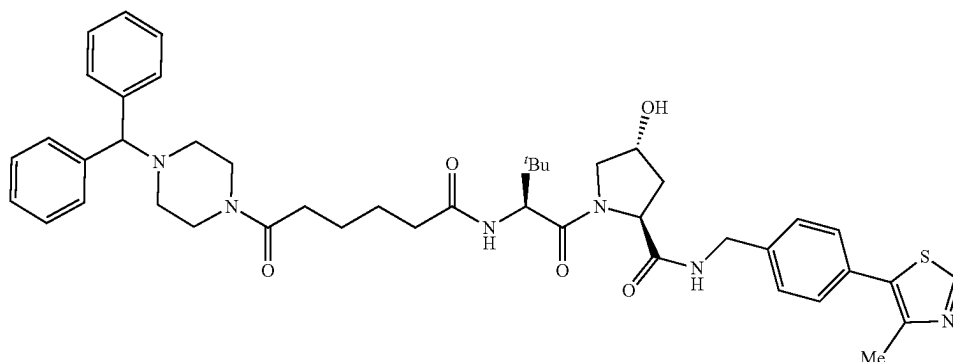
(2S,4R)-1-((S)-19-(4-benzhydrylpiperazin-1-yl)-2-(tert-butyl)-4,19-dioxo-7,10,13,16-tetraoxa-3-azanonadecanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide (32)

[0090] General Procedure 3 then 2. Reaction scale: 15.3 mg (21.65 μmol) of 15 and 6.01 mg (23.81 μmol) of benzhydrylpiperazine. Purified by pTLC (10% MeOH/DCM) to afford 32 as a white solid (7.0 mg, 35%). $R_f=0.47$ (10% MeOH/ CH_2Cl_2 , UV-active); LC-MS (ESI+) calc'd for $\text{C}_{51}\text{H}_{68}\text{N}_6\text{O}_9\text{S}$ 940.5, found $[\text{M}+\text{H}]^+$ 941.4. ^1H NMR (400 MHz, CDCl_3) δ 8.68 (s, 1H), 7.43-7.31 (m, 9H), 7.27 (d, $J=5.3$ Hz, 5H), 7.23-7.15 (m, 2H), 7.03 (d, $J=8.2$ Hz, 1H), 4.73 (t, $J=8.0$ Hz, 1H), 4.57 (dd, $J=14.9$, 6.6 Hz, 1H), 4.52-4.43 (m, 2H), 4.33 (dd, $J=14.9$, 5.2 Hz, 1H), 4.22 (s, 1H), 4.14 (d, $J=7.1$ Hz, 1H), 3.78-3.70 (m, 5H), 3.66-3.58 (m, 15H), 3.46 (dd, $J=11.2$, 6.2 Hz, 2H), 2.61-2.57 (m, 2H), 2.52 (d, $J=1.7$ Hz, 3H), 2.50-2.46 (m, 2H), 2.39-2.30 (m, 4H), 0.93 (s, 9H).

(2S,4R)-1-((S)-2-(4-(4-benzhydrylpiperazin-1-yl)-4-oxobutanamido)-3,3-dimethylbutanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide (33)

[0091] General Procedure 3 then 2. Reaction scale: 6.7 mg (12.63 μmol) of 16 and 3.5 mg (13.89 μmol) of benzhydrylpiperazine. Purified by pTLC (10% MeOH/DCM) to afford 33 as a pale yellow solid (4.2 mg, 44%). $R_f=0.58$ (10% MeOH/ CH_2Cl_2 , UV-active); LC-MS (ESI+) calc'd for $\text{C}_{43}\text{H}_{52}\text{N}_6\text{O}_5\text{S}$ 764.4, found $[\text{M}+\text{H}]^+$ 766.1. ^1H NMR (400 MHz, CDCl_3) δ 8.67 (s, 1H), 7.42-7.32 (m, 12H), 7.27 (s, 3H), 7.20 (d, $J=7.3$ Hz, 2H), 4.75 (t, $J=8.1$ Hz, 2H), 4.57 (dd, $J=14.9$, 6.6 Hz, 1H), 4.49 (d, $J=8.3$ Hz, 2H), 4.33 (dd, $J=14.9$, 5.1 Hz, 1H), 4.22 (s, 1H), 4.12 (d, $J=6.8$ Hz, 1H), 3.65 (d, $J=7.6$ Hz, 2H), 3.56 (d, $J=13.7$ Hz, 4H), 3.49 (s, 1H), 3.41 (d, $J=5.5$ Hz, 3H), 2.51 (s, 3H), 2.35 (d, $J=4.9$ Hz, 4H), 0.92 (s, 9H).



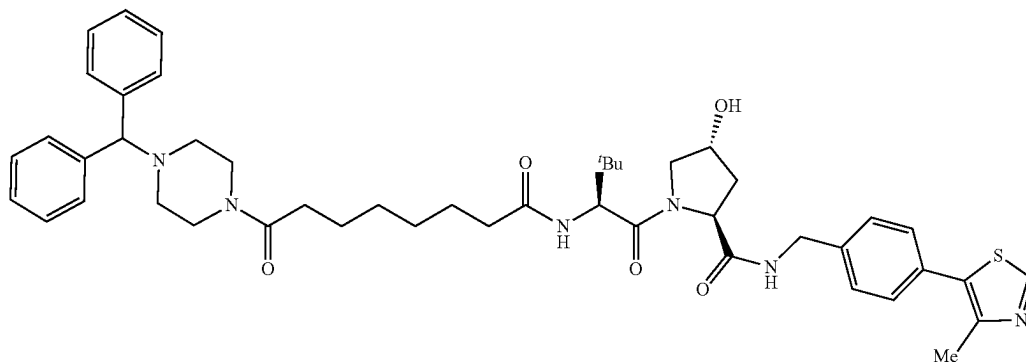


(2S,4R)-1-((S)-2-(6-(4-benzhydrylpiperazin-1-yl)-6-oxohexanamido)-3,3-dimethylbutanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide (34)

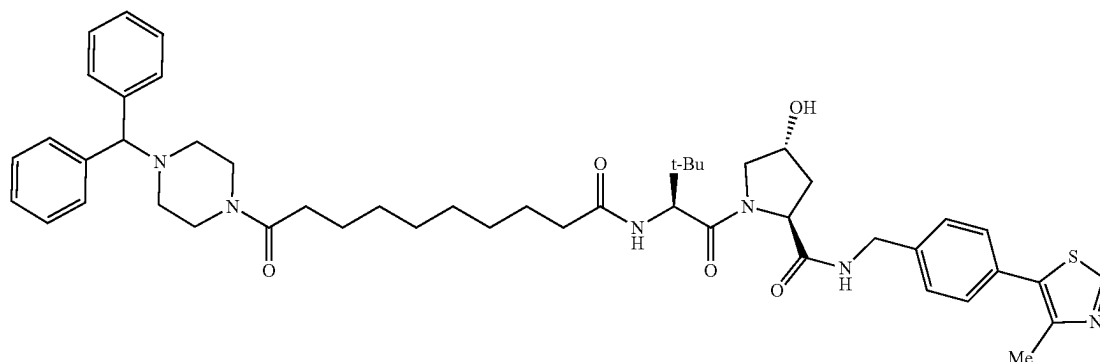
[0092] General Procedure 3 then 2. Reaction scale: 9 mg (16.11 μmol) of 17 and 4.47 mg (17.72 μmol) of benzhydrylpiperazine. Purified by pTLC (10% MeOH/DCM) to afford 21 as a pale yellow solid (3.3 mg, 26%). $R_f=0.54$ (10% MeOH/ CH_2Cl_2 , UV-active); LC-MS (ESI+) calc'd for $\text{C}_{45}\text{H}_{56}\text{N}_6\text{O}_5\text{S}$ 792.4, found $[\text{M}+\text{H}]^+$ 793.9. ^1H NMR (400 MHz, CDCl_3) δ 8.86 (br. s, 1H), 7.73 (dd, $J=8.4$, 7.4 Hz, 1H), 7.54 (dd, $J=7.4$, 0.6 Hz, 1H), 7.50 (br. t, 2H), 7.39 (d, $J=7.3$ Hz, 5H), 7.30 (s, 1H), 7.25 (br. t, 2H), 7.20-7.17 (m, 3H), 4.72 (t, 2H), 4.58 (s, 1H) 4.50 (s, 1H), 4.60-4.30 (dd, $J=7$ Hz, 1H), 4.20 (s, 1H), 4.15-4.10 (d, $J=7$ Hz, 2H), 3.62 (q, $J=5.4$ Hz, 4H), 3.47 (q, $J=6.7$, 4.9 Hz, 4H), 2.60-2.35 (m, 13H), 2.20-2.10 (m, 1H), 2.05 (s, 1H), 0.93 (s, 9H).

(2S,4R)-1-((S)-2-(8-(4-benzhydrylpiperazin-1-yl)-8-oxooctanamido)-3,3-dimethylbutanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide (35)

[0093] General Procedure 5 then 2. Reaction scale: 8.5 mg (14.49 μmol) of 19 and 4.02 mg (15.94 μmol) of benzhydrylpiperazine. Purified by pTLC (10% MeOH/DCM) to afford 23 as a pale yellow solid (6.4 mg, 54%). $R_f=0.53$ (10% MeOH/ CH_2Cl_2 , UV-active); LC-MS (ESI+) calc'd for $\text{C}_{47}\text{H}_{60}\text{N}_6\text{O}_5\text{S}$ 820.4, found $[\text{M}+\text{H}]^+$ 821.0. ^1H NMR (400 MHz, CDCl_3) δ 8.86 (br. s, 1H), 7.73 (dd, $J=8.4$, 7.4 Hz, 1H), 7.54 (dd, $J=7.4$, 0.6 Hz, 1H), 7.50 (br. t, 2H), 7.39 (d, $J=7.3$ Hz, 5H), 7.30 (s, 1H), 7.25 (br. t, 2H), 7.20-7.17 (m, 3H), 4.72 (t, 2H), 4.58 (s, 1H) 4.50 (s, 1H), 4.60-4.30 (dd, $J=7$ Hz, 1H), 4.20 (s, 1H), 4.15-4.10 (d, $J=7$ Hz, 2H), 3.62 (q, $J=5.4$ Hz, 4H), 3.47 (q, $J=6.7$, 4.9 Hz, 4H), 2.60-2.35 (m, 17H), 2.20-2.10 (m, 1H), 2.05 (s, 1H), 0.93 (s, 9H).



36



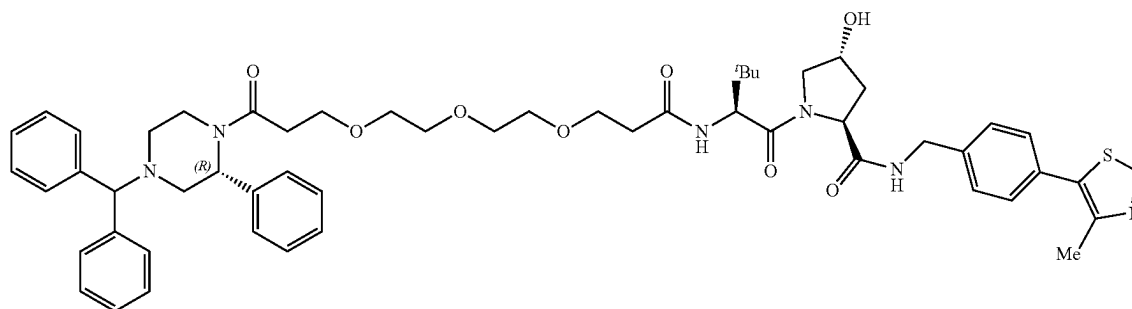
(2S,4R)-1-((S)-2-(10-(4-benzhydrylpiperazin-1-yl)-10-oxodecanamido)-3,3-dimethylbutanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide (36)

[0094] General Procedure 5 then 2. Reaction scale: 10 mg (16.27 μmol) of 20 and 4.52 mg (17.89 μmol) of benzhydrylpiperazine. Purified by pTLC (10% MeOH/DCM) to afford 36 as a pale yellow solid (8.5 mg, 62%). $R_f=0.62$ (10% MeOH/ CH_2Cl_2 , UV-active); LC-MS (ESI+) calc'd for $\text{C}_{49}\text{H}_{64}\text{N}_6\text{O}_5\text{S}$ 848.5, found $[\text{M}+\text{H}]^+$ 849.3. ^1H NMR (400 MHz, CDCl_3) δ 8.86 (br. s, 1H), 7.73 (dd, $J=8.4$, 7.4 Hz, 1H), 7.54 (dd, $J=7.4$, 0.6 Hz, 1H), 7.50 (br. t, 2H), 7.39 (d, $J=7.3$ Hz, 5H), 7.30 (s, 1H), 7.25 (br. t, 2H), 7.20-7.17 (m, 3H), 4.72 (t, 2H), 4.58 (s, 1H) 4.50 (s, 1H), 4.60-4.30 (dd, $J=7$ Hz, 1H), 4.20 (s, 1H), 4.15-4.10 (d, $J=7$ Hz, 2H), 3.62 (q, $J=5.4$ Hz, 4H), 3.47 (q, $J=6.7$, 4.9 Hz, 4H), 2.60-2.35 (m, 21H), 2.20-2.10 (m, 1H), 2.05 (s, 1H), 0.93 (s, 9H).

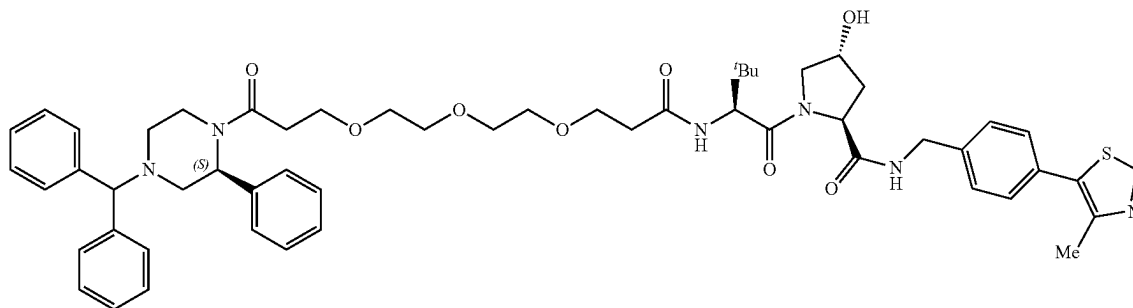
(2S,4R)-1-((S)-16-((R)-4-benzhydryl-2-phenylpiperazin-1-yl)-2-(tert-butyl)-4,16-dioxo-7,10,13-trioxo-3-azahexadecanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide (37)

[0095] General Procedure 3 then 1. Reaction scale: 15 mg (22.6 μmol , 1.0 eq.) of 12 and 8.2 mg (24.9 μmol , 1.1 eq.) of (4R)-benzhydrylpiperazine. Purified by pTLC (8% MeOH/ CH_2Cl_2) to afford 37 as an off-white solid (8.3 mg, 38%). $R_f=0.55$ (8% MeOH/ CH_2Cl_2 , UV-active); LC-MS (ESI+) calc'd for $\text{C}_{55}\text{H}_{70}\text{N}_6\text{O}_8\text{S}$ $[\text{M}+\text{H}]^+$: 972.5, found 974.0; ^1H NMR (600 MHz, CDCl_3) δ 8.68 (s, 1H), 7.34 (s, 13H), 7.19 (s, 6H), 7.01 (d, $J=8.3$ Hz, 1H), 4.73 (s, 1H), 4.47 (d, $J=8.3$ Hz, 2H), 4.28-4.02 (m, 3H), 3.79-3.03 (m, 19H), 2.50 (s, 8H), 2.17-2.03 (m, 4H), 0.91 (s, 9H). Note: presence of rotamers.

37



38

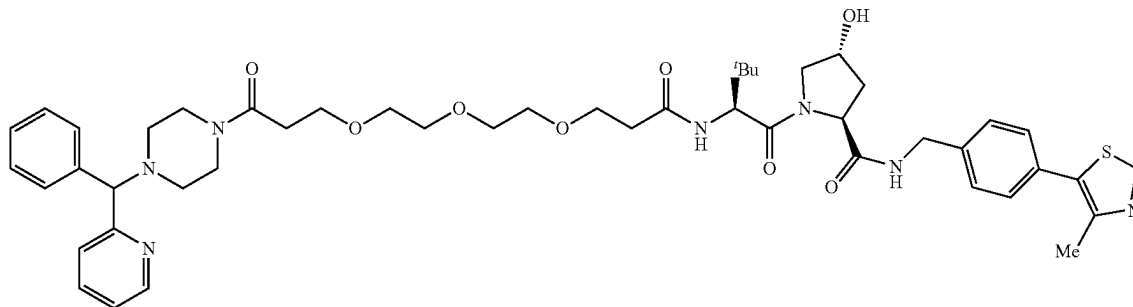


(2S,4R)-1-((S)-16-((S)-4-benzhydryl-2-phenylpiperazin-1-yl)-2-(tert-butyl)-4,16-dioxo-7,10,13-trioxa-3-azahexadecanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide (38)

[0096] General Procedure 1 and 3. Reaction scale: 21.6 mg (30.0 μ mol, 1.00 equiv) of 12 and 14.9 mg (35.0 μ mol, 1.17 equiv) of (4S)-phenylbenzhydrylpiperazine derivative. Purified by PTLC (10% MeOH/EtOAc) to afford 38 as a clear oil (5.5 mg, 19%). R_f =0.47 (10% MeOH/EtOAc, UV-active); LC-MS (ESI+) calc'd for $C_{55}H_{70}N_6O_8S$ $[M+2H]^{2+}$: 487.2, found 487.1; 1H NMR (600 MHz, $CDCl_3$) δ 8.67 (s, 1H), 7.41-7.27 (m, 10H), 7.23-7.13 (m, 9H), 6.34 (d, J =8.5 Hz, 1H), 5.76 (br s, 1H), 4.59 (m, 1H), 4.51-4.46 (m, 2H), 4.27-4.18 (m, 3H), 3.83-3.30 (m, 19H), 2.91 (m, 1H), 2.60-2.46 (m, 6H), 2.41 (m, 2H), 2.31-2.19 (m, 2H), 2.07 (m, 1H), 0.87-0.76 (m, 9H). Note: presence of rotamers.

MeOH/ CH_2Cl_2) to afford 39 as white foam (18.2 mg, 67%). R_f =0.62 (8% MeOH/ CH_2Cl_2 , UV-active); LC-MS (ESI+) calc'd for $C_{48}H_{63}N_7O_8S$ $[M+H]^+$: 897.5, found 898.7; 1H NMR (600 MHz, $CDCl_3$) δ 8.59 (s, 1H), 8.44-8.40 (m, 1H), 7.55 (td, J =7.7, 1.8 Hz, 1H), 7.45 (dd, J =7.9, 1.0 Hz, 1H), 7.43-7.40 (m, 1H), 7.38 (dt, J =8.1, 1.5 Hz, 2H), 7.28-7.24 (m, 4H), 7.20 (dd, J =8.3, 6.9 Hz, 2H), 7.18 (s, 1H), 7.14-7.10 (m, 1H), 7.03 (ddt, J =7.3, 4.9, 1.2 Hz, 1H), 6.95 (dd, J =8.3, 1.6 Hz, 1H), 4.63 (td, J =8.0, 1.4 Hz, 1H), 4.47 (dd, J =15.0, 6.7 Hz, 1H), 4.43-4.38 (m, 2H), 4.34 (s, 1H), 4.24 (dd, J =15.0, 5.3 Hz, 1H), 4.06-3.99 (m, 2H), 3.68-3.59 (m, 4H), 3.55-3.48 (m, 12H), 3.41-3.37 (m, 2H), 2.49 (qd, J =6.8, 1.6 Hz, 2H), 2.42 (s, 3H), 2.39 (ddd, J =8.7, 6.4, 3.9 Hz, 3H), 2.28-2.23 (m, 2H), 0.85 (s, 9H). ^{13}C NMR (151 MHz, $CDCl_3$) δ 172.20, 171.74, 171.19, 170.95, 169.49, 161.47,

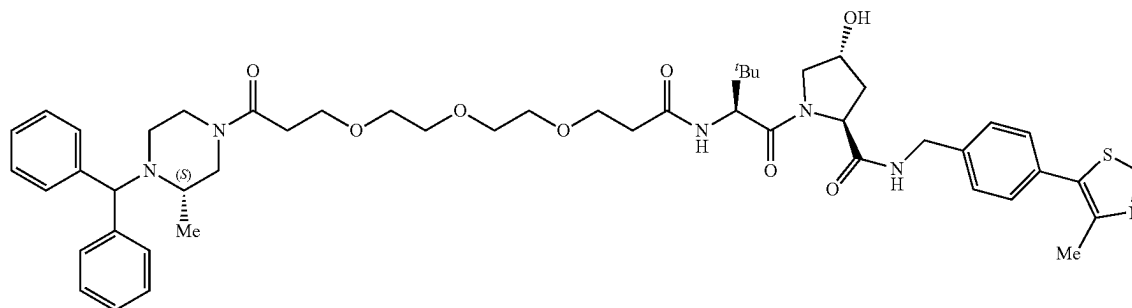
39



(2S,4R)-1-((2S)-2-(tert-butyl)-4,16-dioxo-16-(4-(phenyl(pyridin-2-yl)methyl) piperazin-1-yl)-7,10,13-trioxa-3-azahexadecanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide (39)

[0097] General Procedure 3 then 1. Reaction scale: 20 mg (30.2 μ mol, 1.0 eq.) of 12 and 8.4 mg (33.2 μ mol, 1.1 eq.) of phenylpyridin-methylpiperazine. Purified by pTLC (8%

150.33, 149.30, 148.43, 140.48, 138.28, 136.93, 131.66, 130.85, 129.48, 128.71, 128.23, 128.11, 127.60, 122.29, 122.25, 77.54, 77.30, 77.26, 77.05, 76.92, 76.84, 70.41, 70.32, 70.27, 70.25, 70.15, 70.10, 67.16, 67.14, 60.42, 58.44, 57.75, 56.86, 56.70, 56.00, 51.97, 51.55, 45.59, 43.18, 41.59, 36.58, 36.06, 36.03, 34.87, 33.34, 31.94, 29.71, 29.37, 29.33, 26.66, 26.42, 22.70, 21.07, 19.25, 18.17, 16.06, 14.21, 14.14.

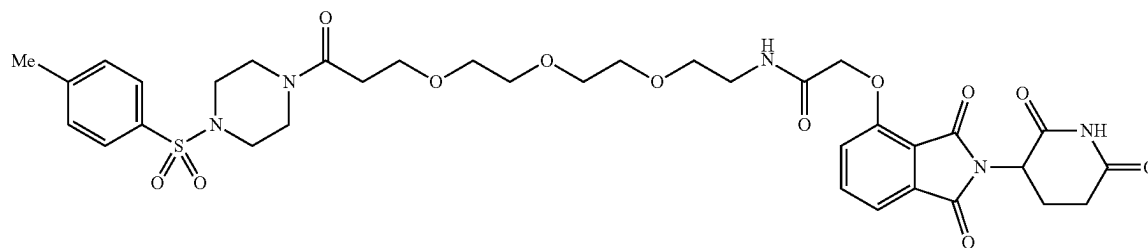


(2S,4R)-1-((S)-16-((S)-4-benzhydryl-3-methylpiperazin-1-yl)-2-(tert-butyl)-4,16-dioxo-7,10,13-trioxahexadecanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide (40)

[0098] General Procedure 1 and 3. Reaction scale: 18.4 mg (26.0 μ mol, 1.00 equiv) of 12 and 23.6 mg (64.0 μ mol, 2.46 equiv) of (4S)-methyl benzhydrylpiperazine. Purified by PTLC (10% MeOH/EtOAc) to afford 40 as a clear oil (12.5 mg, 53%). R_f =0.39 (10% MeOH/EtOAc, UV-active); LC-MS (ESI+) calc'd for $C_{50}H_{68}N_6O_8S$ $[M+2H]^{2+}$: 456.2, found 456.4; 1H NMR (600 MHz, $CDCl_3$) δ 8.67 (s, 1H), 7.44-7.41 (m, 2H), 7.39-7.32 (m, 7H), 7.28-7.24 (m, 5H), 7.21-7.13 (m, 3H), 4.72 (m, 1H), 4.55-4.46 (m, 3H), 4.35-4.31 (m, 1H), 4.16 (m, 1H), 4.10 (d, J =11.6 Hz, 1H), 3.76-3.54 (m, 16H), 3.46-3.37 (m, 1H), 3.00-2.91 (m, 2H), 2.63-2.39 (m, 12H), 2.16 (m, 1H), 0.94-0.91 (m, 9H).

tosylpiperazine. Purified by pTLC (10% MeOH/ CH_2Cl_2) to afford 41 as a clear oil (4.8 mg, 28%). R_f =0.65 (8% MeOH/ CH_2Cl_2 , UV-active); LC-MS (ESI+) calc'd for $C_{35}H_{43}N_5O_{12}S_2$ 757.3, found $[M+H]^+$ 758.2. 1H NMR (600 MHz, $CDCl_3$) δ 8.95 (s, 1H), 7.74 (dd, J =8.4, 7.4 Hz, 1H), 7.63 (s, 1H), 7.62-7.60 (m, 3H), 7.55 (d, J =7.3 Hz, 1H), 7.33 (d, J =8.0 Hz, 3H), 7.19 (d, J =8.4 Hz, 1H), 4.94 (dd, J =12.4, 5.4 Hz, 1H), 3.73 (td, J =6.6, 1.4 Hz, 3H), 3.68-3.66 (m, 3H), 3.64 (dd, J =4.7, 2.7 Hz, 2H), 3.01-2.95 (m, 5H), 2.86-2.68 (m, 4H), 2.43 (s, 4H), 1.27 (s, 9H). ^{13}C NMR (151 MHz, $CDCl_3$) δ 171.09, 169.63, 168.16, 166.83, 166.67, 165.83, 154.47, 144.10, 137.01, 133.69, 132.33, 129.86, 127.79, 119.35, 118.06, 117.33, 77.24, 77.03, 76.82, 70.41, 70.28,

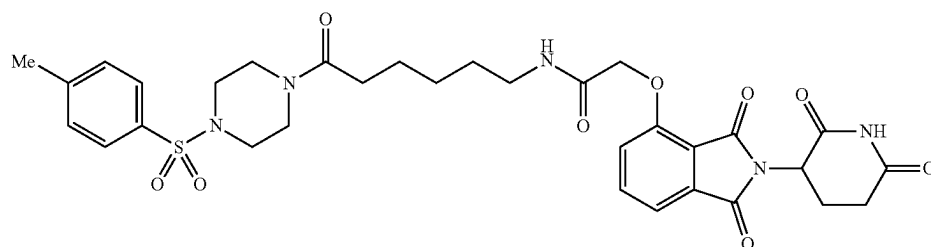
41



2-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)oxy)-N-(2-(2-(2-(3-oxo-3-(4-tosylpiperazin-1-yl)propoxy)ethoxy)ethoxy)ethyl)acetamide (41)

[0099] General Procedure 3 then 1. Reaction scale: 12 mg (22.4 μ mol, 1.0 eq.) of 3 and 5.9 mg (24.7 μ mol, 1.1 eq.) of

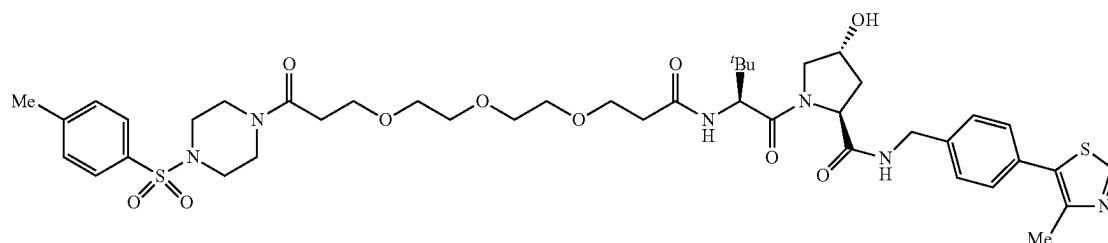
70.26, 70.24, 69.49, 67.91, 67.12, 60.42, 56.00, 53.45, 49.32, 46.16, 45.82, 45.10, 41.25, 40.90, 39.11, 38.63, 35.95, 33.71, 33.39, 31.94, 31.92, 31.47, 30.19, 29.79, 29.72, 29.54, 29.49, 29.38, 29.34, 29.25, 27.23, 25.55, 23.19, 22.70, 21.57, 21.08, 14.22, 14.14, 14.07.



2-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)oxy)-N-(6-oxo-6-(4-tosylpiperazin-1-yl) hexyl) acetamide (42)

[0100] General Procedure 3 then 1. Reaction scale: 8 mg (18.0 μ mol, 1.0 eq.) of 3 and 4.8 mg (19.8 μ mol, 1.1 eq.) of tosylpiperazine. Purified by pTLC (10% MeOH/CH₂Cl₂) to afford 42 as an off-white solid (7.1 mg, 28%). R_f =0.75 (8% MeOH/CH₂Cl₂, UV-active); LC-MS (ESI+) calc'd for C₃₂H₃₇N₅O₉S₂ 667.2, found [M+H]⁺ 668.3. ¹H NMR (600 MHz, CDCl₃) δ 9.38 (s, 1H), 7.75 (d, J=1.1 Hz, 1H), 7.63-7.61 (m, 2H), 7.59 (s, 1H), 7.56 (d, J=7.4 Hz, 1H), 7.34 (s, 2H), 7.20 (d, J=8.3 Hz, 1H), 4.94 (dt, J=8.0, 4.0 Hz, 1H), 4.79-4.55 (m, 4H), 3.74-3.65 (m, 5H), 3.53 (t, J=5.2 Hz, 3H), 3.41-3.32 (m, 5H), 2.97 (dd, J=12.1, 5.3 Hz, 5H), 2.87 (d, J=3.1 Hz, 1H), 2.79-2.68 (m, 3H), 2.44 (s, 4H). ¹³C NMR (151 MHz, CDCl₃) δ 171.94, 171.21, 168.30, 166.82, 166.54, 166.29, 154.79, 144.17, 137.11, 133.55, 132.23, 129.90, 127.78, 120.27, 118.47, 117.67, 77.24, 77.03, 76.82, 68.66, 56.00, 49.31, 46.19, 45.87, 45.05, 40.93, 39.13, 33.71, 32.95, 31.95, 31.39, 30.18, 29.72, 29.38, 29.34, 28.84, 27.23, 26.65, 25.12, 23.19, 22.84, 22.75, 22.71, 21.58, 14.21, 14.14.

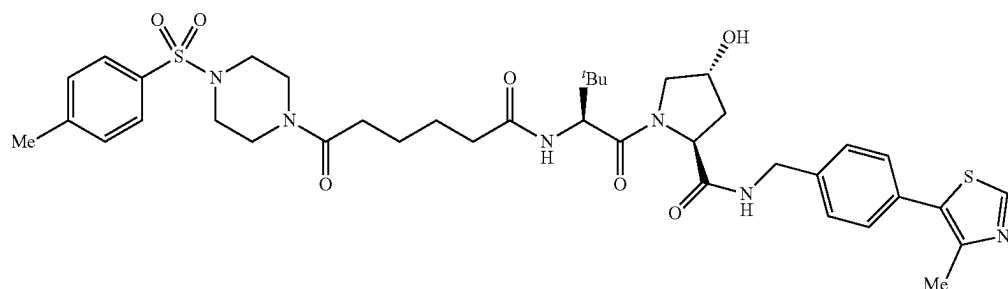
zine. Purified by pTLC (10% MeOH/DCM) to afford 43 as a white solid (3.3 mg, 41%). R_f =0.65 (8% MeOH/CH₂Cl₂, UV-active); LC-MS (ESI+) calc'd for C₄₃H₆₀N₆O₁₀S₂ 884.4, found [M+HCOO]⁻ 931.2. ¹H NMR (600 MHz, CDCl₃) δ 8.68 (s, 1H), 7.64-7.60 (m, 2H), 7.46-7.41 (m, 1H), 7.36-7.30 (m, 6H), 6.95 (d, J=8.3 Hz, 1H), 4.73 (t, J=8.1 Hz, 1H), 4.62-4.56 (m, 1H), 4.51-4.45 (m, 2H), 4.35-4.29 (m, 1H), 3.77-3.64 (m, 7H), 3.65-3.51 (m, 13H), 3.06-2.88 (m, 4H), 2.54-2.50 (m, 6H), 2.15-2.11 (m, 1H), 0.94 (s, 12H). ¹³C NMR (151 MHz, CDCl₃) δ 172.15, 171.80, 170.90, 169.80, 150.31, 148.47, 144.15, 138.25, 132.46, 132.28, 131.65, 130.89, 129.88, 129.49, 128.82, 128.11, 127.79, 77.24, 77.03, 76.82, 70.39, 70.23, 70.17, 70.13, 70.09, 68.17, 67.20, 67.06, 58.42, 57.78, 56.77, 56.00, 46.13,



(2S,4R)-1-((S)-2-(tert-butyl)-4,16-dioxo-16-(4-tosylpiperazin-1-yl)-7,10,13-trioxa-3-azahexadecanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl) pyrrolidine-2-carboxamide (43)

[0101] General Procedure 3 then 2. Reaction scale: 6.0 mg (9.05 μ mol) of 12 and 2.39 mg (9.96 μ mol) of tosylpiperazine.

45.81, 45.04, 43.21, 40.82, 38.74, 37.11, 36.54, 35.95, 34.76, 33.71, 33.45, 33.41, 32.77, 31.95, 30.37, 30.18, 30.05, 29.72, 29.50, 29.38, 29.34, 28.94, 27.23, 26.72, 26.43, 23.75, 23.19, 23.00, 22.71, 21.58, 16.08, 14.20, 14.14, 14.07, 10.97, 7.49.



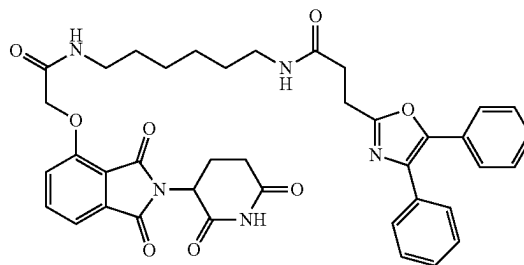
44

(2S,4R)-1-((S)-3,3-dimethyl-2-(6-oxo-6-(4-tosylpiperazin-1-yl) hexanamido) butanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl) pyrrolidine-2-carboxamide (44)

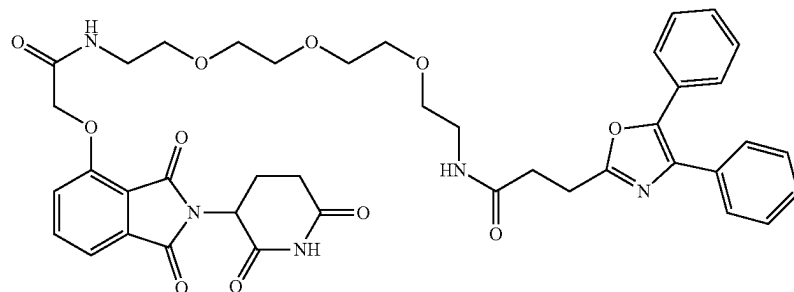
4.61 (s, 2H), 3.66-3.50 (m, 16H), 3.19 (m, 2H), 2.85-2.81 (m, 1H), 2.78-2.65 (m, 4H), 2.11-2.07 (m, 1H).

46

[0102] General Procedure 3 then 1. Reaction scale: 10 mg (17.9 μ mol, 1.0 eq.) of 3 and 4.7 mg (19.7 μ mol, 1.1 eq.) of tosylpiperazine. Purified by pTLC (10% MeOH/ CH_2Cl_2) to afford 41 as a white solid (4.0 mg, 29%). R_f =0.65 (8% MeOH/ CH_2Cl_2 , UV-active); LC-MS (ESI+) calc'd for $\text{C}_{39}\text{H}_{52}\text{N}_6\text{O}_7\text{S}_2$ 780.3, found $[\text{M}+\text{H}]^+$ 781.8. ^1H NMR (600 MHz, CDCl_3) δ 8.69 (s, 1H), 7.61 (d, J =8.4 Hz, 2H), 7.37-7.31 (m, 7H), 6.25 (d, J =8.5 Hz, 1H), 4.76-4.71 (m, 2H), 4.59 (dd, J =14.9, 6.7 Hz, 3H), 4.52-4.47 (m, 4H), 4.40-4.26 (m, 5H), 4.15-4.08 (m, 3H), 3.35 (s, 13H), 2.97 (dd, J =7.0, 3.9 Hz, 1H), 0.92 (s, 9H). Note: presence of rotamers.



45

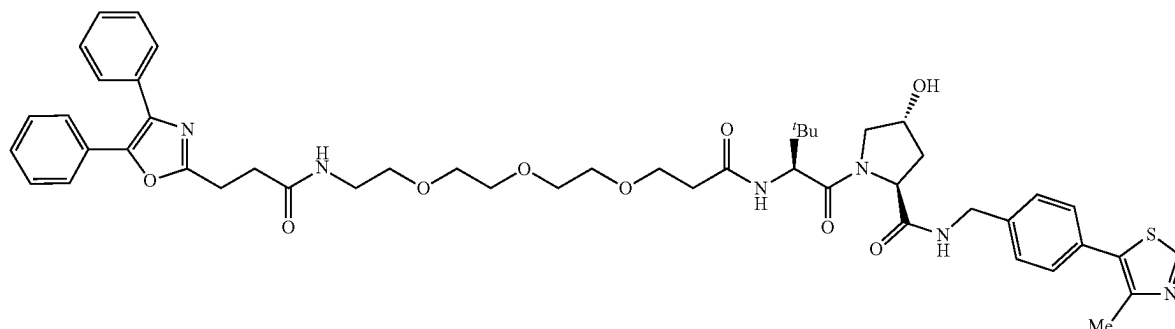


N-(1-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)oxy)-2-oxo-6,9,12-trioxa-3-azatetradecan-14-yl)-3-(4,5-diphenyloxazol-2-yl) propenamide (45)

N-(6-(2-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)oxy) acetamido) hexyl)-3-(4,5-diphenyloxazol-2-yl) propenamide (46)

[0103] General Procedure 1 and 3. Reaction scale: 13.4 mg (22.0 μ mol, 1.00 equiv) of 4 and 7.6 mg (26.0 μ mol, 1.20 equiv) of oxaprozin. Purified by PTLC (10% MeOH/EtOAc) to afford 45 as a clear oil (9.7 mg, 57%). R_f =0.31 (10% MeOH/EtOAc, UV-active); LC-MS (ESI+) calc'd for $\text{C}_{41}\text{H}_{44}\text{N}_5\text{O}_{11}$ $[\text{M}+\text{H}]^+$: 782.3, found 782.1; ^1H NMR (600 MHz, CDCl_3) δ 9.27 (br s, 1H), 7.71-7.68 (m, 2H), 7.61-7.59 (m, 2H), 7.55-7.51 (m, 3H), 7.36-7.29 (m, 6H), 7.14 (d, J =7.8 Hz, 1H), 6.93 (m, 1H), 4.92 (dd, J =12.4, 5.5 Hz, 1H),

[0104] General Procedure 1 and 3. Reaction scale: 8.7 mg (16.0 μ mol, 1.00 equiv) of 7 and 7.0 mg (24.0 μ mol, 1.50 equiv) of oxaprozin. Purified by PTLC (10% MeOH/ CH_2Cl_2) to afford 46 as a white foam (5.7 mg, 50%). LC-MS (ESI-) calc'd for $\text{C}_{39}\text{H}_{38}\text{N}_5\text{O}_8$ $[\text{M}-\text{H}]^-$: 704.3, found 704.0; ^1H NMR (600 MHz, CDCl_3) δ 9.34 (br s, 1H), 7.71 (m, 1H), 7.60-7.58 (m, 2H), 7.55-7.52 (m, 3H), 7.46 (m, 1H), 7.36-7.29 (m, 5H), 7.17 (d, J =8.3 Hz, 1H), 6.36 (m, 1H), 4.95 (m, 1H), 4.62 (m, 2H), 3.37 (m, 1H), 3.29 (m, 2H), 3.21-3.16 (m, 3H), 2.85 (m, 1H), 2.78-2.70 (m, 4H), 2.13-2.09 (m, 1H), 1.55-1.45 (m, 4H), 1.37-1.29 (m, 4H).

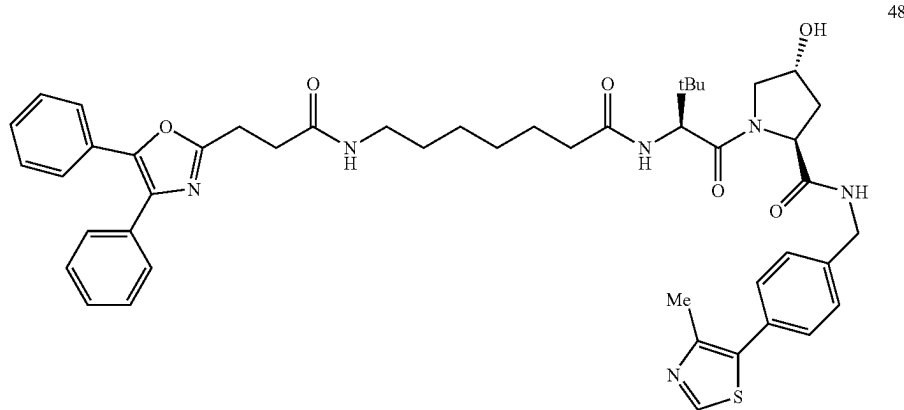


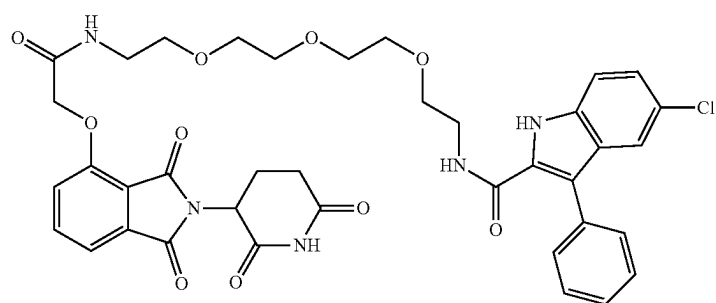
(2S,4R)-1-((S)-2-(tert-butyl)-19-(4,5-diphenyloxazol-2-yl)-4,17-dioxo-7,10,13-trioxa-3,16-diazanona-decanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide (47)

[0105] General Procedure 3 then 1. Reaction scale: 8 mg (12.6 μmol , 1.0 eq.) of 14 and 4.1 mg (13.9 μmol , 1.1 eq.) of oxaprozin. Purified by pTLC (10% MeOH/ CH_2Cl_2) to afford 47 as white powder (6.2 mg, 54%). $R_f=0.58$ (10% MeOH/ CH_2Cl_2 , UV-active); LC-MS (ESI $^-$) calc'd for $\text{C}_{49}\text{H}_{60}\text{N}_6\text{O}_9\text{S}$ 908.4, found $[\text{M}+\text{HCOO}]^-$ 954.2. ^1H NMR (400 MHz, CDCl_3) δ 8.67 (s, 1H), 7.55 (ddt, $J=14.2, 6.2, 1.9$ Hz, 5H), 7.37-7.30 (m, 10H), 7.15 (s, 1H), 7.09 (d, $J=8.4$ Hz, 1H), 4.67 (t, $J=8.2$ Hz, 1H), 4.56-4.48 (m, 2H), 4.42 (s, 1H), 4.32 (dd, $J=15.0, 5.4$ Hz, 1H), 4.13-4.07 (m, 1H), 3.74 (q, $J=5.2$ Hz, 1H), 3.68-3.64 (m, 1H), 3.62-3.53 (m, 11H), 3.43 (tt, $J=8.8, 4.7$ Hz, 2H), 3.18 (t, $J=7.6$ Hz, 2H), 2.77 (t, $J=7.7$ Hz, 2H), 2.46 (t, $J=5.6$ Hz, 2H), 2.36 (ddd, $J=13.1, 8.5, 4.3$ Hz, 2H), 2.06 (d, $J=14.3$ Hz, 3H), 0.94 (s, 9H). ^{13}C NMR (151 MHz, CDCl_3) δ 172.16, 171.66, 171.62, 171.11, 162.85, 150.35, 148.44, 145.47, 138.26, 134.85, 133.32, 132.40, 130.85, 129.55, 129.47, 128.87, 128.73, 128.70, 128.66, 128.63, 128.50, 128.43, 128.22, 128.17, 128.12, 128.06, 127.90, 126.51, 126.33, 77.24, 77.03, 76.82, 70.51, 70.40, 70.30, 70.08, 69.92, 68.86, 67.23, 58.55, 57.82, 57.72, 56.87, 56.00, 45.33, 43.28, 43.13, 39.78, 39.46, 36.54, 36.40, 35.82, 34.91, 33.71, 32.74, 31.95, 30.18, 30.06, 29.72, 29.68, 29.63, 29.38, 29.34, 27.69, 27.23, 26.72, 26.43, 26.38, 23.95, 23.20, 22.72, 21.51, 16.05, 14.21, 14.14.

(2S,4R)-1-((S)-2-(7-(3-(4,5-diphenyloxazol-2-yl)propanamido)heptanamido)-3,3-dimethylbutanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide (48)

[0106] General Procedure 1 and 3. Reaction scale: 15.2 mg (23.0 μmol , 1.00 equiv) of 18 and 9.9 mg (34.0 μmol , 1.20 equiv) of oxaprozin. Purified by PTLC (10% MeOH/EtOAc) to afford 48 as a clear oil (14.0 mg, 73%). $R_f=0.23$ (10% MeOH/EtOAc, UV-active); LC-MS (ESI $^+$) calc'd for $\text{C}_{47}\text{H}_{57}\text{N}_6\text{O}_6\text{S}$ $[\text{M}+\text{H}]^+$: 833.4, found 833.4; ^1H NMR (600 MHz, CDCl_3) δ 8.65 (s, 1H), 7.58-7.49 (m, 5H), 7.36-7.30 (m, 10H), 6.49 (m, 1H), 6.28 (d, $J=8.7$ Hz, 1H), 4.67 (t, $J=8.1$ Hz, 1H), 4.55-4.50 (m, 2H), 4.45 (m, 1H), 4.31 (dd, $J=15.0, 5.4$ Hz, 1H), 4.05 (d, $J=11.4$ Hz, 1H), 3.58 (dd, $J=11.3, 3.6$ Hz, 1H), 3.22-3.12 (m, 4H), 2.71 (t, $J=7.4$ Hz, 2H), 2.49 (s, 3H), 2.38-2.34 (m, 1H), 2.21-2.16 (m, 1H), 2.13-2.07 (m, 2H), 1.60-1.42 (m, 4H), 1.29-1.20 (m, 5H), 0.93 (s, 9H).





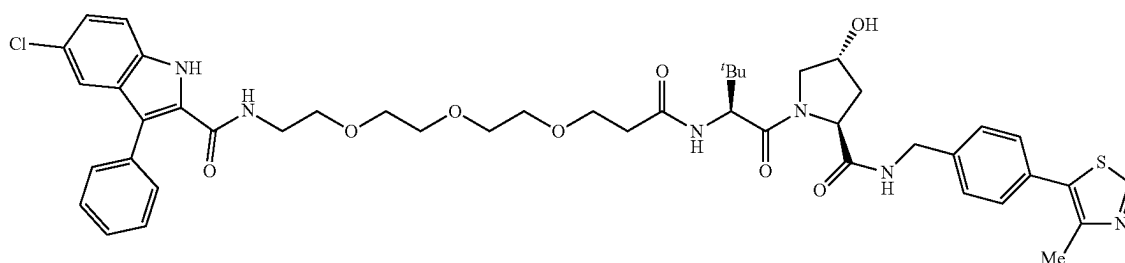
49

5-chloro-N-(1-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)oxy)-2-oxo-6,9,12-trioxa-3-azatetradecan-14-yl)-3-phenyl-1H-indole-2-carboxamide (49)

[0107] General Procedure 1 and 3. Reaction scale: 9.5 mg (18.0 μ mol, 1.00 equiv) of 4 and 6.0 mg (22.0 μ mol, 1.20 equiv) of SM2. Purified by PTLC (10% MeOH/EtOAc) to afford 49 as a yellow foam (9.2 mg, 75%). R_f =0.55 (10% MeOH/EtOAc, UV-active); LC-MS (ESI⁻) calc'd for $C_{36}H_{33}N_5O_7Cl$ [M-H]⁻: 682.2, found 682.0; ¹H NMR (600 MHz, CDCl₃) δ 10.22 (s, 1H), 9.76 (br s, 1H), 7.71 (m, 1H), 7.54-7.45 (m, 7H), 7.40-7.37 (m, 2H), 7.22-7.17 (m, 2H), 6.02 (m, 1H), 5.01-4.98 (m, 1H), 4.62 (m, 2H), 3.41-3.17 (m, 4H), 2.90-2.73 (m, 3H), 2.13 (m, 1H), 1.52 (m, 2H), 1.43-1.30 (m, 4H), 1.22-1.18 (m, 2H).

5-chloro-N-(6-(2-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)oxy) acetamido) hexyl)-3-phenyl-1H-indole-2-carboxamide (50)

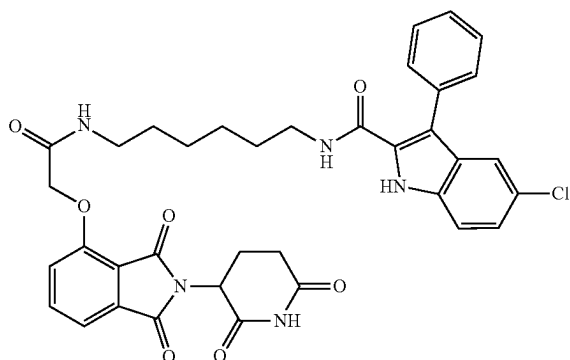
[0108] General Procedure 1 and 3. Reaction scale: 9.5 mg (18.0 μ mol, 1.00 equiv) of 7 and 6.0 mg (22.0 μ mol, 1.20 equiv) of SM2. Purified by PTLC (10% MeOH/EtOAc) to afford 50 as a yellow foam (9.2 mg, 75%). R_f =0.55 (10% MeOH/EtOAc, UV-active); LC-MS (ESI⁻) calc'd for $C_{36}H_{33}N_5O_7Cl$ [M-H]⁻: 682.2, found 682.0; ¹H NMR (600 MHz, CDCl₃) δ 10.22 (s, 1H), 9.76 (br s, 1H), 7.71 (m, 1H), 7.54-7.45 (m, 7H), 7.40-7.37 (m, 2H), 7.22-7.17 (m, 2H), 6.02 (m, 1H), 5.01-4.98 (m, 1H), 4.62 (m, 2H), 3.41-3.17 (m, 4H), 2.90-2.73 (m, 3H), 2.13 (m, 1H), 1.52 (m, 2H), 1.43-1.30 (m, 4H), 1.22-1.18 (m, 2H).



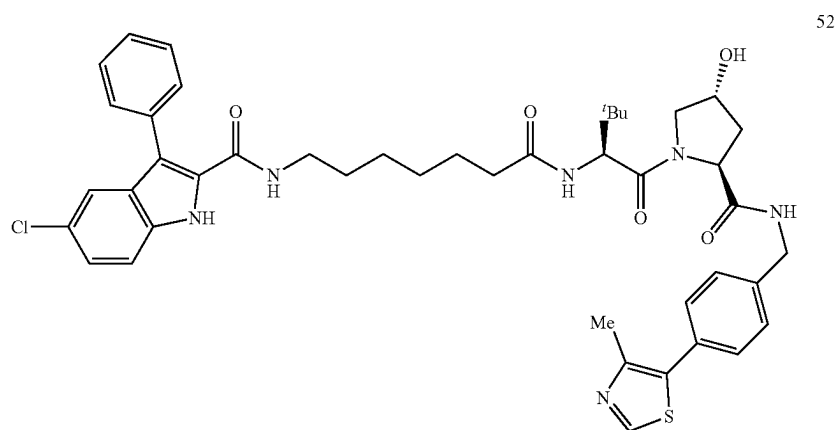
51

50

5-chloro-N-((S)-14-((2S,4R)-4-hydroxy-2-((4-(4-methylthiazol-5-yl)benzyl) carbamoyl) pyrrolidine-1-carbonyl)-15,15-dimethyl-12-oxo-3,6,9-trioxa-13-azahexadecyl)-3-phenyl-1H-indole-2-carboxamide (51)



[0109] General Procedure 3 then 1. Reaction scale: 8 mg (12.6 μ mol, 1.0 eq.) of 14 and 4.9 mg (13.9 μ mol, 1.1 eq.) of chlorophenylindole. Purified by pTLC (10% MeOH/CH₂Cl₂) to afford 51 as yellow powder (5.7 mg, 51%). R_f =0.57 (10% MeOH/CH₂Cl₂, UV-active); LC-MS (ESI⁻) calc'd for $C_{46}H_{55}ClN_6O_8S$ 886.4, found [M-H]⁻ 885.5. ¹H NMR (400 MHz, CDCl₃) δ 10.22 (s, 1H), 8.67 (s, 1H), 7.55-7.47 (m, 4H), 7.46-7.37 (m, 3H), 7.33 (s, 4H), 7.21 (dd, J=8.7, 2.0 Hz, 1H), 7.11 (d, J=8.6 Hz, 1H), 6.37 (s, 1H), 4.81 (t, J=8.2 Hz, 1H), 4.60-4.54 (m, 2H), 4.50 (s, 1H), 4.37 (dd, J=15.0, 5.3 Hz, 1H), 4.17 (d, J=11.4 Hz, 1H), 3.65-3.56 (m, 5H), 3.47 (d, J=6.5 Hz, 10H), 2.48 (s, 3H), 2.38 (dd, J=14.4, 8.3 Hz, 2H), 2.17 (s, 3H), 0.97 (s, 9H).

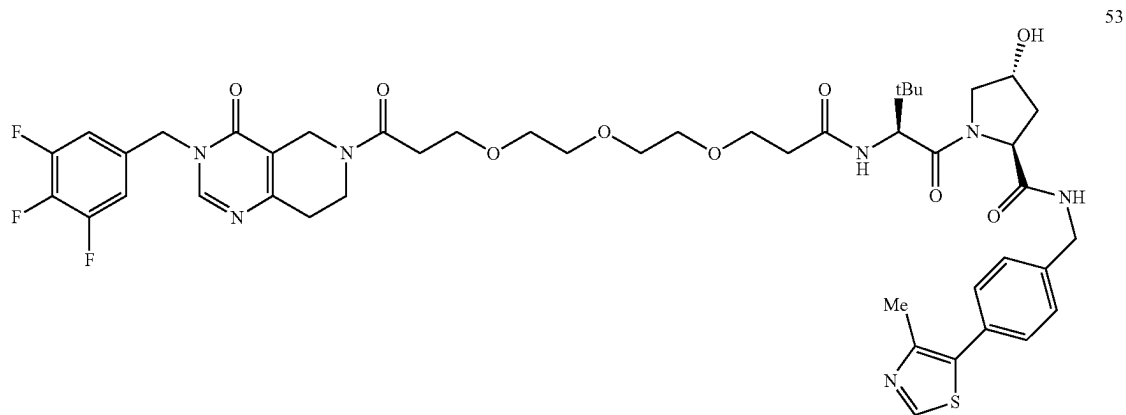


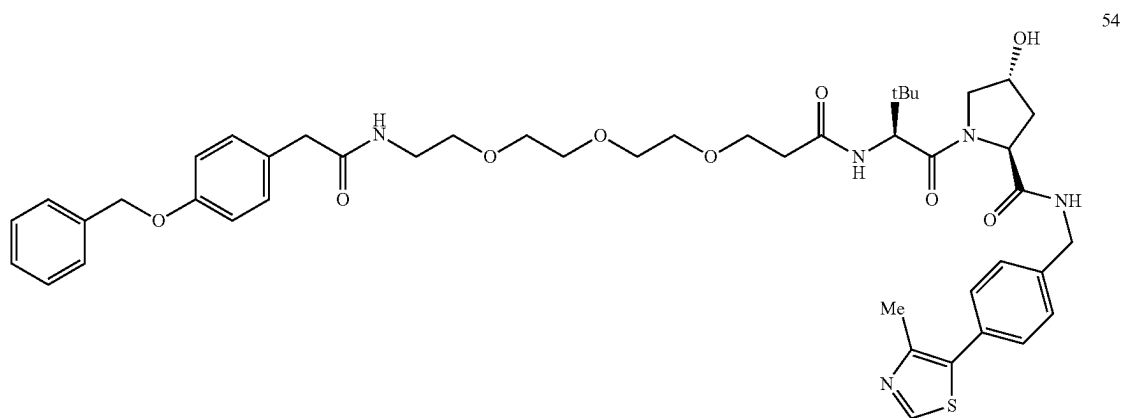
5-chloro-N-(7-((((S)-1-(((2S,4R)-4-hydroxy-2-((4-(4-methylthiazol-5-yl)benzyl) carbamoyl) pyrrolidin-1-yl)-3,3-dimethyl-1-oxobutan-2-yl)amino)-7-oxoheptyl)-3-phenyl-1H-indole-2-carboxamide (52)

[0110] General Procedure 1 and 3. Reaction scale: 18.3 mg (28.0 μ mol, 1.00 equiv) of 18 and 11.6 mg (43.0 μ mol, 1.54 equiv) of SM2. Purified by PTLC (10% MeOH/EtOAc) to afford 52 as a yellow solid (17.9 mg, 79%). R_f =0.38 (10% MeOH/EtOAc, UV-active); LC-MS (ESI+) calc'd for $C_{44}H_{52}N_6O_5SCl$ $[M+H]^+$: 811.3, found 811.1; 1H NMR (600 MHz, $CDCl_3$) δ 10.40 (m, 1H), 8.65 (s, 1H), 7.49-7.28 (m, 12H), 7.16 (m, 1H), 6.57 (m, 1H), 5.91 (m, 1H), 4.74 (m, 1H), 4.59 (d, J =9.0 Hz, 1H), 4.51-4.47 (m, 2H), 4.35 (dd, J =15.0, 5.6 Hz, 1H), 4.09 (d, J =11.3 Hz, 1H), 3.62 (m, 1H), 3.16 (m, 2H), 2.46 (s, 3H), 2.38 (m, 1H), 2.19-2.06 (m, 3H), 1.53-1.46 (m, 2H), 1.27-1.22 (m, 3H), 1.17 (m, 2H), 1.06 (m, 2H), 0.96 (s, 9H).

(2S,4R)-1-(((S)-2-(tert-butyl)-4,16-dioxo-16-(4-oxo-3-(3,4,5-trifluorobenzyl)-3,5,7,8-tetrahydropyrido[4,3-d]pyrimidin-6(4H)-yl)-7,10,13-trioxo-3-azahexadecanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide (53)

[0111] General Procedure 1 and 3. Reaction scale: 14.2 mg (20.0 μ mol, 1.00 equiv) of 12 and 15.2 mg (38.0 μ mol, 1.90 equiv) of SM2. Purified by PTLC (100% EtOAc) then plate was dried and run again (10% MeOH/ CH_2Cl_2) to afford 53 as a white foam (4.3 mg, 23%). 1H NMR (400 MHz, $CDCl_3$) δ 8.86 (s, 1H), 8.09-8.05 (m, 1H), 7.53-7.45 (m, 1H), 7.36 (m, 4H), 7.07-6.96 (m, 3H), 5.03-4.95 (m, 2H), 4.74 (m, 1H), 4.62-4.42 (m, 5H), 4.33 (dd, J =15.0, 5.2 Hz, 1H), 4.12 (m, 1H), 3.85-3.76 (m, 3H), 3.74-3.68 (m, 3H), 3.63-3.55 (m, 10H), 2.77 (m, 1H), 2.72-2.69 (m, 3H), 2.54-2.51 (m, 4H), 2.49-2.44 (m, 2H), 2.19-2.10 (m, 1H), 0.93 (s, 9H).



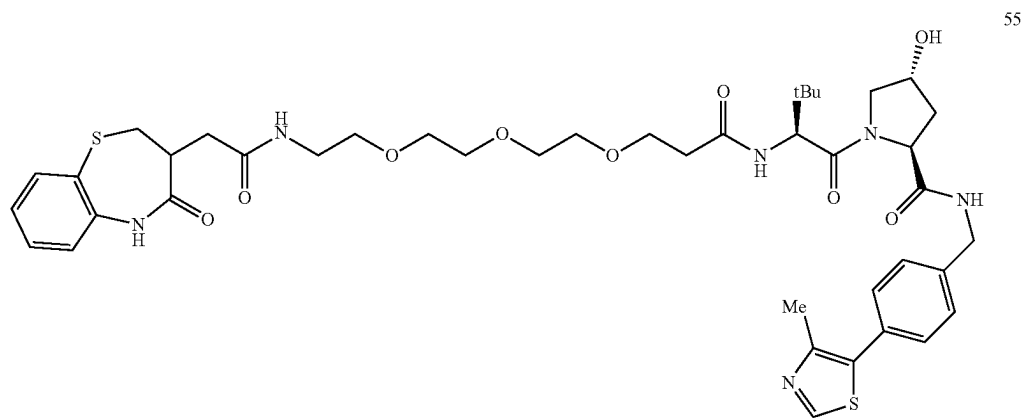


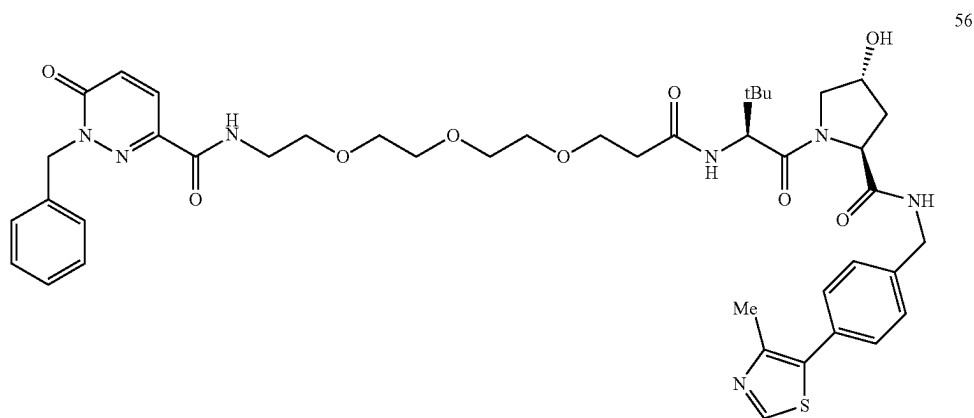
(2S,4R)-1-((S)-1-(4-(benzyloxy)phenyl)-17-(tert-butyl)-2,15-dioxo-6,9,12-trioxa-3,16-diazaoctadecan-18-oyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide (54)

[0112] General Procedure 1 and 3. Reaction scale: 8.6 mg (12.0 μ mol, 1.00 equiv) of 14 and 7.5 mg (31.0 μ mol, 2.58 equiv) of SM2. Purified by PTLC (10% MeOH/EtOAc) to afford 54 as a white foam (8.5 mg, 84%). ¹H NMR (400 MHz, CDCl₃) δ 8.69 (s, 1H), 7.43-7.30 (m, 11H), 7.17 (d, J=8.58 Hz, 2H), 6.92 (d, J=8.58 Hz, 2H), 5.04 (s, 2H), 4.70 (m, 1H), 4.58-4.48 (m, 3H), 4.33 (dd, J=14.8, 5.2 Hz, 1H), 4.11 (d, J=11.3 Hz, 1H), 3.74-3.35 (m, 19H), 2.50-2.44 (m, 6H), 1.42 (d, J=5.7 Hz, 1H), 0.93 (s, 9H).

(2S,4R)-1-((S)-19-(tert-butyl)-4,17-dioxo-1-(4-oxo-3,5,6,7-tetrahydro-4H-cyclopenta[4,5]thieno[2,3-d]pyrimidin-2-yl)-8,11,14-trioxa-2-thia-5,18-diazaicosan-20-oyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide (55)

[0113] General Procedure 1 and 3. Reaction scale: 10.4 mg (14.0 μ mol, 1.00 equiv) of 14 and 4.2 mg (18.0 μ mol, 1.29 equiv) of SM2. Purified by PTLC (10% MeOH/CH₂Cl₂) to afford 55 as a yellow oil (9.0 mg, 77%). LC-MS (ESI+) calc'd for C₄₂H₅₇N₆O₉S₂ [M+H]⁺: 853.4, found 853.7.



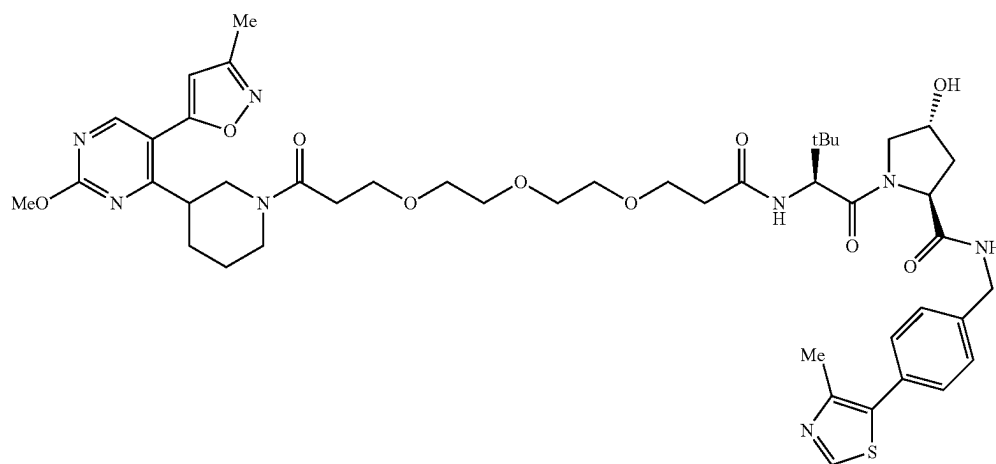


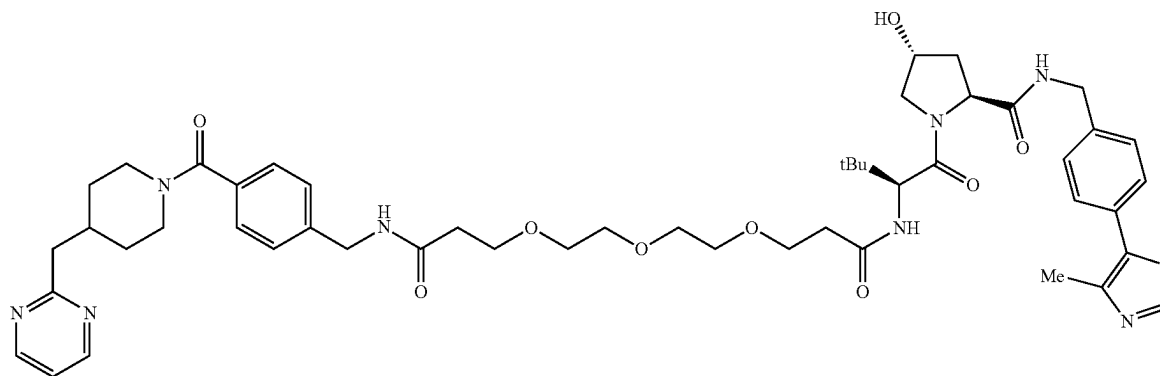
1-benzyl-N-((S)-14-((2S,4R)-4-hydroxy-2-((4-(4-methylthiazol-5-yl)benzyl) carbamoyl) pyrrolidine-1-carbonyl)-15,15-dimethyl-12-oxo-3,6,9-trioxa-13-azahexadecyl)-6-oxo-1,6-dihydropyridazine-3-carboxamide (56)

[0114] General Procedure 1 and 3. Reaction scale: 10.3 mg (14.0 μ mol, 1.00 equiv) of 14 and 5.0 mg (21.0 μ mol, 1.50 equiv) of SM2. Purified by PTLC (10% MeOH/ CH_2Cl_2) to afford 56 as a white solid (8.4 mg, 71%). ^1H NMR (400 MHz, CDCl_3) δ 8.67 (s, 1H), 7.89 (d, $J=9.6$ Hz, 1H), 7.52 (m, 1H), 7.41 (m, 2H), 7.36-7.30 (m, 7H), 6.97 (d, $J=9.6$ Hz, 1H), 6.81 (d, $J=8.4$ Hz, 1H), 5.35 (s, 2H), 4.68 (m, 1H), 4.54-4.49 (m, 3H), 4.35 (dd, $J=14.9, 5.6$ Hz, 1H), 4.15-4.09 (m, 2H), 3.67-3.57 (m, 16H), 2.52-2.50 (m, 6H), 2.42 (m, 1H), 0.94 (s, 9H).

(2S,4R)-1-((2S)-2-(tert-butyl)-16-(3-(2-methoxy-5-(3-methylisoxazol-5-yl)pyrimidin-4-yl)piperidin-1-yl)-4,16-dioxo-7,10,13-trioxa-3-azahexadecanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide (57)

[0115] General Procedure 1 and 3. Reaction scale: 7.0 mg (10.0 μ mol, 1.00 equiv) of 12 and 6.0 mg (15.0 μ mol, 1.50 equiv) of SM2. Purified by PTLC (10% MeOH/ CH_2Cl_2) to afford 57 as a white foam (6.8 mg, 74%). $R_f=0.68$ (10% MeOH/ CH_2Cl_2 , UV-active); LC-MS (ESI-) calc'd for $\text{C}_{47}\text{H}_{63}\text{N}_8\text{O}_{12}\text{S}$ $[\text{M}+\text{HCO}_2]^-$: 963.4, found 963.0; ^1H NMR (600 MHz, CDCl_3) δ 8.67 (s, 1H), 8.66 (d, $J=2.7$ Hz), 7.54-7.44 (m, 1H), 7.36-7.32 (m, 4H), 7.06-7.00 (s, 1H), 6.43 (s, 1H), 4.73-4.69 (m, 1H), 4.59-4.54 (m, 1H), 4.51-4.45 (m, 2H), 4.36-4.29 (m, 1H), 4.10 (br d, $J=11.5$ Hz, 1H), 4.07-4.05 (m, 3H), 4.00-3.90 (m, 1H), 3.80-3.66 (m, 5H), 3.64-3.55 (m, 11H), 3.11-3.05 (m, 1H), 2.67-2.63 (m, 1H), 2.62-2.57 (m, 1H), 2.51 (s, 3H), 2.49-2.44 (m, 3H), 2.38-2.36 (m, 3H), 2.16-2.10 (m, 1H), 1.97-1.76 (m, 5H), 0.93 (s, 9H).



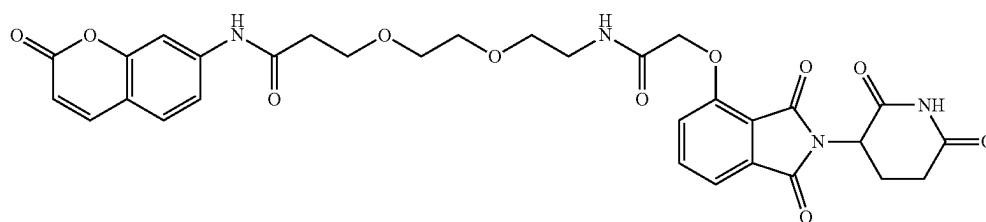


(2S,4R)-1-((S)-17-(tert-butyl)-3,15-dioxo-1-(4-(4-(pyrimidin-2-ylmethyl) piperidine-1-carbonyl)phenyl)-6,9,12-trioxa-2,16-diazaoctadecan-18-oyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide (58)

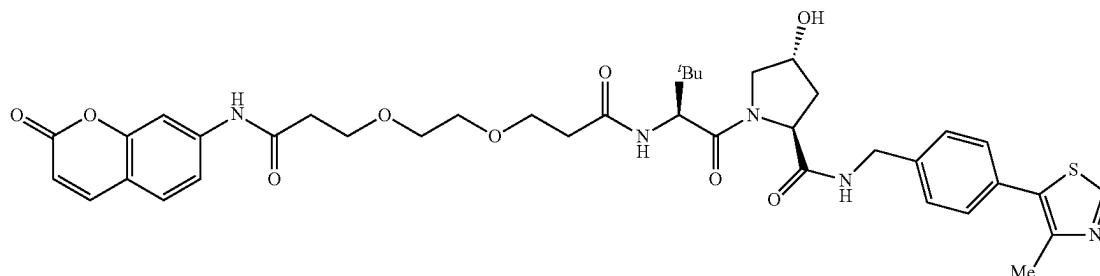
[0116] General Procedure 1 and 3. Reaction scale: 8.0 mg (11.0 μ mol, 1.00 equiv) of 12 and 9.2 mg (22.0 μ mol, 2.00 equiv) of SM2. Purified by PTLC (100% EtOAc) then plate was dried and run again (10% MeOH/CH₂Cl₂) to afford 58 as a yellow oil (1.3 mg, 13%). LC-MS (ESI+) calc'd for C₅₀H₆₇N₈O₉S [M+H]⁺: 956.0, found 956.4; ¹H NMR (600 MHz, CDCl₃) δ 8.66 (m, 3H), 7.38-7.27 (m, 9H), 7.14 (t, J=4.9 Hz, 1H), 6.98 (d, J=8.4 Hz, 1H), 4.63 (m, 1H), 4.54-4.45 (m, 4H), 4.34 (td, J=15.7, 5.4 Hz, 2H), 3.98 (d, J=11.4 Hz, 1H), 3.79-3.52 (m, 14H), 3.10 (q, J=7.4 Hz, 1H), 2.99-2.87 (m, 3H), 2.76 (m, 1H), 2.54-2.44 (m, 5H), 2.40-2.32 (m, 2H), 2.24 (m, 1H), 1.48-1.39 (m, 8H), 0.94 (s, 9H).

3-(2-(2-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)oxy) acetamido) ethoxy) ethoxy)-N-(2-oxo-2H-chromen-7-yl) propenamide (59)

[0117] General Procedure 2. Reaction scale: 8.0 μ mol of 2 and 1.4 mg (8.8 μ mol) of coumarin. Purified by pTLC (EtOAc, then 10% MeOH/CH₂Cl₂) to afford 59 as a glassy solid (4.3 mg, 85%). Molecular formula: C₃₁H₃₀N₄O₁₁ ¹H NMR (400 MHz, CDCl₃) δ_H 2.14-2.25 (1H, m, CL1, diastereotopic proton), 2.65 (2H, t, J 5.6, C(O)CH₂CH₂O), 2.71-2.94 (2H, m, CL1 diastereotopic proton, other proton), 3.51-3.66 (8H, m, O(CH₂)₂O(CH₂)₂NH), 3.85 (2H, t, J 5.6, C(O)CH₂CH₂O), 4.56 (2H, s, C(O)CH₂O), 4.96 (1H, dd, J 5.5, 12.4, CL1 stereocenter proton), 6.36 (1H, d, J 9.5, coumarin double bond nearest carbonyl), 7.13 (1H, d, J 8.7, ortho-linker), 7.20 (1H d, J 8.9, coumarin proton), 7.41 (1H, dd, J 2.4, 8.9, coumarin proton nearest linker), 7.51 (1H, d, J 7.4, CL1, para-linker), 7.56 (1H, br t, J 5.6, linker NH), 7.64 (1H, J 9.5, coumarin double bond farthest from carbonyl), 7.72 (1H dd, J 8.7, 7.4, CL1, meta-linker), 8.06 (1H, d, J 2.4, coumarin proton no neighbours), 8.86 (1H, br. s, NH).



60

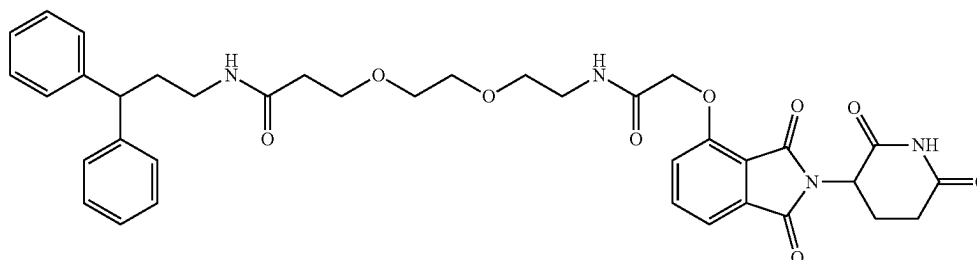


(2S,4R)-1-((S)-3,3-dimethyl-2-(3-(2-(3-oxo-3-((2-oxo-2H-chromen-7-yl)amino) propoxy) ethoxy) propanamido) butanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl) pyrrolidine-2-carboxamide (60)

[0118] General Procedure 3 then 2. Reaction scale: 11.1 μ mol of 2 and 2.0 mg (12.2 μ mol) of coumarin. Purified by pTLC (EtOAc, then 10% MeOH/CH₂Cl₂) to afford 60 as a translucent glass (5.1 mg, 60%). Molecular formula:

C₃₉H₄₇N₅O₉S. ¹H NMR (400 MHz, CDCl₃) δ 9.27 (s, 1H), 8.68 (s, 1H), 7.92 (d, J=2.2 Hz, 1H), 7.75-7.51 (m, 2H), 7.44 (d, J=8.7 Hz, 1H), 7.37 (q, J=8.3 Hz, 4H), 7.21-7.09 (m, 2H), 6.33 (d, J=9.6 Hz, 1H), 4.79 (t, J=8.3 Hz, 1H), 4.71-4.55 (m, 3H), 4.32 (dd, J=15.1, 5.2 Hz, 1H), 4.20 (d, J=11.3 Hz, 1H), 3.84 (td, J=9.9, 3.2 Hz, 1H), 3.79-3.44 (m, 7H), 2.78 (ddd, J=14.2, 9.9, 4.2 Hz, 1H), 2.52 (s, 3H), 2.49-2.30 (m, 2H), 2.30-2.13 (m, 3H), 1.02 (s, 9H).

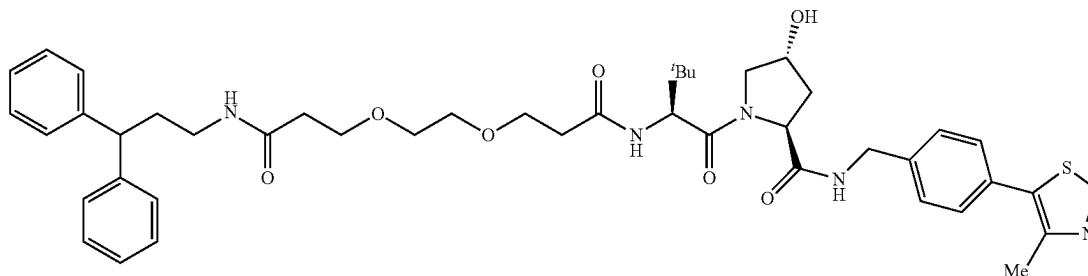
61



3-(2-(2-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)oxy) acetamido) ethoxy) ethoxy)-N-(3,3-diphenylpropyl) propanamide (61)

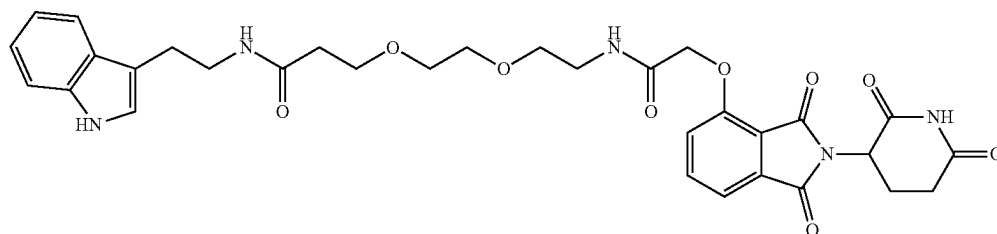
[0119] General Procedure 3 then 2. Molecular formula: C₃₇H₄₀N₄O₉. Yield 61: 4.3 mg (78%). ¹H NMR (400 MHz, CDCl₃) δ 8.68 (s, 1H), 7.42-7.30 (m, 4H), 7.30-7.19 (m, 11H), 7.19-7.13 (m, 2H), 6.98 (d, J=8.5 Hz, 1H), 6.51 (t, J=5.8 Hz, 1H), 4.62 (t, J=8.1 Hz, 1H), 4.56 (dd, J=15.0, 6.7 Hz, 1H), 4.50 (d, J=8.6 Hz, 2H), 4.31 (dd, J=15.0, 5.2 Hz, 1H), 4.04 (d, J=11.3 Hz, 1H), 3.94 (t, J=7.8 Hz, 1H), 3.75-3.45 (m, 10H), 3.16 (dt, J=8.1, 6.1 Hz, 2H), 2.51 (s, 3H), 2.48-2.27 (m, 4H), 2.23 (td, J=8.0, 6.3 Hz, 2H), 2.15-2.03 (m, 1H), 0.94 (s, 9H).

62



(2S,4R)-1-((S)-2-(tert-butyl)-4,13-dioxo-17,17-diphenyl-7,10-dioxo-3,14-diazaheptadecanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl) pyrrolidine-2-carboxamide (62)

[0120] General Procedure 3 then 2. Molecular formula: $C_{45}H_{57}N_5O_7S$. Yield 62:7.7 mg (80%).



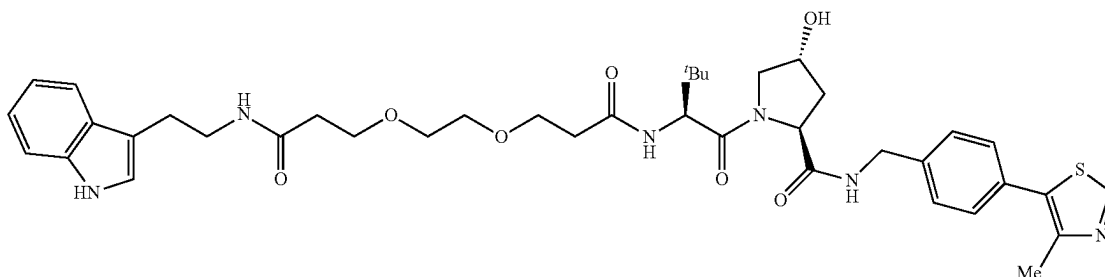
63

N-(2-(1H-indol-3-yl)ethyl)-3-(2-(2-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)oxy)acetamido)ethoxy)ethoxy)propenamide (63)

[0121] General Procedure 3 then 2. Reaction scale: 17.6 μ mol of 2 and 3.2 mg (20 μ mol) of indole amine. Purified by pTLC (EtOAc, then 10% MeOH/ CH_2Cl_2) to afford 63 as a yellow translucent glass (2.8 mg, 25%). Molecular formula: $C_{32}H_{35}N_5O_9$. 1H NMR (400 MHz, $CDCl_3$) δ_H 2.04-2.12 (1H, m, CL1, diastereotopic proton), 2.41 (2H, t, J 5.9, tryptamine $NHCH_2CH_2$), 2.57-2.84 (3H, m, CL1 diastereotopic proton,

imide- CH_2), 2.95 (2H, t, J 6.5, $C(O)CH_2CH_2O$), 3.46-3.64 (10H, m, $O(CH_2)_2O(CH_2)_2NH$, tryptamine CH_2NH), 3.70 (2H, t, J 5.8, $C(O)CH_2CH_2O$), 4.58 (2H, d, J 2.2, $C(O)CH_2O$), 4.90 (1H, dd, J 5.4, 12.2, CL1 stereocenter proton), 6.36 (1H, t, J 5.5, tryptamine amide $NHC(O)$), 7.04-7.18 (3H, m, indole), 7.13 (1H, d, J 8.7, ortho-linker), 7.32 (1H, d, J 8.1, indole), 7.52 (1H, d, J 7.4, CL1, para-linker), 7.56 (1H, d, J 7.7, indole), 7.62 (1H, br t, J 5.6, linker NH), 7.72 (1H dd, J 8.7, 7.4, CL1, meta-linker), 8.41 (1H, br s, imide NH), 8.82 (1H, br s, indole NH).

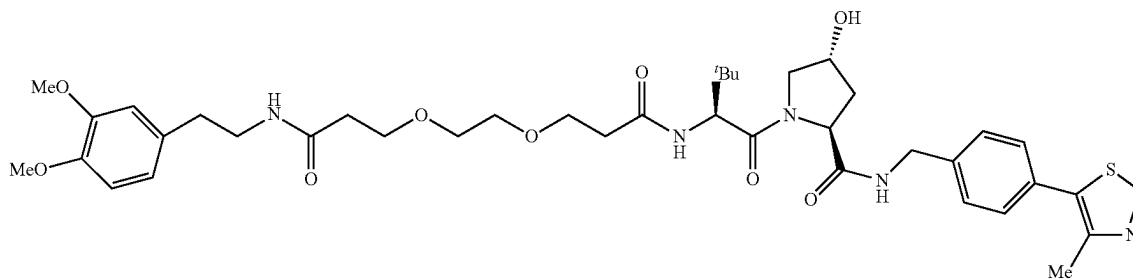
64



(2S,4R)-1-((S)-2-(tert-butyl)-16-(1H-indol-3-yl)-4,13-dioxo-7,10-dioxo-3,14-diazahexadecanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl) pyrrolidine-2-carboxamide (64)

[0122] General Procedure 3 then 2. Molecular formula: $C_{40}H_{52}N_6O_7S$. Yield 64:4.3 mg (45%).

65

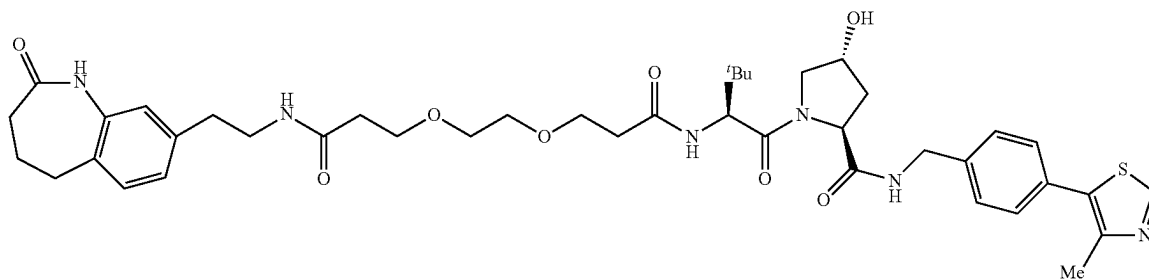


(2S,4R)-1-((S)-2-(tert-butyl)-16-(3,4-dimethoxyphenyl)-4,13-dioxo-7,10-dioxo-3,14-diazahexadecanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide (65)

[0123] General Procedure 3 then 2. Molecular formula: $C_{40}H_{55}N_5O_9S$. Yield 65:7.9 mg (78%). 1H NMR (400 MHz,

$CDCl_3$) δ 8.68 (s, 1H), 7.36 (s, 3H), 6.98 (d, $J=8.4$ Hz, 1H), 6.85-6.65 (m, 4H), 6.53 (d, $J=5.9$ Hz, 1H), 4.60 (dt, $J=23.2$, 7.4 Hz, 2H), 4.49 (d, $J=8.5$ Hz, 2H), 4.33 (dd, $J=15.0$, 5.1 Hz, 1H), 4.08 (d, $J=11.4$ Hz, 1H), 3.86 (s, 3H), 3.84 (s, 3H), 3.76-3.38 (m, 12H), 2.74 (q, $J=7.4$ Hz, 2H), 2.54-2.30 (m, 9H), 2.19-2.09 (m, 1H), 0.94 (s, 9H).

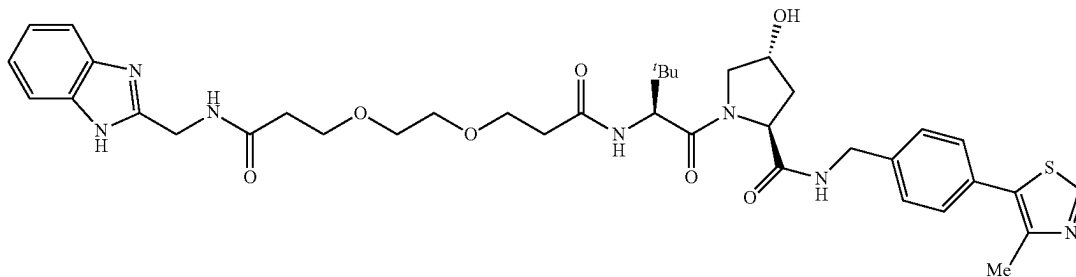
66



(2S,4R)-1-((S)-2-(tert-butyl)-4,13-dioxo-16-(2-oxo-2,3,4,5-tetrahydro-1H-benzo[b]azepin-8-yl)-7,10-dioxo-3,14-diazahexadecanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide (66)

[0124] General Procedure 3 then 2. Molecular formula: $C_{42}H_{56}N_6O_8S$. Yield 66:5.0 mg (50%).

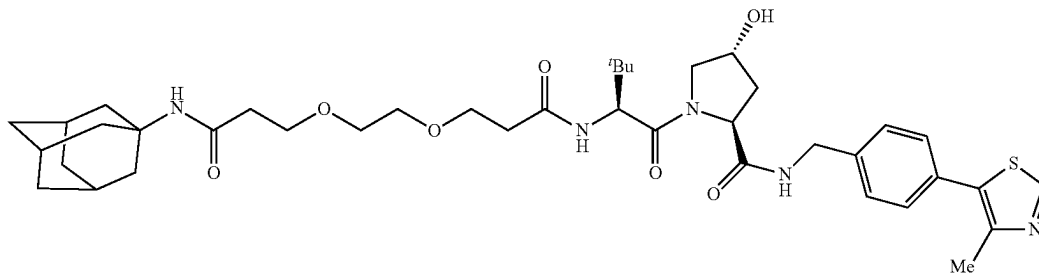
67



(2S,4R)-1-((S)-1-(1H-benzof[d]imidazol-2-yl)-14-(tert-butyl)-3,12-dioxo-6,9-dioxo-2,13-diazapentadecan-15-oyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide (67)

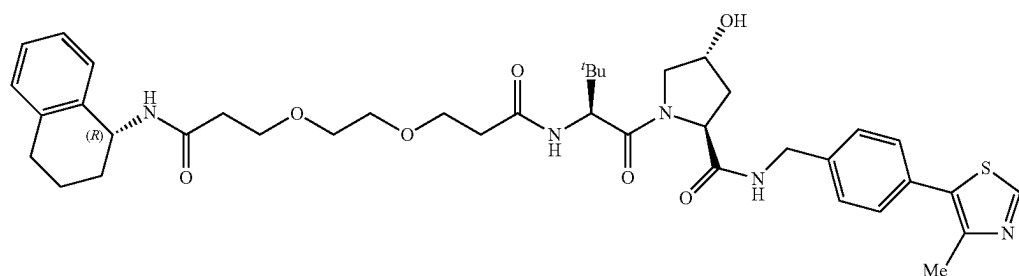
[0125] General Procedure 3 then 2. Molecular formula: $C_{38}H_{49}N_7O_7S$. Yield 67:4.3 mg (45%).

68



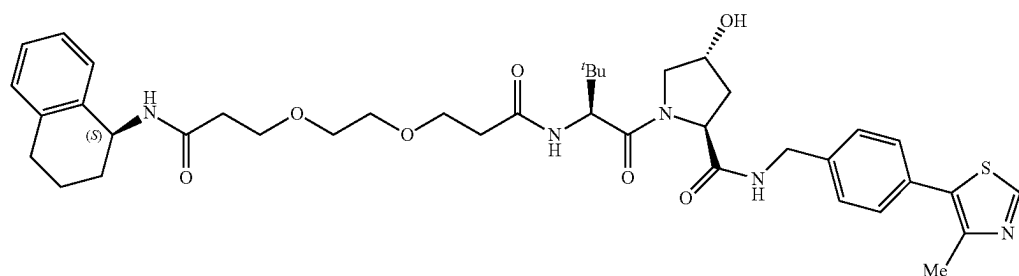
(2S,4R)-1-((S)-2-(3-(2-(3-(((3s,5s,7s)-adamantan-1-yl)amino)-3-oxopropoxy) ethoxy) propanamido)-3,3-dimethylbutanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl) pyrrolidine-2-carboxamide (68)

[0126] General Procedure 3 then 2. Molecular formula: $C_{40}H_{57}N_5O_7S$. Yield 68:4.2 mg (43%).



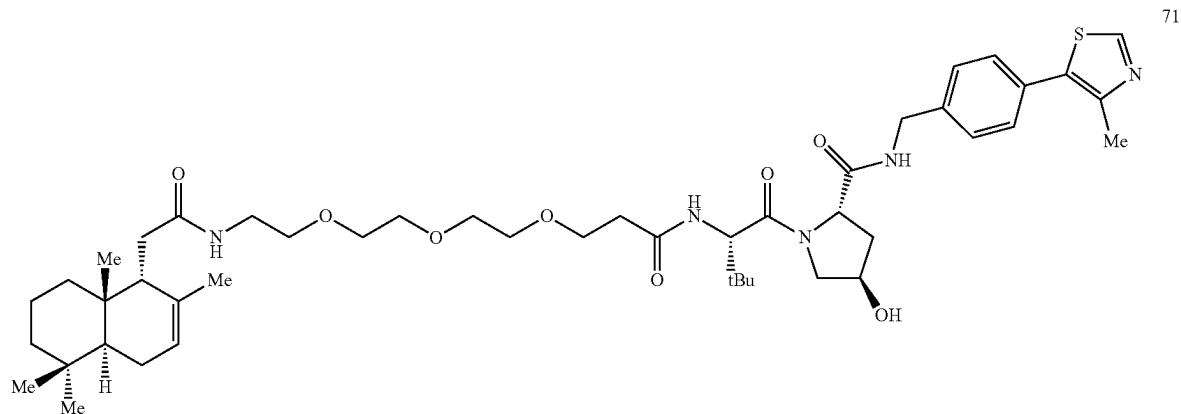
(2S,4R)-1-((S)-3,3-dimethyl-2-(3-(2-(3-oxo-3-(((R)-1,2,3,4-tetrahydronaphthalen-1-yl)amino) propoxy) ethoxy) propanamido) butanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl) pyrrolidine-2-carboxamide (69)

[0127] General Procedure 3 then 2. Molecular formula: $C_{40}H_{53}N_5O_7S$. Yield 69:5.2 mg (41%).



(2S,4R)-1-((S)-3,3-dimethyl-2-(3-(2-(3-oxo-3-(((S)-1,2,3,4-tetrahydronaphthalen-1-yl)amino) propoxy) ethoxy) propanamido) butanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl) pyrrolidine-2-carboxamide (70)

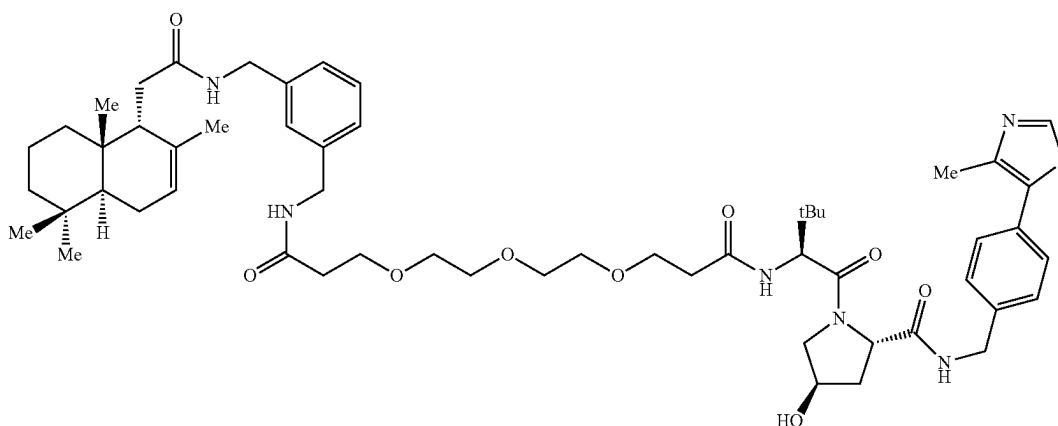
[0128] General Procedure 3 then 2. Molecular formula: $C_{40}H_{53}N_5O_7S$. Yield 70:8.2 mg (96%).



(2S,4R)-1-((S)-16-(tert-butyl)-1,14-dioxo-1-((1S,4S)-4,7,7-trimethyl-3-oxo-2-oxabicyclo[2.2.1]heptan-1-yl)-5,8,11-trioxa-2,15-diazaheptadecan-17-oyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide (71)

[0129] General Procedure 1 and 3. Reaction scale: 17.0 mg (23.0 μmol , 1.00 equiv) of 14 and 11.8 mg (47.0 μmol , 2.04 equiv) of (+)-sclereolide. Purified by PTLC (50%

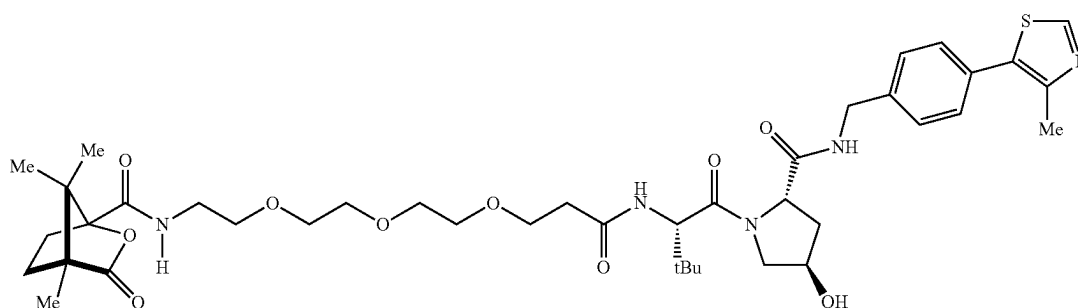
EtOAc/hexanes) to afford 71 as a yellow oil (16.8 mg, 84%).
 ^1H NMR (400 MHz, CDCl_3) δ 8.67 (s, 1H), 7.40-7.32 (m, 5H), 6.97 (d, $J=8.4$ Hz, 1H), 6.33 (m, 1H), 5.31 (m, 1H), 4.70 (t, $J=8.1$ Hz, 1H), 4.58-4.48 (m, 3H), 4.33 (dd, $J=15.1$, 5.3 Hz, 1H), 3.76-3.56 (m, 13H), 3.53-3.49 (m, 2H), 3.45-3.38 (m, 2H), 2.50-2.44 (m, 7H), 2.35-2.29 (m, 1H), 2.17-2.06 (m, 3H), 1.65 (m, 3H), 1.42-1.36 (m, 4H), 1.24-1.07 (m, 4H), 0.93 (s, 9H), 0.92 (s, 3H), 0.87 (s, 3H), 0.84 (s, 3H).



72

(2S,4R)-1-((S)-17-(tert-butyl)-3,15-dioxo-1-(3-((2-((1R,4aS,8aS)-2,5,5,8a-tetramethyl-1,4,4a,5,6,7,8,8a-octahydronaphthalen-1-yl) acetamido)methyl)phenyl)-6,9,12-trioxa-2,16-diazaoctadecan-18-oyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide (72)

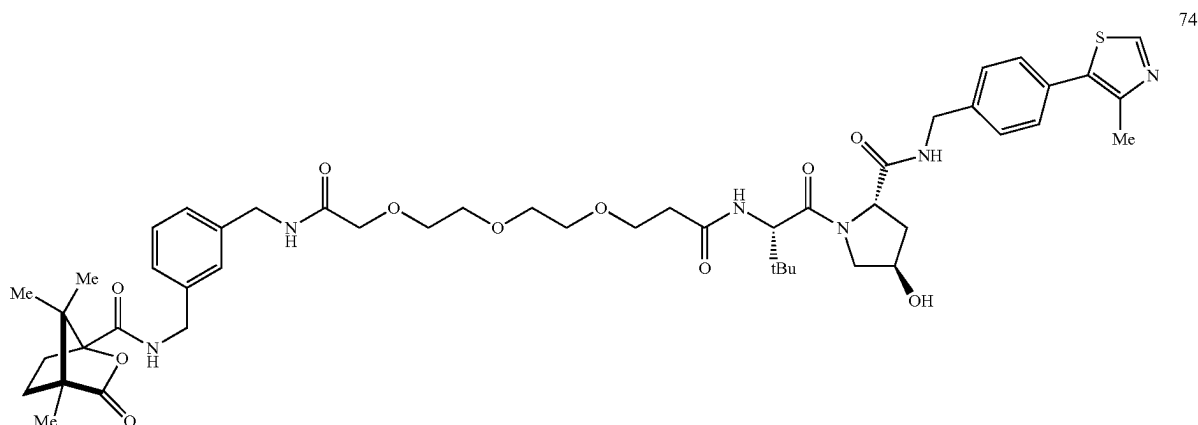
[0130] Reaction scale: 12.3 mg (17.0 μmol , 1.00 equiv) of 12 and 10.8 mg (23.0 μmol , 1.35 equiv) of (+)-sclereolide phenyl derivative. Purified by PTLC (100% EtOAc) to afford 72 as a yellow oil (3.8 mg, 22%).



73

(2S,4R)-1-((S)-16-(tert-butyl)-1,14-dioxo-1-((1S,4S)-4,7,7-trimethyl-3-oxo-2-oxabicyclo[2.2.1]heptan-1-yl)-5,8,11-trioxa-2,15-diazaheptadecan-17-oyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide (73)

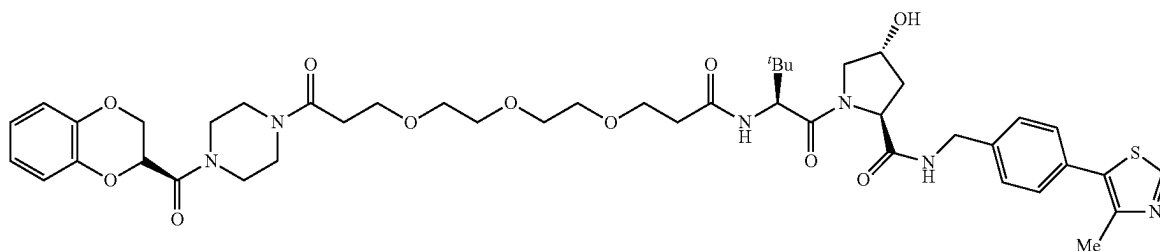
[0131] General Procedure 1 and 3. Reaction scale: 14.4 mg (20.0 μ mol, 1.00 equiv) of 14 and 7.0 mg (35.0 μ mol, 1.75 equiv) of (S)-camphanic acid. Purified by PTLC (50% EtOAc/hexanes) to afford 73 as a yellow oil (13.1 mg, 80%). ¹H NMR (400 MHz, CDCl₃) δ 8.67 (s, 1H), 7.45 (t, J=5.9 Hz, 1H), 7.37-7.32 (m, 4H), 7.02-6.96 (m, 2H), 4.72 (t, J=8.1 Hz, 1H), 4.55 (dd, J=15.0, 6.6 Hz, 1H), 4.51-4.45 (m, 2H), 4.33 (dd, J=15.1, 5.3 Hz, 1H), 4.11 (m, 1H), 3.71 (m, 2H), 3.63-3.44 (m, 14H), 2.52-2.47 (m, 7H), 2.16-2.11 (m, 1H), 1.95-1.84 (m, 2H), 1.68-1.62 (m, 1H), 1.08 (s, 3H), 1.08 (s, 3H), 0.94 (s, 9H), 0.88 (s, 3H).



(2S,4R)-1-((S)-18-(tert-butyl)-3,16-dioxo-1-(3-(((1S,4S)-4,7,7-trimethyl-3-oxo-2-oxabicyclo[2.2.1]heptane-1-carboxamido)methyl)phenyl)-7,10,13-trioxa-2,17-diazanonadecan-19-oyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide (74)

[0132] General Procedure 1 and 3. Reaction scale: 15.4 mg (21.0 μ mol, 1.00 equiv) of 12 and 12.5 mg (30.0 μ mol, 1.43 equiv) of (S)-camphanic acid phenyl derivative. Puri-

fied by PTLC (100% EtOAc) to afford 74 as a yellow oil (13.9 mg, 68%). ¹H NMR (400 MHz, CDCl₃) δ 8.66 (s, 1H), 7.36-7.28 (m, 6H), 7.19-7.13 (m, 3H), 7.08-6.98 (m, 2H), 6.91 (d, J=8.6 Hz, 1H), 4.61 (t, J=8.1 Hz, 1H), 4.56-4.38 (m, 8H), 4.32 (dd, J=15.1, 5.3 Hz, 1H), 4.06 (d, J=11.5 Hz, 1H), 3.74 (m, 2H), 3.68-3.50 (m, 12H), 2.51-2.49 (m, 6H), 2.44-2.38 (m, 2H), 2.12-2.07 (m, 2H), 1.97-1.87 (m, 3H), 1.70-1.63 (m, 1H), 1.11 (s, 3H), 1.09 (s, 3H), 0.93 (s, 9H), 0.89 (s, 3H).

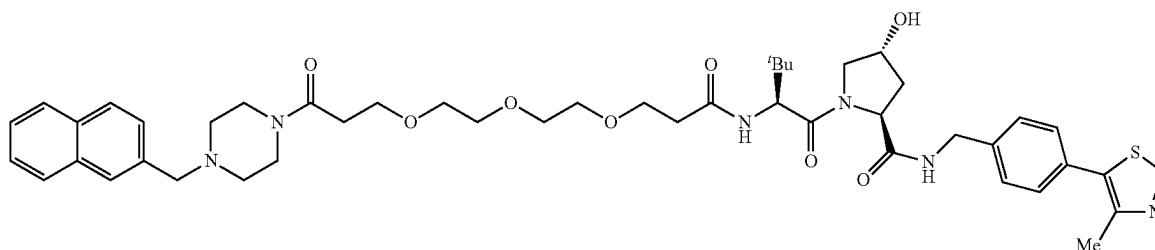


(2S,4R)-1-((S)-2-(tert-butyl)-16-(4-((R)-2,3-dihydrobenzo[b][1,4]dioxine-2-carbonyl) piperazin-1-yl)-4,16-dioxo-7,10,13-trioxa-3-azahexadecanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide (75)

[0133] General Procedure 3 then 2. Reaction scale: 6.0 mg (9.1 μ mol, 1.0 eq.) of 12 and 2.4 mg (10.0 μ mol, 1.1 eq.) of (S)-benzodioxanpiperazine methanone. Purified by pTLC (10% MeOH/CH₂Cl₂) to afford 75 as a white solid (3.3 mg,

41%). $R_f=0.49$ (8% MeOH/CH₂Cl₂, UV-active); LC-MS (ESI+) calc'd for C₄₅H₆₀N₆O₁₁S 892.4, found [M+H]⁺ 893.9. ¹H NMR (400 MHz, CDCl₃) δ 8.86 (br. s, 1H), 7.83-7.23 (m, 4H), 7.50 (br. t, 2H), 7.30 (s, 1H), 7.25 (br. t, 2H), 5.04 (t, 1H), 4.72 (t, 2H), 4.63 (d, J=3.1 Hz, 2H), 4.50 (s, 1H), 4.60-4.30 (m, 3H), 4.20 (s, 1H), 4.15-4.10 (d, J 7 Hz, 2H), 3.78 (ddd, J=9.1, 7.1, 2.4 Hz, 2H), 3.62 (q, J=5.4 Hz, 4H), 3.66-3.51 (m, 8H), 3.47 (q, J=6.7, 4.9 Hz, 4H), 2.60-2.35 (m, 9H), 2.20-2.10 (m, 1H), 2.05 (s, 1H), 0.93 (s, 9H).

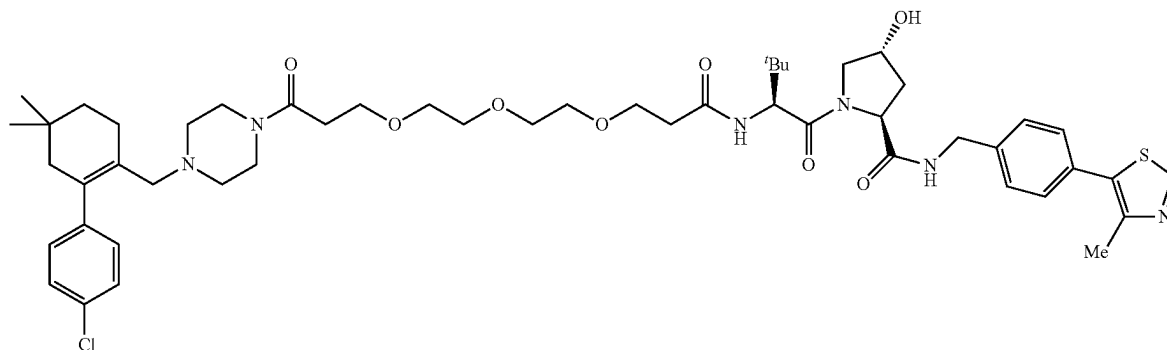
76



(2S,4R)-1-((S)-2-(tert-butyl)-16-(4-(naphthalen-2-ylmethyl) piperazin-1-yl)-4,16-dioxo-7,10,13-trioxa-3-azahexadecanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide (76)

[0134] General Procedure 3 then 2. Reaction scale: 6.0 mg (9.1 μ mol, 1.0 eq.) of 12 and 2.25 mg (10.0 μ mol, 1.1 eq.) of naphthalene piperazine. Purified by pTLC (10% MeOH/CH₂Cl₂) to afford 76 as a slightly yellow solid (3.1 mg, 40%). $R_f=0.55$ (8% MeOH/CH₂Cl₂, UV-active); LC-MS (ESI+) calc'd for C₄₇H₆₂N₆O₈S 870.4, found [M+NH₄]⁺ 888.8. ¹H NMR (400 MHz, CDCl₃) δ 8.86 (br. s, 1H), 7.80-7.35 (m, 7H), 7.50 (br. t, 2H), 7.30 (s, 1H), 7.25 (br. t, 2H), 5.04 (t, 1H), 4.72 (t, 2H), 4.63 (d, J=3.1 Hz, 2H), 4.50 (s, 1H), 4.60-4.30 (m, 3H), 4.20 (s, 1H), 4.15-4.10 (d, J 7 Hz, 2H), 3.92 (s, 2H), 3.78 (ddd, J=9.1, 7.1, 2.4 Hz, 2H), 3.62 (q, J=5.4 Hz, 4H), 3.66-3.51 (m, 16H), 3.47 (q, J=6.7, 4.9 Hz, 4H), 2.20-2.10 (m, 1H), 2.05 (s, 1H), 0.93 (s, 9H).

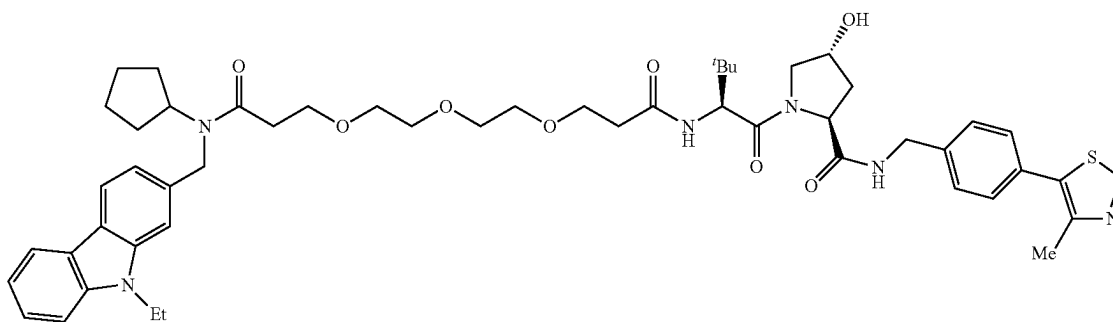
77



(2S,4R)-1-((S)-2-(tert-butyl)-16-(4-((4'-chloro-5,5-dimethyl-3,4,5,6-tetrahydro-[1,1'-biphenyl]-2-yl)methyl) piperazin-1-yl)-4,16-dioxo-7,10,13-trioxa-3-azaheptadecanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl) pyrrolidine-2-carboxamide (77)

[0135] General Procedure 3 then 2. Reaction scale: 6.5 mg (9.8 μ mol, 1.0 eq.) of 12 and 3.4 mg (10.8 μ mol, 1.1 eq.) of chlorophenyldimethylcyclohexenpiperazine. Purified by pTLC (10% MeOH/CH₂Cl₂) to afford 77 as a white solid (3.1 mg, 33%). R_f =0.61 (8% MeOH/CH₂Cl₂, UV-active);

LC-MS (ESI-) calc'd for C₅₁H₇₁ClN₆O₈S 962.5, found [M+HCOO]⁻: 1007.3. ¹H NMR (400 MHz, CDCl₃) δ 8.86 (br. s, 1H), 7.80-7.35 (m, 4H), 7.50 (br. t, 2H), 7.30 (s, 1H), 7.25 (br. t, 2H), 5.04 (t, 1H), 4.72 (t, 2H), 4.63 (d, J=3.1 Hz, 2H), 4.50 (s, 1H), 4.60-4.30 (m, 3H), 4.20 (s, 1H), 4.15-4.10 (d, J 7 Hz, 2H), 3.78 (ddd, J=9.1, 7.1, 2.4 Hz, 2H), 3.62 (q, J=5.4 Hz, 4H), 3.66-3.51 (m, 8H), 3.47 (q, J=6.7, 4.9 Hz, 4H), 3.10 (s, 2H), 2.60-2.35 (m, 8H), 2.20-2.10 (m, 5H), 2.05 (s, 1H), 1.53 (m 1H), 0.93 (s, 9H), 0.88 (m, 6H).

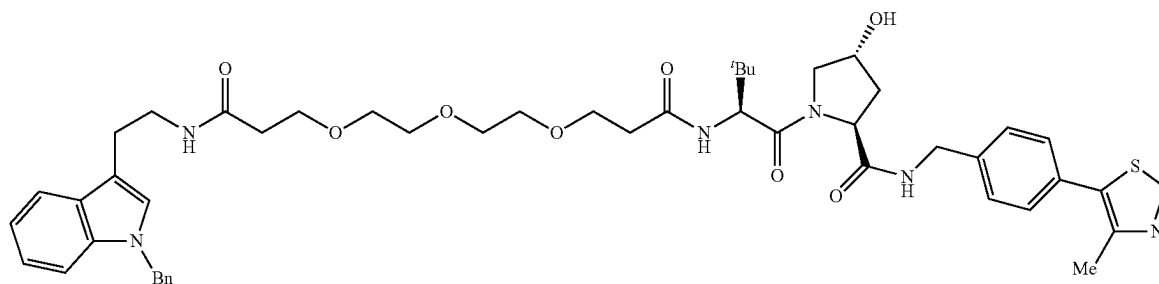


78

(2S,4R)-1-((S)-17-(tert-butyl)-2-cyclopentyl-1-(9-ethyl-9H-carbazol-2-yl)-3,15-dioxo-6,9,12-trioxa-2,16-diazaoctadecan-18-oyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl) pyrrolidine-2-carboxamide (78)

[0136] General Procedure 3 then 2. Reaction scale: 6.0 mg (9.05 μ mol) of 12 and 5.29 mg (18.11 μ mol) of carbazole-cyclopentanamine. Purified by pTLC (8% MeOH/DCM) to afford 78 as a white solid (5.3 mg, 63%). R_f =0.53 (8% MeOH/CH₂Cl₂, UV-active); LC-MS (ESI+) calc'd for C₅₂H₆₈N₆O₈S 936.5, found [M+H]⁺ 938.2. ¹H NMR (400 MHz, CDCl₃) δ 8.86 (br. s, 1H), 7.80-7.35 (m, 8H), 7.50 (br. t, 2H), 7.30 (s, 1H), 7.25 (br. t, 2H), 5.04 (t, 1H), 4.72 (t, 2H), 4.63 (d, J=3.1 Hz, 2H), 4.50 (s, 1H), 4.60-4.30 (m, 5H), 4.20 (s, 1H), 4.15-4.10 (d, J=7 Hz, 2H), 3.78 (ddd, J=9.1, 7.1, 2.4 Hz, 2H), 3.66-3.51 (m, 11H), 2.60-2.35 (m, 6H), 2.20-2.10 (m, 1H), 2.05 (s, 1H), 1.60-1.87 (8H), 1.26 (t, 2H), 0.93 (s, 9H).

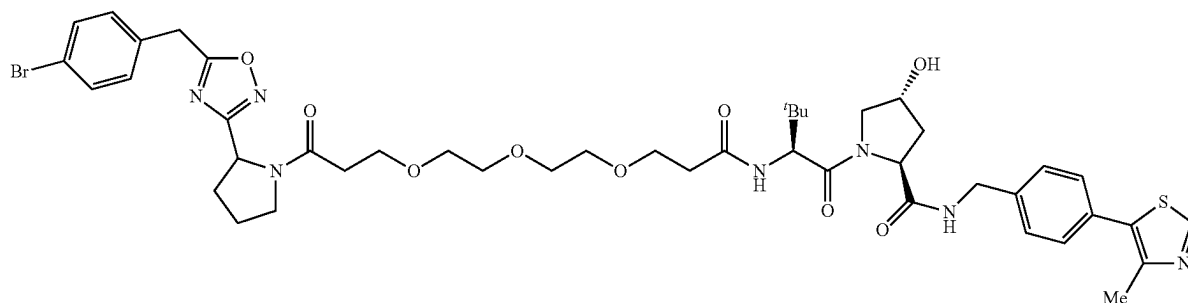
79



(2S,4R)-1-((S)-19-(1-benzyl-1H-indol-3-yl)-2-(tert-butyl)-4,16-dioxo-7,10,13-trioxa-3,17-diazanonadecanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide (79)

[0137] General Procedure 2. Reaction scale: 6.0 mg (9.05 μ mol) of 12 and 4.79 mg (18.11 μ mol) of benzyl indole amine. Purified by pTLC (8% MeOH/DCM) to afford 79 as a slightly orange solid (mg, %). Molecular formula:

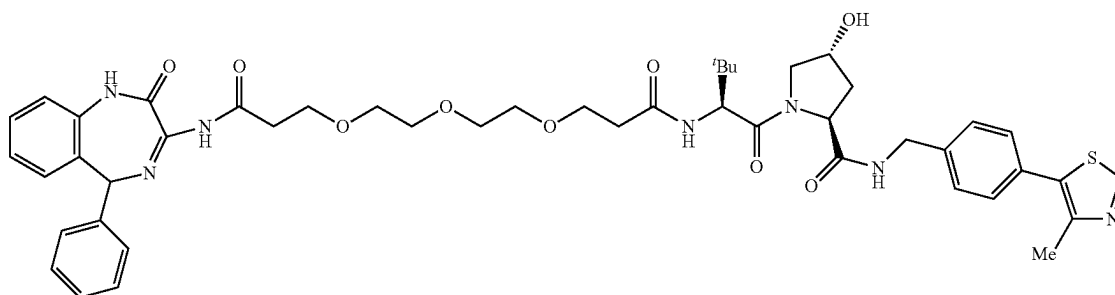
$C_{50}H_{64}N_6O_8S$. MS calc'd 908.5, found $[M+H]^+$ 909.7. 1H NMR (400 MHz, $CDCl_3$) δ 8.86 (br. s, 1H), 7.80-7.35 (m, 9H), 7.50 (br. t, 2H), 7.30 (s, 1H), 7.25 (br. t, 2H), 5.04 (t, 1H), 4.72 (t, 2H), 4.63 (d, $J=3.1$ Hz, 2H), 4.50 (s, 1H), 4.60-4.30 (m, 3H), 4.20 (s, 1H), 4.15-4.10 (d, $J=7$ Hz, 2H), 3.78 (ddd, $J=9.1, 7.1, 2.4$ Hz, 2H), 3.66-3.51 (m, 12H), 2.60-2.35 (m, 8H), 2.20-2.10 (m, 1H), 2.13 (s, 3H), 2.05 (s, 1H), 0.93 (s, 9H).



80

(2S,4R)-1-((2S)-16-(2-(5-(4-bromobenzyl)-1,2,4-oxadiazol-3-yl)pyrrolidin-1-yl)-2-(tert-butyl)-4,16-dioxo-7,10,13-trioxa-3-azahexadecanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide (80)

[0138] General Procedure 3 then 2. Reaction scale: 6.0 mg (9.05 μ mol) of 12 and 5.58 mg (18.11 μ mol) of bromobenzyl oxadiazolepyrrolidine. Purified by pTLC (8% MeOH/DCM) to afford 80 as a slightly yellow solid (mg, %). Molecular formula: $C_{45}H_{58}BrN_7O_9S$. MS calc'd 951.3, found $[M+H]^+$ 952.4. 1H NMR (400 MHz, $CDCl_3$) δ 8.86 (br. s, 1H), 7.80-7.35 (m, 4H), 7.50 (br. t, 2H), 7.30 (s, 1H), 7.25 (br. t, 1H), 4.72 (t, 2H), 4.63 (d, $J=3.1$ Hz, 2H), 4.50 (s, 1H), 4.60-4.30 (m, 4H), 4.20 (s, 1H), 4.15-4.10 (d, $J=7$ Hz, 2H), 3.78 (ddd, $J=9.1, 7.1, 2.4$ Hz, 2H), 3.66-3.51 (m, 12H), 2.60-2.35 (m, 10H), 2.20-2.10 (m, 1H), 2.05 (s, 1H), 0.93 (s, 9H).

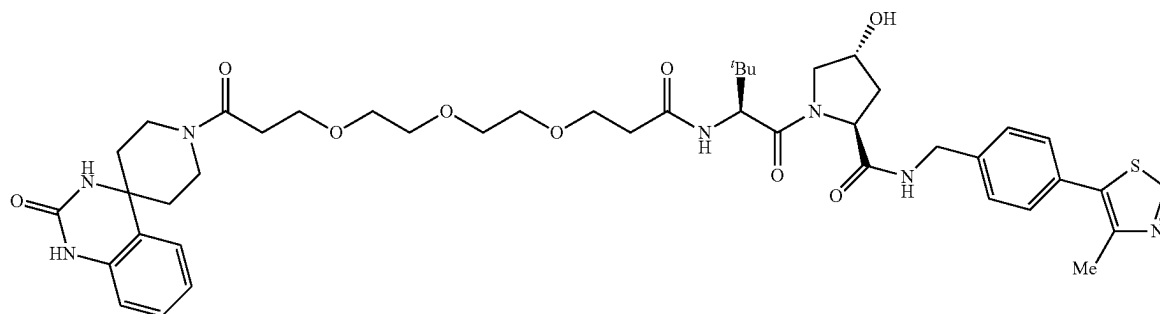


81

(2S,4R)-1-((2S)-2-(tert-butyl)-4,16-dioxo-16-((2-oxo-5-phenyl-2,5-dihydro-1H-benzo[e][1,4]diazepin-3-yl)amino)-7,10,13-trioxa-3-azahexadecanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl) pyrrolidine-2-carboxamide (81)

[0139] General Procedure 3 then 1. Reaction scale: 8 mg (12.1 μ mol, 1.0 eq.) of 12 and 3.3 mg (13.3 μ mol, 1.1 eq.) of benzodiazepam. Purified by pTLC (10% MeOH/CH₂Cl₂) to afford 81 as a white solid (7 mg, 68%). *R*_f=0.57 (10% MeOH/CH₂Cl₂, UV-active); LC-MS (ESI+) calc'd for C₄₇H₅₇N₇O₉S 895.4, found [M+H]⁺ 896.2. ¹H NMR (400 MHz, CDCl₃) δ 9.96 (s, 1H), 8.67 (d, *J*=6.2 Hz, 2H), 8.42 (d, *J*=7.8 Hz, 1H), 7.58-7.49 (m, 5H), 7.37 (s, 6H), 7.13 (t, *J*=8.6 Hz, 4H), 5.56-5.46 (m, 2H), 4.71-4.62 (m, 3H), 4.56-4.49 (m, 3H), 4.36-4.29 (m, 1H), 4.22 (d, *J*=11.5 Hz, 2H), 4.12 (q, *J*=7.2 Hz, 4H), 3.86 (q, *J*=4.4 Hz, 2H), 3.67 (d, *J*=5.0 Hz, 7H), 2.50 (s, 5H), 0.92 (s, 9H). Note: presence of rotamers.

82



(2S,4R)-1-((S)-2-(tert-butyl)-4,16-dioxo-16-(2'-oxo-2',3'-dihydro-1'H-spiro[piperidine-4,4'-quinazolin]-1-yl)-7,10,13-trioxa-3-azahexadecanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl) pyrrolidine-2-carboxamide (82)

[0140] General Procedure 3 then 1. Reaction scale: 8 mg (12.1 μ mol, 1.0 eq.) of 12 and 2.88 mg (13.3 μ mol, 1.1 eq.) of spiroquinazoline. Purified by pTLC (10% MeOH/CH₂Cl₂) to afford 82 as a yellow solid (5 mg, 48%). *R*_f=0.40

(10% MeOH/CH₂Cl₂, UV-active); LC-MS (ESI+) calc'd for C₄₄H₅₉N₇O₉S 861.4, found [M+H]⁺ 862.4. ¹H NMR (400 MHz, CDCl₃) δ 8.65 (d, *J*=16.1 Hz, 1H), 7.55 (s, 1H), 7.41-7.30 (m, 5H), 7.22-7.08 (m, 2H), 6.99 (d, *J*=5.5 Hz, 1H), 6.74 (d, *J*=7.8 Hz, 1H), 4.71 (t, *J*=8.6 Hz, 2H), 4.59-4.47 (m, 3H), 4.38-4.24 (m, 1H), 4.18-4.05 (m, 2H), 3.92-3.73 (m, 5H), 3.67-3.44 (m, 12H), 2.76 (dt, *J*=13.4, 6.3 Hz, 1H), 2.58 (dd, *J*=15.2, 5.5 Hz, 1H), 2.44 (s, 3H), 2.26 (dd, *J*=14.0, 5.5 Hz, 2H), 1.94 (q, *J*=14.0 Hz, 6H), 0.88 (s, 9H). Note: presence of rotamers.

TABLE 3

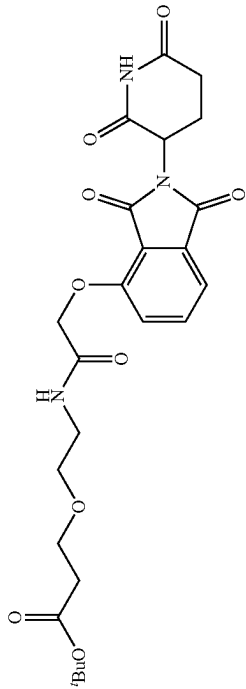
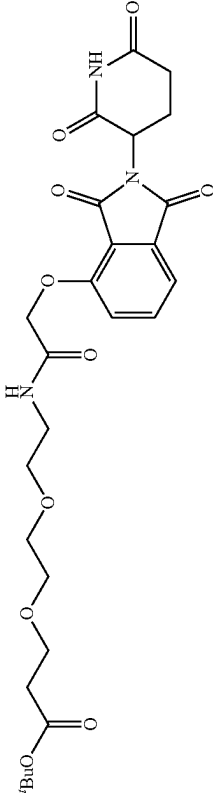
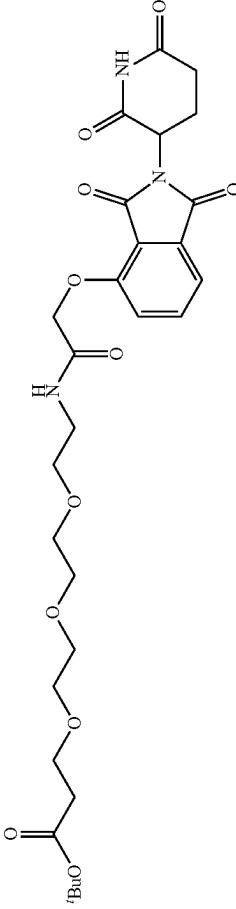
Non-limiting Examples of FragTAC Probes and Intermediates		
Compound Example Number	Chemical Name	Structure
1	Tert-butyl 3-(2-(2-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)oxy)acetamido)ethoxy)propanoate	
2	Tert-butyl 3-(2-(2-(2-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)oxy)acetamido)ethoxy)ethoxy)propanoate	
3	Tert-butyl 1-(2-(2-(2-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)oxy)-2-oxo-6,9,12-trioxa-3-azapentadecan-15-oate	

TABLE 3-continued

Non-limiting Examples of FragTAC Probes and Intermediates		
Compound Example Number	Chemical Name	Structure
4 IF-092	Tert-butyl 1-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)oxy)-2-oxo-6,9, 12-trioxa-3-azatetradecan-14-yl)carbamate	
5 IF-015	Tert-butyl 1-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)oxy)-2-oxo-6,9,12,15-tetraoxa-3-azaoctadecan-18-olate	
6 IF-025	Tert-butyl 6-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)oxy)acetamido)hexanoate	
7 CMC-6-194	Tert-butyl (6-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)oxy)acetamido)hexyl)carbamate	

TABLE 3-continued

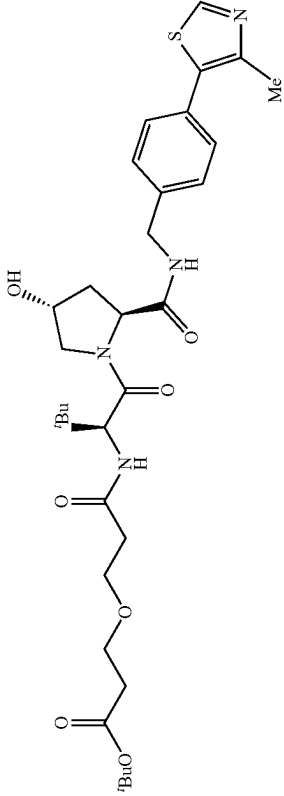
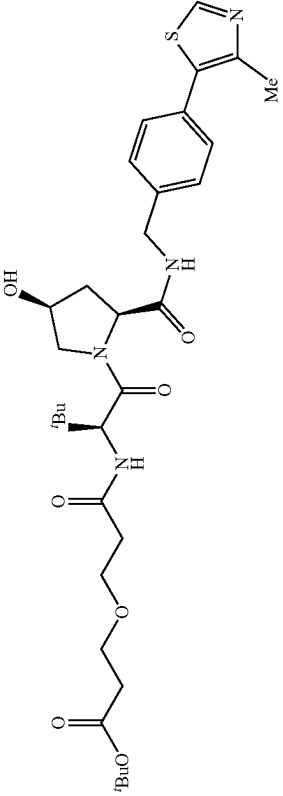
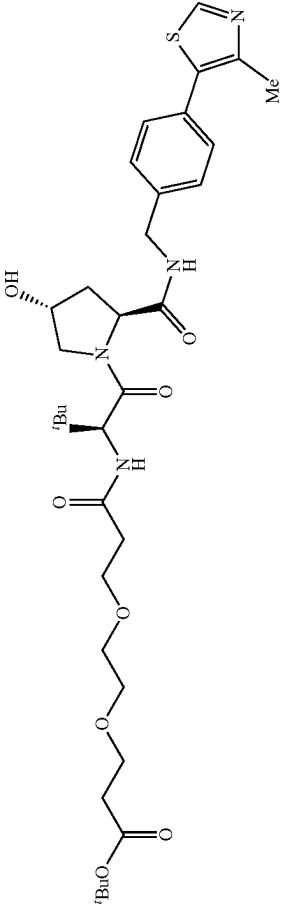
Non-limiting Examples of FragTAC Probes and Intermediates			
Compound Example Number	Compound Code	Structure	Chemical Name
8	IF-081		Tert-butyl 3-(3-((S)-1-((2S,4R)-4-hydroxy-2-((4-(4-methylthiazol-5-yl)benzyl)carbamoyl)pyrrolidin-1-yl)-3,3-dimethyl-1-oxobutan-2-yl)amino)-3-oxopropoxy)propanoate
9	LPC115		Tert-butyl 3-(3-((S)-1-((2S,4S)-4-hydroxy-2-((4-(4-methylthiazol-5-yl)benzyl)carbamoyl)pyrrolidin-1-yl)-3,3-dimethyl-1-oxobutan-2-yl)amino)-3-oxopropoxy)propanoate
10	LPC013		Tert-butyl 3-(2-(3-((S)-1-((2S,4R)-4-hydroxy-2-((4-(4-methylthiazol-5-yl)benzyl)carbamoyl)pyrrolidin-1-yl)-3,3-dimethyl-1-oxobutan-2-yl)amino)-3-oxopropoxy)ethoxy)propanoate

TABLE 3-continued

Non-limiting Examples of FragTAC Probes and Intermediates	
Compound Example Number	Chemical Name
11	Tert-butyl 3-(2-(3-((S)-1-((2S,4S)-4-hydroxy-2-((4-(4-methylthiazol-5-yl)benzyl)carbamoyl)pyrrolidin-1-yl)-3,3-dimethyl-1-oxobutan-2-yl)amino)-3-oxopropoxy)ethoxy)propanoate
12	Tert-butyl (S)-15-((2S,4R)-4-hydroxy-2-((4-(4-methylthiazol-5-yl)benzyl)carbamoyl)pyrrolidine-1-carbonyl)-16,16-dimethyl-13-oxo-4,7,10-trioxo-14-azaheptadecanoate
13	Tert-butyl (S)-15-((2S,4S)-4-hydroxy-2-((4-(4-methylthiazol-5-yl)benzyl)carbamoyl)pyrrolidine-1-carbonyl)-16,16-dimethyl-13-oxo-4,7,10-trioxo-14-azaheptadecanoate

TABLE 3-continued

Non-limiting Examples of FragTAC Probes and Intermediates	
Compound Example Number	Chemical Name
14	Tert-butyl ((S)-14-((2S,4R)-4-hydroxy-2-((4-(4-methylthiazol-5-yl)benzyl)carbamoyl)pyrrolidine-1-carbonyl)-15,15-dimethyl-12-oxo-3,6,9-trioxo-13-azahexadecyl)carbamate
15	Tert-butyl (S)-18-((2S,4R)-4-hydroxy-2-((4-(4-methylthiazol-5-yl)benzyl)carbamoyl)pyrrolidine-1-carbonyl)-19,19-dimethyl-16-oxo-4,7,10,13-tetraoxa-17-azaiicosanoate
16	Tert-butyl 4-(((S)-1-((2S,4R)-4-hydroxy-2-((4-(4-methylthiazol-5-yl)benzyl)carbamoyl)pyrrolidine-1-yl)-3,3-dimethyl-1-oxobutan-2-yl)amino)-4-oxobutanoate

TABLE 3-continued

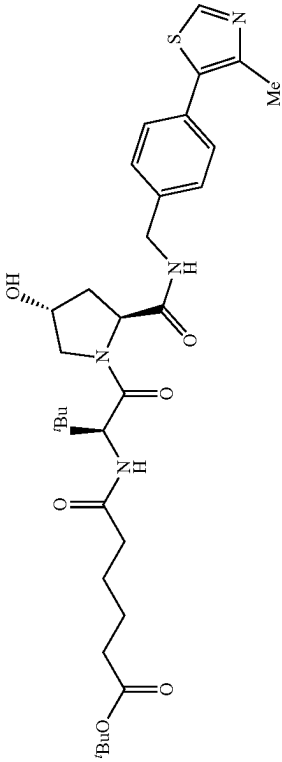
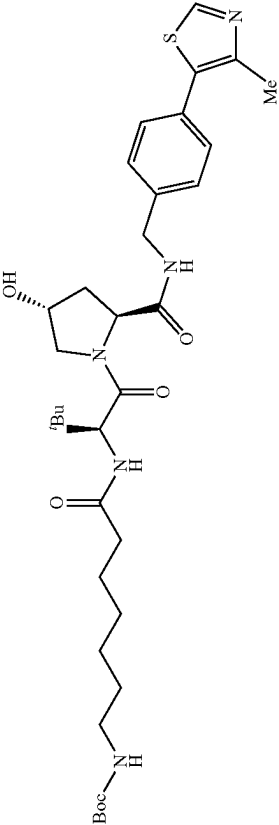
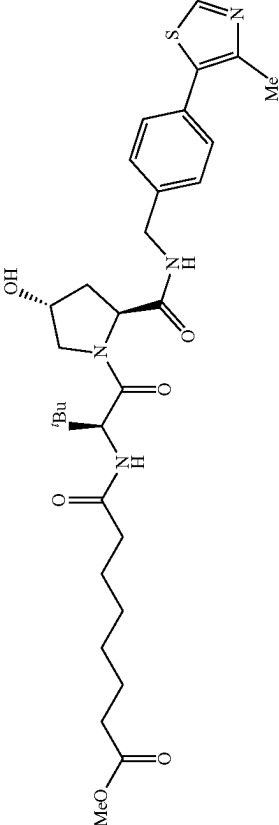
Non-limiting Examples of FragTAC Probes and Intermediates		
Compound Example Number	Chemical Name	Structure
17	Tert-butyl 6-(((S)-1-(2S,4R)-4-hydroxy-2-((4-(4-methylthiazol-5-yl)benzyl)carbamoyl)pyrrolidin-1-yl)-3,3-dimethyl-1-oxobutan-2-yl)amino)-6-oxohexanoate	
18	Tert-butyl (7-(((S)-1-(2S,4R)-4-hydroxy-2-((4-(4-methylthiazol-5-yl)benzyl)carbamoyl)pyrrolidin-1-yl)-3,3-dimethyl-1-oxobutan-2-yl)amino)-7-oxoheptyl)carbamate	
19	Methyl 8-(((S)-1-(2S,4R)-4-hydroxy-2-((4-(4-methylthiazol-5-yl)benzyl)carbamoyl)pyrrolidin-1-yl)-3,3-dimethyl-1-oxobutan-2-yl)amino)-8-oxooctanoate	

TABLE 3-continued

Non-limiting Examples of FragTAC Probes and Intermediates	
Compound Example Number	Chemical Name
20	Methyl 10-((S)-1-((2S,4R)-4-hydroxy-2-((4-(4-methylthiazol-5-yl) benzyl) carbamoyl)pyrrolidin-1-yl)-3,3-dimethyl-1-oxobutan-2-yl)amino)-10-oxodecanoate
21	N-(2-(3-(4-benzhydryl)piperazin-1-yl)-3-oxopropoxyethyl)-2-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)oxy)acetamide
22	N-(2-(2-(3-(4-benzhydryl)piperazin-1-yl)-3-oxopropoxyethoxyethyl)-2-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)oxy)acetamide

TABLE 3-continued

Non-limiting Examples of FragTAC Probes and Intermediates	
Compound Example Number	Chemical Name
23	N-(2-(2-(2-(3-(4-benzhydrylpiperazin-1-yl)-3-oxopropoxy)ethoxy)ethyl)-2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)oxy)acetamide
24	N-(15-(4-benzhydrylpiperazin-1-yl)-15-oxo-3,6,9,12-tetraoxapentadecyl)-2-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)oxy)acetamide
25	N-(6-(4-benzhydrylpiperazin-1-yl)-6-oxohexyl)-2-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)oxy)acetamide

TABLE 3-continued

Non-limiting Examples of FragTAC Probes and Intermediates	
Compound Example Number	Chemical Name
26	(2S,4R)-1-(S)-2-(3-(4-benzylhydriylpiperazin-1-yl)-3-oxopropoxy)propanamido)-3,3-dimethylbutanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide
27	(2S,4S)-1-(S)-2-(3-(4-benzylhydriylpiperazin-1-yl)-3-oxopropoxy) propanamido)-3,3-dimethylbutanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide

TABLE 3-continued

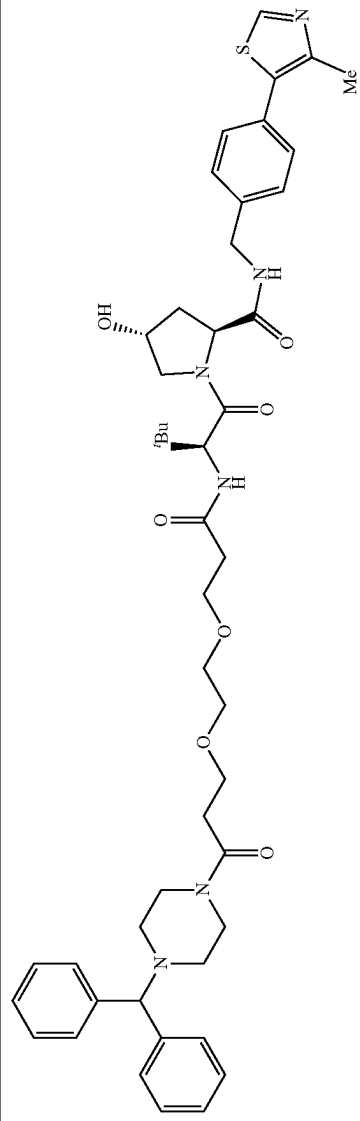
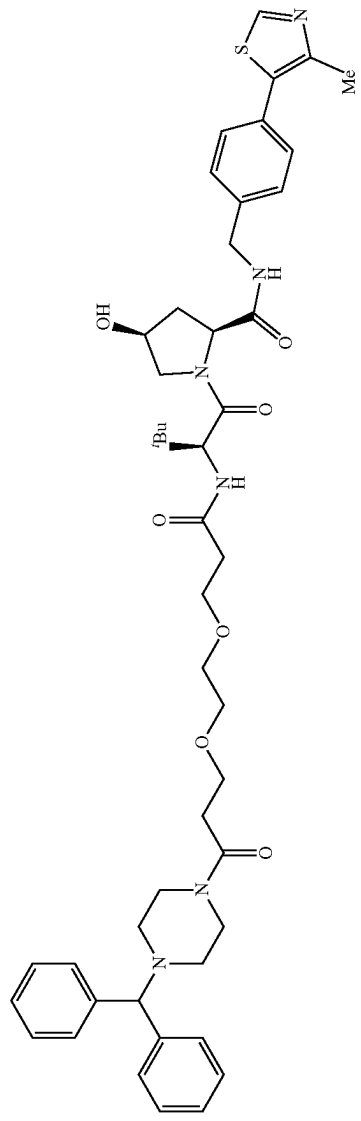
Non-limiting Examples of FragTAC Probes and Intermediates		
Compound Example Number	Chemical Name	Structure
28 LPC020	(2S,4R)-1-(S)-2-(3-(2-(3-(4-benzhydryl)piperazin-1-yl)-3-oxopropoxy)ethoxy)propanamido)-3,3-dimethylbutanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide	
29 LPC117	(2S,4S)-1-(S)-2-(3-(2-(3-(4-benzhydryl)piperazin-1-yl)-3-oxopropoxy)ethoxy)propanamido)-3,3-dimethylbutanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide	

TABLE 3-continued

Non-limiting Examples of FragTAC Probes and Intermediates	
Compound Example Number	Chemical Name
30	(2S,4R)-1-((S)-16-(4-benzhydrylpiperazin-1-yl)-2-(tert-butyl)-4,16-dioxo-7,10,13-trioxo-3-azahexadecanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide
31	(2S,4S)-1-((S)-16-(4-benzhydrylpiperazin-1-yl)-2-(tert-butyl)-4,16-dioxo-7,10,13-trioxo-3-azahexadecanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide
32	(2S,4R)-1-((S)-19-(4-benzhydrylpiperazin-1-yl)-2-(tert-butyl)-4,19-dioxo-7,10,13,16-tetraoxo-3-azanonadecanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide

TABLE 3-continued

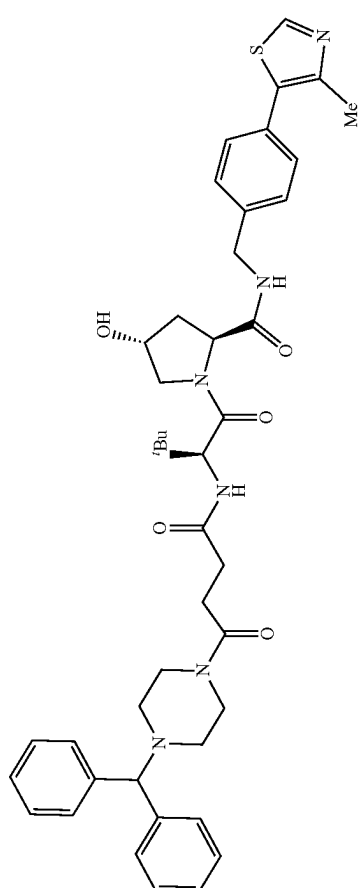
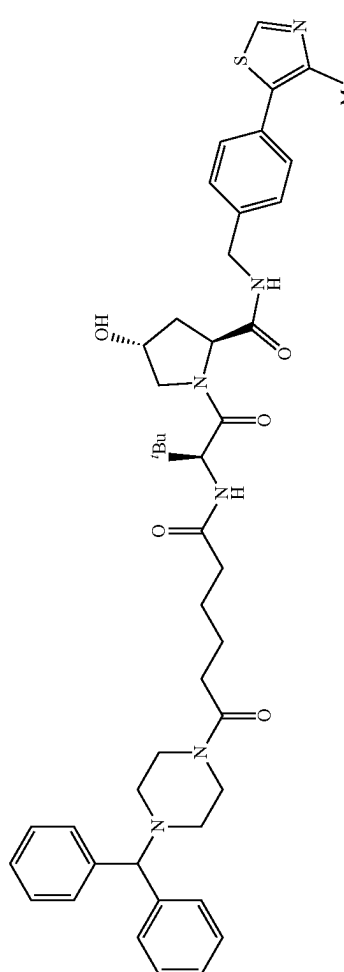
Non-limiting Examples of FragTAC Probes and Intermediates	
Compound Example Number	Chemical Name
Structure	
33	(2S,4R)-1-((S)-2-(4-(4-benzhydryl)piperazin-1-yl)-4-oxobutanimido)-3,3-dimethylbutanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide
	
34	(2S,4R)-1-((S)-2-(6-(4-benzhydryl)piperazin-1-yl)-6-oxohexanimido)-3,3-dimethylbutanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide
	

TABLE 3-continued

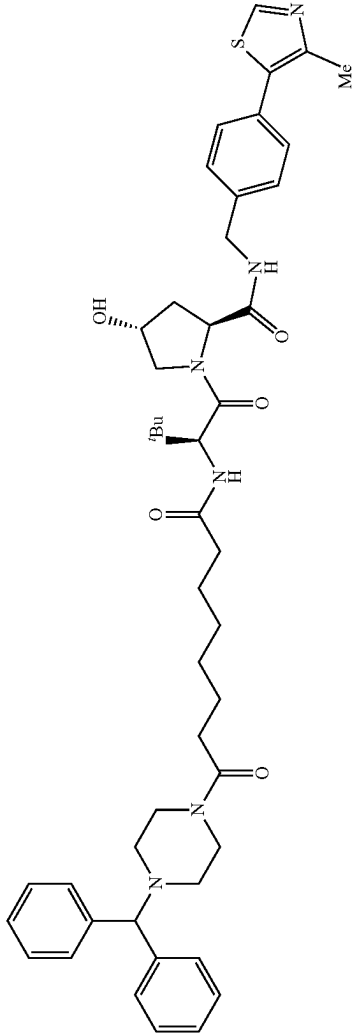
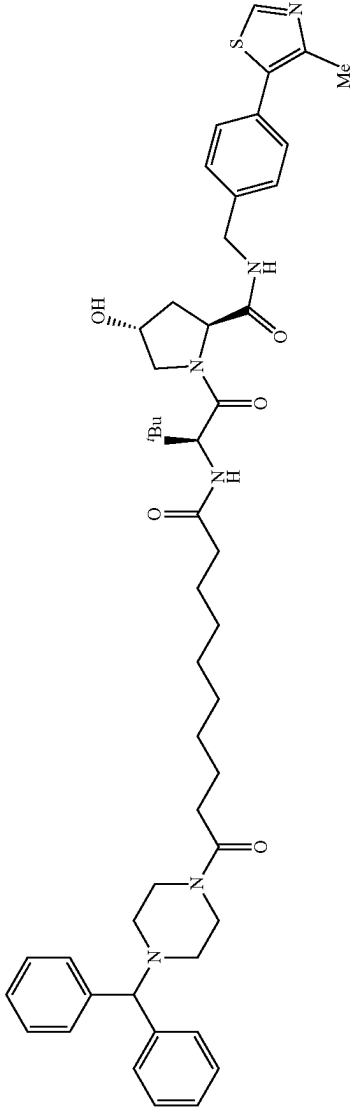
Non-limiting Examples of FragTAC Probes and Intermediates		
Compound Example Number	Chemical Name	Structure
35	(2S,4R)-1-((S)-2-(8-(4-benzylhydriylpiperazin-1-yl)-8-oxooctanamido)-3,3-dimethylbutanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide	
36	(2S,4R)-1-((S)-2-(10-(4-benzylhydriylpiperazin-1-yl)-10-oxodecanamido)-3,3-dimethylbutanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide	

TABLE 3-continued

Non-limiting Examples of FragTAC Probes and Intermediates	
Compound Example Number	Chemical Name
37	(2S,4R)-1-((S)-16-((R)-4-benzhydryl-2-phenylpiperazin-1-yl)-2-(tert-butyl)-4,16-dioxo-7,10,13-trioxa-3-azahexadecanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide
38	(2S,4R)-1-((S)-16-((S)-4-benzhydryl-2-phenylpiperazin-1-yl)-2-(tert-butyl)-4,16-dioxo-7,10,13-trioxa-3-azahexadecanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide
39	(2S,4R)-1-((2S)-2-(tert-butyl)-4,16-dioxo-16-((4-phenyl(pyridin-2-yl)methyl)piperazin-1-yl)-7,10,13-trioxa-3-azahexadecanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide

Non-limiting Examples of FragTAC Probes and Intermediates		
Compound Example Number	Chemical Structure	Chemical Name
40		(2 <i>S</i> ,4 <i>R</i>)-1-((<i>S</i>)-16-(<i>S</i>)-4-benzhydryl-3-methylpiperazin-1-yl)-2-(tert-butyl)-4,16-dioxo-7,10,13-trioxa-3-(4-hydroxy-N-(benzylpyrrolidin-2-yl)carboxamide
41		2-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)oxy)-N-(2-(2-(2-(3-oxo-3-(4-tosylpiperazin-1-yl) propoxy)ethoxy)ethoxy)ethyl)acetamide
42		2-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)oxy)-N-(6-oxo-6-(4-tosylpiperazin-1-yl)hexyl)acetamide

TABLE 3-continued

Non-limiting Examples of FragTAC Probes and Intermediates	
Compound Example Number	Chemical Name
Structure	
	(2S,4R)-1-((S)-2-(tert-butyl)-4,16-dioxo-16-(4-tosylpiperazin-1-yl)-7,10,13-trioxa-3-azahexadecanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide
	(2S,4R)-1-((S)-3,3-dimethyl-2-(6-oxo-6-(4-tosylpiperazin-1-yl)hexanamido)butanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide
	N-(1-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)oxy)-2-oxo-6,9,12-trioxa-3-azatetradecan-14-yl)-3-(4,5-diphenyloxazol-2-yl)propanamide

TABLE 3-continued

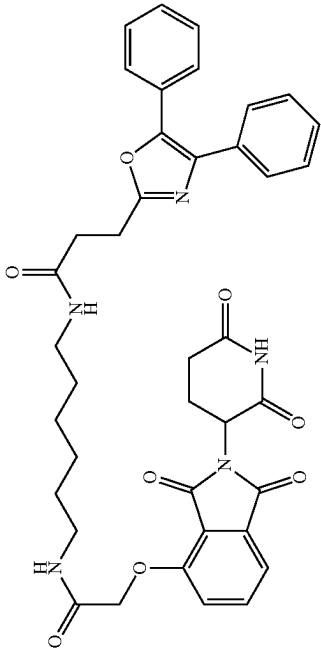
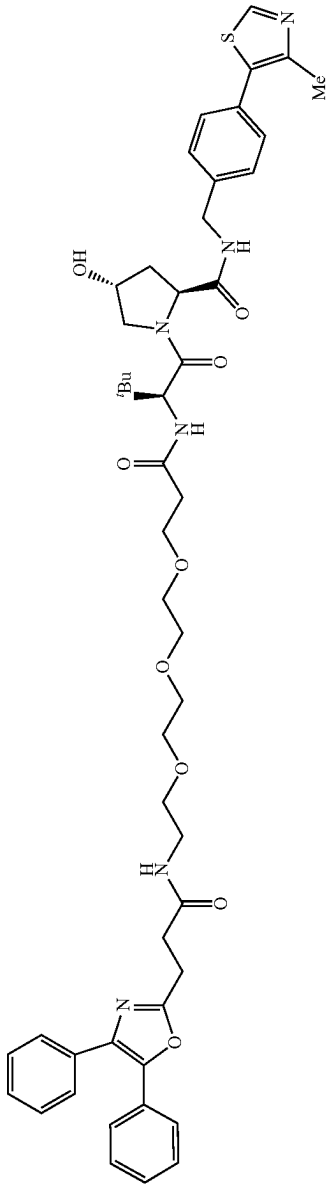
Non-limiting Examples of FragTAC Probes and Intermediates		
Compound Example Number	Chemical Name	Structure
46	CMC-6-100 N-(6-(2-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)oxy)acetamido)hexyl)-3-(4,5-diphenyloxazol-2-yl)propanamide	
47	IF-072 (2S,4R)-1-(S)-2-(tert-butyl)-19-(4,5-diphenyloxazol-2-yl)-4,17-dioxo-7,10,13-trioxa-3,16-diazanonadecanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide	

TABLE 3-continued

Non-limiting Examples of FragTAC Probes and Intermediates	
Compound Example Number	Chemical Name
48	(2S,4R)-1-((S)-2-(7-(3-(4,5-diphenyloxazol-2-yl)propanamido)heptanamido)-3,3-dimethylbutanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide
49	5-chloro-N-(1-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)oxy)-2-oxo-6,9,12-trioxa-3-azatetradecan-14-yl)-3-phenyl-1H-indole-2-carboxamide

TABLE 3-continued

Non-limiting Examples of FragTAC Probes and Intermediates		
Compound Example Number	Chemical Name	Structure
50	CMC-6-102 5-chloro-N-(6-(2-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)oxy)acetamido)hexyl)-3-phenyl-1H-indole-2-carboxamide	
51	IF-073 5-chloro-N-(S)-14-((2S,4R)-4-hydroxy-2-((4-(4-methylthiazol-5-yl)benzyl)carbamoyl)pyrrolidine-1-carboxyl)-15,15-dimethyl-13-oxo-3,6,9-trioxadecyl-3-phenyl-1H-indole-2-carboxamide	

TABLE 3-continued

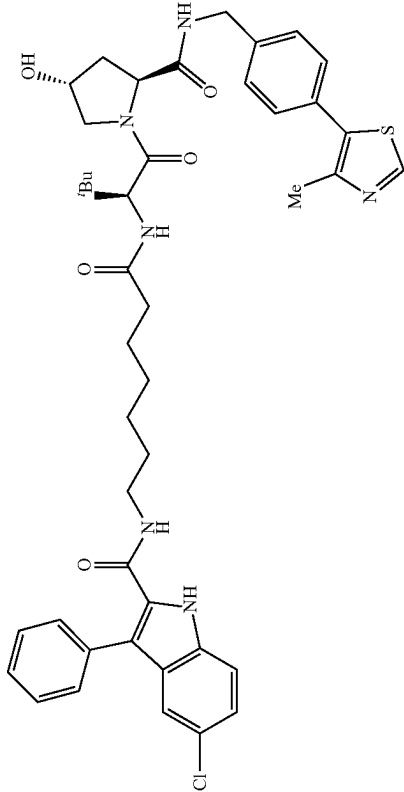
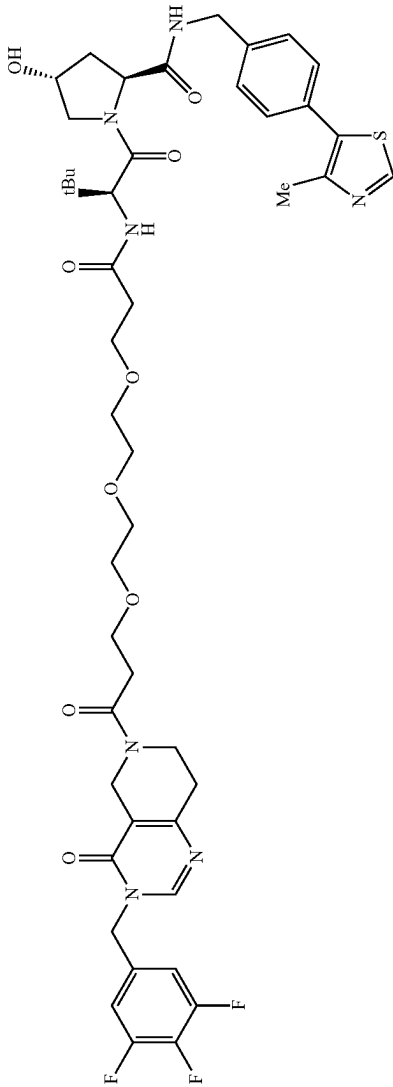
Non-limiting Examples of FragTAC Probes and Intermediates			
Compound Example Number	Compound Code	Structure	Chemical Name
52	CMC-6-112		5-chloro-N-(7-(((S)-1-((2S,4R)-4-hydroxy-2-((4-(4-methylthiazol-5-yl) benzyl) carbanoyl)pyrrolidin-1-yl)-3,3-dimethyl-1-oxobutan-2-yl)amino)-7-oxoheptyl)-3-phenyl-1H-indole-2-carboxamide
53	CMC-6-27		(2S,4R)-1-((S)-2-(tert-butyl)-4,16-dioxo-16-(4-oxo-3-(3,4,5-trifluorobenzyl)-3,5,7,8-tetrahydropyrido[4,3-d]pyrimidin-6(4H)-yl)-7,10,13-trioxo-3-azahexadecanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide

TABLE 3-continued

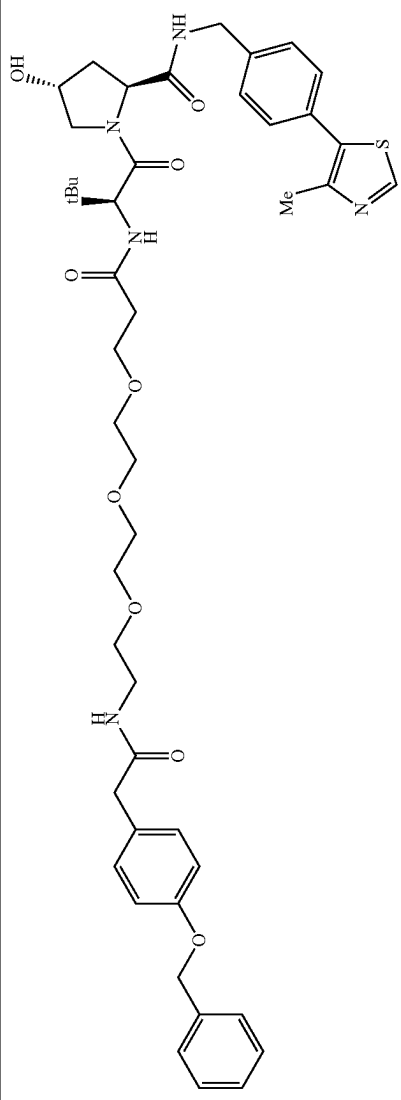
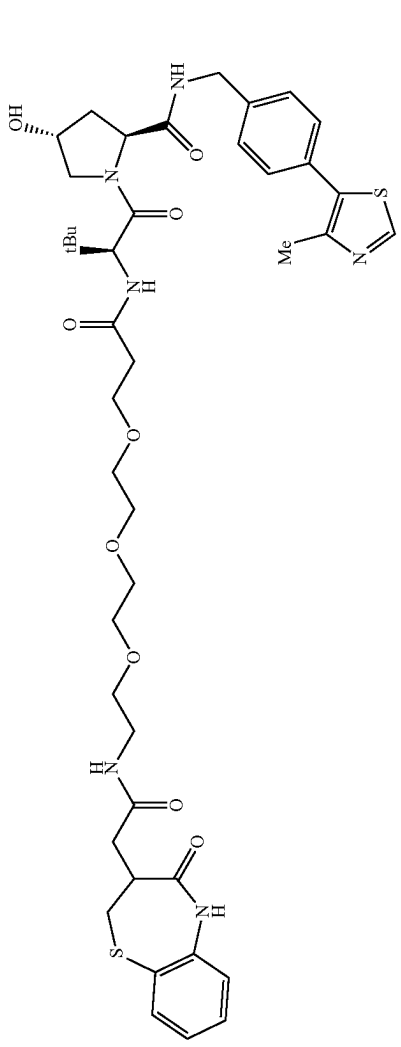
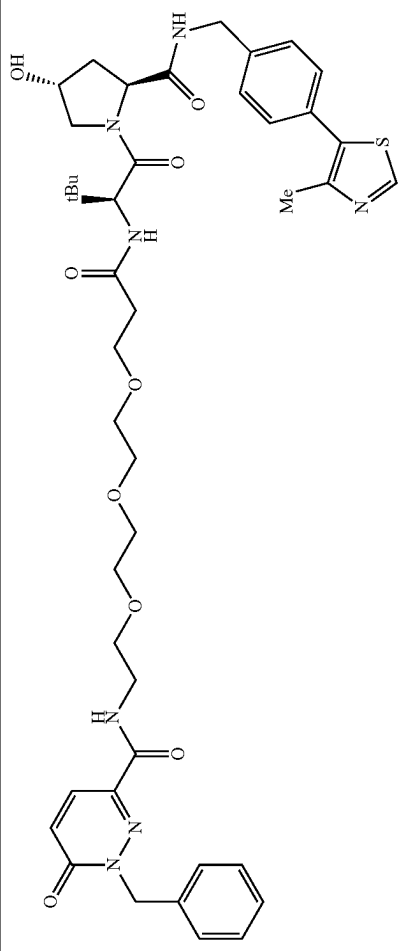
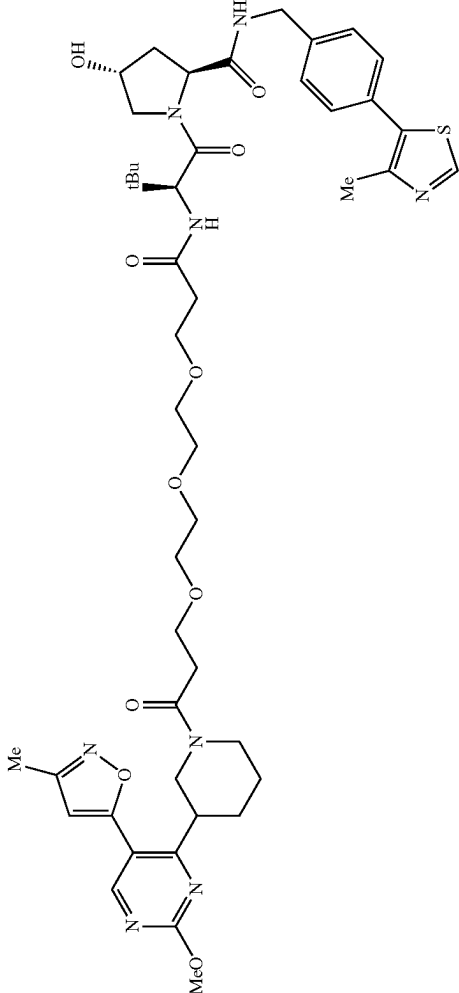
Non-limiting Examples of FragTAC Probes and Intermediates		
Compound Example Number	Chemical Name	Structure
54 CMC-6-35	(2S,4R)-1-((S)-1-(4-(benzyloxy)phenyl)-17-(tert-butyl)-2,15-dioxo-6,9,12-trioxo-3,16-diazaoctadecan-18-yl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide	
55 CMC-6-40	(2S,4R)-1-((1-(17S)-17-(tert-butyl)-2,15-dioxo-1-(4-oxo-2,3,4,5-tetrahydrobenzo[b][1,4]thiazepin-3-yl)-6,9,12-trioxo-3,16-diazaoctadecan-18-yl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide	

TABLE 3-continued

Non-limiting Examples of FragTAC Probes and Intermediates		
Compound Example Number	Compound Code	Chemical Name
56	CMC-6-46	1-benzyl-N-(S)-14-((2S,4R)-4-hydroxy-2-((4-(4-methylthiazol-5-yl)benzyl)carbamoyl)pyrrolidine-1-carbonyl)-15,15-dimethyl-12-oxo-3,6,9-trioxa-13-azahexadecyl)-6-oxo-1,6-dihydropyridazine-3-carboxamide
		
57	CMC-6-73	(2S,4R)-1-((2S)-2-(tert-butyl)-16-(3-(2-methoxy-5-(3-methylisoxazol-5-yl)pyrimidin-4-yl)piperidin-1-yl)-4,16-dioxo-7,10,13-trioxa-3-azahexadecanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide
		

Non-limiting Examples of FragTAC Probes and Intermediates		
Compound Example Number	Compound Code	Chemical Name
58	CMC-6-74	(2S,4R)-1-((S)-17-(tert-butyl)-3,15-dioxo-1-(4-(4-(pyrimidin-2-ylmethyl)piperidine-1-carbonyl)phenyl)-6,9,12-trioxo-2,16-diazaoctadecan-18-yl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide
59	LPC004	3-(2-(2-(2-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)oxy)acetamido)ethoxy)ethoxy)-N-(2-oxo-2H-chromen-7-yl)propanamide
60	LPC019	(2S,4R)-1-((S)-3,3-dimethyl-2-(3-(2-(3-oxo-3-(2-oxo-2H-chromen-7-yl)lamino)propoxy)ethoxy)propanamido)butanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide

Non-limiting Examples of FragTAC Probes and Intermediates

Compound Example Number	Chemical Name
61	LPC006 3-(2-(2-(2-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yloxy)acetamido)ethoxy)ethoxy)-N-(3,3-diphenylpropyl)propanamide
62	LPC022 (2S,4R)-1-(S)-2-(tert-butyl)-4,13-dioxo-17,17-diphenyl-7,10-dioxo-3,14-diazaheptadecanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide
63	LPC015 N-(2-(1H-indol-3-yl)ethyl)-3-(2-(2-(2-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yloxy)acetamido)ethoxy)ethoxy)propanamide

TABLE 3-continued

Non-limiting Examples of FragTAC Probes and Intermediates			
Compound Example Number	Compound Code	Structure	Chemical Name
64	LPC024		(2S,4R)-1-((S)-2-(tert-butyl)-16-((1H-indol-3-yl)-4,13-dioxo-7,10-dioxo-3,14-diazahexadecanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide
65	LPC033		(2S,4R)-1-((S)-2-(tert-butyl)-16-((3,4-dimethoxyphenyl)-4,13-dioxo-7,10-dioxo-3,14-diazahexadecanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide
66	LPC034		(2S,4R)-1-((S)-2-(tert-butyl)-4,13-dioxo-16-((2-oxo-2,3,4,5-tetrahydro-1H-benzo[b]azepin-8-yl)-7,10-dioxo-3,14-diazahexadecanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide

TABLE 3-continued

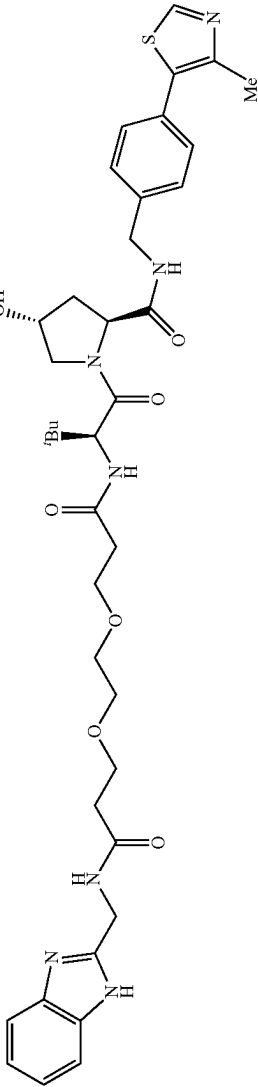
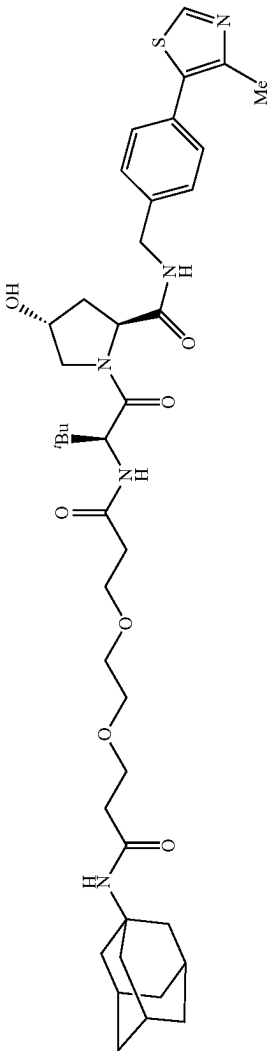
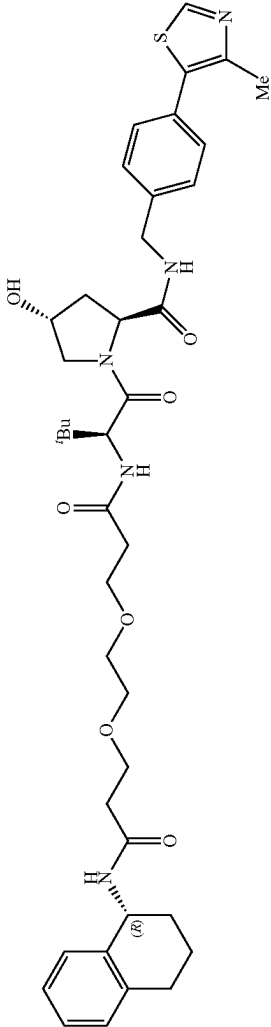
Non-limiting Examples of FragTAC Probes and Intermediates			
Compound Example Number	Compound Code	Structure	Chemical Name
67	LPC036		(2S,4R)-1-((S)-1-((H-benzol[d]imidazol-2-yl)-1,4-(tert-butyl)-3,12-dioxo-6,9-dioxo-2,13-diazapentadecan-15-oyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide
68	LPC042		(2S,4R)-1-((S)-2-(3-(3-(((3s,5s,7s)-adamantan-1-yl)amino)-3-oxopropoxy)ethoxy)propanamido)-3,3-dimethylbutanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide
69	LPC039		(2S,4R)-1-((S)-3,3-dimethyl-2-(3-(2-(3-oxo-3-(((R)-1,2,3,4-tetrahydronaphthalen-1-yl)amino)propoxy)ethoxy)propanamido)butanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide

TABLE 3-continued

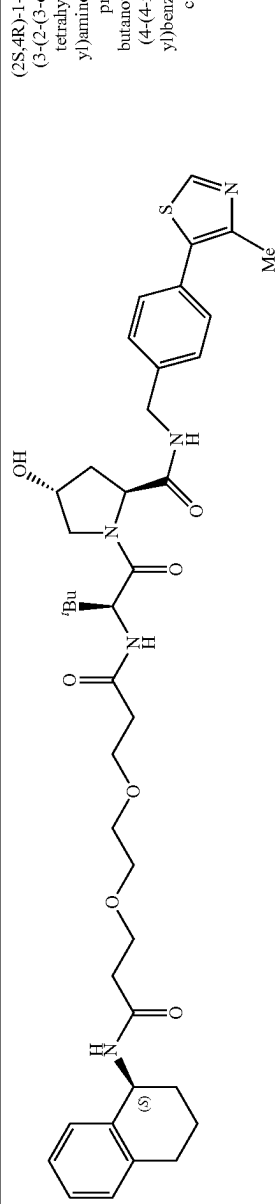
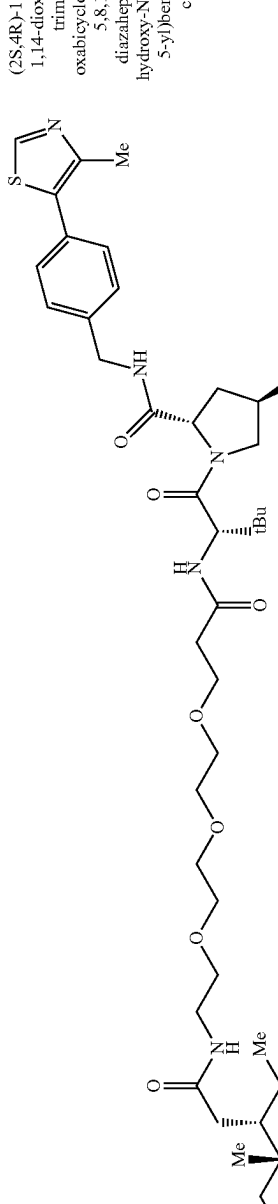
Non-limiting Examples of FragTAC Probes and Intermediates		
Compound Example Number	Chemical Name	Structure
70	(2S,4R)-1-((S)-3,3-dimethyl-2-(3-(2-(3-oxo-3-(((S)-1,2,3,4-tetrahydronaphthalen-1-yl)amino)propoxy)ethoxy)butanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide	
71	(2S,4R)-1-((S)-16-(tert-butyl)-1,14-dioxo-1-((1S,4S)-4,7,7-trimethyl-3-oxo-2-oxabicyclo[2.2.1]heptan-1-yl)-5,8,11-trioxa-2,15-diazaheptadecan-17-yl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide	

TABLE 3-continued

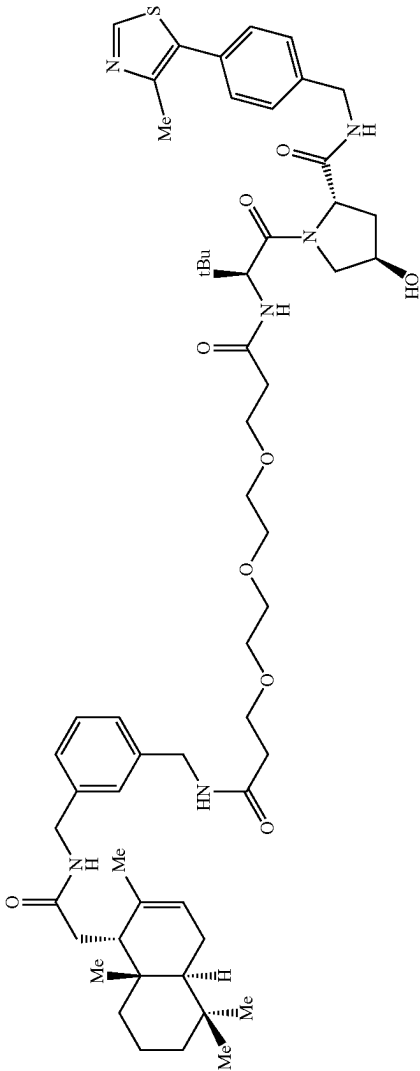
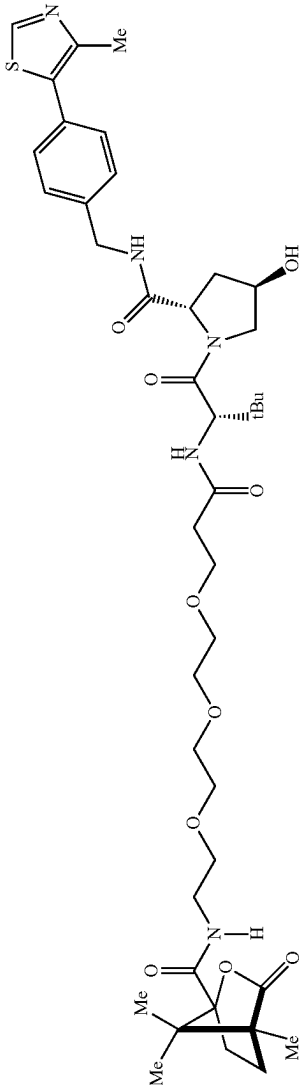
Non-limiting Examples of FragTAC Probes and Intermediates	
Compound Example Number	Compound Code
72	CMC-6-182
	(2S,4R)-1-((S)-17-(tert-butyl)-3,15-dioxo-1-(3-((2-((1R,4aS,8aS)-2,5,5,8a-tetramethyl-1,4,4a,5,6,7,8,8a-octahydronaphthalen-1-yl)acetamido)methyl)phenyl)-6,9,12-trioxa-2,16-diazaoctadecan-18-yl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide
73	CMC-6-177
	(2S,4R)-1-((S)-16-(tert-butyl)-1,14-dioxo-1-((1S,4S)-4,7,7-trimethyl-3-oxo-2-oxabicyclo[2.2.1]heptan-1-yl)-5,8,11-trioxa-2,15-diazaheptadecan-17-yl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide

TABLE 3-continued

Non-limiting Examples of FragTAC Probes and Intermediates		
Compound Example Number	Chemical Structure	Chemical Name
74 CMC-6-181		(2S,4R)-1-(S)-18-(tert-butyl)-3,16-dioxo-1-(3-(((1S,4S)-4,7,7-trimethyl-3-oxo-2-oxabicyclo[2.2.1]heptane-1-carboxamido)methyl)phenyl)-7,10,13-trioxa-2,17-diazaanonadecan-19-oyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide
75 IF-057		(2S,4R)-1-(S)-2-(tert-butyl)-16-(4-((R)-2,3-dihydrobenzo[b][1,4]dioxine-2-carbonyl)piperazin-1-yl)-4,16-dioxo-7,10,13-trioxa-3-dioxo-7,10,13-trioxa-3-azahexadecanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide
76 IF-059		(2S,4R)-1-(S)-2-(tert-butyl)-16-(4-(naphthalen-2-ylmethyl)piperazin-1-yl)-4,16-dioxo-7,10,13-trioxa-3-azahexadecanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide

TABLE 3-continued

Non-limiting Examples of FragTAC Probes and Intermediates	
Compound Example Number	Chemical Name
77	(2S,4R)-1-(S)-2-(tert-butyl)-16-((4-(4'-chloro-5,5-dimethyl-3,4,5,6-tetrahydro-[1,1'-biphenyl]-2-yl)methyl)piperazin-1-yl)-4,16-dioxo-7,10,13-trioxa-3-azahexadecanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide
78	(2S,4R)-1-(S)-17-(tert-butyl)-2-cyclopentyl-1-(9-ethyl-9H-carbazol-2-yl)-3,15-dioxo-6,9,12-trioxa-2,16-diazaododecan-18-oyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide
79	(2S,4R)-1-(S)-19-(1-benzyl-1H-indol-3-yl)-2-(tert-butyl)-4,16-dioxo-7,10,13-trioxa-3,17-diazaonadecanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide

TABLE 3-continued

Non-limiting Examples of FragTAC Probes and Intermediates	
Compound Example Number	Chemical Name
Structure	
80 IF-064	(2S,4R)-1-((2S)-16-(2-(5-(4-bromobenzyl)-1,2,4-oxadiazol-3-yl)pyrrolidin-1-yl)-2-(tert-butyl)-4,16-dioxo-7,10,13-trioxo-3-azahexadecanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide
81 IF-065	(2S,4R)-1-((2S)-2-(tert-butyl)-4,16-dioxo-16-(2-(2-oxo-5-phenyl-2,5-dihydro-1H-benzo[e][1,4]diazepin-3-yl)amino)-7,10,13-trioxo-3-azahexadecanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide
82 IF-066	(2S,4R)-1-((S)-2-(tert-butyl)-4,16-dioxo-16-(2'-oxo-2',3'-dihydro-1'H-spiro[piperidine-4,4'-quinazolin]-1-yl)-7,10,13-trioxo-3-azahexadecanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide

[0141] The present invention is directed to designed bifunctional small molecules that modulate protein modifications via proximity-induced effects, such as ubiquitination-inducing small molecules. These represent a transformative therapeutic strategy. The strategy formulates FragTAC heterobifunctional compositions which integrate powerful chemical proteomic platforms with robust chemical biology tools to expedite the discovery of proteins amenable to these approaches as well as their corresponding bifunctional chemical probes. The techniques and chemical tools of the present invention provide a broad utility of powerful targeted protein degradation (TPD) methods. The new E3 ligase systems and corresponding ligands that are capable of supporting TPD enable modification and treatment of such maladies as autoimmune disease and neoplastic disease both benign and malignant, and especially malignant such as cancer.

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SUMMARY STATEMENTS

[0158] The inventions, examples, biological assays and results described and claimed herein have many attributes and embodiments include, but not limited to, those set forth or described or referenced in this application.

[0159] All patents, publications, scientific articles, web sites and other documents and material references or mentioned herein are indicative of the levels of skill of those skilled in the art to which the invention pertains, and each such referenced document and material is hereby incorporated by reference to the same extent as if it had been incorporated verbatim and set forth in its entirety herein. The right is reserved to physically incorporate into this specification any and all materials and information from any such patent, publication, scientific article, web site, electronically available information, textbook or other referenced material or document.

[0160] The written description of this patent application includes all claims. All claims including all original claims are hereby incorporated by reference in their entirety into the written description portion of the specification and the right is reserved to physically incorporate into the written description or any other portion of the application any and all such claims. Thus, for example, under no circumstances may the patent be interpreted as allegedly not providing a written description for a claim on the assertion that the precise wording of the claim is not set forth in haec verba in written description portion of the patent.

[0161] While the invention has been described in conjunction with the detailed description thereof, the foregoing description is intended to illustrate and not limit the scope of the invention, which is defined by the scope of the appended claims. Thus, from the foregoing, it will be appreciated that, although specific nonlimiting embodiments of the invention have been described herein for the purpose of illustration, various modifications may be made without deviating from the spirit and scope of the invention. Other aspects, advantages, and modifications are within the scope of the following claims and the present invention is not limited except as by the appended claims.

[0162] The specific methods and compositions described herein are representative of preferred nonlimiting embodiments and are exemplary and not intended as limitations on the scope of the invention. Other objects, aspects, and embodiments will occur to those skilled in the art upon consideration of this specification and are encompassed within the spirit of the invention as defined by the scope of the claims. It will be readily apparent to one skilled in the art that varying substitutions and modifications may be made to

the invention disclosed herein without departing from the scope and spirit of the invention. The invention illustratively described herein suitably may be practiced in the absence of any element or elements, or limitation or limitations, which is not specifically disclosed herein as essential. Thus, for example, in each instance herein, in nonlimiting embodiments or examples of the present invention, the terms “comprising”, “including”, “containing”, etc. are to be read expansively and without limitation. The methods and processes illustratively described herein suitably may be practiced in differing orders of steps, and that they are not necessarily restricted to the orders of steps indicated herein or in the claims.

[0163] The terms and expressions that have been employed are used as terms of description and not of limitation, and there is no intent in the use of such terms and expressions to exclude any equivalent of the features shown and described or portions thereof, but it is recognized that various modifications are possible within the scope of the invention as claimed. Thus, it will be understood that although the present invention has been specifically disclosed by various nonlimiting embodiments and/or preferred nonlimiting embodiments and optional features, any and all modifications and variations of the concepts herein disclosed that may be resorted to by those skilled in the art are considered to be within the scope of this invention as defined by the appended claims.

[0164] Unless defined otherwise, all technical and scientific terms used herein have the same meaning as commonly understood by a person of ordinary skill in the art.

[0165] The term “about” as used herein, when referring to a numerical value or range, allows for a degree of variability in the value or range, for example, within 10%, or within 5% of a stated value or of a stated limit of a range.

[0166] All percent compositions are given as weight-percentages, unless otherwise stated.

[0167] All average molecular weights of polymers are weight-average molecular weights, unless otherwise specified.

[0168] The term “may” in the context of this application means “is permitted to” or “is able to” and is a synonym for the term “can.” The term “may” as used herein does not mean possibility or chance.

[0169] It is also to be understood that as used herein and in the appended claims, the singular forms “a,” “an,” and “the” include plural reference unless the context clearly dictates otherwise, for example, the term “X and/or Y” means “X” or “Y” or both “X” and “Y”, and the letter “s” following a noun designates both the plural and singular forms of that noun. In addition, where features or aspects of the invention are described in terms of Markush groups, it is intended, and those skilled in the art will recognize, that the invention embraces and is also thereby described in terms of any individual member and any subgroup of members of the Markush group, and the right is reserved to revise the application or claims to refer specifically to any individual member or any subgroup of members of the Markush group.

[0170] The term and/or means both as well as one or the other as in A and/or B means A alone, B alone and A and B together.

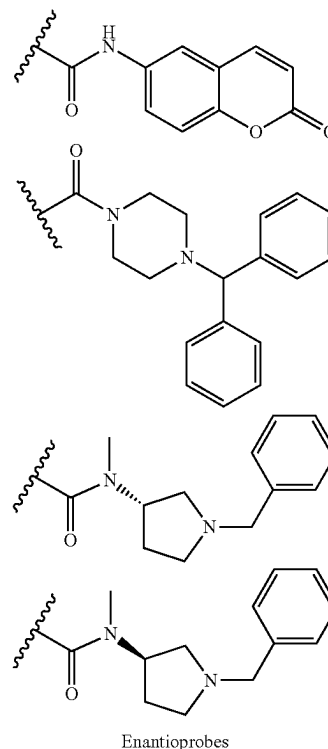
1. A heterobifunctional FragTAC composition comprising a compound of Formula I,

PBF-L-RBF

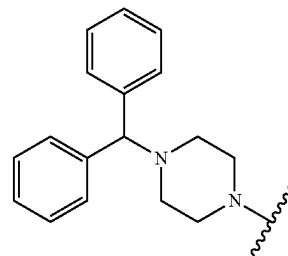
(Formula I)

wherein PBF is a protein binding fragment of a hetero-organic molecule which is capable of binding an endogenous protein, L is an organic linker and RBF is a recruiter binding fragment of a hetero-organic molecule which is capable of binding a ligase of the polyubiquitin system.

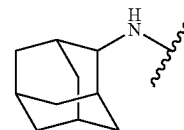
2. A FragTAC composition according to claim 1, wherein the protein binding fragment is selected from a group consisting of



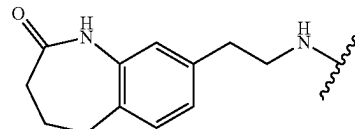
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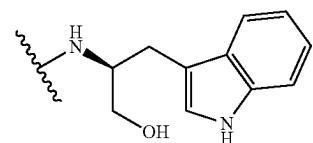
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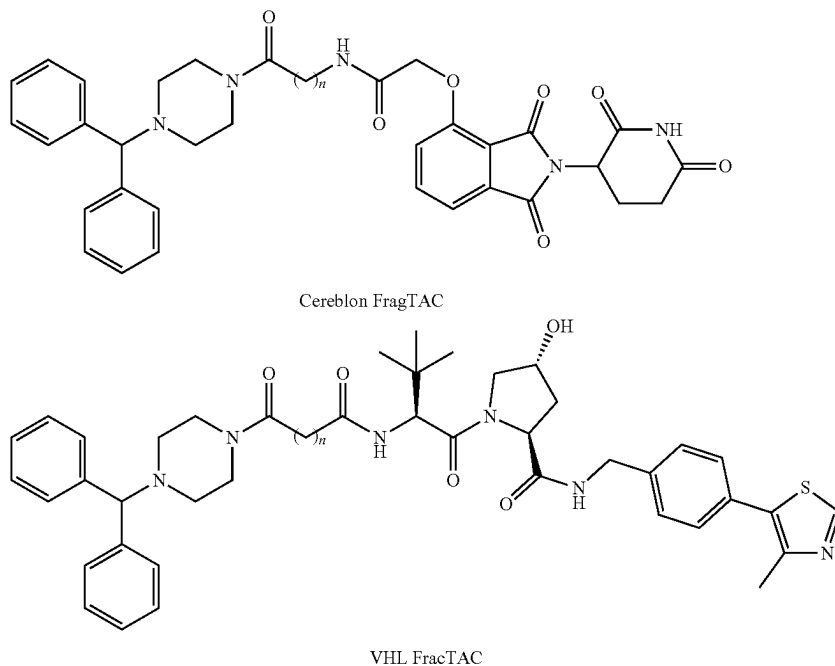


4



3. A FragTAC composition according to claim 1, wherein the recruiter binding fragment is a fragment of thalidomide, a thalidomide derivative or a VHL ligand.

4. A FragTAC composition according to claim 1, wherein Formula I is a cereblon ligase binding molecule or a VHL ligase binding molecule of the Cereblon FragTAC structure or the VHL FragTAC structure wherein n is an integer of 1 to 10:



5. A composition according to claim 1, wherein the linker is a dimer, trimer, tetramer, or oligomer of an multiether, multi-PEG, multi-amide, or alkyl moiety.

6. A method for cleaving an endogenous protein comprising contacting the protein with a heterobifunctional composition of claim 1 and an E3 ligase.

7. A method according to claim 6, comprising conducting the contacting step in an aqueous medium.

8. A method according to claim 7, wherein the aqueous medium is cytoplasm.

9. A method according to claim 8, wherein the contacting is conducted in a viable cell.

10. A method according to claim 9, wherein the viable cell is a cell culture.

11. A method according to claim 9, wherein the viable cell is within a living organism.

* * * * *